

CENTER FOR DRUG EVALUATION AND RESEARCH

APPLICATION NUMBER:

761139Orig1s000

MULTI-DISCIPLINE REVIEW

Summary Review

Office Director

Cross Discipline Team Leader Review

Clinical Review

Non-Clinical Review

Statistical Review

Clinical Pharmacology Review

NDA/BLA Multi-disciplinary Review and Evaluation

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant, which do not necessarily reflect the positions of the FDA.

Application Type	Biologics License Application (BLA) 351(a)
Application Number	BLA 761139
Priority or Standard	Priority
Submit Date(s)	August 29, 2019
Received Date(s)	August 29, 2019
PDUFA Goal Date	April 29, 2020
Division/Office	DO1/OOD/OND/CDER
Review Completion Date	<i>Electronic Stamp Date</i>
Established Name	fam-trastuzumab deruxtecan-nxki
(Proposed) Trade Name	ENHERTU
Pharmacologic Class	HER2-directed antibody and topoisomerase inhibitor conjugate
Code name	
Applicant	Daiichi Sankyo, Inc
Formulation(s)	100 mg lyophilized powder
Dosing Regimen	5.4 mg IV every 3 weeks
Applicant Proposed Indication(s)/Population(s)	(b) (4)
Recommendation on Regulatory Action	<i>Accelerated Approval</i>
Recommended Indication(s)/Population(s) (if applicable)	Treatment of adult patients with unresectable or metastatic HER2-positive breast cancer who have received two or more prior anti-HER2-based regimens in the metastatic setting.

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

Table of Contents

Reviewers of Multi-Disciplinary Review and Evaluation	10
Additional Reviewers of Application.....	10
Glossary.....	12
1 Executive Summary	17
1.1. Product Introduction.....	17
1.2. Conclusions on the Substantial Evidence of Effectiveness	17
1.3. Benefit-Risk Assessment (BRA)	20
1.4. Patient Experience Data.....	23
2 Therapeutic Context.....	26
2.1. Analysis of Condition.....	26
2.2. Analysis of Current Treatment Options	27
3 Regulatory Background	33
3.1. U.S. Regulatory Actions and Marketing History.....	33
3.2. Summary of Presubmission/Submission Regulatory Activity	33
4 Significant Issues from Other Review Disciplines Pertinent to Clinical Conclusions on Efficacy and Safety.....	35
4.1. Office of Scientific Investigations (OSI)	35
4.2. Product Quality	38
4.3. Clinical Microbiology	39
4.4. Devices and Companion Diagnostic Issues	39
5 Nonclinical Pharmacology/Toxicology.....	40
5.1. Executive Summary	40
5.2. Referenced NDAs, BLAs, DMFs.....	43
5.3. Pharmacology.....	43
5.4. ADME/PK.....	48
5.5. Toxicology.....	53
5.5.1. General Toxicology.....	53
5.5.2. Genetic Toxicology.....	62

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

5.5.3. Carcinogenicity.....	65
5.5.4. Reproductive and Developmental Toxicology	66
5.5.5. Other Toxicology Studies	66
6 Clinical Pharmacology.....	75
6.1. Executive Summary	75
6.2. Summary of Clinical Pharmacology Assessment.....	76
6.2.1. Pharmacology and Clinical Pharmacokinetics	76
6.2.2. General Dosing and Therapeutic Individualization.....	79
6.2.2.1. General Dosing	79
6.2.2.2. Therapeutic Individualization	81
6.2.2.3. Outstanding Issues	82
6.3. Comprehensive Clinical Pharmacology Review	82
6.3.1. General Pharmacology and Pharmacokinetic Characteristics.....	82
6.3.2. Clinical Pharmacology Questions.....	93
6.3.2.1. Does the clinical pharmacology program provide supportive evidence of effectiveness?	93
6.3.2.2. Is the proposed dosing regimen appropriate for the general patient population for which the indication is being sought?	95
6.3.2.3. Is an alternative dosing regimen or management strategy required for subpopulations based on intrinsic patient factors?	96
6.3.2.4. Are there clinically relevant food-drug or drug-drug interactions, and what is the appropriate management strategy?	98
7 Sources of Clinical Data	101
7.1. Table of Clinical Studies.....	101
8 Statistical and Clinical Evaluation	109
8.1. Review of Relevant Individual Trials Used to Support Efficacy.....	109
8.1.1. Studies DS8201-A-U201 and DS8201-A-J101	109
8.1.1.1. Study DS8201-A-U201	109
8.1.1.2. Study DS8201-A-J101.....	123
8.1.1.3. Study DS8201-A-U201	126
8.1.1.4. Study DS8201-A-J101.....	144

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

8.1.2. Assessment of Efficacy Across Trials.....	146
8.1.3. Integrated Assessment of Effectiveness	157
8.2. Review of Safety	159
8.2.1. Safety Review Approach	160
8.2.2. Review of the Safety Database	161
8.2.3. Adequacy of Applicant’s Clinical Safety Assessments	167
8.2.4. Safety Results.....	170
8.2.5. Analysis of Submission-Specific Safety Issues.....	193
8.2.5.1. Interstitial Lung Disease	193
8.2.5.2. Left Ventricular Ejection Fraction Decrease	197
8.2.5.3. QT Prolongation.....	199
8.2.5.4. Infusion-Related Reactions.....	200
8.2.6. Clinical Outcome Assessment (COA) Analyses Informing Safety/Tolerability.....	201
8.2.7. Safety Analyses by Demographic Subgroups.....	201
8.2.8. Specific Safety Studies/Clinical Trials.....	203
8.2.9. Additional Safety Explorations.....	204
8.2.10. Integrated Assessment of Safety	207
SUMMARY AND CONCLUSIONS	211
8.3. Statistical Issues	211
8.4. Conclusions and Recommendations	212
9 Advisory Committee Meeting and Other External Consultations.....	213
10 Pediatrics	213
11 Labeling Recommendations	214
12 Risk Evaluation and Mitigation Strategies (REMS)	220
13 Postmarketing Requirements and Commitment	221
14 Division Director (DHOT) (NME ONLY)	225
15 Division Director (OCP)	226
16 Division Director (OB)	227

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

17	Division Director (Clinical)	228
18	Office Director (or designated signatory authority)	229
19	Appendices	230
19.1.	References	230
19.2.	Financial Disclosure	232
19.3.	Nonclinical Pharmacology/Toxicology.....	234
19.4.	OCP Appendices (Technical documents supporting OCP recommendations)	234
	Pharmacometrics Review	246
19.5.	Additional Safety Analyses Conducted by FDA	255

Table of Tables

Table 1: Current Treatment Options for Previously Treated HER2-positive Advanced or Metastatic Breast Cancer.....	30
Table 2: Key Regulatory Interactions Related To BLA 761139.....	33
Table 3: IC ₅₀ values of DS-8201a, MAAL-9001, and MAAA-9001b in select tumor cell lines.	44
Table 4: In vitro plasma protein binding of MAAA-1181a in mouse, rat, monkey, and human plasma.....	51
Table 5: Concentrations of release MAAA-1181a at two DS-8201a concentrations in mouse, rat, monkey, and human plasma.....	51
Table 6: Toxicokinetic Parameters in the 6-Week Intermittent Intravenous Dose Toxicity Study of DS-8201a in rats.....	51
Table 7: Toxicokinetic Parameters in the 3-Month Intermittent Intravenous Dose Toxicity Study of DS-8201a in Cynomolgus Monkeys.	52
Table 8: Key Clinical Pharmacology Review Issues by FDA.....	75
Table 9: Pooled dose-response analysis for ORR.....	81
Table 10: Summary of statistical analyses for assessment of PK similarity of DS-8201a, Total Atni-HER2 antibody and MAAA-1181a (or DXd).....	86
Table 11: Highlights of Clinical Pharmacology for fam-trastuzumab deruxtecan	87
Table 12: Dose proportionality of DS-8201a analysis of PK parameters in dose escalation cohort of Study J101 following a single dose	89
Table 13: Dose proportionality of MAAA-1181a analysis of PK parameters in dose escalation cohort of Study J101 following a single dose	89
Table 14: Dose proportionality of DS-8201a analysis of PK parameters in dose escalation cohort of Study J101 at steady state	90
Table 15: Dose proportionality of MAAA-1181a analysis of PK parameters in dose escalation cohort of Study J101 at steady state	90
Table 16: Summary of Immunogenicity by Study.....	91
Table 17: Summary of statistical analysis of assessment of PK parameters with or without inhibitors	92
Table 18: Summary of statistical analysis of assessment of PK parameters with or without inhibitors	100
Table 19: Listing of Clinical Trials Relevant to this BLA	102
Table 20: Subject Disposition (Enrolled Analysis Set).....	126
Table 21: Major Protocol Deviations (Enrolled Analysis Set)	129
Table 22: Key Demographic Characteristics (Enrolled Analysis Set)	130
Table 23: Baseline Disease Characteristics (Enrolled Analysis Set)	132
Table 24: Prior Anti-cancer Therapy	134
Table 25: Number of Prior Anti-Cancer Regimens in the Advanced/Metastatic Setting in DS8201-A-U201 Study (5.4 mg/kg).....	135
Table 26: Objective Response Rate in the 5.4 mg/kg Dose Cohort (Enrolled Analysis Set). 136	

Table 27: Discordance Rate of Response Assessment between Independent Central Review and Investigator	140
Table 28: FDA Subgroup Analyses of ORR per ICR assessment	141
Table 29: Confirmed Objective Response Rate and Best Objective Response by Independent Central Review (Intent-to-Treat Analysis Set)	148
Table 30: Duration of Confirmed Response by Independent Central Review	150
Table 31: Summary of Duration of Exposure (Safety Analysis Set)	162
Table 32: Summary of Demographics.....	164
Table 33: Summary of Deaths in the Primary Studies	171
Table 34: Summary of Deaths in Other Completed Studies.....	171
Table 35: Serious Adverse Events in the Primary Studies, by Preferred Term.....	174
Table 36: Serious Adverse Events in Other Completed Studies, by Preferred Term	175
Table 37: Adverse Events Associated with Study Drug Discontinuation, by Preferred Term or Grouped Term.....	176
Table 38: Adverse Events Associated with Dose Interruption, by Preferred or Grouped Term	177
Table 39: Adverse Events Associated with Dose Reduction, by Preferred Term	179
Table 40: Selected Adverse Events Reported in the HER2-positive Breast Cancer 5.4 mg/kg Pool	181
Table 41: Most Common Adverse Events, by Preferred Term	183
Table 42: Adverse Events of at Least Grade 3, by Preferred Term	185
Table 43: All Adverse Drug Reactions in HER2-positive Breast Cancer 5.4 mg/kg Pool (N = 234), by MedDRA System Organ Class and Preferred Term/Grouped Term.....	186
Table 44: New or Worsening Laboratory Abnormalities in the HER2-positive Breast Cancer 5.4 mg/kg Pool	188
Table 45: Selected Laboratory Abnormalities in Patients with Unresectable or Metastatic HER2-positive Breast Cancer Treated with ENHERTU	189
Table 46: Events Adjudicated as Drug-related ILD, by Tumor Type/Dose, and Adjudicated Grade	194
Table 47: ILD Events, by Country and Adjudicated Grade in the HER2-positive Breast Cancer 5.4 mg/kg Pool	195
Table 48: Odds Ratio of Risk Factors from the Five Clinical Studies.....	196
Table 49: Summary method performance of a bioanalytical method to measure trastuzumab deruxtecan in human serum	235
Table 50: Summary method performance of a bioanalytical method to measure total anti-HER2 antibody in human serum.....	238
Table 51: Summary method performance of a bioanalytical method to measure MAAA-1181a in human serum	241
Table 52: Summary of Studies with PK Sampling Included in Population PK Analysis.....	247
Table 53: Summary of Baseline Demographic Covariates for Analysis	249
Table 54: Parameters of the final intact DS-8201a model	251
Table 55: Parameters of the final released drug model.....	252

Table of Figures

Figure 1: Activity of DS-8201a and MAAL-9001 in a HER2-positive breast cancer KPL-4 xenograft model in CAnN.Cg-Foxn1 ^{nu} /CrlCrlj mice.	45
Figure 2: Activity of DS-8201a in patient-derived xenograft models of HER2-expressing breast cancer, CTG-0708 and CTG-2308, in Hsd:athymic nude-Foxn1 ^{nu} mice in comparison with T-DM1 and MAAL-9002b.	46
Figure 3: Activity of DS-8201a in a HER2-positive breast cancer JIMT-1 xenograft model in CAnN.Cg-Foxn1nu/CrlCrlj mice in comparison with T-DM1 and MAAL-9002b.	47
Figure 4: Comparison of MAAA-1181a (DXd) and patients with hepatic impairment (NCI-ODWG criteria).....	97
Figure 5: DS8201-A-U201 Study Design	110
Figure 6: Schedule of Events for Part 1 Pharmacokinetic Stage	114
Figure 7: Schedule of Events for Part 1 Dose Finding Stage and Part 2	117
Figure 8: Study DS8201-A-J101 study design	124
Figure 9: Forest Plot of Objective Response Rate by Independent Central Review for the Primary 5.4 mg/kg Dose Cohort by Subgroup (Enrolled Analysis Set)	138
Figure 10: Kaplan-Meier Plot of Duration of Response by Independent Central Review for the Primary 5.4 mg/kg Dose Cohort (Enrolled Analysis Set).....	143
Figure 11: Waterfall Plot of Best (Minimum) Percent Change from Baseline in Sum of Diameters of target Lesions Based on Independent Central Review (Intent-to-Treat Analysis Set)	149
Figure 12: Kaplan-Meier Plot of Pooled Duration of Response Based on Independent Central Review	151
Figure 13: Forest Plot of Objective Response Rate by Independent Central Review by Subgroup for Pooled Population at 5.4 mg/kg Fam-Trastuzumab Deruxtecan	153
Figure 14: Baseline HER2 ECD levels and confirmed best overall response by Independent Central Review in a subset of patients receiving trastuzumab deruxtecan 5.4 mg/kg enrolled in study DS8201-A-U201 (N=177*)	246
Figure 15: Confirmed central ORR exposure-response relationship	254

Reviewers of Multi-Disciplinary Review and Evaluation

[FDA will complete this section.]

Regulatory Project Manager	Sherry Hou, PharmD
Pharmacology/Toxicology Reviewer	Haw-Jyh Chiu, PhD
Pharmacology/Toxicology Team Leader	Tiffany Ricks, PhD
Office of Clinical Pharmacology Reviewers	
Clinical Pharmacology Reviewer	Edwin Chiu Yuen Chow, PhD
Pharmacometrics Reviewer	Junshan Qiu, PhD
Genomics Reviewer	Jielin Sun, PhD
Office of Clinical Pharmacology Team Leaders	
Clinical Pharmacology Team Leader	Pengfei Song, PhD
Pharmacometrics Team Leader	Jingyu Yu, PhD
Genomics Team Leader	Rosane Charlab Orbach, PhD
Clinical Reviewer	Preeti Narayan, MD
Clinical Team Leader	Christy Osgood, MD
Safety Analyst	Yutao Gong, PhD
Statistical Reviewer	Wei Zhang, PhD
Statistical Team Leader	Shenghui Tang, PhD
Associate Director for Labeling	William Pierce, PharmD, MPH
Cross-Disciplinary Team Leader	Christy Osgood, MD
Division Director (DHOT)	John Leighton, PhD, DABT
Division Director (OCP)	Nam Atiqur Rahman, PhD
Division Director (OB)	Rajeshwari Sridhara, PhD
Division Director (DO 1)	Julia Beaver, MD
Acting Deputy Director (OOD)	Marc R. Theoret, MD

Additional Reviewers of Application

OPQ	Maria Guterrez-Lugo, PhD Frances Namuswe, PhD Jun Liu, PhD Zhenzhen Liu, PhD Suong Tran, PhD Ben Zhang, PhD Jessica Hankins, PhD Steven Fong, PhD
------------	--

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	Viviana Matta, PhD Patricia Hughes, PhD
OBP Labeling Reviewer	Vicky BordersHemphill, PharmD
RBPM	Marquita Burnett, MPH
OPDP	Maritsa Serlemitsos-Day, PharmD, BCPS
OPDP Team Leader	Kevin Wright, PharmD
Patient Labeling Reviewer	Susan Redwood, MPH, BSN, RN
Patient Labeling Team Leader	Barbara Fuller, RN, MSN, CWOCN
Associate Director for Patient Labeling	LaShawn Griffiths, MSHS-PH, BSN, RN
OSI	Yang-min (Max) Ning, MD, PhD Aisha Johnson, MD, MPH, MBA Kassa Ayalew, MD, MPH
Safety Team	Felicia Diggs, RN, BSN, MSN Shanthi Marur, MBBS MD
OSE/DEPI	Steven Bird, PhD, PharmD, MS Scott (Richard) Swain, PhD, MPH
OSE/DMEPA	Tingting Gao, PharmD Alice (Chi-Ming) Tu, PharmD
OSE/DRISK	Brad Moriyama, PharmD Naomi Boston, PharmD
OSE/ DPV	Peter Waldron, MD Afrouz Nayernama, PharmD Neha Gada, PharmD
OSE Project Manager	Frances Fahnbulleh, PharmD, RPh, GWCPM

OPQ=Office of Pharmaceutical Quality
 OPDP=Office of Prescription Drug Promotion
 OSI=Office of Scientific Investigations
 OSE= Office of Surveillance and Epidemiology
 DEPI= Division of Epidemiology
 DMEPA=Division of Medication Error Prevention and Analysis
 DRISK=Division of Risk Management
 DPV=Division of Pharmacovigilance

Glossary

AC	advisory committee
ADA	antidrug-antibody
ADC	antibody-drug conjugate
ADME	absorption, distribution, metabolism, excretion
ADR	adverse drug reaction
AE	adverse event
AESI	adverse event of special interest
ALT	alanine aminotransferase
AST	aspartate aminotransferase
AUC	area under the serum concentration-time curve
AUC _{ss}	area under the serum concentration-time curve at steady-state
AUC _{tau}	area under the serum concentration-time curve during the dosing interval
BCRP	breast cancer resistance protein
BI	before infusion
BID	twice daily
BLA	Biologics License Application
BOR	best overall response
BPCA	Best Pharmaceuticals for Children Act
BRF	Benefit Risk Framework
BSEP	bile salt export pump
BTB	Breakthrough Therapy Designation
C ₀	initial plasma concentration
C _{avg}	average concentration
C _{avg} ORR	average concentration from beginning of treatment to the time of ORR
CBER	Center for Biologics Evaluation and Research
CBR	clinical benefit rate
CDER	Center for Drug Evaluation and Research
CDRH	Center for Devices and Radiological Health
CDTL	Cross-Discipline Team Leader
CFR	Code of Federal Regulations
CHF	congestive heart failure
CHO	Chinese hamster ovary
CI	confidence interval
CL	total body clearance
C _{max}	maximum observed serum concentration
C _{max,ss}	maximum observed serum concentration at steady-state
C _{min,ss}	minimum observed serum concentration at steady-state

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

CMC	chemistry, manufacturing, and controls
COSTART	Coding Symbols for Thesaurus of Adverse Reaction Terms
CR	complete response
CRF	case report form
CRO	contract research organization
CRT	clinical review template
CSR	clinical study report
CSS	Controlled Substance Staff
CT	computed tomography
CTCAE	Common Terminology Criteria for Adverse Events
CYP	cytochrome P450
ΔQTcF	change from baseline in QTcF
D	day
DCO	data cut-off
DCR	disease control rate
DDI	drug-drug interaction
DMC	data monitoring committee
DMF	drug master file
DMSO	dimethyl sulfoxide
DoR	duration of response
DP	drug product
DS	drug substance
DS-8201a	product code of fam-trastuzumab deruxtecan
Dxd	released drug, derivative of exatecan, also known as “MAAA-1181a”
EAS	Enrolled Analysis Set (same as ITT)
ECG	electrocardiogram
ECHO	echocardiogram
ECOG PS	Eastern Cooperative Oncology Group performance status
eCRF	electronic case report form
eCTD	electronic common technical document
EDC	electronic data capture
ELISA	enzyme-linked immunosorbent assay
EOI	end of infusion
EOP	End of Phase
EOT	end of treatment
ER	exposure-response
ETASU	elements to assure safe use
FDA	Food and Drug Administration
FDAAA	Food and Drug Administration Amendments Act of 2007
FDASIA	Food and Drug Administration Safety and Innovation Act
FIH	first in human
FL-DP1	frozen liquid-drug product 1

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

FL-DP2	frozen liquid-drug product 2
F/U	follow-up
GCP	Good Clinical Practice
GEJ	gastroesophageal junction
GI	gastrointestinal
GLP	good laboratory practice
GRMP	good review management practice
HER2	human epidermal growth factor receptor 2
HER2 ECD	human epidermal growth factor receptor 2 extracellular domain
HIV	human immunodeficiency virus
HNSTD	highest non-severely toxic dose
IC ₅₀	50% inhibitory concentration
ICF	informed consent form
ICH	International Council for Harmonisation
ICR	independent central review
IE	immature erythrocyte
IHC	immunohistochemistry
ILD	interstitial lung disease
ILD AC	Interstitial Lung Disease Adjudication Committee
IND	Investigational New Drug
INR	international normalized ratio
IP	investigational product
IRR	infusion-related reaction
ISE	integrated summary of effectiveness
ISH	in situ hybridization
ISS	integrated summary of safety
ITT	intent-to-treat
IV	intravenous(ly)
LVEF	left ventricular ejection fraction
Lyo-DP	lyophilized powder-drug product
mAb	monoclonal antibody
MAAA-1181a	released drug, derivative of exatecan, also known as “Dxd”
MAAL-9001	code for anti-HER2 antibody
MATE1	multidrug and toxin extrusion protein 1
MATE2-K	multidrug and toxin extrusion protein 2-K
MedDRA	Medical Dictionary for Regulatory Activities
mITT	modified intent to treat
MNIE	micronucleated immature erythrocyte
MRI	magnetic resonance imaging
MRP1	multidrug resistance protein 1
MTD	maximum tolerated dose
MUGA	multigated acquisition

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

NCI	National Cancer Institute
NCI-CTCAE	National Cancer Institute-Common Terminology Criteria for Adverse Event
NCT	clinicaltrial.gov identifier
NDA	New Drug Application
NME	new molecular entity
OAT1	organic anion transporter 1
OAT3	organic anion transporter 3
OATP	organic anion transporting peptide
OCS	Office of Computational Science
OCT1	organic cation transporter 1
OCT2	organic cation transporter 2
ODWG	Organ Dysfunction Working Group
OPQ	Office of Pharmaceutical Quality
ORR	objective response rate
OS	overall survival
OSE	Office of Surveillance and Epidemiology
OSI	Office of Scientific Investigation
PBRER	Periodic Benefit-Risk Evaluation Report
PD	progressive disease
PFS	progression-free survival
P-gp	P-glycoprotein
PI	prescribing information
PK	pharmacokinetics
PMC	postmarketing commitment
PMR	postmarketing requirement
PopPK	population pharmacokinetics
PP	per protocol
PPI	patient package insert
PR	partial response
PREA	Pediatric Research Equity Act
PRO	patient reported outcome
PSUR	Periodic Safety Update report
PT	preferred term
Q3M	every 3 months
Q3W	every 3 weeks
Q6W	every 6 weeks
QD	once daily
QTc	QT interval corrected for heart rate
QTcF	QT interval corrected for heart rate using Fridericia's formula
RECIST v1.1	Response Evaluation Criteria for Solid Tumors version 1.1
REMS	risk evaluation and mitigation strategy
RES	Response Evaluable Set

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

RP2D	recommended Phase 2 dose
SAE	serious adverse event
SAP	statistical analysis plan
SCR	screening
SEER	Surveillance, Epidemiology, and End Results
SGE	special government employee
SI	standard international
SMQ	Standardised MedDRA Queries
SoC	standard of care
SOC	system organ class
SpO2	peripheral oxygen saturation
Std Dev	standard deviation
t _{1/2}	apparent terminal elimination half-life
TEAE	treatment-emergent adverse event
TBL	total bilirubin
T-DM1	ado-trastuzumab emtansine, KADCYLA [®]
TEAE	treatment emergent adverse event
TK	toxicokinetics
TKI	tyrosine-kinase inhibitor
TL	tumor lesion
T _{max}	time to maximum serum concentration
TPC	treatment of physician's choice
TTR	time to response
ULN	upper limit of normal
US	United States
UVA	ultraviolet A
V _c	central compartment volume
vs.	versus
V _{ss}	volume of distribution at steady state
WBC	white blood cell

1 Executive Summary

1.1. Product Introduction

Fam-trastuzumab deruxtecan (DS8201a) is a HER2-directed antibody and topoisomerase I inhibitor conjugate. The anti-HER2 component, MAAL-9001, is a humanized immunoglobulin G1 (IgG1) monoclonal antibody (mAb). The released drug, MAAA-1181a, is a topoisomerase I inhibitor derivative of exatecan. The mAb is covalently conjugated to a drug-linker, MAAA-1162a, that is composed of a cleavable maleimide tetrapeptide linker and the released drug. Fam-trastuzumab deruxtecan has not been approved for any indication worldwide.

The applicant proposed the following indication for BLA 761139:

(b) (4)

The recommended indication for accelerated approval is:

ENHERTU is a HER2-directed antibody and topoisomerase inhibitor conjugate indicated for the treatment of adult patients with unresectable or metastatic HER2-positive breast cancer who have received two or more prior anti-HER2-based regimens in the metastatic setting.

The recommended dose of fam-trastuzumab deruxtecan is 5.4mg/kg given as an intravenous infusion once every 3 weeks (21-day cycle), administered over 90 minutes for the first cycle and over 30 min in subsequent cycles if prior infusions were well tolerated.

1.2. Conclusions on the Substantial Evidence of Effectiveness

The recommendation for the accelerated approval of fam-trastuzumab deruxtecan, according to 21 Code of Federal Regulations (CFR), Part 601.41, Subpart E of the Biological Licensing Regulations, is based on efficacy and safety data from studies DS8201-A-U201 (DESTINY Breast-01) and DS8201-A-J101. Efficacy was studied in DS8201-A-U201, a phase 2, multicenter, single-arm, trial that enrolled 184 female patients with HER2-positive, unresectable and/or metastatic breast cancer who had previously received at least two lines of HER2-based regimens in the metastatic setting. The major efficacy outcomes were confirmed objective response rate (ORR) assessed by independent central review (ICR) using RECIST v1.1 and duration of response (DOR). The ORR was 60.3% (95% CI: 52.9, 67.4), with a 4.3% complete response rate and a 56% partial response rate. Median response duration was 14.8 months (95% CI: 13.8, 16.9).

Fam-trastuzumab demonstrated acceptable tolerability for the indicated population with a serious and life-threatening disease. The most common adverse reactions (frequency $\geq 20\%$) were nausea, fatigue, vomiting, alopecia, constipation, decreased appetite, anemia, neutropenia, diarrhea, leukopenia, cough, and thrombocytopenia. Serious adverse reactions occurred in 20% of patients receiving fam-trastuzumab deruxtecan, including interstitial lung disease, pneumonia, vomiting, nausea, cellulitis, hypokalemia, and intestinal obstruction. Fatalities due to adverse reactions occurred in 4.3% of patients including interstitial lung disease (2.6%), and the following events occurred in one patient each (0.4%): acute hepatic failure/acute kidney injury, general physical health deterioration, pneumonia, and hemorrhagic shock.

ILD/pneumonitis is an important safety signal related to fam-trastuzumab deruxtecan. Of the 234 patients with HER2-positive breast cancer who received 5.4mg/kg of fam-trastuzumab deruxtecan, ILD (adjudicated by the independent ILD committee) occurred in 9.4% of patients. Fatal outcomes due to ILD and/or pneumonitis occurred in 6 (2.6%) patients treated with fam-trastuzumab deruxtecan. The prescribing information includes a Boxed Warning to advise health professionals of the risk of interstitial lung disease (ILD). Median time to first onset was 4.1 months (range: 1.2 to 8.3).

Overall, the benefit-risk profile based upon results from study DS8201-A-U201 was favorable. This Biologics License Application (BLA) represents a new treatment option for men and women with HER2-positive advanced/metastatic breast cancer whose disease has progressed through at least 2 HER2-based therapies in the metastatic setting. The ORR of 60.3% with a median duration of is a surrogate endpoint that is reasonably likely to predict clinical benefit (i.e., improved survival or progression free survival) in patients with HER2-positive advanced/metastatic breast cancer. The standard of care for first-line metastatic HER2-positive breast cancer in the US is the combination of trastuzumab, pertuzumab, and taxane based on the CLEOPATRA trial. T-DM1 is the preferred second-line option based on the EMILIA study. Beyond the second -line, treatment options are more limited and could include lapatinib and capecitabine, or trastuzumab combined with a chemotherapeutic agent. Even though these therapies have improved the disease outcomes for patients with metastatic HER2-positive breast cancer, there remains an unmet need for further effective therapies in these patients as their disease will eventually progress. There also is a significant unmet need for treatment options for men with breast cancer. Male patients were eligible for study DS8201-A-U201; however, no male patients were treated. HER-2 positive breast cancer is not expected to behave differently in male patients and therefore Fam-trastuzumab will be indicated for males as well as females.

Therefore, the FDA review team recommends granting accelerated approval to ENHERTU (fam-trastuzumab deruxtecan) for the following indication:

ENHERTU is a HER2-directed antibody and topoisomerase inhibitor conjugate indicated for the

treatment of adult patients with unresectable or metastatic HER2-positive breast cancer who have received two or more prior anti-HER2-based regimens in the metastatic setting.

1.3. Benefit-Risk Assessment (BRA)

Benefit-Risk Summary and Assessment

Fam-trastuzumab deruxtecan (DS8201a), a HER2-directed antibody linked to a topoisomerase I inhibitor conjugate, is recommended for accelerated approval for the treatment of adult patients with unresectable or metastatic HER2-positive breast cancer who have received two or more prior anti-HER2-based regimens in the metastatic setting.

In the United States (US), breast cancer is the most common cancer in women, with more than 260,000 new cases and 40,000 deaths annually. Metastatic breast cancer is categorized into different histopathological subtypes based on expression of ER, PR, and HER2. Approximately 20% of patients with breast cancer have HER2-positive tumors. HER2-positive disease is associated with a more aggressive tumor and a younger patient population. Breast cancer in male patients is rare, with fewer than 1% of breast cancer diagnosed in male patients. Men with breast cancer tend to present at a higher stage than female patients, at least in part due to the lack of mammographic screening in men. The standard of care for first-line metastatic HER2-positive breast cancer in the US is the combination of trastuzumab, pertuzumab, and taxane based on results from the CLEOPATRA trial. T-DM1 is the preferred second-line option based on results from the EMILIA trial. Beyond the second-line, treatment options are more limited and could include lapatinib and capecitabine, or trastuzumab combined with a chemotherapeutic agent. Metastatic HER2-positive breast cancer remains an incurable disease. Although treatment with anti-HER2 therapies has improved the disease outcomes for patients with unresectable or metastatic HER2-positive breast cancer, patients invariably progress. There remains an unmet need for further effective therapies for metastatic HER2-positive breast cancer, especially in later lines of therapy.

The efficacy of fam-trastuzumab was evaluated in study DS8201-A-U201 (DESTINY-Breast01), a phase 2, multicenter, single-arm, trial that enrolled 184 female patients with HER2-positive, unresectable and/or metastatic breast cancer previously treated at least two HER2-based regimens in the metastatic patients. All patients had previously received TDM1 and trastuzumab. Patients received fam-trastuzumab deruxtecan 5.4mg/kg given as an intravenous infusion once every 3 weeks (21-day cycle), administered over 90 minutes for the first cycle and over 30 min in subsequent cycles if prior infusions when well tolerated. The major efficacy outcomes were confirmed objective response rate (ORR) assessed by ICR using RECIST v1.1 and duration of response (DOR). Tumor assessments were performed at screening and every 6 weeks thereafter. The ORR was 60.3% (95% CI: 52.9, 67.4), with a 4.3% complete response rate and a 56% partial response rate. Median DOR was 14.8 months (95% CI: 13.8, 16.9). This ORR and DOR represent a clear improvement over ORR and DOR of limited options in this setting and are

reasonably likely to predict a clinical benefit.

Fam-trastuzumab deruxtecan demonstrated acceptable tolerability for the indicated population with a serious and life-threatening disease. The safety of fam-trastuzumab deruxtecan was evaluated in a pooled 234 patients with HER2-positive breast cancer from studies DS8201-A-U201 and DS8201-A-J101 (phase 1, multicenter, nonrandomized, open-label, multiple-dose, FIH study of fam-trastuzumab deruxtecan, with study sites in Japan and the US). The most common adverse reactions were nausea, fatigue, vomiting, alopecia, constipation, decreased appetite, anemia, neutropenia, diarrhea, leukopenia, cough, and thrombocytopenia. The prescribing information includes a Boxed Warning to advise health professionals of the risk of interstitial lung disease (ILD) and embryo-fetal toxicity. Fatal outcomes due to ILD/pneumonitis occurred in 6 (2.6%) of patients. The most common adverse reactions (frequency $\geq 20\%$) were nausea, fatigue, vomiting, alopecia, constipation, decreased appetite, anemia, neutropenia, diarrhea, leukopenia, cough, and thrombocytopenia. Serious adverse reactions occurred in 20% of patients receiving fam-trastuzumab deruxtecan, including interstitial lung disease, pneumonia, vomiting, nausea, cellulitis, hypokalemia, and intestinal obstruction. Fatalities due to adverse reactions occurred in 4.3% of patients including interstitial lung disease (2.6%), and the following events occurred in one patient each (0.4%): acute hepatic failure/acute kidney injury, general physical health deterioration, pneumonia, and hemorrhagic shock.

Overall, the benefit-risk profile based upon results from studies DS8201-A-U201 and DS8201-A-J101 was favorable. This Biologics License Application (BLA) represents a new treatment option for men and women with HER2-positive advanced/metastatic breast cancer whose disease has progressed through at least two HER2-based therapies in the metastatic setting. The durable objective response rate of 60.3% represent a surrogate endpoint that is reasonably likely to predict clinical benefit in patients with HER2-positive advanced/metastatic breast cancer whose disease has progressed through at least two HER2-based therapies which is an area of unmet need. Although men were not included on the trial, they were eligible, and there is no biologic or mechanistic rationale that men and women would have differential response, therefore the indication was extended to adults.

ENHERTU (fam-trastuzumab deruxtecan) is recommended for accelerated approval for the following indication:

ENHERTU is a HER2-directed antibody and topoisomerase inhibitor conjugate indicated for the treatment of adult patients with unresectable or metastatic HER2-positive breast cancer who have received two or more prior anti-HER2-based regimens in the metastatic setting.

Dimension	Evidence and Uncertainties	Conclusions and Reasons
Analysis of Condition	- Breast cancer is the most common cancer in women, with more than 260,000 new cases and 40,000 deaths annually. Approximately 20% of patients with breast cancer have HER2-positive tumors. - Breast cancer in male patients is rare and presents at a higher stage - Advanced or metastatic breast cancer is incurable.	Advanced or metastatic HER2-positive breast cancer is a serious and life-threatening condition with ongoing unmet medical need in both female and male patients.
Current Treatment Options	- Metastatic HER2-positive breast cancer is not curable with treatment goals palliative in nature to delay disease progression, prolong survival, and reduced cancer-related symptoms. - The combination of trastuzumab, pertuzumab, and taxane is an FDA approved treatment for patients with HER2-positive metastatic breast cancer who have not received prior anti-HER2 therapy or chemotherapy for metastatic disease. - T-DM1 is FDA approved for the treatment of patients with HER2-positive, metastatic breast cancer who previously received trastuzumab and a taxane. - Other options beyond 2 lines of HER-based therapies can include lapatinib and capecitabine, or trastuzumab combined with either lapatinib, capecitabine or another chemotherapeutic agent.	There is an unmet medical need to improve the outcomes of female and male patients with HER2-positive advanced or metastatic breast cancer whose disease has progressed on available HER2-directed therapies.
Benefit	- The efficacy of fam-trastuzumab was evaluated in study DS8201-A-U201, a multicenter, single-arm, trial that enrolled 184 female patients with HER2-positive, unresectable and/or metastatic breast cancer who had received two or more prior anti-HER2 therapies. - ORR was 60.3% (95% CI: 52.9, 67.4), with a 4.3% complete response rate and a 56% partial response rate. Median response duration was 14.8 months (95% CI: 13.8, 16.9).	Study DS8201-A-U201 resulted in durable ORR that represents an improvement compared to that of available therapies and is reasonably likely to predict clinical benefit.

Dimension	Evidence and Uncertainties	Conclusions and Reasons
Risk and Risk Management	- The most common adverse reactions were nausea, fatigue, vomiting, alopecia, constipation, decreased appetite, anemia, neutropenia, diarrhea, leukopenia, cough, and thrombocytopenia. - Serious adverse reactions occurred in 20% of patients receiving fam-trastuzumab deruxtecan, including interstitial lung disease, pneumonia, vomiting, nausea, cellulitis, hypokalemia, and intestinal obstruction. - Fatalities due to adverse reactions occurred in 4.3% of patients including interstitial lung disease (2.6%), and the following events occurred in one patient each (0.4%): acute hepatic failure/acute kidney injury, general physical health deterioration, pneumonia, and hemorrhagic shock. - Interstitial lung disease (ILD) is an important safety signal identified during the clinical development program for fam-trastuzumab deruxtecan. An independent ILD adjudication committee was established to evaluate all suspected cases of ILD in all ongoing clinical trials with fam-trastuzumab deruxtecan. Fatal outcomes due to ILD occurred in 2.6% of patients. - Labeling includes a Boxed Warning to advise health professionals of the risk of interstitial lung disease (ILD), and embryo-fetal toxicity.	The safe use of fam-trastuzumab deruxtecan can be managed through appropriate labeling, including boxed warnings for interstitial lung disease, and embryo-fetal toxicity. No REMS is indicated.

1.4. Patient Experience Data

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Patient Experience Data Relevant to this Application (check all that apply)

☐	The patient experience data that was submitted as part of the application, include:	Section where discussed, if applicable
☐	Clinical outcome assessment (COA) data, such as	[e.g., Section 6.1 Study endpoints]
☐	Patient reported outcome (PRO)	
☐	Observer reported outcome (ObsRO)	
☐	Clinician reported outcome (ClinRO)	
☐	Performance outcome (PerfO)	
☐	Qualitative studies (e.g., individual patient/caregiver interviews, focus group interviews, expert interviews, Delphi Panel, etc.)	
☐	Patient-focused drug development or other stakeholder meeting summary reports	[e.g., Section 2.1 Analysis of Condition]
☐	Observational survey studies designed to capture patient experience data	
☐	Natural history studies	
☐	Patient preference studies (e.g., submitted studies or scientific publications)	
☐	Other: (Please specify)	
X	Patient experience data that was not submitted in the application, but was considered in this review.	

X

Christy Osgood, MD
Cross-Disciplinary Team Leader

2 Therapeutic Context

2.1. Analysis of Condition

The Applicant's Position

Breast cancer is the most commonly diagnosed female cancer and the second-most leading cause of cancer death in women ([Bray et al 2018](#)). According to data from the United States (US) Surveillance, Epidemiology, and End Results (SEER) program, it is estimated that in 2019, there will be 268,600 new breast cancer cases in females in the US, which constitutes around 15.2% of all new cancer cases ([Howlader et al 2019](#)). In 2018, there were an estimated 983,907 women living with breast cancer in the US who had been diagnosed in the last 5 years (5-year prevalence rate of 596.3 per 100,000).

Breast cancer is mainly a female disease, with only 1% of cases occurring in males worldwide, ([Ottini et al 2010](#)) and occurs more frequently in women over 40 years of age. Known risk factors for breast cancer include older age, family history, genetic alterations, hormone therapy, and obesity ([US FDA The Voice of the Patient 2015](#)). In the US, the median age at diagnosis is 62 years, with 50.0% of patients diagnosed between the ages of 55 years and 74 years ([Howlader et al 2019](#)).

Approximately 20% of patients with breast cancer have human epidermal growth factor receptor 2 (HER2)-positive tumors (ie, tumors scored as 3+ by immunohistochemistry [IHC] or that demonstrate gene amplification by in situ hybridization [ISH]) ([Mitri et al 2012](#)). HER2 positivity is associated with a more aggressive disease and a younger patient population ([Mitri et al 2012](#), [Taucher et al 2003](#)). Based on 2010 SEER data collected in the US ([Howlader et al 2014](#)), at diagnosis, patients with HER2-positive breast cancer are less likely to be above 65 years old, and more likely to present at Stage III or IV than either hormone receptor-positive/HER2-negative or triple-negative breast cancer, with tumors that are 6.4-fold to 16.8-fold more likely to be high-grade compared to hormone receptor-positive/HER2-negative tumors.

Metastatic HER2-positive breast cancer remains a fatal disease. Although treatment with anti-HER2 therapies has improved the disease outcomes for patients with unresectable or metastatic HER2-positive breast cancer, patients invariably progress. The development of novel therapies, especially options that minimize side effects, remains a focus of patients' perspectives on treatments for breast cancer ([US FDA The Voice of the Patient 2015](#)). Targeted therapy, including antibody-drug conjugates (ADCs), provides the potential to expand the therapeutic window by having strong efficacy while limiting systemic exposure to toxic agents.

Development of such agents are required due to the continued unmet medical need in patients with HER2-positive metastatic breast cancer.

The FDA's Assessment:

The FDA generally agrees with the applicant's assessment of HER2-positive breast cancer. While antibody-drug conjugates may theoretically be associated with potentially limiting systemic exposure to toxic agents, there are still important associated toxicities with fam-trastuzumab deruxtecan. Refer to the safety section of this review for the detailed analysis of toxicities associated with fam-trastuzumab deruxtecan in this BLA application.

2.2. Analysis of Current Treatment Options

Several anti-HER2 therapies have been approved for the treatment of HER2-positive breast cancer. HERCEPTIN® (trastuzumab), a monoclonal antibody (mAb) targeting HER2, was the first approved anti-HER2 therapy. The approved ADC KADCYLA® (ado-trastuzumab emtansine, T-DM1), which targets HER2, consists of the mAb trastuzumab linked to a drug component, emtansine, that inhibits tubulin polymerization. Other HER2-targeting agents have also been approved for use in patients with HER2-positive breast cancer: the tyrosine-kinase inhibitor (TKI) NERLYNX® (neratinib) as monotherapy, and the mAb PERJETA™ (pertuzumab) and the TKI TYKERB® (lapatinib) each in combination with other chemotherapeutic agents.

The current standard of care (SoC) for newly-diagnosed metastatic HER2-positive breast cancer in the US ([NCCN Guidelines 2019](#)) is a combination of pertuzumab, trastuzumab, and taxane, followed by ado-trastuzumab emtansine (T-DM1, KADCYLA®). The triplet regimen of pertuzumab, trastuzumab, and taxane was established based on the results of the CLEOPATRA study ([Baselga et al 2012](#)), which randomized patients to receive trastuzumab plus docetaxel with or without the addition of pertuzumab as first-line therapy for metastatic HER2-positive breast cancer. The addition of pertuzumab improved the median progression-free survival (PFS) from 12.4 months to 18.5 months and the median overall survival (OS) from 40.8 months to 56.5 months.

In patients with metastatic HER2-positive breast cancer who progressed after a trastuzumab-containing therapy, subsequent anti-HER2 therapy with single-agent KADCYLA® is the current SoC as established based on the EMILIA study ([NCCN Guidelines 2019](#), [Verma et al 2012](#)). In the EMILIA study ([Table 1](#); [Verma et al 2012](#)), T-DM1 was compared with combined therapy with lapatinib plus capecitabine in patients who had progressed after treatment with the original trastuzumab plus taxane combination. T-DM1 improved the median PFS from 6.4 months (lapatinib plus capecitabine arm) to 9.6 months (T-DM1 arm) and the median OS from 25.1 months (lapatinib plus capecitabine arm) to 30.9 months (T-DM1 arm). It is important to note that this trial was conducted before the pertuzumab-based combination was SoC in first-line therapy.

Response rates to T-DM1 appear to be lower when T-DM1 is given after pertuzumab-containing regimens. In a retrospective study conducted in 3 US institutions ([Dzimitrowicz et al 2016](#)), treatment sequencing and outcomes for 78 patients with metastatic HER2-positive breast cancer who received T-DM1 after previous therapy with pertuzumab between 01 Mar 2013 and 15 Jul 2015 were available for analysis. The objective response rate (ORR) on T-DM1 therapy was 17.9% (14/78; 95% confidence interval [CI]: 9.4, 26.4), compared to 43.6% for patients in the EMILIA study. Findings were similar in a recently reported retrospective study conducted in Japan ([Noda-Narita et al 2019](#)), which showed a lower ORR in patients who had received both trastuzumab and pertuzumab prior to T-DM1 compared to patients who had received trastuzumab only prior to T-DM1 (ORR in trastuzumab plus pertuzumab arm, 11.1%; ORR in trastuzumab-only arm, 25.0%).

Recent approvals of pertuzumab, neratinib, and T-DM1 for early-stage breast cancer have resulted in a heterogeneous patient population at the time of diagnosis of metastatic disease. Based on results of the APHINITY trial ([Von Minckwitz et al 2017](#)), pertuzumab is available in combination with trastuzumab and chemotherapy for the adjuvant treatment of patients with HER2-positive breast cancer at high risk of recurrence. Through the ExteNET trial, neratinib obtained approval to prevent recurrence in patients with early-stage HER2-positive breast cancer who have finished at least 1 year of post-surgery trastuzumab ([Chan et al 2016](#)). Most recently, in the KATHERINE trial ([Von Minckwitz et al 2019](#)), T-DM1 led to an improvement compared to trastuzumab when used as an adjuvant treatment of patients with HER2-positive early breast cancer who have residual invasive disease following neoadjuvant trastuzumab and chemotherapy. Therefore, the current treatment paradigm, as well as the level of recurrence risk, vary depending on the therapies already given. The addition of multiple agents to the armamentarium for treatment of early-stage HER2-positive breast cancer means that at the time of diagnosis of metastatic disease, patients may have already received anywhere from 0 to 3 prior anti-HER2 therapies. Treatment decisions must therefore consider the history of treatment exposure as opposed to the current “line of therapy.”

Treatment options for advanced/metastatic disease developed prior to the above-mentioned approvals of T-DM1, pertuzumab, and neratinib include lapatinib plus capecitabine, trastuzumab plus capecitabine, trastuzumab plus lapatinib, or trastuzumab plus other agents. Reported ORRs for these regimens when given after a single line of anti-HER2 therapy range from 9% to 22% ([Bartsch et al 2007](#), [Blackwell et al 2010](#), [Geyer et al 2006](#)).

The only large randomized study with full results published to date in patients who had received 2 or more prior anti-HER2 therapies was the TH3RESA study. In TH3RESA ([Krop et al 2014](#)), efficacy outcomes from treatment with T-DM1 (N = 404) were compared with those from treatment of physician’s choice (TPC, N = 198). All patients had received trastuzumab and lapatinib in the advanced setting and previous taxane therapy in any setting. Treatment of physician's choice consisted of combination regimens including 1 or more HER2-directed agents in 83% of patients randomized to this arm (with 80% of patients receiving trastuzumab) and

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

single-agent chemotherapy in 17% of patients. The ORR for patients with measurable disease at baseline and the median PFS for the T-DM1 arm were 31% (95% CI: 26, 36; N = 345) and 6.2 months (95% CI: 5.59, 6.87), respectively, compared to 9% (95% CI: 4.6, 13.4; N = 163) and 3.3 months (95% CI: 2.89, 4.14), respectively, for TPC ([Table 1](#)). Please note that the above-mentioned 95% CIs for ORR were not reported for the TH3RESA study but were approximated by the Sponsor based on available published data, using normal approximation for binomial data.

(b) (4)



Table 1: Current Treatment Options for Previously Treated HER2-positive Advanced or Metastatic Breast Cancer

Products Name	Trial/Reference	Relevant Patient Population	Efficacy Information	Important Safety and Tolerability Issues
FDA Approved Treatment				
Ado-trastuzumab emtansine (KADCYLA®) (approved 22 Feb 2013) versus Lapatinib + Capecitabine	EMILIA ^a (Verma et al 2012)	Patients with HER2-positive advanced breast cancer who had previously been treated with trastuzumab and a taxane	<u>T-DM1:</u> • ORR: 43.6% (95% CI: 38.6, 48.6) • Median DoR: 12.6 months (95% CI: 8.4, 20.8) • Median PFS by ICR: 9.6 months (265 events) <u>Lapatinib + Capecitabine:</u> • ORR: 30.8% (95% CI: 26.3, 35.7) • Median DoR: 6.5 months (95% CI: 5.5, 7.2) • Median PFS by ICR: 6.4 months (304 events)	<u>T-DM1:</u> • Grade ≥3 AEs: 40.8% • SAEs: 15.5% <u>Lapatinib + Capecitabine:</u> • Grade ≥3 AEs: 57% • SAEs: 18.0%
Other Treatments				
Studies After 2 Lines of Anti-HER2 Therapies				
Ado-trastuzumab emtansine (KADCYLA®) versus Treatment of Physician's Choice	TH3RESA ^a (Krop et al 2014)	Patients with progressive HER2-positive advanced breast cancer after 2 or more HER2-directed regimens in the advanced setting, including trastuzumab and lapatinib and previous	<u>T-DM1:</u> • ORR: 31% (108/345 with measurable disease at baseline) • Median DoR: 9.7 months (95% CI: 6.6, 10.5) • Median PFS: 6.2 months (95% CI: 5.59, 6.87) <u>TPC:</u> • ORR: 9% (14/163 with measurable disease at baseline) • Median DoR: not reached at DCO in the 14 patients	<u>T-DM1:</u> • Grade ≥3 AEs: 32% • SAEs: 18% • AEs leading to treatment discontinuation: 7% • Grade 5 AEs: 1% <u>TPC:</u> • Grade ≥3 AEs: 43% • SAEs: 21% • AEs leading to treatment discontinuation:

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Products Name	Trial/ Reference	Relevant Patient Population	Efficacy Information	Important Safety and Tolerability Issues
		taxane therapy in any setting	with objective response Median PFS: 3.3 months (95% CI: 2.89, 4.14)	11% Grade 5 AEs: 2%

(b) (4)

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

DCO = data cut-off

^arandomized controlled trial

The Applicant's Position:

Although anti-HER2 therapies have improved the disease outcomes for patients with metastatic HER2-positive breast cancer, there is no clearly defined SoC for patients with metastatic breast cancer after 2 or more anti-HER2 therapies.

Metastatic HER2-positive breast cancer remains an incurable disease. The lack of clearly preferential treatment options after 2 or more anti-HER2 therapies for metastatic HER2-positive breast cancer, as reflected by the absence of guideline-specific recommendations and low reported ORRs, highlights the unmet medical need for these patients and the opportunity to bring new HER2-targeted agents in this particular setting.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment of available therapies for HER2-positive breast cancer. The standard of care for first line metastatic HER2-positive breast cancer in the US is the combination of trastuzumab, pertuzumab, and taxane based on the CLEOPATRA trial which showed a median overall survival of 56.5 months in patients treated with trastuzumab, pertuzumab, and docetaxel compared to 40.8 months in those treated with placebo, trastuzumab, and docetaxel, [HR=0.68, (95% CI 0.56, 0.84)]. T-DM1 is the preferred second-line option based on the EMILIA study (Table 1). Beyond the second-line, treatment options are more limited and include lapatinib and capecitabine, or trastuzumab combined with a chemotherapeutic agent. Even though these therapies have improved the disease outcomes for patients with metastatic HER2-positive breast cancer, there remains an unmet need for further effective therapies in these patients as their disease will eventually progress.

The data from the (b) (4) trial have not been reviewed by the FDA and (b) (4) are not approved for the treatment of for metastatic HER2-positive breast cancer who have received prior HER2-based regimens. Therefore, FDA does not consider the data from the (b) (4) to be current treatment options as these as not approved by the FDA.

3 Regulatory Background

3.1. U.S. Regulatory Actions and Marketing History

The Applicant's Position:

Fam-trastuzumab deruxtecan is not currently marketed in the US.

The FDA's Assessment:

FDA agrees that fam-trastuzumab deruxtecan is currently not approved in any country.

3.2. Summary of Presubmission/Submission Regulatory Activity

The Applicant's Position:

The key US pre-submission regulatory activities are outlined in [Table 2](#). Fam-trastuzumab deruxtecan has been under development under IND 127553 since 2016 and is being investigated in clinical development programs as a single agent for the treatment of breast cancer as well as other tumor types.

Table 2: Key Regulatory Interactions Related To BLA 761139

Date	Regulatory Interaction
27 Jan 2016	Original IND 127553 submitted to support initiation of a Phase 1 study (DS8201-A-J101)
10 Nov 2016	Fast Track designation granted for the treatment of patients with HER2-positive, unresectable and/or metastatic breast cancer subjects who progressed after prior HER2-targeted therapies, including T-DM1.
28 Mar 2017	Type B End of Phase 1 (EOP1) meeting to discuss study DS8201-A-U201 design, dose selection strategy and clinical plan to evaluate QT prolongation.
25 Aug 2017	Breakthrough Therapy Designation (BTD) granted for the treatment of patients with HER2-positive, locally advanced or metastatic breast cancer who have been treated with trastuzumab and pertuzumab and have disease progression after T-DM1.
12 May 2017	Receipt of Type C CMC preliminary meeting comments regarding the adequacy of the overall CMC process development for DS-8201a to support

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	the introduction of frozen liquid drug product 2 (FL-DP2) material into planned clinical studies.
07 Sep 2017	Type C CMC Meeting/Written Responses Only feedback received on the adequacy of an analytical comparability protocol to compare two dosage forms, FL-DP2 and lyophilized drug product (Lyo-DP).
12 Dec 2017	Type B Guidance (BTD Initial Comprehensive) meeting to discuss the overall development plan of DS-8201a in HER2-positive breast cancer to support the initial BLA for accelerated approval consideration.
08 Feb 2018	Receipt of Type B meeting preliminary comments regarding the design of Phase 3 study DS8201-A-U302.
03 Apr 2018	Type B (EOP) CMC meeting to review the overall CMC development plan of DS-8201a to support consideration for an accelerated BLA approval opportunity based on future clinical results from the Phase 1 DS8201-A-J101 and Phase 2 DS8201-A-U201 trials.
10 Oct 2018	Type B CMC meeting to discuss a) the proposed commercial specifications with respect to the list of tests for mAb, drug substance (DS) and drug product (DP) as well as microbial control strategy for DS, b) the details of the stability plans for DS and DP, and extractable/leachable studies for DS and DP.
09 Nov 2018	Receipt of Type B pre-BLA preliminary meeting comments to discuss format, structure, and content of planned BLA.
31 Jan 2019	Receipt of Type C CMC preliminary meeting comments regarding commercial specifications for drug-linker, the strategy of change of Host Cell Protein detection method to a custom kit and batch records required to be submitted to the BLA.
10 Jun 2019	Type B pre-BLA meeting to discuss results from studies DS8201-A-J101 and DS8201-A-U201 in support of accelerated approval.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment of regulatory intractions in reference to this BLA application. FDA also discussed the monitoring of interstitial lung disease with the applicant during some of these meetings, an important safety signal associated with fam-trastuzumab deruxtecan.

4 Significant Issues from Other Review Disciplines Pertinent to Clinical Conclusions on Efficacy and Safety

4.1. Office of Scientific Investigations (OSI)

The FDA's Assessment:

Inspections by OSI were requested by the Division of Oncology 1. The objectives of the inspections included the following:

- Verify efficacy endpoint of ORR
- Evaluate identification, documentation and reporting of adverse events
- Assess general compliance with investigational plan

Three clinical investigator study sites (#1002, 1004, and 8108) were selected for clinical inspection. These sites had a relatively high number of subjects enrolled and/or responders reported among all clinical investigator sites. Relative to other sites, one site (Site 1002) had fewer serious adverse events and deaths reported.

The inspection findings verified the applicant's reported clinical data with source records at the three study sites. There were no substantial Good Clinical Practice (GCP) compliance deficiencies identified at these sites.

Based on the results of these inspections, the data reported by these investigator sites appear acceptable and supportive of this BLA. Details of these inspections are below.

Site #1002: Dr. Ian Krop

Reason selected: Fewer number of deaths and SAEs reported relative to number of patients enrolled.

This clinical investigator site was inspected between August 8 and August 26, 2019 as a data audit for the Study DS8201-A-U201. For the investigator, this was the initial FDA inspection. The site enrolled 8 of the 12 screened subjects. Seven subjects received DS8201a at 5.4 mg/kg and one at 7.4 mg/kg. As of the data cutoff date, four subjects remained on study treatment, one came off due to disease progression, and three (Subjects (b) (6) and (b) (6) at the 5.4 mg/kg dose level, Subject (b) (6) at the 7.4 mg/kg dose level) were discontinued due to toxicity.

The inspection reviewed study-related documents and source records for all enrolled subjects and compared the records with the site-level data listings submitted to the BLA. The reviewed documents and records included but were not limited to: Institutional Review Board (IRB)

documentation and communications, investigator's agreements, financial disclosures, training records, delegation of authority logs, informed consent forms (ICFs), adverse events, protocol deviation logs, source records, electronic case report forms (eCRFs), study drug accountability, dosing, monitoring logs, and monitoring reports.

The inspection found that the study was adequately conducted at this site. For all the enrolled subjects, the submitted data listings were verifiable with the site's source records, with no discrepancies reported. Note that the ICR reported tumor response in the data listings were not available at the site. There were no evidence of underreporting of adverse events and protocol deviations.

There was one inspectional observation reported, which led to a Form FDA 483 issued to the investigator. This observation related to nurses' collection of research blood samples outside of the protocol-required time frames in four study subjects at the infusion facility. For Subject (b) (6), the high sensitivity troponin sample was not drawn 2-3 hours after the end of infusion (EOI) as specified in the protocol, but rather "within 15 minutes of EOI" on Day 1 of Cycle 1 and "4 minutes after the EOI" on Day 1 of Cycle 7. For Subjects (b) (6), their pharmacokinetic (PK) samples were drawn 5 minutes before the EOI on Day 1 of Cycle 1. This is not consistent with the protocol-specified time frame of "within 15 minutes after EOI" for PK sampling.

In the written response to the FDA 483, the Investigator acknowledged the observation and stated that corrective and preventive actions have been implemented. These include:

- Immediate Corrective Action: "All staff infusion nurses are required to attend a clinical research course and training on how to read an investigational protocol plan";
- Ongoing Corrective Action, which entails "Investigational plan education, a clinical research course (including but not limited to good clinical documentation practice), and one to one re-training continues to occur for all staff infusion nurses". In addition, standardized investigational plan flowsheets along with training for documentation on flowsheets will be used at the Institute of this study site.

OSI Classification: No action indicated (NAI).

Site #1004: Dr. Shanu Modi

Reason selected: Highest enrollment and high efficacy.

This clinical investigator site was inspected on September 9-12, 2019 as a data audit for the Study DS8201-A-U201. This was the first FDA inspection for Dr. Shanu Modi at Memorial Sloan Kettering Cancer Center (MSKCC). The study was initiated at the site on 11/14/2017 and was ongoing at the time of inspection. As of the data cutoff date, the site enrolled 15 of the 28

screened subjects. Eight subjects received DS8201a at 5.4 mg/kg: two were discontinued due to disease progression and six remained on study treatment.

The inspection reviewed all subjects' source documents and compared them with the data listings submitted to the BLA by the Applicant for the site. The reviewed source documents consisted of records such as data collection worksheets, informed consent forms, medical progress notes, laboratory reports, adverse events, and concomitant medications. At the site, study documentation for study subjects were scanned and uploaded into the hospital's electronic medical record (EMR) system per the MSKCC written procedures and maintained within the EMR system. The inspection also reviewed the conduct of this study at the site, including the IRB approvals and oversight, clinical site training, delegation of authority, signed financial disclosure and Form FDA 1572s, sponsor's monitoring practices, drug accountability records, and reports to the sponsor.

The inspection found no significant deficiencies, with no Form FDA 483 issued. The submitted data listings for the site were verified with all subjects' source documents, with no discrepancies reported. Note that the inspection only verified the investigator-assessed best overall response since the ICR results were not available at the site for review. The site remained blinded to the central review assessments. A few issues were noted in source documents and were discussed with the study team at the closeout meeting:

- There was no clear documentation of medical review of some abnormal laboratory results given lack of signatures;
- The Subject RECIST Forms (for local use) were not signed by the radiologist and the investigator in a timely manner (e.g., 5-9 months after the scans were performed);
- The investigator instructed her administrative assistant via email to insert her wet signature image into Notes to File.

The investigator and her study team provided explanations and relevant records, and the issues were addressed. Regarding the use of the wet signature image, the investigator stated that she would implement "a better method for electronic signatures that are more compliant with electronic records and signatures regulations (21 CFR Part 11)".

OSI Classification: No action indicated (NAI).

Site #8108: Dr. Toshinari Yamashita

Reason selected: higher efficacy among Asian sites, represents population of patients (Asian) that were highly enrolled in study.

This foreign clinical investigator site was inspected on October 7-10, 2019 as a data audit for the Study DS8201-A-U201. The inspection report is not currently available.

Based on the inspector's preliminary summary, the site enrolled 10 of the 11 screened subjects. As of the data cutoff date, six subjects received DS8201a at 5.4 mg/kg and four received DS8201a at 6.4 mg/kg or 7.4 mg/kg. Four of the six subjects at the 5.4 mg/kg dose level remained on study treatment and two were discontinued (Subject (b) (6) due to adverse event and Subject (b) (6) at the Investigator's discretion). At the time of inspection, two subjects continued receiving DS8201a at 5.4 mg/kg (note that the ID of the two subjects was not provided in the preliminary summary).

The inspection audited source documents for all study subjects at the site and found that the efficacy and safety data were verifiable with the source documents. There was no evidence of underpotting of adverse events.

The inspection had no significant observations reported. No Form FDA 483 was issued to the investigator at the end of this inspection. The discussed items were related to documentation at the site:

- Documenting the number of the study kits administered to subjects to ensure that it matches the IWRS-assigned number
- Signing off each out-of range laboratory value to show that the investigator reviewed and assessed the result as opposed to only signing the cover page
- Documenting the reason for crossing out any adverse events.

OSI Classification: No action indicated (NAI).

4.2. Product Quality

The Office of Pharmaceutical Quality (OPQ), CDER recommends approval of this application for Enhertu manufactured by Daiichi Sankyo, Inc. The data and information submitted in this application are sufficient to support the conclusion that the manufacture of Enhertu is well controlled and leads to a product that is pure and potent for the duration of the product shelf life. OPQ recommends that this product be approved for human use under the conditions specified in the package insert.

Based on the inspections conducted by the product quality review team, the following issues were discussed with the applicant. All these issues have been resolved. Refer to the detailed review by the product quality team for further details. Refer to Section 13 of this Assessment Aid for the Postmarketing commitments for product quality.

1. Satisfactory evaluation of the manufacturing facilities is required for BLA approval.

2. Inadequate information provided for analytical method transfer of the analytical methods used for Drug Substance Intermediate, Drug Substance, and Drug Product in process, release and stability testing.
3. Most raw materials used for manufacture of the Drug Substance Intermediate comply with Japanese Industrial Standards (JIS). Data and information are needed to support that these raw materials are suitable for pharmaceutical use.

4.3. Clinical Microbiology

Based on the inspections conducted by the clinical microbiology review team, the following issues were discussed with the applicant. All these issues have been resolved. Refer to the detailed review by the clinical microbiology team for further details. Refer to Section 13 of this Assessment Aid for the Postmarketing commitments for clinical microbiology.

4.4. Devices and Companion Diagnostic Issues

The Sponsor is not proposing the use of a companion diagnostic or FDA-approved test in their indication.

All subjects enrolled in DS8201-A-J101 and DS8201-A-U201 had to have documented HER2-expressing tumors based on a validated testing method prior to study entry.

In DS8201-A-J101, the tumor HER2 status was determined locally either on archival tissue samples or on fresh tissue samples obtained after the last HER2-targeting treatment for metastatic disease. In DS8201-A-U201, the tumor HER2 status was confirmed from the most recent adequate tumor sample by central laboratory according to American Society of Clinical Oncology-College of American Pathologists (ASCO/CAP) 2013 guidelines.

The sponsor is proposing to treat HER2-positive metastatic breast cancer patients with third line DS-8201a (following two prior HER2-targeted therapies) without the use of a companion diagnostic to screen for HER2 amplification. CDRH had no issue with this population of patients being selected for fam-trastuzumab deruxtecan for treatment without a companion diagnostic device since patients will have been screened for HER2 amplification for previous HER2-directed therapies. Refer to the CDRH consult review by Dr. Abukhdeir, PhD.

5 Nonclinical Pharmacology/Toxicology

5.1. Executive Summary

The FDA's Assessment:

Fam-trastuzumab deruxtecan-nxki (DS-8201a) is a HER2-directed antibody and topoisomerase inhibitor conjugate, composed of a humanized anti-HER2 IgG1 monoclonal antibody (MAAL-9001), covalently linked to a topoisomerase inhibitor (MAAA-1181a, DXd) via a tetrapeptide-based cleavable linker, with a drug-to-antibody ratio (DAR) of approximately 8. In vitro, MAAL-9001 bound to human recombinant HER2 protein but not to recombinant EGFR, HER3, or HER4 proteins. DS-8201a and MAAL-9001 bound to cynomolgus monkey HER2 protein but not to mouse or rat HER2 proteins, showing that the cynomolgus monkey is an appropriate species in which to conduct toxicity evaluation with DS-8201a, based on its binding to HER2. MAAA-1181a, an exatecan derivative, inhibited the activity of topoisomerase 1, in vitro, as shown by inhibition of supercoiled DNA relaxation in a dose-dependent manner. DS-8201a and MAAA-1181a, but not MAAL-9001, induced phosphorylation of Chk1 (24 hours post-treatment), phosphorylation of histone H2A.X (48 hours post-treatment), and cleavage of poly-(adenosine diphosphate-ribose) polymerase (PARP) (72 hours post-treatment), in vitro. These changes also correlated with increased levels of phosphorylated and total Akt protein in SK-BR-3 cells, 24 hours after treatment with DS-8201a. In vitro, DS-8201a exhibited cell growth inhibitory activity towards HER2-expressing cells but not HER2-negative cells. DS-8201a and MAAL-9001 showed antibody-dependent cellular cytotoxic activity when evaluated using human peripheral blood mononuclear cells derived from 3 healthy donors as effector cells and SK-BR-3 cells as target cells. DS-8201a showed anti-tumor activity in HER2-positive or HER2-low-expressing breast cancer mouse tumor models. Based on the nonclinical pharmacology studies provided in the BLA submission, the Established Pharmacological Class (EPC) of "HER2-directed antibody and topoisomerase inhibitor conjugate" is appropriate based on its pharmacological activity, prior EPC for related approved products, and the approved EPC for topoisomerase inhibitors.

In vitro, MAAA-1181a did not have any effects on the hERG channel current up to 10 µmol/L ($IC_{50} > 10 \mu M$). In a safety pharmacology study, DS-8201a did not have any significant effects on blood pressure, heart rate, ECG parameters, body temperature, or neurobehavioral function in conscious and unrestrained telemetered male cynomolgus monkeys following a single 78.8 mg/kg dose. However, in a repeat-dose toxicology study with DS-8201a in cynomolgus monkeys, shortening of the PR interval and QTc prolongation of up to 50 msec was observed in one male animal at 78.8 mg/kg (approximately 17 times the human recommended dose of 5.4 mg/kg based on AUC) on dosing Day 26.

In cynomolgus monkeys, C_{max} and AUC_{0-21d} values increased dose-dependently, and there were no accumulation or sex-related differences in toxicokinetic parameters following administration of 3, 10, or 30 mg/kg DS-8201a IV, once every 3 weeks for 3 months. After a single administration of 6.4 mg/kg ^{14}C -DS-8201a IV, the highest radioactivity was observed in the large intestinal contents, followed by the blood, lung, adrenal gland, and liver (all tissue-to-blood ratios were less than 1). No significant distribution of DS-8201a into the central nervous system was observed (maximum tissue-to-blood ratio of 0.02 and 0.01 for cerebellum and cerebrum, respectively). In vitro plasma binding studies with MAAA-1181a showed that the highest mean plasma protein binding was observed in human plasma (96.8 – 98.0%), followed by rat, mouse, and monkey plasma. MAAA-1181a showed a low release rate from DS-8201a following incubation with mouse, rat, monkey, and human plasma (2.1% and 1.5% in human plasma at 10 and 100 mg/mL, respectively) at 37°C for 21 days. No human-specific metabolites were observed following incubation of DS-8201a with cryopreserved hepatocytes, in vitro. The predominant route of excretion for DS-8201a was through the hepatobiliary pathway, with 67.3% and 18.7% of the dose observed in the feces and urine, respectively, following a single administration of ^{14}C -DS-8201a to male cynomolgus monkeys.

The Applicant conducted GLP-compliant, repeat-dose general toxicology studies to evaluate the toxicity of DS-8201a, MAAA-1181a, or MAAL-9001 in rats and/or monkeys. The major observed target organs of toxicity following administration of 30 mg/kg DS-8201a once every 3 weeks for up to 3 months to rats or monkeys included the bone marrow (decreases in reticulocyte count correlated to slight to moderate decreases in erythroblasts), skin (abnormal skin color and black foci correlated to very slight hair follicle single cell necrosis and slight epidermal pigmentation), lung (white foci correlated to very slight to slight foamy alveolar macrophage aggregation and focal interstitial and alveolus inflammation), testis (very slight Stage V-VI round spermatid decreases and sperm stasis, and slight tubular degeneration/atrophy), kidney (very slight proximal tubule anisokaryosis), and the gastrointestinal system (very slight single cell necrosis of epithelial cells). Except for the findings in the skin and kidneys, all other histological findings were reversible by the end of the recovery period. Overall, the toxicity findings in animals administered DS-8201a were consistent with clinical adverse reactions observed in clinical trials with fam-trastuzumab deruxtecan-nxki. Intravenous administration of up to 30 mg/kg or 12 mg/kg MAAA-1181 once-weekly to rats and monkeys, respectively, resulted in similar target organs of toxicity to those observed in toxicology studies with DS-8201a, with additional findings of single cell corneal epithelial necrosis in the eye in both species administered MAAA-1181. Intravenous administration of up to 78.8 mg/kg MAAL-9001 once every 3 weeks (3 total doses) to rats and monkeys, did not result in significant toxicity.

MAAA-1181a was clastogenic in both an in vivo rat bone marrow micronucleus assay and an in vitro Chinese hamster lung chromosome aberration assay but was not mutagenic in an in vitro bacterial reverse mutation assay. The Applicant did not conduct any carcinogenicity studies

with DS-8201a. Carcinogenicity studies are not warranted to support the licensing application of fam-trastuzumab deruxtecan-nxki for the proposed indication.

The Applicant did not conduct any reproductive toxicology studies with DS-8201a. As recommended by ICH S9 guidance, embryo-fetal toxicity studies were not conducted based on the justification that these studies are not warranted to support the proposed indication because MAAA-1181a is genotoxic and targets rapidly dividing cells, and the use of a HER2-directed antibody during pregnancy resulted in cases of oligohydramnios. Fertility and early embryonic development and pre- and postnatal toxicology studies are not warranted to support marketing of pharmaceuticals intended for the treatment of patients with cancer. An assessment of the potential for fam-trastuzumab deruxtecan-nxki to cause impairment of fertility was provided based on results from general toxicology studies with DS-8201a. In a six-week repeat-dose toxicology study in rats, intravenous administration of DS-8201a resulted in spermatid retention at 20 mg/kg and 60 mg/kg (4 and 9 times the exposure at the human recommended dose of 5.4 mg/kg based on AUC, respectively) and decreased testes and epididymides weights, tubular atrophy/degeneration in testes, and reduced sperm count in epididymides at a dose of 197 mg/kg (19 times the exposure at human recommended dose of 5.4 mg/kg based on AUC). In a three-month repeat-dose toxicity study in monkeys, intravenous administration of DS-8201a decreased numbers of round spermatids in the testes at seminiferous tubule stages V to VI at doses ≥ 30 mg/kg (≥ 7 times the exposure at the human recommended dose of 5.4 mg/kg based on AUC). The animal to human exposure margins were calculated based on toxicokinetic data from nonclinical toxicology studies and the population PK analysis of DS-8201a exposure at steady state following multiple doses in patients with HER2-positive breast cancer (geometric mean $AUC_{\tau} = 735 \mu\text{g}\cdot\text{d/mL}$).

Based on its mechanism of action, a Warning and Precaution for embryo-fetal toxicity is recommended to be included in the label. Due to the potential of Enhertu to cause fetal harm, effective contraception use is recommended for female patients during treatment with Enhertu and for at least 7 months following the last dose, based on the half-life of fam-trastuzumab deruxtecan-nxki of approximately 6 days (i.e., $5 \times T_{1/2} + 6$ months). Based on the finding of genotoxicity with MAAA-1181a, effective contraception use is recommended for male patients with female partners of reproductive potential during treatment with Enhertu and for at least 4 months following the last dose (i.e., $5 \times T_{1/2} + 3$ months). The recommendations for duration of contraception are based on the FDA guidance, "Oncology Pharmaceuticals: Reproductive Toxicity Testing and Labeling Recommendations", for drugs that are genotoxic. There is no data regarding the presence of fam-trastuzumab deruxtecan-nxki in human milk, the effects on the breastfed child, or the effects on milk production. Because of the potential for serious adverse events in a breastfed child, the applicant recommended that women are advised not to breastfeed during treatment with Enhertu and for 7 months after the final dose, which covers the drug washout period.

Overall, the nonclinical pharmacology, ADME/PK, and toxicology data submitted to this BLA are

adequate to support the approval of Enhertu for the proposed indication.

5.2. Referenced NDAs, BLAs, DMFs

The Applicant's Position:

There are no referenced NDAs, BLAs, or DMFs related to nonclinical pharmacology or toxicology for fam-trastuzumab deruxtecan (DS-8201a).

The FDA's Assessment:

FDA agrees.

5.3. Pharmacology

The FDA's Assessment:

Primary pharmacology

DS-8201a (Process 1) and MAAL-9001 (unconjugated antibody, Process 1) bound to human HER2-His protein (K_d [ED_{50}] = 7.33 ng/mL and 7.84 ng/mL, respectively) but not to other related human recombinant proteins in the HER family (EGFR-His, HER3-His, or HER4-His proteins), when evaluated by enzyme-linked immunosorbent assay (ELISA). In the same assay, DS-8201a and MAAL-9001 bound to cynomolgus monkey HER2-His protein (K_d [ED_{50}] = 7.46 ng/mL and 6.77 ng/mL, respectively) but not to mouse or rat HER2-His proteins. (Study CD13-H0072-R01)

The Applicant proposed MAAA-1181a to be a derivative of DX-8951f, a camptothecin analog. In order to show similarity in the mechanism of action, MAAA-1181a was evaluated in an in vitro topoisomerase inhibition assay by measuring the fluorescent intensity of supercoiled DNA following treatment with MAAA-1181c, an acetonitrile-methanol-water solvate of MAAA-1181. MAAA-1181c inhibited the relaxation of supercoiled DNA (IC_{50} = 3.58 μ M), in vitro, as shown by a dose-dependent increase in the amount of supercoiled DNA. (Study CD13-H0072-R05)

The ability of DS-8201a, MAAL-9001, and MAAA-1181a to induce DNA damage and apoptosis was evaluated by measurement of Chk1 and histone H2A.X or cleaved PARP, respectively, following treatment of KPL-4 cells for 24, 48, and 72 hours. DS-8201a and MAAA-1181a, but not MAAL-9001, induced phosphorylation of Chk1 (24 hours post-treatment), phosphorylation of histone H2A.X (48 hours post-treatment), and PARP cleavage (72 hours post-treatment). (Study CD13-H0072-R04) These changes also correlated with increased levels of phosphorylated and total Akt protein in SK-BR-3 cells, 24 hours after treatment with DS-8201a. (Study CD13-H0072-R04)

DS-8201a treatment to HER2-expressing Calu-3 (human adenocarcinoma), KPL-4 (human breast cancer), NCI-N87 (human gastric carcinoma), and SK-BR-3 (human breast adenocarcinoma) cell

lines resulted in cell growth inhibition. DS-8201a did not have cell growth inhibitory activity against HER2-negative MDA-MB-468 (human breast adenocarcinoma) cell line. MAAL-9001 showed cell growth inhibitory activity against NCI-N87 and SK-BR-2 cells but at lower potencies when compared to DS-8201a. No cell growth inhibitory activity was observed in cells treated with MAAA-9001b, a control human IgG ADC conjugated to MAAA-1181. Cell viability was evaluated 6 days after treatment by measuring adenosine triphosphate. (Study CD13-H0072-R02)

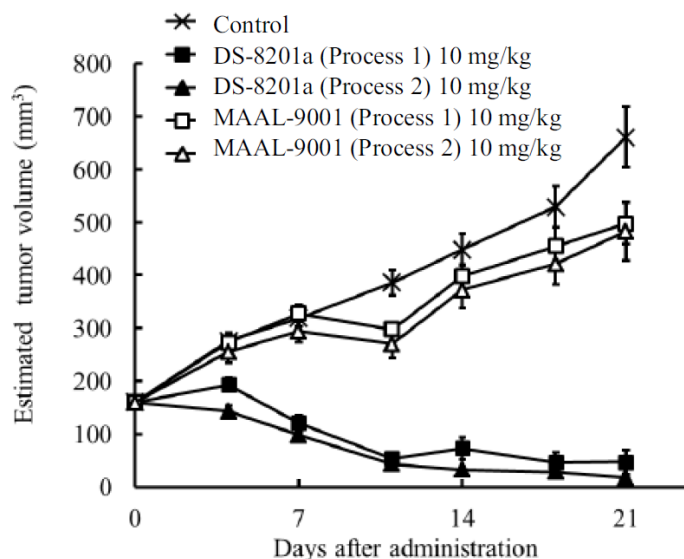
Table 3: IC₅₀ values of DS-8201a, MAAL-9001, and MAAA-9001b in select tumor cell lines.

Cell Line	IC ₅₀ (ng/mL)			HER2 Expression
	DS-8201a	MAAL-9001	MAAA-9001b	
Calu-3	9.29	> 10000	> 10000	Positive
KPL-4	26.8	> 10000	> 10000	Positive
NCI-N87	25.4	204	> 10000	Positive
SK-BR-3	6.65	65.9	> 10000	Positive
MDA-MB-468	> 10000	> 10000	> 10000	Negative

DS-8201a and MAAL-9001 showed antibody-dependent cellular cytotoxic activity when evaluated using human peripheral blood mononuclear cells derived from 3 healthy donors as effector cells and SK-BR-3 cells as target cells, with EC₅₀ values of 3.8 – 5.9 ng/mL and 2.1 – 2.7 ng/mL, respectively. (Study P140118)

The anti-tumor activity of DS-8201a was evaluated in several mouse models of breast cancer. In a mouse xenograft model of HER2-positive breast cancer, the activity of a single dose of 10 mg/kg DS-8201a or MAAL-9001 was evaluated 13 days after inoculation of KPL-4 cells to female CAnN.Cg-Foxn1^{nu}/CrIcrIj nude mice (Study CR16-H0018-R06). DS-8201a significantly inhibited tumor growth (97.4% [p < 0.0001] for DS-8201a [Process 2]) when compared to vehicle control on Day 21. There were no significant differences in anti-tumor activity in this model between DS-8201a (Process 2) and DS-8201a (Process 1), under the conditions tested. MAAL-9001 (Process 1) and MAAL-9001 (Process 2) inhibited tumor growth by 24.7% and 26.9%, respectively, on Day 21.

Figure 1: Activity of DS-8201a and MAAL-9001 in a HER2-positive breast cancer KPL-4 xenograft model in CAnN.Cg-Foxn1^{nu}/CrIcrlj mice.

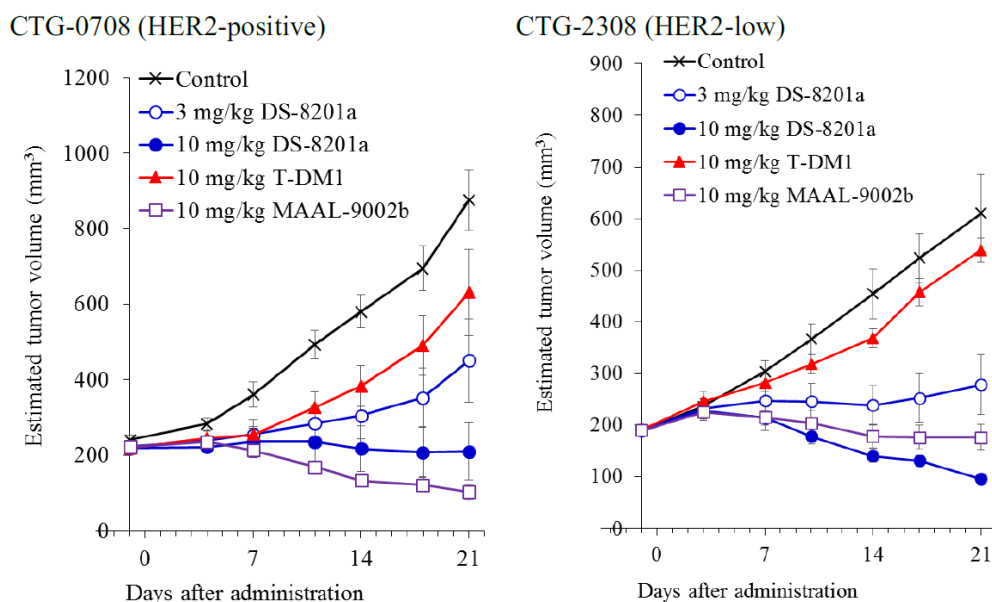


The data are expressed as the mean $\pm$ standard error (n = 12).
Mice were subcutaneously inoculated with KPL-4 cells.
Estimated tumor volume of each mouse was calculated using the following equation:
Estimated tumor volume (mm³) = $1/2 \times \text{length} \times \text{width}^2$

(Excerpted from Applicant's submission)

The anti-tumor activities of DS-8201a (DAR = 7 – 8), T-DM1 (DAR = 3.5), and MAAL-9002b (a related ADC to DS-8201a with a lower DAR [3.7]) were evaluated in female Hsd:athymic nude-Foxn1nu mice implanted with tumor samples from HER2-positive (CTG-0708) and HER2-low-expressing (CTG-2308) breast cancers (Study CR16-H0018-R28). In the CTG-0708 model, 3 and 10 mg/kg DS-8201a and 10 mg/kg MAAL-9002b significantly inhibited tumor growth by tumor growth inhibition (TGI) of 48%, 76%, and 88%, respectively, compared with the vehicle control on Day 21 (P = 0.0044, P < 0.0001, and P < 0.0001, respectively, Dunnett's test). T-DM1 at 10 mg/kg did not show significant activity in the CTG-0708 model (P = 0.1599, Dunnett's test). In the CTG-2308 model, 3 and 10 mg/kg DS-8201a and 10 mg/kg MAAL-9002b significantly inhibited tumor growth by TGI of 54%, 84%, and 71%, respectively, compared with the vehicle control on Day 21 (P < 0.0001, P < 0.0001, and P < 0.0001, respectively, Dunnett's test). Similar to the CTG-0708 model, T-DM1 at 10 mg/kg did not show significant activity (P = 0.6413, Dunnett's test). The results show that DS-8201a exhibits more potent antitumor activity than T-DM1 in both evaluated HER2-positive and HER2-low-expressing breast cancer tumor models, under the conditions tested.

Figure 2: Activity of DS-8201a in patient-derived xenograft models of HER2-expressing breast cancer, CTG-0708 and CTG-2308, in Hsd:athymic nude-*Foxn1*^{nu} mice in comparison with T-DM1 and MAAL-9002b.

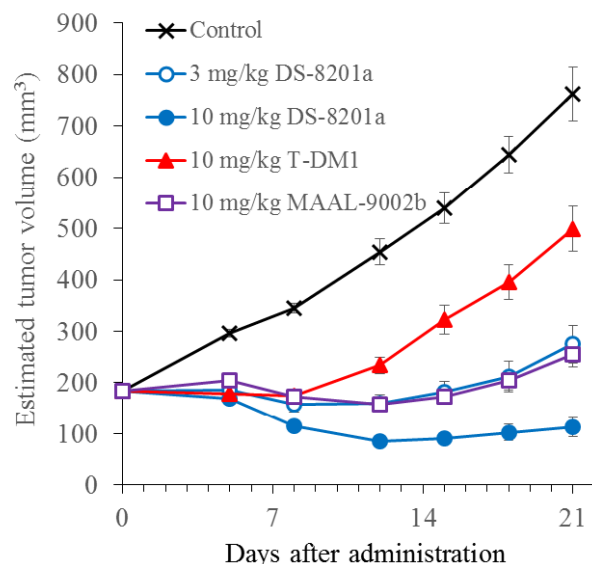


The data are expressed as the mean $\pm$ standard error (n = 10).

(Excerpted from Applicant's submission)

The activity of DS-8201a was evaluated in a model of trastuzumab-resistant HER2-positive breast cancer xenograft model in nude mice. Briefly, JIMT-1 cells (derived from a trastuzumab-resistant patient and known to be refractory to T-DM1) were implanted into female CAnN.Cg-Foxn1nu/CrlCrlj mice. Single dose of DS-8201a, T-DM1, or MAAL-9002b were administered 10 days after tumor cell inoculation. Administration of 3 or 10 mg/kg DS-8201a, 10 mg/kg T-DM1, or 10 mg/kg MAAL-9002b significantly inhibited tumor growth by 64%, 85%, 35%, and 66%, respectively, compared with the vehicle control on Day 21 ($P < 0.0001$, $P < 0.0001$, $P < 0.0001$, and $P < 0.0001$, respectively, Dunnett's test). DS-8201a at 10 mg/kg showed significantly greater efficacy than T-DM1 and MAAL-9002b ($P < 0.0001$ and $P = 0.0003$, respectively, Student's *t*-test). The results suggest that DS-8201a has anti-tumor activity in T-DM1-refractory, HER2-positive breast cancer tumors, under the conditions tested.

Figure 3: Activity of DS-8201a in a HER2-positive breast cancer JIMT-1 xenograft model in CAnN.Cg-Foxn1nu/CrlCrlj mice in comparison with T-DM1 and MAAL-9002b.



The data are expressed as the mean $\pm$ standard error (n = 10).
Mice were subcutaneously inoculated with JIMT-1 cells.
Estimated tumor volume of each mouse was calculated using the following equation:
Estimated tumor volume (mm³) = $1/2 \times \text{length} \times \text{width}^2$

(Excerpted from Applicant's submission)

Secondary Pharmacology

The Applicant's Position:

No secondary pharmacodynamics studies have been conducted.

The FDA's Assessment:

FDA agrees.

Safety Pharmacology

Data:

Cardiovascular system

In vitro: In hERG transfected Chinese hamster ovary (CHO) cells, MAAA-1181a had no effect on the hERG channel current at 1, 3, and 10 $\mu\text{mol/L}$.

In vivo: A safety pharmacology study of DS-8201a on the cardiovascular system was conducted in 8 conscious and unrestrained male cynomolgus monkeys using a telemetry system. DS-8201a was intravenously administered at single doses of 30 or 78.8 mg/kg. The control

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

article was administered to all 8 animals on Day 1 and then the test article was administered to the 30 and 78.8 mg/kg groups (4 animals/group) on Day 22. No test-article related changes were observed in blood, heart rate, electrocardiogram (ECG) parameters.

Central nervous and respiratory systems

In the safety pharmacology study noted above, male cynomolgus monkeys administered DS-8201a were evaluated for respiratory rate, blood gas parameters (arterial blood pH, arterial oxygen tension, arterial carbon dioxide tension, and hemoglobin oxygen saturation), intra-abdominal body temperature, presence or absence of arrhythmia, general behavior and neurobehavioral function were evaluated. No test-article related changes were observed in the examinations at single doses of 30 or 78.8 mg/kg.

The Applicant's Position:

DS-8201a had no effect on the cardiovascular, respiratory, or central nervous systems at doses of up to 78.8 mg/kg when administered once intravenously to male monkeys. In addition, MAAA-1181a had no effect on hERG currents at concentrations up to 10 µmol/L (approximately 5000 ng/mL) in hERG transfected CHO cells, at which there was a sufficient margin of exposure compared to the C_{max} of MAAA-1181a (7.2 ng/mL) in patients administered at 5.4 mg/kg of DS-8201a in the clinical study DS8201-A-U201.

The FDA's Assessment:

FDA agrees with the Applicant's conclusions of the hERG assay and single-dose safety pharmacology studies described above. However, in a repeat-dose toxicity study with DS-8201a in cynomolgus monkeys, shortening of the PR interval and QTc prolongation of up to 50 msec was observed in one male animal at 78.8 mg/kg (approximately 17 times the exposure at human recommended dose of 5.4 mg/kg based on AUC) on dosing Day 26.

5.4. ADME/PK

Data:

Pharmacokinetics

A Pharmacokinetic Study of DS-8201a after Single Intravenous Administration to Male Cynomolgus Monkeys: SBL315-043

In cynomolgus monkeys, after single intravenous administration of DS-8201a (0.1 mg/kg to 3 mg/kg), the plasma DS-8201a concentration decreased exponentially and the AUC of DS-8201a increased in a greater than dose-proportional manner. The CL of DS-8201a (14.0 mL/d/kg to 55.7 mL/d/kg) was much lower than the hepatic blood flow, and the CL decreased as the dose increased. The V_{ss} (30.6 mL/kg to 55.4 mL/kg) was close to the plasma volume. Both DS-8201a and the total antibody, the sum of conjugated and

unconjugated antibody, exhibited similar pharmacokinetic profiles. The plasma levels of MAAA-1181a, the released drug from DS-8201a, were quite low (molar ratio: about 1/400) compared with those of DS-8201a and total antibody. No anti-DS-8201a antibodies were detected in any animals.
Distribution Plasma Concentrations of Radioactivity, Excretion of Radioactivity in Urine and Feces, and Quantitative Whole-body Autoradiography after a Single Intravenous Administration of [¹⁴C]DS-8201a to Male Monkeys: AE-7542-G
After single intravenous administration of 6.4 mg/kg of ¹⁴C-labeled DS-8201a (¹⁴C-DS-8201a), the drug component of which was ¹⁴C-labeled, to cynomolgus monkeys, the highest radioactivity was observed in the blood during the study period, excluding in the large intestinal contents (an excretion site). Following the blood, radioactivity was distributed in highly perfused organs, such as kidney, lung, and liver, however, the tissue/blood ratios of the radioactivity of these organs were <1. The radioactivity in all tissues declined in proportion to the decline in the blood radioactivity, suggesting no retention in specific tissues.
Metabolism Structure elucidation of metabolites of DS-8201a with cryopreserved hepatocytes from rats, monkeys and humans: AM13-H0073-R01
No human-specific metabolites were detected in cryopreserved hepatocytes in vitro. The most representative peak, MAAA-1181a, which was released from DS-8201a after a 6-hour incubation with human cryopreserved hepatocytes, was estimated to account for roughly 0.1% of the total amount of MAAA-1181a that is conjugated to DS-8201a.
Excretion Plasma Concentrations of Radioactivity, Excretion of Radioactivity in Urine and Feces, and Quantitative Whole-body Autoradiography after a Single Intravenous Administration of [¹⁴C]DS-8201a to Male Monkeys: AE-7542-G
In monkeys after single intravenous administration of ¹⁴C-DS-8201a (6.4 mg/kg), the predominant route of excretion was fecal, with 67.3% and 18.7% of the dose found in feces and urine, respectively and MAAA-1181a was the only detectable catabolite in urine and feces.
TK data from general toxicology studies Intermittent Dose Toxicity Study in Monkeys Treated Intravenously with DS-8201a Once per 3 Weeks for 3 Months Followed by a 3-Month Recovery Period: SBL315-526
The C₀ and AUC_{21d} values of DS-8201a and total antibody generally increased along with the dose ranged from 3 mg/kg to 30 mg/kg. The C_{max} and AUC_{21d} values of MAAA-1181a also generally increased along with the dose ranged from 3 mg/kg to 30 mg/kg. Anti-DS-8201a antibodies were not detected in any animals during the dosing or recovery period.

TK data from reproductive toxicology studies
NA
TK data from carcinogenicity studies
NA

The Applicant's Position:

The CL of DS-8201a was much lower than the hepatic blood flow, and the CL decreased as the dose increased, suggesting non-linear elimination in animal species. The Vss was close to the plasma volume. Both DS-8201a and the total antibody, the sum of conjugated and unconjugated antibody, exhibited similar pharmacokinetic profiles. The plasma levels of MAAA-1181a, the released drug from DS-8201a, were low (molar ratio: about 1/400) compared with those of DS-8201a and total antibody. DS-8201a is considered to be metabolically stable in rat, monkey, and human cryopreserved hepatocytes in vitro. Intravenous (IV) administration of ¹⁴C-labeled MAAA-1181a to rats indicated that the major excretion pathway of radioactivity was feces via the biliary route. Neither apparent change after repeated dosing nor apparent sex differences were noted in the TK parameters. Anti-DS-8201a antibodies were not detected in any animals during the dosing or recovery period.

The FDA's Assessment:

FDA agrees with the Applicant's review of the ADME/PK data described above. Additional comments of the studies reviewed above, and FDA review of additional ADME/PK data is provided below.

Distribution

Based on whole-body autoradiography, there was no significant distribution of DS-8201a into the central nervous systems. The cerebellum and cerebrum had the lowest radioactivity of all tissues evaluated (maximum tissue-to-blood ratio of 0.02 or 0.01 for cerebellum and cerebrum, respectively, at 24 h). (Study AE-7542-G)

Studies evaluating the in vitro binding of MAAA-1181a and stability of DS-8201a in mouse, rat, monkey, or human plasma were reviewed under IND 127553. To evaluate in vitro plasma binding of MAAA-1181a, MAAA-1181c was added to plasma from different species at final concentrations of 10, 30, and 100 ng/mL as MAAA-1181a and the concentration of MAAA-1181a was determined by LC-MS/MS. The highest protein binding of MAAA-1181a was observed in human plasma, followed by rats, mice, and monkeys. (Study B131141) The results from the study is summarized in Table 4 below.

Table 4: In vitro plasma protein binding of MAAA-1181a in mouse, rat, monkey, and human plasma.

Species	MAAA-1181a (ng/mL)	Unbound fraction (%)	Protein binding ratio (%)
Mouse	10	7.5 ± 0.6	92.5 ± 0.6
	30	8.5 ± 0.6	91.5 ± 0.6
	100	9.7 ± 0.7	90.3 ± 0.7
Rat	10	3.3 ± 0.2	96.7 ± 0.2
	30	3.9 ± 0.2	96.1 ± 0.2
	100	5.8 ± 1.4	94.2 ± 1.4
Monkey	10	10.9 ± 0.3	89.1 ± 0.3
	30	13.1 ± 1.3	86.9 ± 1.3
	100	13.5 ± 0.8	86.5 ± 0.8
Human	10	2.0 ± 0.2	98.0 ± 0.2
	30	2.6 ± 0.3	97.4 ± 0.3
	100	3.2 ± 0.1	96.8 ± 0.1

Data represent the mean ± S.D. of three samples.

To evaluate the release rate of MAAA-1181a from DS-8201a in mouse, rat, monkey, and human plasma, 10 or 100 µg/mL DS-8201a was incubated in plasma in 1% BSA buffer for 21 days and MAAA-1181a concentrations were measured on Days 0, 1, 3, 7, 14, and 21. Unbound MAAA-1181a increased with time in plasma from all species. In human plasma, unbound MAAA-1181a appeared to plateau by Day 14, under the conditions tested. (Study PRD14-316)

Table 5: Concentrations of release MAAA-1181a at two DS-8201a concentrations in mouse, rat, monkey, and human plasma.

DS-8201a (µg/mL)	Time (days)	MAAA-1181a (ng/mL)			
		Mouse	Rat	Monkey	Human
10	0	0.06	0.06	0.06	0.04
	1	0.21	0.32	0.47	0.36
	3	0.5	1.07	2.13	1.29
	7	1.01	1.85	4.65	2.80
	14	1.9	3.55	9.18	5.49
	21	3.06	4.58	9.74	5.12
100	0	0.53	0.62	0.68	0.41
	1	2.2	3.49	5.42	4.24
	3	5.39	10.1	18.3	14.4
	7	11.1	11.9	16.9	20.5
	14	21.0	32.0	66.2	59.1
	21	29.0	42.8	85.8	37.4

TK Data From General Toxicology Studies

Table 6: Toxicokinetic Parameters in the 6-Week Intermittent Intravenous Dose Toxicity Study of DS-8201a in rats.

	Males (mg/kg)	Females (mg/kg)
--	---------------	-----------------

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	20 (n = 4)	60 (n = 4)	197 (n = 4)	20 (n = 4)	60 (n = 4)	197 (n = 4)
DS-8201a						
C₀ (µg/mL)						
Day 1	439	1400	4300	497	1510	5070
Day 22	613	1740	6410	615	1890	5730
AUC_{0-21d} (µg.d/mL)						
Day 1	1776	4903	13824	1869	4871	15096
Day 22	2676	6756	14264	2339	7391	19818
T_{1/2} (d)						
Day 1	8.07	8.52	5.78	7.61	8.67	6.88
Day 22	8.47	8.98	5.20	7.63	9.56	8.43
Total Antibody						
C₀ (µg/mL)						
Day 1	459	1530	4980	514	1470	4700
Day 22	611	2030	6560	632	1780	5970
AUC_{0-21d} (µg.d/mL)						
Day 1	2095	6017	17481	2182	6067	17036
Day 22	3035	8069	17895	2736	8448	22533
T_{1/2} (d)						
Day 1	9.79	10.4	6.12	9.76	10.4	7.86
Day 22	11.0	9.64	5.13	9.04	11.7	10.3
MAAA-1181a						
C₀ (µg/mL)						
Day 1	0.819	2.49	9.89	1.21	3.54	15.0
Day 22	1.41	4.2	13.4	1.19	4.36	14.0
AUC_{0-21d} (µg.d/mL)						
Day 1	2.88	8.57	31.9	2.45	6.86	28.1
Day 22	4.68	14.4	40.3	3.02	10.0	32.4
T_{1/2} (d)						
Day 1	4.24	3.90	3.57	3.16	3.46	5.34
Day 22	6.64	6.14	3.23	5.73	6.90	6.37

Values are expressed as the mean ± standard deviation.

Table 7: Toxicokinetic Parameters in the 3-Month Intermittent Intravenous Dose Toxicity Study of DS-8201a in Cynomolgus Monkeys.

	Males (mg/kg)			Females (mg/kg)		
	3 (n = 3)	10 (n = 5)	30 (n = 5)	3 (n = 3)	10 (n = 5)	30 (n = 5)
DS-8201a						
C₀ (µg/mL)						
Day 1	101	295	877	95.3	339	899
Day 64	78.2	320	898	98.4	352	1090
AUC_{0-21d} (µg.d/mL)						
Day 1	317	1220	4090	268	1080	3770
Day 64	286	1390	5030	285	1110	4910
T_{1/2} (d)						
Day 1	3.95	5.56	7.71	3.85	5.13	6.53
Day 64	4.32	6.08	9.02	3.84	5.05	7.40
Total Antibody						
C₀ (µg/mL)						

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Day 1	113	280	845	96.9	312	886
Day 64	78.4	295	949	106	320	1090
AUC_{0-21d} (µg.d/mL)						
Day 1	357	1280	4300	287	1130	3790
Day 64	297	1450	5960	319	1150	5500
T_{1/2} (d)						
Day 1	4.33	6.25	9.25	4.09	5.89	7.95
Day 64	4.73	7.04	9.56	4.18	5.80	8.31
MAAA-1181a						
C₀ (ng/mL)						
Day 1	0.242	0.656	2.71	0.248	1.02	3.9
Day 64	0.223	0.842	2.33	0.186	1.28	4.86
AUC_{0-21d} (µg.d/mL)						
Day 1	0.779	2.52	9.91	0.874	2.75	12.0
Day 64	0.634	3.06	11.0	0.294	2.39	13.5
T_{1/2} (d)						
Day 1	NC	3.33	7.15	NC	2.35	4.07
Day 64	NC	3.61	6.15	NC	2.36	3.72

Values are expressed as the mean ± standard deviation.

NC: Not calculated

5.5. Toxicology

5.5.1. General Toxicology

Data:

The binding profile of DS-8201a using recombinant protein-based enzyme-linked immunosorbent assay (ELISA) showed that it binds to the extracellular domain of humanized HER2 in cynomolgus monkeys and in humans but did not bind to the extracellular domain of HER2 in mice and rats. As the relevant species for safety assessments, cynomolgus monkeys were chosen as the cross-reactive species for DS-8201a, and rats (a non-cross-reactive species) were chosen to evaluate the target-independent effects. The intermittent dose studies were conducted in these animal species. An intravenous administration once every 3 weeks (q3w) was employed for the toxicity study of DS-8201a in correspondence to the clinical administration route and regimen. No single dose toxicity studies of DS-8201a were conducted, since the acute toxicity information is available from the repeated dose toxicity studies.

6-week Intermittent Intravenous Dose Toxicity Study of DS-8201a (Process-1) in Rats with a 9-week Recovery Period/ SBL315-030:

Key Study Findings

- No animals died or became moribund at dose levels up to 197 mg/kg.
- The target organs/tissues of DS-8201a included intestines, lymphatic/hematopoietic organs, kidneys, testes, skin, and teeth.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

- The histopathological changes that were observed at the end of the dosing period, excluding those in the testes and teeth, showed reversibility at the end of the 9-week recovery period.

Conducting laboratory and location: (b) (4)

GLP compliance: Yes

Methods

Dose and frequency of dosing: 20, 60, 197 mg/kg, once every 3 weeks (Days 1, 22, and 43), 3 times in total

Route of administration: Intravenous

Formulation/Vehicle: (b) (4)

Species/Strain: Crl:CD(SD) rats

Number/Sex/Group: 10M, 10F (recovery: 5M, 5F, [vehicle, 60 and 197 mg/kg])

Age: 7 weeks at start of dosing

Satellite groups/ unique design: toxicokinetic evaluation groups (4M, 4F for all doses)

Deviation from study protocol affecting interpretation of results: No

Observations and Results: changes from control

Parameters	Major findings
Mortality	No mortality occurred.
Clinical Signs	≥20 mg/kg: trauma and/or crust in the skin, defect of the teeth ≥60 mg/kg: whitening of the teeth 197 mg/kg: sparse fur and/or loss of fur
Body Weights	197 mg/kg: decreased body weight
Food consumption	197 mg/kg: decreased food consumption
Ophthalmoscopy	No test article-related changes
Hematology	≥20 mg/kg: decrease in reticulocyte ratio ≥60 mg/kg: decreases in leukocyte, lymphocyte, basophil, and neutrophil counts, decrease in eosinophil and increase in platelet count, decrease in large unstained cell count 197 mg/kg: decrease in monocyte count
Clinical Chemistry	60 mg/kg: increase in inorganic phosphorus 197 mg/kg : increases in creatinine and potassium, decreases in sodium and chloride, and increases in urea nitrogen
Urinalysis	≥60 mg/kg: increase in protein 197 mg/kg; decrease in sperm (no sperm in 14 of 15 males).
Gross Pathology	197 mg/kg : small-sized thymuses, and small-sized testes and epididymides
Organ Weights	197 mg/kg: low organ weight of thymus, testes, and epididymides

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Histopathology	≥20 mg/kg: single cell necrosis of crypt epithelial cells in small and large intestines, spermatid retention in testes ≥60 mg/kg: decreased erythroblasts in bone marrow, single cell necrosis of lymphocytes in the thymus, atrophy of follicles in the submandibular lymph nodes and ileac Peyer's patches, focal atrophy of villi in the duodenum, tubular basophilia and hyaline cast in kidneys, single cell necrosis in the hair follicle, ulcer/crust/epidermal thickening/fibrosis/inflammatory cell infiltration in the dermis, atrophy of the mammary gland, single cell necrosis in the base area and unifocal/multifocal degeneration of the enamel organ 197 mg/kg: decreased myelocyte in the bone marrow, atrophy of the thymus, inflammatory cell infiltration in the lamina propria and erosion in the mucosa in the duodenum, tubular degeneration/atrophy in the testes, luminal cell debris and reduced sperm in the epididymides, increased and abnormal dentin formation/hemorrhage in the sub-enamel organ tissue, focal lack of cementum and hypoplasia of the dentin, gingivitis Except for the testicular and incisor tooth changes, all changes that were observed at the end of the dosing period showed reversibility at the end of the 9-week recovery period.
Other evaluations	None

6-week Intermittent Intravenous Dose Toxicity Study of DS-8201a (Process-1) in Monkeys with a 6-week Recovery Period/ SBL315-031:

Key Study Findings

- One moribund female was found at 78.8 mg/kg.
- The target organs/tissues of DS-8201a included bone marrow, kidneys, intestines, testes, skin, and lungs.
- Except for pulmonary and skin toxicity (epidermal/dermal pigmentation), the findings showed reversibility or a trend of reversibility after the 6-week recovery period.

Conducting laboratory and location: (b) (4)

GLP compliance: Yes

Methods

Dose and frequency of dosing: 10, 30, 78.8 mg/kg, once every 3 weeks (Days 1, 22, and 43), 3 times in total

Route of administration: Intravenous

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

Formulation/Vehicle: (b) (4)

Species/Strain: Cynomolgus monkeys

Number/Sex/Group: 3M, 3F (recovery: 2M, 2F, [30 and 78.8 mg/kg])

Age: 3-7 years at start of acclimation

Satellite groups/ unique design: None

Deviation from study protocol affecting interpretation of results: No

Observations and Results: changes from control

Parameters	Major findings
Mortality	78.8 mg/kg: 1 female (Day 26) Findings: decreased body weight and food consumption, diarrhea, abnormal skin color (blackish brown), moderate decrease in spontaneous activity, hypothermia, decreased red blood cell parameters (erythrocytes, hemoglobin, hematocrit), increases in aspartate transaminase, alanine transaminase, creatine kinase, and urea nitrogen, decreased erythroblasts and myelocytes in bone marrow, single cell necrosis of crypt epithelial cells in the duodenum, tubular basophilia/ proliferation of the tubular epithelium/anisokaryosis in the proximal tubules/hyaline or cellular casts/ hyaline material/cellular infiltration in the interstitium in the kidneys, single cell necrosis in the hair follicles, and epidermal thickening and pigmentation in the skin and at the injection sites. The cause of moribundity was due to deteriorated condition which resulted from decreased body weight and food consumption as well as bone marrow toxicity and intestinal toxicity.
Clinical Signs	78.8 mg/kg: diarrhea, abnormal skin color (blackish brown)
Body Weights	78.8 mg/kg: decreased body weight
Food consumption	78.8 mg/kg: decreased food consumption
Ophthalmoscopy	No test article-related changes
ECG	78.8 mg/kg: shortening of the PR interval and prolongation of QTc (1 male)
Hematology	78.8 mg/kg: decreased red blood cell parameters (erythrocytes, hemoglobin, hematocrit, reticulocyte ratio)
Clinical Chemistry	≥10 mg/kg: increases in aspartate transaminase and alanine transaminase
Urinalysis	78.8 mg/kg: occult blood in urine
Gross Pathology	78.8 mg/kg: black foci in the skin, white foci in the lungs
Organ Weights	No test article-related changes
Histopathology	≥10 mg/kg: single cell necrosis of the crypt epithelial cells in the small and large intestines ≥30 mg/kg: single cell necrosis in the hair follicles in the skin and at the injection sites, and decreased number of round spermatids in Stage V-VI seminiferous tubule in the testes; focal interstitial inflammation,

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

	aggregation of the foamy alveolar macrophage, and cholesterol cleft in the alveolar macrophage in the lung (At 30 mg/kg, the lung lesions were observed only at the end of recovery period) 78.8 mg/kg: decreased erythroblasts and myelocytes in the sternal bone marrow, epidermal thickening and pigmentation in the skin, tubular basophilia/proliferation of the tubular epithelium/anisokaryosis in the proximal tubules/hyaline or cellular casts/hyaline material/cellular infiltration in the interstitium in the kidneys, edema in the alveolus in the lungs Except for the skin pigmentation and lung lesions, all changes showed reversibility or a trend of reversibility at the end of the 6-week recovery period.
Other evaluations	None

6-week Intermittent Intravenous Dose Toxicity Study of DS-8201a (Process-2) in Monkeys / SBL315-553:

Key Study Findings

- No animal died or became moribund at dose levels up to 30 mg/kg, the highest dose tested.
- Target organs/tissues of DS-8201a included bone marrow, intestines, and skin.

Conducting laboratory and location: (b) (4)

GLP compliance: Yes

Methods

Dose and frequency of dosing: 10, 30 mg/kg, once every 3 weeks (Days 1, 22, and 43), 3 times in total

Route of administration: Intravenous

Formulation/Vehicle: (b) (4) sucrose

Species/Strain: Cynomolgus monkeys

Number/Sex/Group: 3M, 3F

Age: 3-6 years at start of acclimation

Satellite groups/ unique design: None

Deviation from study protocol affecting interpretation of results: No

Observations and Results: changes from control

Parameters	Major findings
Mortality	No mortality occurred.

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

Clinical Signs	No test article-related changes
Body Weights	No test article-related changes
Food consumption	No test article-related changes
Ophthalmoscopy	No test article-related changes
ECG	No test article-related changes
Hematology	30 mg/kg: decreased red blood cell parameters (erythrocytes, hemoglobin, hematocrit)
Clinical Chemistry	≥10 mg/kg: increases in aspartate transaminase, lactate dehydrogenase, and creatine kinase
Urinalysis	No test article-related changes
Gross Pathology	No test article-related changes
Organ Weights	No test article-related changes
Histopathology	≥10 mg/kg: single cell necrosis of the crypt epithelial cells in the small and large intestines 30 mg/kg: single cell necrosis in the hair follicle in the skin and at the injection sites, and decreased erythroblasts in the sternal bone marrow
Other evaluations	None

3-month Intermittent Intravenous Dose Toxicity Study of DS-8201a (Process-2) in Monkeys with a 3-month Recovery Period / SBL315-526:

Key Study Findings

- No animal died or became moribund at up to 30 mg/kg, the highest dose tested.
- The target organs/tissues of DS-8201a included bone marrow, kidneys, intestines, testes, skin, and lungs.
- Except for renal and skin toxicity (epidermal pigmentation), the findings showed reversibility or a trend of reversibility after the 3-month recovery period.

Conducting laboratory and location: (b) (4)

GLP compliance: Yes

Methods

Dose and frequency of dosing: 3, 10, 30 mg/kg, once every 3 weeks (Days 1, 22, 43, 64, and 85), 5 times in total

Route of administration: Intravenous

Formulation/Vehicle: (b) (4) sucrose

Species/Strain: Cynomolgus monkeys

Number/Sex/Group: 4M, 4F (recovery: 2M, 2F, [only at 30 mg/kg])

Age: 3-10 years at start of acclimation

Satellite groups/ unique design: None

Deviation from study protocol affecting interpretation of results: No

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

Observations and Results: changes from control

Parameters	Major findings
Mortality	No mortality occurred.
Clinical Signs	30 mg/kg: abnormal skin color (blackish brown)
Body Weights	No test article-related changes
Food consumption	No test article-related changes
Ophthalmoscopy	No test article-related changes
ECG	No test article-related changes
Hematology	30 mg/kg: a decrease in reticulocyte ratio
Clinical Chemistry	30 mg/kg: increases in aspartate transaminase, lactate dehydrogenase, and creatine kinase
Urinalysis	No test article-related changes
Gross Pathology	30 mg/kg: black foci in the skin, white foci in the lungs
Organ Weights	No test article-related changes
Histopathology	

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Dose (mg/kg)	0 (Control)		3		10		30	
No. of Animals	M:4	F:4	M:4	F:4	M:4	F:4	M:6	F:6
Main (No. examined)	(4)	(4)	(4)	(4)	(4)	(4)	(4)	(4)
Bone marrow (sternal): Decrease, erythroblast	0	0	0	0	0	0	3	3
Bone marrow (sternal): Decrease, myelocyte	0	0	0	0	0	0	1	0
Bone marrow (sternal): Brown pigment deposition, macrophage	1	0	0	1	1	0	4	1
Duodenum: Single cell necrosis, epithelial cell, crypt	0	0	4	1	3	4	4	4
Jejunum: Single cell necrosis, epithelial cell, crypt	0	0	3	0	1	3	4	4
Ileum: Single cell necrosis, epithelial cell, crypt	0	0	4	0	1	2	4	3
Cecum: Single cell necrosis, epithelial cell, crypt	0	0	1	0	1	1	4	2
Colon: Single cell necrosis, epithelial cell, crypt	0	0	0	0	0	0	1	0
Rectum: Single cell necrosis, epithelial cell, crypt	0	0	4	0	0	1	4	2
Kidney: Anisokaryosis, proximal tubule	0	0	0	0	0	0	2	1
Injection site: Single cell necrosis, hair follicle	0	0	0	0	0	0	2	0
Skin: Single cell necrosis, hair follicle	0	0	0	0	0	1	3	2
Skin: Pigmentation, epidermal	0	0	0	0	0	0	2	1
Spleen: Brown pigment deposition, macrophage	0	0	0	0	0	0	2	0
Testis: Decrease, round spermatid, Stage V-VI	0	NA	0	NA	0	NA	1	NA
Lung: Aggregation, foamy alveolar macrophage	0	0	0	0	0	0	3	1
Lung: Inflammation, alveolus, focal	0	0	0	0	0	0	1	0
Lung: Inflammation, interstitial, focal	0	0	0	0	0	0	3	1
Liver: Brown pigment deposition, Kupffer cell/hepatocyte/bile canaliculus	0	0	0	0	0	0	1	0
Recovery (No. examined)	(0)	(0)	(0)	(0)	(0)	(0)	(2)	(2)
Kidney: Anisokaryosis, proximal tubule	NE	NE	NE	NE	NE	NE	1	1
Skin: Pigmentation, epidermal	NE	NE	NE	NE	NE	NE	1	1
Lung: Aggregation, foamy alveolar macrophage	NE	NE	NE	NE	NE	NE	1	0

NA: not applicable; NE: not examined.

The changes observed at the end of the dosing period, except for the skin and kidney lesions, showed recovery or tendency toward recovery.

Other evaluations	No test article-related changes were noted in cardiac function, chest computed tomography examinations, or cardiac troponin I measurement.
--------------------------	--

The Applicant's Position:

In 6-week intermittent dose toxicity studies of DS-8201a (Process-1) in rats (SBL315-030), the highest dose levels selected for these studies were maximum feasible levels, taking into account the tolerable range of dosing volume and formulation concentration. The dose levels for intravenous administration in the rat study were 20, 60, and 197 mg/kg. No deaths or moribund animals were found at dose levels up to 197 mg/kg. The major toxicity was observed in intestines, and testes at ≥ 20 mg/kg, and in lymphatic/hematopoietic organs, kidneys, skin,

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

and incisor tooth at ≥ 60 mg/kg. These changes were all found to recover except for the testicular and incisor tooth changes that were observed at ≥ 60 mg/kg.

In a 6-week intermittent dose (q3w) toxicity study of DS-8201a (Process-1) in cynomolgus monkeys (SBL315-031), the dose levels for intravenous administration in the cynomolgus monkey study were 10, 30, and 78.8 mg/kg. One moribund female was found at 78.8 mg/kg. The major toxicity findings in this moribund animal were intestinal toxicity, hematopoietic organ toxicity, skin toxicity, and renal toxicity. The cause of moribundity appeared to be a result of the deteriorated clinical condition, which was associated with decreased body weight and food consumption, as well as bone marrow toxicity and intestinal toxicity. The major findings in the surviving animals were intestinal toxicity at ≥ 10 mg/kg, and pulmonary, testicular, and skin toxicity at ≥ 30 mg/kg. In addition, hematopoietic organ toxicity and renal toxicity, as well as ECG abnormalities (PR interval shortened and prolongation of QT interval corrected with Bazett's formula [QTc]) were found at 78.8 mg/kg. These findings except for the pulmonary toxicity and skin toxicity (pigmentation) showed reversibility after the 6-week recovery period.

In a 6-week intermittent dose (q3w) toxicity study of DS-8201a (Process-2) in cynomolgus monkeys (SBL315-553, dose levels: 10 and 30 mg/kg), the major findings of toxicity included intestinal toxicity, hematopoietic organ toxicity, and skin toxicity, which were also observed in the previous 6-week intermittent dose (q3w) toxicity study of DS-8201a (Process-1) in cynomolgus monkeys. In a 3-month intermittent dose (q3w) toxicity study of DS-8201a (Process-2) in monkeys (dose levels: 3, 10 and 30 mg/kg), the target organs/tissues of DS-8201a included the bone marrow, kidneys, intestines, testes, skin, and lungs. None of the test article-related changes suggested severe toxicity of DS-8201a (Process 2) at up to 30 mg/kg.

The FDA's Assessment:

Studies SBL315-030 (AN14-C0001-R01) and SBL315-031 (AN14-C0002-R01) were reviewed under IND 127553. Overall, FDA agrees with Applicant's review of these studies described above. In the 6-week repeat dose toxicity study in rats (Study SBL315-030), target organs of toxicity included the testes/epididymides, teeth, kidneys, skin, GI tract, bone marrow, and thymus. The histopathological findings in the testes/epididymis, teeth, and kidneys (hyaline/cellular cast) did not fully recover by the end of the recovery period. In the 6-week repeat dose toxicity study in monkeys (SBL315-031), one mortality occurred at 78.8 mg/kg (euthanized due to moribundity on Day 26). Target organs of toxicity included GI tract, lung, testes, skin, kidney, bone marrow, and thymus. All findings were reversible or trended towards recovery, since all findings were either absent or very slight in severity by the end of the 6-week recovery period.

FDA agrees with Applicant's overall conclusions for the 6-week (Study SBL315-553) and 3-month (Study SBL315-526) repeat-dose toxicity studies with DS-8201a (Process-2) described above. In general, the findings were consistent with those observed in previously conducted 6-week repeat-dose toxicity studies with DS-8201a (Process-1), and no additional toxicity findings

were identified in the 3-month study. Major target organs of toxicity in these studies included the bone marrow (decreases in reticulocyte count correlated to slight to moderate decreases in erythroblasts), skin (abnormal skin color and black foci correlated to very slight hair follicle single cell necrosis and slight epidermal pigmentation), lung (white foci correlated to very slight to slight foamy alveolar macrophage aggregation and focal interstitial and alveolus inflammation), testis (very slight Stage V-VI round spermatid decreases and sperm stasis, and slight tubular degeneration/atrophy), kidney (very slight proximal tubule anisokaryosis), and the gastrointestinal system (very slight single cell necrosis of epithelial cells). Except for the findings in the skin and kidneys, all the other histopathological findings were reversible by the end of the recovery period. Based on the histological findings in the testis at ≥ 30 mg/kg (≥ 7 times the exposure at the human recommended dose of 5.4 mg/kg based on AUC), Enhertu may impair male reproductive function and fertility.

General toxicology; additional studies

The Applicant's Position:

No additional general toxicity studies with DS-8201a have been conducted.

The FDA's Assessment:

FDA agrees that no additional general toxicity studies were conducted or needed to support this BLA submission.

5.5.2. Genetic Toxicology

In Vitro Reverse Mutation Assay in Bacterial Cells (Ames)

Data:

Bacterial Reverse Mutation Study of MAAA-1181a/ SBL315-617

Key Study Finding:

- MAAA-1181a did not induce gene mutation in this test system.

GLP compliance: Yes

Test system:

MAAA-1181a monohydrate dissolved in dimethyl sulfoxide (DMSO) were dosed to each bacterial strain (4 *Salmonella typhimurium* strains [TA100, TA1535, TA98, and TA1537], and an *Escherichia coli* strain [WP2uvrA]) in either the absence or presence of a metabolic activation system (rat liver 9000 g supernatant [S9] mix); the overall dose levels ranged from 313 μ g to 5000 μ g per plate (calculated as MAAA-1181a). In addition, a negative control (DMSO) and

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

positive controls (4-nitroquinoline 1-oxide, sodium azide, 9-aminoacridine hydrochloride monohydrate, or 2-aminoanthracene) were included.

The Applicant's Position:

The in vitro genotoxicity study indicated that MAAA-1181a had no potential to induce gene mutation in bacteria.

The FDA's Assessment:

FDA agrees with the study results described above.

In Vitro Chromosomal Damage Assay in Mammalian Cells (Chromosomal Aberration Assay)
Data:

Chromosome Aberration Study of MAAA-1181a with Mammalian Cultured Cells/ SBL315-618

Key Study Finding:

- MAAA-1181a induced structural chromosome aberrations in this test system.

GLP compliance: Yes

Test system:

MAAA-1181a monohydrate dissolved in DMSO dosed to mammalian Chinese hamster lung cells. The cells were treated with 0.0125 µg to 3 µg/mL of MAAA-1181a in either the absence or presence of a metabolic activation system (S9 mix). In addition, a vehicle control (DMSO) and positive controls (mitomycin C and cyclophosphamide monohydrate) were included.

Treatment Time (h)	Metabolic Activation	Test Article	Dose (µg/mL) ^a	Relative Cell No. (%)	Aberrant Cell No. (%)	Polyploid Cell No. (%)
6	Without Activation	Vehicle	0	100	0.0	0.0
		MAAA-1181a monohydrate	0.05	98.2	0.0	0.3
			0.1	98.5	3.3*	0.0
			0.2	71.3	16.7*	0.0
			0.3	59.6	NE	NE
			0.4	57.1	20.0*	0.0
			0.5	38.2	NE	NE
			0.8	40.7	NE	NE
			1	32.4	NE	NE
		Mitomycin C	0.15	68.4	15.3	0.0
6	With Activation	Vehicle control	0	100	0.3	0.3
		MAAA-1181a monohydrate	0.05	96.9	1.0	0.3
			0.1	103.1	10.3*	0.3
			0.2	94.2	12.7*	0.0

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

			0.4	67.3	12.3*	0.7
			0.8	61.9	NE	NE
			1	52.5	14.3*	1.3
			1.5	46.6	NE	NE
			3	39.9	NE	NE
		Cyclophosphamide monohydrate	10	78.5	16.7	0.3
24	Without Activation	Vehicle control	0	100	0.0	0.7
		MAAA-1181a monohydrate	0.0125	100.0	0.7	0.0
			0.025	106.6	3.7*	0.3
			0.05	103.5	5.3*	0.3
			0.075	85.7	NE	NE
			0.1	73.4	19.7*	0.0
			0.15	64.9	NE	NE
			0.2	51.0	51.3*	0.0
			0.3	46.3	NE	NE
		Mitomycin C	0.05	80.3	12.7	0.7

^a Calculated as MAAA-1181a

NE: not examined; * A statistically significant increase compared with vehicle control by the Fisher's exact test ($p < 0.05$, two-tailed)

The Cochran-Armitage trend test ($p < 0.05$, two-tailed) showed these increase in the incidence of structural aberrations to be dose dependent.

The Applicant's Position:

The in vitro genotoxicity study indicated that MAAA-1181a had the potential to induce structural chromosome aberrations in mammalian cultured cells.

The FDA's Assessment:

FDA agrees with the Applicant's conclusion that MAAA-1181a was clastogenic in the in vitro Chinese hamster lung chromosome aberration assay, under the conditions tested.

In Vivo Chromosomal Damage Assay in Rodent (Micronucleus Assay)

Data:

Micronucleus Study in Bone Marrow of Rats Treated Intravenously with MAAA-1181a/SBL315-756

Key Study Finding:

- MAAA-1181a induced micronuclei in this test system.

GLP compliance: Yes

Test system:

A micronucleus study of MAAA-1181a was performed in male Crl:CD(SD) rats. MAAA-1181a monohydrate dissolved in physiological saline was intravenously administered once at dose levels of 0, 0.025, 0.05, 0.1, and 0.2 mg/kg (calculated as MAAA-1181a). Bone marrow was obtained for the evaluation of micronucleated immature erythrocytes (MNIEs) at 24 hours after

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

administration. Five male animals were used for each group. The number of MNIEs and the proportion of immature erythrocytes (IEs) were examined. In addition, a negative control group (treated with physiological saline) was included, and preserved positive control (cyclophosphamide monohydrate) specimens were used as the positive control.

Test Article	Dose (mg/kg)	No. of Animals	IE% (Mean ± SD)	Number of MNIEs (/20000IEs)	MNIE% (Mean ± SD)
Physiological saline	0	5:M	47.3 ± 4.1	37	0.19 ± 0.04
MAAA-1181a monohydrate	0.025 ^a	5:M	46.4 ± 3.7	50	0.25 ± 0.08
	0.05 ^a	5:M	47.0 ± 0.8	68#	0.34 ± 0.09
	0.1 ^a	5:M	48.6 ± 3.1	107#	0.54 ± 0.10
	0.2 ^a	5:M	42.4 ± 5.2	324#	1.62 ± 0.38
Cyclophosphamide monohydrate	20	5:M	40.5 ± 5.5	488#	2.44 ± 0.78

^a Calculated as MAAA-1181a

IE%, Proportion of immature erythrocytes to total erythrocytes; MNIE%, Incidence of micronucleated immature erythrocytes. Statistics (vs vehicle control): # $P < 0.01$ (Kastenbaum and Bowman's table, one-tailed)

The Applicant's Position:

The in vivo micronucleus study in bone marrow of rats indicated that MAAA-1181a had the potential to induce micronuclei.

The FDA's Assessment:

FDA agrees with the Applicant's conclusion that MAAA-1181a was clastogenic in the in vivo rat bone marrow micronucleus assay, under the conditions tested.

Other Genetic Toxicity Studies

The Applicant's Position:

No other genetic toxicity studies were conducted.

The FDA's Assessment:

FDA agrees.

5.5.3. Carcinogenicity

The Applicant's Position:

No carcinogenicity studies have been conducted in accordance with ICH S9.

The FDA's Assessment:

FDA agrees that carcinogenicity studies are not needed to support this BLA submission.

5.5.4. Reproductive and Developmental Toxicology

The Applicant's Position:

In accordance with ICH S9, no studies were conducted to assess fertility and early embryonic development to implantation, effects on pre and postnatal development, including maternal function, or embryo-fetal developmental toxicity.

Effects on the reproductive system following the intravenous administration of DS-8201a were histopathologically evaluated in the intermittent dose toxicity studies in rats and cynomolgus monkeys. Toxicology studies in rats and monkeys with DS-8201a or MAAA-1181a indicated that both had toxic effects on rapidly dividing cells (lymphatic/hematopoietic organs, intestines, or testes), and MAAA-1181a was genotoxic in the in vitro chromosome aberration study and in vivo micronucleus study. In addition, trastuzumab has been known to cause fetal harm when administered to a pregnant woman. Based on this evidence suggesting the potential of DS-8201a to cause fetal harm and consistent with ICH S9, reproductive and developmental toxicity studies were not conducted.

The FDA's Assessment:

FDA agrees. An assessment of the potential for fam-trastuzumab deruxtecan-nxki to cause impairment of fertility was provided based on results from general toxicology studies with DS-8201a. The results from these studies will be included in Section 13.1 of the PL.

5.5.5. Other Toxicology Studies

Local Tolerance

The Applicant's Position:

No local tolerance studies have been conducted. Local irritation following the intravenous administration of DS-8201a was evaluated in intermittent dose toxicity study in rats and cynomolgus monkeys (SBL315-030, SBL315-031, SBL315-526, SBL315-553). As a result of macro- and microscopic examinations at the injection sites, there were no DS-8201a-related findings suggestive of irritative effects at up to the highest dose levels tested.

The FDA's Assessment:

FDA agrees with the study results described above.

Tissue Cross-reactivity of DS-8201a

The Applicant's Position:

The potential tissue cross-reactivity of DS-8201a was evaluated using a normal human tissue panel (38 tissues) from at least 3 donors at 1 and 10 µg/mL. Syncytiotrophoblasts and decidual cells in the placenta only showed positive reactions on the cellular membrane, which was thought to be the compartment that the antibody has access to in vivo (study 20064734). In a cross-reactivity study of DS-8201a with selected cynomolgus monkey tissues (ie, bone marrow, brain, heart, intestines, kidney, liver, lung, skin, spleen, and testis) from 3 animals at 1 and 10 µg/mL, neither membranous nor cytoplasmic staining was noted. No staining on the cell membrane in the selected cynomolgus monkey tissues was consistent with the findings in the corresponding human tissues (study 20069107).

The FDA's Assessment:

Tissue cross-reactivity studies with human (study no. an14-c0051-r01) and cynomolgus monkey (study no. an14-c0120-r01) tissues were reviewed under IND 127533. FDA agrees with the Applicant's conclusions that no cross-reactivity of DS-8201a was observed in the cryosections of the select panel of cynomolgus monkey tissues and that membrane staining was only observed in placental syncytiotrophoblasts and/or decidual cells in human tissues.

Repeated dose toxicity of MAAA-1181a

Data:

4-week Intermittent Intravenous Dose Toxicity Study of MAAA-1181a in Rats with a 4-week Recovery Period/ SBL315-026:

Key Study Findings

- No animals died or became moribund at dose levels up to 30 mg/kg.
- The target organs/tissues of MAAA-1181a included intestines, lymphatic/hematopoietic organs and cornea.
- Changes observed during the dosing period showed reversibility.

Conducting laboratory and location:

(b) (4)

GLP compliance: Yes

Methods

Dose and frequency of dosing: 3, 10, 30 mg/kg, once weekly (Days 1, 8, 15, 22, and 29), 5 times in total

Route of administration: Intravenous

Formulation/Vehicle: Physiological saline

Species/Strain: CrI:CD(SD) rats

Number/Sex/Group: 10M, 10F (recovery: 5M, 5F, [vehicle and 30 mg/kg])

Age: 7 weeks at start of dosing

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

Satellite groups/ unique design: toxicokinetic evaluation groups (4M, 4F for all doses)
Deviation from study protocol affecting interpretation of results: No

Observations and Results: changes from control

Parameters	Major findings
Mortality	No mortality occurred.
Clinical Signs	No test article-related changes
Body Weights	≥3 mg/kg: transient suppression of body weight gain
Food consumption	≥3 mg/kg: low food consumption
Ophthalmoscopy	No test article-related changes
Hematology	≥3 mg/kg: decreased red blood cell parameters (erythrocyte, hemoglobin, hematocrit, reticulocyte ratio), and decreased white blood cell parameters (leukocyte, neutrophil, lymphocyte, monocyte, eosinophil, basophil)
Clinical Chemistry	≥3 mg/kg: high serum glucose levels
Urinalysis	No test article-related changes
Gross Pathology	No test article-related changes
Organ Weights	≥3 mg/kg: low thymus weight
Histopathology	≥3 mg/kg: decreased erythroblasts and myelocytes in the bone marrow, single cell necrosis of lymphocytes in the thymus, atrophy of follicles in the submandibular lymph nodes and/or ileac Peyer's patches, single cell necrosis of crypt epithelial cells in the small and large intestines, and in the corneal epithelium; and atrophy of villi in the duodenum 30 mg/kg: atrophy of the thymus and regeneration of the crypt epithelial cells with luminal dilatation in the cecum All changes that were observed at the end of the dosing period showed reversibility.
Other evaluations	None

4-week Intermittent Intravenous Dose Toxicity Study of MAAA-1181a in Monkeys with a 4-week Recovery Period/ SBL315-032:

Key Study Findings

- One female died and one male was found moribund at 12 mg/kg.
- The target organs/tissues of MAAA-1181a included heart, intestines, lymphatic/hematopoietic organs and cornea.
- Changes observed in the surviving animals during the dosing period showed reversibility.

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

Conducting laboratory and location:

(b) (4)

GLP compliance: Yes

Methods

Dose and frequency of dosing: 1, 3, 12 mg/kg, once weekly (Days 1, 8, 15, 22, and 29),
5 times in total

Route of administration: Intravenous

Formulation/Vehicle: Physiological saline

Species/Strain: Cynomolgus monkeys

Number/Sex/Group: 3M, 3F (recovery: 2M, 2F, [12 mg/kg])

Age: 3-8 years at start of acclimation

Satellite groups/ unique design: None

Deviation from study protocol affecting interpretation of results: No

Observations and Results: changes from control

Parameters	Major findings
Mortality	12 mg/kg: 1 male (moribund on Day 12), 1 female (died on Day 23) Findings: vomiting, diarrhea, lateral position, hypothermia, pale oral mucosa, and/or suppression of touch response, decreased body weight and food consumption, decreased in hematocrit value and lymphocyte count, increases in aspartate transaminase and alanine transaminase, decreases in sodium and chloride, degeneration/necrosis of cardiac myocytes in the heart (only in the male), decreased erythroblasts and myelocytes in the bone marrow, atrophy of follicles and the periarterial lymphatic sheath in the spleen, atrophy of follicles in the mesenteric lymph nodes and ileac Peyer's patches, single cell necrosis, focal necrosis, and increased mitosis of hepatocytes, dilatation of the bile canaliculus, bile thrombus, brown pigment deposition in Kupffer cells in the liver, and regeneration of crypt epithelial cells, single cell necrosis of the crypt epithelial cells and/or dilatation of the crypt in the small and large intestines. The cause of morbidity was due to deteriorated condition resulting from decreased body weight and food consumption as well as bone marrow toxicity and intestinal toxicity
Clinical Signs	≥3 mg/kg: vomiting 12 mg/kg: diarrhea/soft stool
Body Weights	≥3 mg/kg: decreased body weight
Food consumption	12 mg/kg: decreased food consumption
Ophthalmoscopy	No test article-related changes
Hematology	≥1 mg/kg: decrease in reticulocyte ratio 12 mg/kg: decreases in red blood cell parameters (erythrocyte, hemoglobin, and hematocrit) and

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	lymphocyte count, increase in platelet count, prolongation and shortening of activated partial thromboplastin time
Clinical Chemistry	≥3 mg/kg: increases in aspartate transaminase 12 mg/kg: increase in alanine transaminase, and increases in inorganic phosphorus and potassium
Urinalysis	No test article-related changes
Gross Pathology	No test article-related changes
Organ Weights	No test article-related changes
Histopathology	≥1 mg/kg: atrophy of follicles in the spleen, mesenteric lymph nodes, and ileac Peyer's patches, and single cell necrosis of lymphocytes in the thymus 12 mg/kg: single cell necrosis in the hepatocytes, single cell necrosis of crypt epithelial cells in the small intestine, and single cell necrosis in the corneal epithelium All changes that were observed at the end of the dosing period showed reversibility.
Other evaluations	None

The Applicant's Position:

For the purpose of clarifying the toxicity of the released drug component of DS-8201a, 4-week intermittent dose toxicity studies of MAAA-1181a were conducted. MAAA-1181a was administered intravenously (once weekly) to rats at 0, 3, 10, and 30 mg/kg (SBL315-026) and to cynomolgus monkeys at 0, 1, 3, and 12 mg/kg (SBL315-032). In the 4-week intermittent intravenous dose (once weekly) toxicity study of MAAA-1181a in rats, although lymphatic/hematopoietic organ toxicity, intestinal toxicity, and corneal epithelium toxicity were observed at ≥3 mg/kg, no deaths occurred at dose levels of up to 30 mg/kg. Findings similar to those in rats were observed in cynomolgus monkeys at dose levels of ≥1 mg/kg. However, 1 female monkey died, and 1 male was found moribund at 12 mg/kg. Effects on the heart (myocardial cell degeneration/necrosis) were found in the moribund male in addition to other toxicities, while no cardiotoxicity was reported in the dead female. Both animals exhibited similar worsening clinical condition which was associated with sustained decreases in food consumption, and bone marrow and intestinal toxicity. Therefore, these changes were considered to be the cause of the death and moribundity. In the surviving animals, the following changes that were similar to those noted in the animals that died prematurely included: histopathological changes in the lymphatic/hematopoietic organs (spleen, thymus, and bone marrow) at ≥1 mg/kg, and findings in the small and large intestines, liver, and corneal epithelium of the eye at 12 mg/kg.

The FDA's Assessment:

Toxicology studies with MAAA-1181a were reviewed under 127553. FDA agrees with the

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

conclusions described above.

Repeated dose toxicity of MAAL-9001

Data:

6-week Intermittent Intravenous Dose Toxicity Study of MAAL-9001 in Rats/ SBL315-144:

Key Study Findings

- No test article-related toxic changes were observed.

Conducting laboratory and location: (b) (4)

GLP compliance: Yes

Methods

Dose and frequency of dosing: 197 mg/kg, once every 3 weeks (Days 1, 22, and 43),
3 times in total

Route of administration: Intravenous

Formulation/Vehicle: (b) (4)

Species/Strain: Crl:CD(SD) rats

Number/Sex/Group: 10M, 10F

Age: 7 weeks at start of dosing

Satellite groups/ unique design: toxicokinetic evaluation groups (4M, 4F at 197 mg/kg)

Deviation from study protocol affecting interpretation of results: No

Observations and Results: changes from control

Parameters	Major findings
Mortality	No mortality occurred.
Clinical Signs	No test article-related changes
Body Weights	No test article-related changes
Food consumption	No test article-related changes
Ophthalmoscopy	No test article-related changes
Hematology	No test article-related changes
Clinical Chemistry	No test article-related changes
Urinalysis	No test article-related changes
Gross Pathology	No test article-related changes
Organ Weights	No test article-related changes
Histopathology	No test article-related changes
Other evaluations	None

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

6-week Intermittent Intravenous Dose Toxicity Study of MAAL-9001 in Monkeys/ SBL315-143:

Key Study Findings

- No test article-related toxic changes were observed.

Conducting laboratory and location:

(b) (4)

GLP compliance: Yes

Methods

Dose and frequency of dosing: 78.8 mg/kg, once every 3 weeks (Days 1, 22, and 43),
3 times in total

Route of administration: Intravenous

Formulation/Vehicle:

(b) (4)

Species/Strain: Cynomolgus monkeys

Number/Sex/Group: 3M, 3F

Age: 3-5 years at start of acclimation

Satellite groups/ unique design: None

Deviation from study protocol affecting interpretation of results: No

Observations and Results: changes from control

Parameters	Major findings
Mortality	No mortality occurred.
Clinical Signs	No test article-related changes
Body Weights	No test article-related changes
Food consumption	No test article-related changes
Ophthalmoscopy	No test article-related changes
ECG	No test article-related changes
Hematology	No test article-related changes
Clinical Chemistry	No test article-related changes
Urinalysis	No test article-related changes
Gross Pathology	No test article-related changes
Organ Weights	No test article-related changes
Histopathology	No test article-related changes
Other evaluations	None

The Applicant's Position:

For the purpose of clarifying the toxicity of the unconjugated antibody component of DS-8201a, 6-week intermittent dose toxicity studies of MAAL-9001 were conducted. MAAL-9001 was administered intravenously (once every 3 weeks) to rats at 0 and 197 mg/kg (SBL315-144) and

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

to cynomolgus monkeys at 0 and 78.8 mg/kg (SBL315-143). In these toxicity studies, no test article-related toxic changes were observed.

The FDA's Assessment:

FDA agrees with the conclusions described above. According to the ICH S9 Q&A guidance, studies with the unconjugated antibody are usually not needed.

Phototoxicity of MAAA-1181a

Data:

In Vitro 3T3 NRU Phototoxicity Test with MAAA-1181a/SBL315-101

Key Study Finding:

- MAAA-1181a was phototoxic to Balb/c 3T3 mouse fibroblasts.

GLP compliance: Yes

Test system:

The potential phototoxicity of MAAA-1181a was evaluated using in vitro Balb/c 3T3 mouse fibroblasts. The highest concentration was determined to be 25 µg/mL based on the results of the dose range-finding test, and 8 concentrations of MAAA-1181a ranging from 0.195 µg/mL to 25 µg/mL were tested. For evaluation of phototoxicity, the cells were exposed to 5 J/cm² of ultraviolet A (UVA) irradiation in total. The calculated mean photo effect of 0.432 was >0.15, the criteria for a positive result. Therefore, MAAA-1181a was considered to be phototoxic under these study conditions.

Single Dose Phototoxicity Study in Pigmented Rats Treated Intravenously with MAAA-1181a Monohydrate/ SBL315-450

Key Study Finding:

- MAAA-1181a had no phototoxic potential in pigmented rats.

GLP compliance: Yes

Test system:

In an in vivo phototoxicity study in male lar:Long-Evans pigmented rats (5 rats/group), MAAA-1181a monohydrate was intravenously administered at dose levels of 0, 1, and 3 mg/kg (calculated as MAAA-1181a) followed by a single exposure to UVA irradiation (10 J/cm²) at 0.5 hours after dosing. Under UVA irradiation, the highest dose (3 mg/kg) resulted in plasma concentrations of MAAA-1181a (90.5 ng/mL), which were beyond the C_{max} (7.2 ng/mL) at the recommended clinical dose of 5.4 mg/kg (DS8201-A-U201). No abnormalities related to MAAA-1181a and UVA irradiation were observed in the macroscopic examinations of the skin, auricular thickness measurement, or macroscopic examinations of the eyes.

The Applicant's Position:

In an in vitro 3T3 NRU phototoxicity study, MAAA-1181a was found to be phototoxic to Balb/c 3T3 mouse fibroblasts. However, in an in vivo single dose phototoxicity study of MAAA-1181a in pigmented rats, no phototoxic reaction was noted at 3 mg/kg, the highest dose tested. The plasma concentration of MAAA-1181a in rats at 3 mg/kg was 90.5 ng/mL, which is 13 times higher than the observed C_{max} (7.2 ng/mL) in humans given the recommended clinical dose of 5.4 mg/kg of DS-8201a in the clinical study DS8201-A-U201. Based on the findings from the in vivo study, the risk of phototoxicity following the administration of DS-8201a is considered limited.

The FDA's Assessment:

The in vitro phototoxicity study was reviewed under IND 127553. FDA agrees with the Applicant's conclusions that MAAA-1181a induced cell death of Balb/c 3T3 mouse fibroblasts in the presence of irradiation (IC₅₀ = 2.356 mcg/mL) and that no phototoxic reaction was observed at the highest dose tested of 3 mg/kg (approximately 13 times the exposure at the human recommended dose of 5.4 mg/kg based on C_{max}).

X

X

Haw-Jyh Chiu

Tiffany Ricks

Primary Reviewer

Supervisor

6 Clinical Pharmacology

6.1. Executive Summary

The FDA's Assessment:

Fam-trastuzumab deruxtecan (DS-8201a) is an antibody drug conjugates (ADC) targeting HER2-positive expressing cells. The payload (MAAA-1181a or DXd) is a DNA topoisomerase I inhibitor. The Applicant is seeking approval of fam-trastuzumab deruxtecan (b) (4)

. The proposed dosing regimen is 5.4 mg/kg as IV infusion (90 min) once every 3 weeks (21-days cycle).

The clinical pharmacology review focused on the dose selection, organ dysfunctions, and drug-drug interactions.

Recommendations: The Office of Clinical Pharmacology has reviewed the information submitted in BLA 761139. This BLA is approvable from a clinical pharmacology perspective. The key review issues with specific recommendations/comments are summarized below in Table 8.

Table 8: Key Clinical Pharmacology Review Issues by FDA

Review Issue	Recommendations and Comments
Pivotal and Supportive evidence of effectiveness	The primary evidence of effectiveness comes from 2 studies, Phase 2 study DS8201-A-U201 and Phase 1 study DS8201-A-J101. The proposed dosing regimen is supported by the overall efficacy of fam-trastuzumab deruxtecan monotherapy with an objective response rate (ORR) of 58.3% (51.7, 64.7) at the target dose of 5.4 mg/kg IV Q3W in patients with unresectable or metastatic HER2-positive breast cancer (BC) who have 2 or more prior anti-HER2 therapy.
General dosing instructions	The proposed fam-trastuzumab deruxtecan dosing regimen of 5.4 mg/kg given by intravenous infusion over 90 minutes is acceptable for approval from a clinical pharmacology perspective: • Favorable anti-tumor activity: In pooled analysis, this dosing regimen demonstrated an ORR of 58.3% in patients with HER2-positive BC who have 2 or more prior anti-HER2 therapy. • Lower dose levels not tested: There is insufficient activity and safety information at lower dose levels (< 5.4 mg/kg Q3W). • Positive exposure-response relationships for safety: Higher exposure, higher incidence of adverse events. Interstitial lung disease (ILD) is a significant risk factor, with higher observed incidence at higher dose levels (≥ 6.4 mg/kg).

Review Issue	Recommendations and Comments
Dosing in patient subgroups (intrinsic and extrinsic factors)	- No dose adjustment is recommended for patients with mild or moderate hepatic impairment (NCI-ODWG criteria). Patients with moderate hepatic impairment should be closely monitored for increased toxicities related to potentially increase exposure of MAAA-1181a (DXd), the topoisomerase inhibitor. There is no data available for patients with severe hepatic impairment. - No dose adjustment is recommended for patients with mild or moderate renal impairment (CLcr > 30 mL/min). There is no data available for patients with severe renal impairment (CLcr < 30 mL/min). - Based on PopPK analyses, sex and baseline body weight were noted to be identified as significant covariates on central volume compartment (Vc). - The incidence of ADA is low. Overall, 4 (0.6%) patients were identified as having treatment-emergent ADA. The effect of anti-drug antibody (ADA) on the PK of trastuzumab deruxtecan and MAAA-1181a is unknown due to limited number of patients with positive ADA. The neutralizing ADA assay has not been developed.
Drug-drug interactions	No dose adjustment is needed for patients who are co-administered with CYP3A, OATP1B or P-gp inhibitors.
Labeling	Overall, the proposed labeling recommendations are acceptable upon the Applicant's agreement to the FDA revisions to the label. FDA recommended no dose adjustment for patients with moderate hepatic impairment, based on totality of available PK, efficacy, and safety data. Clinical pharmacology labeling recommendations are detailed in Section 12.3.

6.2. Summary of Clinical Pharmacology Assessment

6.2.1. Pharmacology and Clinical Pharmacokinetics

Data:

Pharmacokinetic (PK) parameters for fam-trastuzumab deruxtecan and total anti-HER2 antibody were comparable. Serum concentrations of MAAA-1181a gradually increased and reached peak concentrations with longer median time to maximum serum concentration (Tmax) (approximately 6.8 hours) compared to fam-trastuzumab deruxtecan (approximately 2 hours). On a molar basis, the fam-trastuzumab deruxtecan maximum observed serum concentration (Cmax) and trough concentrations across the 5.4 mg/kg to 7.4 mg/kg doses were approximately 50-fold to 53-fold and 69-fold to 94-fold higher, respectively, than those for MAAA-1181a.

The exposure of fam-trastuzumab deruxtecan, total anti-HER2 antibody, and MAAA-1181a increased in a dose-related manner, with fam-trastuzumab deruxtecan exposure proportional

to dose in the 3.2 mg/kg to 8.0 mg/kg dose range. In the 0.8 mg/kg to 8.0 mg/kg dose range, the C_{max} increased proportional to dose, while the area under the serum concentration-time curve (AUC) for fam-trastuzumab deruxtecan and total anti-HER2 antibody increased slightly more than proportional to dose. Exposures (C_{max} and AUC) for MAAA-1181a increased proportional to dose in the 0.8 mg/kg to 8.0 mg/kg dose range.

Multiple-dose PK parameters of fam-trastuzumab deruxtecan were evaluated in Study DS8201-A-J102 in Cycle 1 and Cycle 3. The area under the serum concentration-time curve during the dosing interval (AUC_{tau}) of fam-trastuzumab deruxtecan in Cycle 3 after multiple dosing of fam-trastuzumab deruxtecan at 6.4 mg/kg was approximately 35% higher than that in Cycle 1, suggesting there was some accumulation.

Based on population-pharmacokinetics (PopPK) analysis, the clearance of fam-trastuzumab deruxtecan was estimated to be 0.421 L/day. In Study DS8201-A-J101 ([Table 19](#)), the apparent terminal elimination half-life (t_{1/2}) of fam-trastuzumab deruxtecan ranged from 2.2 days to 7.3 days across the 0.8 mg/kg to 8.0 mg/kg dose range.

Effect of Intrinsic Factors:

The effect of intrinsic factors (age, sex, race/ethnicity, body weight, hepatic and renal impairment, baseline albumin, baseline tumor size, and tumor type) on the exposures of fam-trastuzumab deruxtecan was evaluated using PopPK analysis. Based on the PopPK analysis, no dose adjustment is required for subjects with mild or moderate renal impairment and subjects with mild hepatic impairment (based on the National Cancer Institute Organ Dysfunction Working Group [NCI ODWG] criteria). Fam-trastuzumab deruxtecan was not studied in subjects with severe hepatic impairment.

Drug-drug Interactions:

A clinical drug-drug interaction (DDI) study to evaluate the effect of inhibitors of organic anion transporting peptide (OATP)1B, and cytochrome P450 (CYP)3A was conducted (Study DS8201-A-A104; see [Table 19](#)). Results from this study indicated that these inhibitors had no significant effect on the exposures of fam-trastuzumab deruxtecan and MAAA-1181a.

Effects on Human Cardiac Parameters:

Study DS8201-A-J102 (see [Table 19](#)) evaluated the effect of fam-trastuzumab deruxtecan on human cardiac parameters. Results from the study indicated that fam-trastuzumab deruxtecan 6.4 mg/kg administration was not associated with a clinically meaningful (defined as change from baseline >10 milliseconds) prolongation of QT interval corrected for heart rate using Frederica's formula (QTcF). The upper bound of the 90% CI for change from baseline in QTcF (Δ QTcF) at the observed mean C_{max} for each analyte (fam-trastuzumab deruxtecan and MAAA-1181a) in the linear model of concentration versus Δ QTcF for Cycles 1 and 3 was under 10 milliseconds.

Immunogenicity:

The incidence of immunogenicity was low with 3.5% and 1.1% of subjects demonstrating antidrug-antibody (ADA)-positive results at baseline and post-baseline, respectively. Overall, 4 (0.6%) subjects were identified as having treatment-emergent ADA. No relationship was identified between the rate of infusion-site reactions (as a measure of safety) and the incidence of ADA.

The Applicant's Position:

The clinical pharmacology of fam-trastuzumab deruxtecan has been characterized with concentrations of all 3 analytes components of the ADC (ie, intact fam-trastuzumab deruxtecan, total anti-HER2 antibody, and released drug MAAA-1181a) analyzed in all 5 clinical studies included in the submission. The data included in the dossier encompasses the PK properties following single dose and multiple doses of fam-trastuzumab deruxtecan in subjects with metastatic breast cancer, as well as popPK analyses of studies included in the application. The results from both in vitro human biomaterial studies and in vivo clinical pharmacology studies (QT interval corrected for heart rate [QTc] and DDI studies) conducted in subjects with metastatic breast cancer indicated no clinically meaningful effect on QT or QTc interval and no need for dose adjustment when coadministering inhibitors of CYP3A enzymes, and P-glycoprotein (P-gp), breast cancer resistance protein (BCRP), and OATP1B transporters. Effects of intrinsic and extrinsic factors on PK of fam-trastuzumab deruxtecan were assessed using PopPK analyses. Based on the PopPK analysis, no dose adjustment is required for subjects with mild or moderate renal impairment and subjects with mild hepatic impairment (based on the NCI ODWG criteria). In addition, exposure-efficacy and exposure-safety analyses conducted for fam-trastuzumab deruxtecan support the use of the recommended dose of 5.4 mg/kg once every 3 weeks (Q3W) fam-trastuzumab deruxtecan in patients with unresectable or metastatic HER2-positive breast cancer. In general, the incidence of immunogenicity was low with 3.5% and 1.1% of subjects demonstrating ADA-positive results at baseline and post-baseline, respectively.

The FDA's Assessment:

The FDA generally agrees with the applicant's assessment from a clinical pharmacology perspective. Based on the totality of data (DDI results, activity and safety), FDA recommends no dose adjustment in patients with moderate hepatic impairment, but closed safety monitoring due to potentially increased exposure of MAAA-1181a (DXd), the topoisomerase inhibitor.

6.2.2. General Dosing and Therapeutic Individualization

6.2.2.1. General Dosing

Data:

The selection of 5.4 mg/kg fam-trastuzumab deruxtecan Q3W as the recommended dosing regimen for the treatment of metastatic HER2-positive breast cancer occurred in multiple steps as described below.

The first-in-human (FIH) Phase 1 study DS8201-A-J101 included: a Dose Escalation portion (Part 1) evaluating a dose range of 0.8 mg/kg to 8.0 mg/kg Q3W to identify the maximum tolerated dose (MTD) and select doses for the Phase 2 study; and a Dose Expansion portion (Part 2) to further evaluate the safety, tolerability, and efficacy of the dose range to be evaluated in Phase 2 (see [Table 19](#)). The FIH dose of 0.8 mg/kg was selected based on one-twelfth of the human equivalent dose of the highest non-severely toxic dose (HNSTD) in cynomolgus monkeys in toxicology studies and was also estimated to provide trough concentration in humans comparable to the minimum effective concentrations in nonclinical studies. Following the FIH dose, dose escalation was conducted to evaluate doses up to 8.0 mg/kg in Part 1 of this study. Based on the balance between efficacy and safety across the dose range evaluated in Part 1 of Study DS8201-A-J101 and on the exposure-response (ER) analyses of safety and efficacy, doses of 5.4 mg/kg and 6.4 mg/kg Q3W were selected for the Dose Expansion portion (Part 2) of this study.

The Phase 2 Study, DS8201-A-U201, investigated the safety and efficacy of fam-trastuzumab deruxtecan in subjects with HER2-positive unresectable and/or metastatic breast cancer previously treated with T-DM1. This study included 2 parts: Part 1 included a PK stage and a Dose Finding stage designed to identify a single dose level to be studied in Part 2. Part 2 was designed to evaluate the efficacy and safety of fam-trastuzumab deruxtecan in subjects with HER2-positive, unresectable and/or metastatic breast cancer previously treated with T-DM1 or those who discontinued T-DM1 for reasons other than progressive disease (PD).

The selection of doses for each part of Study DS8201-A-U201 followed the steps outlined below:

- The PK stage of Part 1 evaluated 3 dose levels of fam-trastuzumab deruxtecan: 5.4 mg/kg, 6.4 mg/kg, and 7.4 mg/kg Q3W. These dosages were selected based on the efficacy and safety data, and on the ER analyses for safety and efficacy from Part 1 of the FIH study that support the positive benefit/risk profile provided by the 5.4 mg/kg and 6.4 mg/kg doses. This study also enabled the human PK comparison of fam-trastuzumab deruxtecan formulations frozen liquid-drug product 1 (FL-DP1) and frozen liquid-drug product 2 (FL-DP2) used in the FIH study and this Phase 2 study, respectively. The 7.4 mg/kg dose was selected to compensate for a potential reduction (estimated at

possibly ~20%) in fam-trastuzumab deruxtecan exposures with the FL-DP2 formulation compared to the FL-DP1 formulation based on nonclinical PK comparison between the 2 formulations.

- Two dose regimens (5.4 mg/kg and 6.4 mg/kg Q3W) were chosen for the Dose Finding stage of Part 1 based on similar human fam-trastuzumab deruxtecan and released drug (MAAA-1181a) PK (or exposures) for the FL-DP1 and FL-DP2 formulations. This Dose Finding stage was designed to provide additional PK, safety, and efficacy data with the 5.4 mg/kg and 6.4 mg/kg dosages for the FL-DP2 formulation.
- A dosage of 5.4 mg/kg Q3W was chosen for Part 2 of Study DS8201-A-U201 and for the ongoing Phase 3 studies in metastatic breast cancer (see [Table 19](#)). This was based on comparison of safety and efficacy data at 5.4 mg/kg and 6.4 mg/kg Q3W evaluated in Study DS8201-A-J101 (Parts 1 and 2) and Part 1 of Study DS8201-A-U201; on ER analyses for efficacy and safety using data from Study DS8201-A-J101 (Parts 1 and 2); and on available interim data from Part 1 of Study DS8201-A-U201. The 6.4 mg/kg dose regimen showed a numerical trend for an increase in ORR, as well as a numerical increase in the incidence of AEs, compared to the 5.4 mg/kg dose.

The Applicant's Position:

The proposed dosing regimen of fam-trastuzumab deruxtecan in patients with metastatic HER2-positive breast cancer is 5.4 mg/kg Q3W administered intravenously (IV) over 90 minutes in the first cycle and over 30 minutes in subsequent cycles if the previous infusion was well tolerated.

The recommended dosing regimen of fam-trastuzumab deruxtecan 5.4 mg/kg administered Q3W was evaluated in the FIH Phase 1 study (DS8201-A-J101) and the Phase 2 study (DS8201-A-U201) in subjects with metastatic breast cancer. While both 5.4 mg/kg and 6.4 mg/kg Q3W showed a positive benefit/risk profile, 5.4 mg/kg was chosen for further investigation in subjects with metastatic breast cancer due to its more positive safety profile. The cumulative efficacy data across these 2 studies and safety data across all 5 clinical studies included in this application (DS8201-A-J101, DS8201-A-J102, DS8201-A-A103, DS8201-A-A104, and DS8201-A-U201; see [Table 19](#)) support a positive benefit/risk profile for fam-trastuzumab deruxtecan at the target dose in the target indication. Additionally, the ER analyses (as described in Section [6.3.2.1](#)) for efficacy and key safety variables from the clinical studies further support the positive benefit/risk profile for the 5.4 mg/kg dose.

The FDA's Assessment:

The FDA generally agrees with the applicant's assessment on dosing. FDA agrees that the dose of 5.4 mg/kg Q3W may support positive benefit/risk profile of fam-trastuzumab deruxtecan based on currently available data. Due to a lack of available patients with HER2+ expressing tumors to evaluate efficacy and safety at doses between 1.6 and 5.4 mg/kg (Table 9), it is

unclear whether lower doses may maintain efficacy while reducing the risks of fam-trastuzumab deruxtecan.

Table 9: Pooled dose-response analysis for ORR

Dose (mg/kg)	0.8	1.6	3.2	5.4	6.4	7.4	8.0
# patients with HER2+	2	1	0	211	100	21	2
# patients with response	0	1	0	132	67	17	1
ORR (%)	0	100	0	63	67	81	50

Source: 5.3.3.5: DS8201-PMx003, Exposure-response analysis report (PMx003); Table A

6.2.2.2. Therapeutic Individualization

Data:

The effect of intrinsic factors (age, sex, race/ethnicity, body weight, baseline albumin, baseline tumor size, and tumor type) was evaluated using PopPK analysis (described in Section 6.2.1). In addition, the effect of hepatic and renal impairment on exposures of released drug were also evaluated using PopPK.

The drug interaction potential was evaluated in a clinical study (described in Section 6.2.1) and also using PopPK analysis. Strong inhibitors of CYP3A and OATP1B can be administered without fam-trastuzumab deruxtecan dose adjustment.

The Applicant's Position:

No therapeutic individualization for intrinsic or extrinsic factors is recommended. The results of the PopPK analysis indicate no clinically meaningful effects of any of the intrinsic factors on the exposures of fam-trastuzumab deruxtecan and released MAAA-1181a. Based on the PopPK analysis, no dose adjustment is required for patients with mild or moderate renal impairment and for patients with mild hepatic impairment. Limited data preclude a recommendation for patients with moderate hepatic impairment and severe renal impairment. Fam-trastuzumab deruxtecan was not studied in subjects with severe hepatic impairment.

Strong inhibitors of CYP3A and OATP1B can be administered without fam-trastuzumab deruxtecan dose adjustment. Although a dedicated study with a P-gp inhibitor was not conducted, since itraconazole and ritonavir have been reported to be P-gp inhibitors, it can be inferred from the results from DS8201-A-A104 that P-gp inhibitors are also not expected to cause clinically meaningful increases in exposure of fam-trastuzumab deruxtecan or MAAA-1181a. Similarly, although MAAA-1181a was identified in vitro as a substrate of the efflux transporter BCRP, no dedicated study was conducted to evaluate the effects of a BCRP inhibitor. Ritonavir has been reported to be an inhibitor of BCRP (Bierman et al 2010, Gupta et al 2004), supporting the conclusion that BCRP inhibitors will also have no clinically meaningful effects on exposures of fam-trastuzumab deruxtecan or MAAA-1181a.

The FDA's Assessment:

The FDA agrees with the applicant's position that no therapeutic individualization is needed based on demographic factors (body weight, age, race, sex, baseline albumin, baseline tumor size, and tumor type), in patients with mild (based on NCI ODWG criteria), or in patients with mild or moderate renal impairment. Based on totality of data, including DDI, safety, and nonclinical, FDA recommends that no dose adjustment is needed for patients with moderate hepatic impairment.

Based on DDI results from Study A104, FDA agrees that there are no clinically meaningful changes with MAAA-11981a exposure. FDA also agrees that the DDI results may indirectly infer that the risk of DDI with P-gp inhibitors is low since itraconazole and ritonavir can inhibit P-gp.

6.2.2.3. Outstanding Issues

The Applicant's Position:

None

The FDA's Assessment:

The FDA has no outstanding issue as PMR or PMC. However, FDA recommends that the Applicant collect safety data in at least 10 patients with moderate hepatic impairment in ongoing trials and submit those data in the final analysis reports.

6.3. Comprehensive Clinical Pharmacology Review

6.3.1. General Pharmacology and Pharmacokinetic Characteristics

Data:

Population Pharmacokinetic Analysis:

Based on PopPK analysis, the clearance of fam-trastuzumab deruxtecan was estimated to be 0.421 L/day. In Study DS8201-A-J101 (see [Table 19](#)), the t_{1/2} of fam-trastuzumab deruxtecan ranged from 2.2 days to 7.3 days across the 0.8 mg/kg to 8.0 mg/kg dose range.

Based on the PopPK analysis, fam-trastuzumab deruxtecan central compartment volume (V_c) was estimated to be 2.77 L. In vitro, the binding of MAAA-1181a to human plasma proteins was approximately 97%.

The fam-trastuzumab deruxtecan data were best described by a 2-compartment model with linear clearance from the central compartment. In this analysis, baseline body weight, baseline

albumin, sex, baseline tumor size, and country (Japan and non-Japan) were identified as statistically significant covariates on clearance; sex was identified as a significant covariate on V_c; and country was identified as a significant covariate on peripheral compartment volume. The model suggests that fam-trastuzumab deruxtecan clearance increased with increasing baseline body weight, baseline tumor size, and being a male; and the clearance decreased with increasing albumin and in subjects from Japan. Central volume of distribution increased with increasing body weight and being male, and peripheral volume of distribution decreased with being in Japan.

The MAAA-1181a PK data were described by a time-varying 1-compartment model with first-order elimination. In this analysis, baseline body weight, baseline aspartate aminotransferase (AST), total bilirubin (TBL), and use of ritonavir or itraconazole were identified as significant covariates on the clearance of MAAA-1181a; and age and fam-trastuzumab deruxtecan formulation were identified as significant covariates on volume of distribution. The model suggests that clearance of MAAA-1181a is increased with increasing body weight and that clearance is decreased with increasing bilirubin, AST, and coadministration of itraconazole or ritonavir. Volume of distribution increased with increasing age or with use of the FL-DP2 formulation.

The effect of the covariates on the exposures (area under the serum concentration-time curve at steady-state [AUC_{ss}]) indicates that no covariate had an impact of more than 25% on the AUC_{ss} ratio for fam-trastuzumab deruxtecan or MAAA-1181a, except for baseline body weight for which subjects with body weight at 95th percentile of the distribution (86 kg) had AUC_{ss} approximately 30% higher than those with median body weight (57.8 kg).

Drug Product Formulations:

Three drug products (DPs) of fam-trastuzumab deruxtecan, FL-DP1, FL-DP2, and lyophilized powder-drug product (Lyo-DP, the to-be-marketed formulation), have been administered across the clinical development program. Between FL-DP1 and FL-DP2, there was a change in

(b) (4)

Between FL-DP2 and Lyo-DP, there was only a change in product presentation: from frozen liquid to lyophilized powder.

The exposures to fam-trastuzumab deruxtecan, total anti-HER2 antibody, and MAAA-1181a were similar across clinical studies irrespective of the fam-trastuzumab deruxtecan product based on within-study PK comparison as well as integrated PK analysis across studies. These observations were supported by the PopPK analysis, where fam-trastuzumab deruxtecan product was not a significant covariate on human exposure. The rate of ADAs was low, with no apparent difference in the rate of ADAs between FL-DP2 and Lyo-DP.

Metabolism and Elimination:

Metabolites of MAAA-1181a were difficult to identify due to the small quantity of MAAA-1181a release in the incubation. After IV administration of ¹⁴C-labeled MAAA-1181a to rats and monkeys, the major excretion pathway of radioactivity was feces via the biliary route and MAAA-1181a was the most abundant component of radioactivity in urine, feces, and bile.

Results of experiments with CYP-expressing microsomes showed that CYP1A2, CYP2D6, CYP3A4, and CYP3A5 were involved in the rapid metabolism of MAAA-1181a.

Effect of Extrinsic Factors:

Human biomaterial studies indicated that MAAA-1181a is a substrate of P-gp, multidrug and toxin extrusion protein 2-K (MATE2-K), OATP1B1, OATP1B3, BCRP, multidrug resistance protein 1 (MRP1), and CYP3A4.

A clinical DDI study to evaluate the effect of inhibitors of OATP1B, and CYP3A (Study DS8201-A-A104) indicated that these inhibitors had no significant effect on the exposures of fam-trastuzumab deruxtecan and MAAA-1181a.

Effect of Fam-Trastuzumab Deruxtecan on Pharmacokinetics of Other Drugs:

Based on in vitro data, MAAA-1181a is not a reversible or time-dependent inhibitor of CYP isoforms, and also did not show induction potential on mRNA expression or metabolic activity of CYP isoforms.

In human biomaterial studies of transporters, MAAA-1181a inhibited organic anion transporter 1 (OAT1) and OATP1B (50% inhibitory concentration [IC₅₀] values 12.7 µmol/L and 14.4 µmol/L, respectively). MAAA-1181a did not inhibit organic anion transporter 3 (OAT3), organic cation transporter 1 (OCT1), organic cation transporter 2 (OCT2), OATP1B3, multidrug and toxin extrusion protein 1 (MATE1), MATE2-K, P-gp, BCRP, or bile salt export pump (BSEP). No clinically meaningful DDI is expected with drugs that are substrates of OAT1 or OATP1B1 transporters as the mean C_{max} at the maximum dose used in clinical studies (5.4 mg/kg) for MAAA-1181a (9.65 ng/mL, 0.019 µmol/L) is very low compared with the IC₅₀ values.

Immunogenicity:

Subjects were tested for ADAs against fam-trastuzumab deruxtecan in all 5 clinical studies. The immunogenicity assay developed was sensitive and drug tolerant. In general, the rate of baseline and post-baseline immunogenicity was low with 3.5% and 1.1% incidence, respectively, of subjects demonstrating ADA-positive results. The rate of treatment-emergent ADA (defined as subjects who were ADA-negative at baseline and became ADA-positive post-treatment; or subjects who were ADA-positive at baseline and post-treatment, but had an increase in ADA titer from baseline to post-treatment; or subjects who had missing ADA data at

baseline and were ADA-positive post-treatment) continued to be low, with a total of 4 (0.6%) subjects having treatment emergent ADA. Given the low rate of ADA-positive data, no obvious effect of immunogenicity on PK and efficacy was observed. No relationship was identified between the rate of infusion-site reactions (as a measure of safety) and the incidence of ADA.

The Applicant's Position:

Fam-trastuzumab deruxtecan Vc was similar to plasma volume, suggesting that fam-trastuzumab deruxtecan is not widely distributed outside of the central compartment. The magnitude of the covariate effects based on the PopPK analysis suggests that no dose adjustment is required for any of the identified covariates. Based on the PopPK analysis, no dose adjustment is required for subjects with mild or moderate renal impairment and subjects with mild hepatic impairment (based on the NCI ODWG criteria). Limited data preclude a recommendation for subjects with moderate hepatic impairment or with severe renal impairment. Fam-trastuzumab deruxtecan was not studied in subjects with severe hepatic impairment.

Overall, on the basis of similar PK and low rate of ADA generation, it can be concluded that the 3 DPs used in the fam-trastuzumab deruxtecan development program can be considered comparable.

Results of in vitro studies indicate that fam-trastuzumab deruxtecan is metabolically stable in the in vitro hepatocyte incubation system. Experiments in human liver microsomes with specific inhibitors of CYP enzymes indicated that CYP3A4 was the primary CYP isoform involved in the metabolism of MAAA-1181a.

Based on the results of a clinical DDI study, strong inhibitors of CYP3A4 and inhibitors of OATP1B, P-gp, and BCRP can be administered without dose adjustment of fam-trastuzumab deruxtecan.

In general, the incidence of immunogenicity was low with 3.5% and 1.1% of subjects demonstrating ADA-positive results at baseline and post-baseline, respectively.

The FDA's Assessment:

The FDA generally agrees with the applicant's assessments on general pharmacology and pharmacokinetic characteristics of fam-trastuzumab deruxtecan and MAAA-1181a.

FDA agrees that sex and baseline body weight were noted to be identified as significant covariates on central volume compartment (Vc).

FDA assessment on the geometric mean ratio and the 90% confidence interval between different drug product formulations at the 5.4 mg/kg dose are shown in Table 10. Three formulations, FL-D1, FL-D2, and FL-Lyo have been used across clinical development program.

Formulation FL-D1 has been used in Study J101; Formulation FL-D2 has been used in Study J101 and U201; Formulation FL-Lyo has been used in Study J102, A103, A104, and U201. Cross-study comparisons suggested that there are no clinically meaningful differences in the PK of fam-trastuzumab deruxtecan and MAAA-1181a between drug product formulations.

Table 10: Summary of statistical analyses for assessment of PK similarity of DS-8201a, Total Atni-HER2 antibody and MAAA-1181a (or DXd)

	FL-D1 vs. FL-DP2 (N=144)	FL-DP2 vs. Lyo DP (N=184)
DS-8201a		
C_{max}	1.05 (0.97 – 1.13)	1.00 (0.92 – 1.08)
AUC _{0-21d}	0.97 (0.89 – 1.06)	0.98 (0.90 – 1.07)
Total Anti-HER2 antibody		
C_{max}	1.04 (0.96 – 1.11)	0.96 (0.89 – 1.04)
AUC _{0-21d}	0.97 (0.86 – 1.09)	0.95 (0.84 – 1.07)
MAAA-1181a (DXd)		
C_{max}	1.18 (1.02 – 1.38)	0.88 (0.75 – 1.02)
AUC _{0-21d}	1.10 (0.97 – 1.26)	0.96 (0.79 – 1.17)

Source: 5.3.5.3 IPK Analysis - Integrated PK report; Table 14.1.3.1, 14.1.3.2, and 14.1.3.3

The general overview of fam-trastuzumab deruxtecan and MAAA-1181a ADME and clinical PK information as assessed by FDA are presented in Table 11:

Table 11: Highlights of Clinical Pharmacology for fam-trastuzumab deruxtecan

Physiochemical properties	
Chemical structure and molecular weight	Chemical Structure of DS-8201a (fam-trastuzumab deruxtecan), consisting of MAAAL-9001 (anti-HER2 antibody), MAAA-1162a polypeptide linker, MAAA-1181a (small molecule payload or DXd) Molecular weight of DS-8201a: (b) (4) Da Target drug-to-antibody ratio = 8 Molecular weight of MAAA-1181a: (b) (4) g/mol
Pharmacology	
Mechanism of Action	Fam-trastuzumab deruxtecan is an antibody drug conjugates targeting HER-2 positive expressing cells. The payload (MAAA-1181a) is a DNA topoisomerase I inhibitor. The mechanism of action (MOA) involves the MAAAL-9901 binding to HER2 receptors on the tumor target cell membrane, and where the DS-8201a gets internalized into the target cells. MAAL-1162a is metabolized by lysosomal enzymes where it will release the payload, MAAA-1181a, which inhibits the DNA topoisomerase I enzyme.
Active Moieties	MAAA-1181a or DXd
QT/QTc Prolongation	The effect of fam-trastuzumab deruxtecan, total anti-HER2 antibody, and MAAA-118a was evaluated in Study A102. Fifty-one patients with HER2-positive expressing BC were dosed with 6.4 mg/kg Q3W fam-trastuzumab deruxtecan. Multiple ECGs were collected on Cycle to Cycle 3. An analysis of clinical ECG data demonstrates the absence of large effect (i.e. >10 ms) on QTcF prolongation at the dose of 6.4 mg/kg Q3W.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

General Information	
Bioanalysis	In the pivotal Study U201, three validated PK assays were used to measure fam-trastuzumab deruxtecan, total anti-HER2 antibody (MAAL-9001), and MAAA-1181a. Plasma fam-trastuzumab deruxtecan and MAAL-9001 concentrations were measured using a validated electrochemiluminescent method. Total plasma MAAA-1181a (not attached to DS-8201a) concentrations were measured using a validated LC-MS/MS method. A summary of the method validation reports is included in Appendix 19.4.
Healthy Volunteers vs. Patients	All studies are conducted in patients with cancer.
Drug exposure at steady state following the therapeutic dosing regimen	In adult patients with HER2+ expressing BC who received fam-trastuzumab deruxtecan 5.4 mg/kg Q3W, popPK approach derived geometric steady-state (CV%) C_{max} was 122 (20%) µg/mL and AUC_{0-21days} was 735 (31%) µg*h/mL for fam-trastuzumab deruxtecan and geometric steady-state (CV%) C_{max} was 4.4 (40%) ng/mL and AUC_{0-21days} was 28 (37%) ng*h/mL for MAAA-1181a. Noncompartmental pooled PK analysis was not used as there is discrepancies in the PK sampling time points between studies.
Minimal effective dose or exposure	Not determined. There is insufficient anticancer activity and safety information at doses < 5.4 mg/kg to determine the minimal effective dose.
Maximal tolerated dose or exposure	In Study J101, the MTD of fam-trastuzumab deruxtecan as a single-agent treatment in patients with HER2+ expressing BC was not reached at the dose range of 0.8 to 8.0 mg/kg Q3W.

Dose Proportionality	The dose proportionality analysis was assed using a power model with PK parameters obtained following a single dose administration on Day 1 of Cycle 1 in Study J101.		
	The estimation of the slope of the dose proportionality analysis was approximately 1 for doses 3.2-8.0 mg/kg group for DS-8201a and total anti-HER2 antibody. DS-8201a C _{max} was dose proportional across the dose range of 0.8 to 8.0 mg/kg (Table 12). AUC _{last} and AUC _{tau} increased dose proportionally at doses at above 3.2 mg/kg.		
	Table 12: Dose proportionality of DS-8201a analysis of PK parameters in dose escalation cohort of Study J101 following a single dose		
	Parameter	Dose	
		3.2-8.0 mg/kg	
		Estimation	95% CI
	C _{max} (µg/mL)		
	Intercept	2.91	(2.40, 3.43)
	Slope	1.19	(0.903, 1.48)
	AUC _{last} (µg*d/mL)		
Intercept	4.04	(3.13, 4.95)	
Slope	1.39	(0.882, 1.90)	
AUC _{tau} (µg*d/mL)			
Intercept	4.06	(3.17, 4.96)	
Slope	1.38	(0.878, 1.89)	
	Source: 5.3.3.2 DS8201-a-J101; Study Report Body; Table 9.4		
	Estimation (95% CI) of the slope for AUC _{last} for MAAA-1181 was 0.507 (-0.0255, 1.04) for the ≥3.2 mg/kg dose range, due to the constant exposure over 5.4 mg/kg (Table 13).		
	Table 13: Dose proportionality of MAAA-1181a analysis of PK parameters in dose escalation cohort of Study J101 following a single dose		
	Parameter	Dose	
		3.2-8.0 mg/kg	
		Estimation	95% CI
	C _{max} (ng/mL)		
	Intercept	1.33	(0.201, 2.46)
	Slope	0.411	(-0.225, 1.05)
	AUC _{last} (ng*d/mL)		
	Intercept	2.61	(1.66, 3.56)
	Slope	0.507	(-0.0255, 1.04)
	AUC _{tau} (ng*d/mL)		
	Intercept	2.60	(1.66, 3.55)
	Slope	0.510	(-0.0255, 1.04)
	Source: 5.3.3.2 DS8201-a-J101; Study Report Body; Table 9.6		

The following tables show the results of dose proportionality analysis conducted with a power model, similar to the analysis done using Cycle 1 data in Study DS8201-A-J101, using PK parameters obtained following multiple dose administration of trastuzumab deruxtecan in Cycle 3 of Study DS8201-A-J101. In the 3.2 to 8.0 mg/kg dose range (Table 14), a more than dose proportional increase in intact fam-trastuzumab deruxtecan C_{max} was observed, while for AUC_{last} and AUC_{tau} a slightly more than dose proportional increase was observed. It should be noted that the 95 % CI of the slope estimate for AUC included 1 although the intervals were wide.

Table 14: Dose proportionality of DS-8201a analysis of PK parameters in dose escalation cohort of Study J101 at steady state

Parameter	Dose	
	3.2-8.0 mg/kg	
	Estimation	95% CI
C_{max} (µg/mL)		
Intercept	0.731	(-1.08, 2.54)
Slope	2.29	(1.29, 3.28)
AUC_{last} (µg*d/mL)		
Intercept	4.52	(3.32, 5.73)
Slope	1.35	(0.700, 2.01)
AUC_{tau} (µg*d/mL)		
Intercept	4.52	(3.32, 5.73)
Slope	1.35	(0.700, 2.01)

Source: Response to FDA information request on 11/4/2019; Table 1.2

MAAA-1181a C_{max} , AUC_{last} and AUC_{tau} were 0.50 for the ≥ 3.2 mg/kg dose range likely due to the comparable AUC_{last} and AUC_{tau} values for dose levels 5.4 mg/kg and above (Table 15).

Table 15: Dose proportionality of MAAA-1181a analysis of PK parameters in dose escalation cohort of Study J101 at steady state

Parameter	Dose	
	3.2-8.0 mg/kg	
	Estimation	95% CI
C_{max} (ng/mL)		
Intercept	-2.36	(-4.68, -0.036)
Slope	2.15	0.855, 3.44)
AUC_{last} (ng*d/mL)		
Intercept	2.86	(1.38, 4.33)
Slope	0.50	(-0.301, 1.29)
AUC_{tau} (ng*d/mL)		
Intercept	2.86	(1.38, 4.33)
Slope	0.50	(-0.301, 1.29)

Source: Response to FDA information request on 11/4/2019; Table 1.3

Version date: June 11, 2019 (ALL NDA/BLA reviews)

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Accumulation	The mean accumulation ratios (R_{acc}) following fam-trastuzumab deruxtecan 5.4 mg/kg Q3W was approximately 1.3-fold based on population PK analysis.							
Variability	Based on population PK analysis, the inter-subject variability (CV%) of steady state fam-trastuzumab deruxtecan AUC_{tau} and C_{max} was 31% and 20% and MAAA-1181a AUC_{tau} and C_{max} was 37% and 40%, respectively.							
Immunogenicity	Across all doses evaluated in clinical studies, only 0.6% (4/640) of evaluable patients developed antibodies against fam-trastuzumab deruxtecan following treatment as shown in Table 16.							
	Table 16: Summary of Immunogenicity by Study							
		Study	J101	J102	A103	A104	U201	Overall
	# of subjects with baseline or post-baseline	N	289	51	12	40	253	645
	Baseline Assessment	N	284	51	12	40	250	637
	ADA positive at baseline	N	5	3	0	2.5	5.2	3.5
	ATA Titer at baseline	N	5	3	NC	1	13	22
	Post-baseline assessment	N	289	51	12	39	249	640
	Total ADA positive post-baseline	N	1	1	0	0	5	7
	ADA positive at baseline and post-baseline	N	1	1	0	0	3	5
Treatment-Emergent ADA positive	N	1	0	0	0	3	4 (0.6%)	
Source: 5.3.5.3 IPK Analysis - Integrated PK report; Table 14.2.1.1								
The PK data of four patients with positive ADA did not appear to affect the exposure of fam-trastuzumab deruxtecan. However, no conclusion can be made as the data is limited to four patients.								
All patients enrolled across studies had prior treatment of trastuzumab and/or ado-trastuzumab emtansine (T-DM1). Incidence of ADA against fam-trastuzumab deruxtecan without prior trastuzumab therapy(ies) is unknown. No neutralizing ADA assay has been developed.								
Absorption								
Bioavailability	Not applicable.							

T _{max}	The median time to reach peak plasma concentration (T _{max}) ranged between 1.8 to 2.1 hours for fam- trastuzumab deruxtecan and 3.8 to 23.8 hours for MAAA-1181a at the dose of 5.4 mg/kg Q3W.		
Ritonavir and Itraconazole (DDI)	Study A104 is an open-label, single sequence study in patients with HER2-expressing advanced solid malignant tumors. The study comprised of Cohort 1 and Cohort 2. Cohort 1 was a non-randomized cohort to evaluate the effect of ritonavir (dual inhibitor of CYP3A and OATP1B inhibitor) on fam-trastuzumab deruxtecan and MAAA-1181a PK. Cohort 2 was a non-randomized cohort to evaluate the effect of itraconazole (strong CYP3A inhibitor) on fam-trastuzumab deruxtecan and MAAA-1181a PK. The PK results show no significant clinically meaningful changes (Table 17).		
	Table 17: Summary of statistical analysis of assessment of PK parameters with or without inhibitors		
		+/- Ritonavir (GMR, 90% CI) N=8 (Cmax), 12 (AUC)	+/- Itraconazole (GMR, 90% CI) N=14
	Fam-trastuzumab deruxtecan		
	Ratio Cmax	1.05 (0.98-1.13)	1.02 (0.96-1.09)
	Ratio AUC _{0-17days}	1.19 (1.13-1.25)	1.10 (1.07-1.15)
MAAA-1181a			
	Ratio Cmax	0.99 (0.85-1.14)	1.03 (0.92-1.18)
	Ratio AUC _{0-17days}	1.22 (1.08-1.37)	1.18 (1.11-1.25)
Source: 5.3.3.4 DS8201-a-A104; Study Report Body; Tables 9.7 and 9.9			
Distribution			
Volume of distribution	The popPK approach derived mean (%CV) apparent central volume of distribution of fam-trastuzumab at steady-state is 2.77 L.		
Plasma protein binding	In vitro, protein binding of MAAA-1181a is 97%.		
Blood to plasma ratio	The mean blood-to-plasma ratio was 0.6 for MAAA-1181a.		
Elimination			
Half-life	Following a single 5.4 mg/kg dose of fam-trastuzumab deruxtecan, the median elimination half-life (t _{1/2}) of fam-trastuzumab deruxtecan was approximately 5.7 days based on noncompartmental analysis.		
	Following a single 5.4 mg/kg dose of fam-trastuzumab deruxtecan, the median apparent elimination t _{1/2} of MAAA-1181a was approximately 5.8 days based on noncompartmental analysis.		

Clearance	The popPK approach derived mean (%CV) clearance of fam-trastuzumab deruxtecan is 0.42 L/day.
Metabolism	
Primary metabolic pathway(s)	In vitro, MAAA-1181a is primarily metabolized by CYP3A4.
Inhibitor/Inducer	- In vitro, MAAA-1181a does not inhibit CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, CYP2D6 and CYP3A nor induce CYP1A2, CYP2B6, or CYP3A. - At clinically relevant concentrations (steady-state C_{max} of ~0.2 µmol/L), MAAA-1181a has a low potential to inhibit OAT1 (IC₅₀ value of 12.7 µmol/L), OAT3, OCT1, OCT2, OATP1B1 (IC₅₀ value of 14.4 µmol/L), OATP1B3, MATE1, MATE2-K, P-gp, BCRP, or BSEP transporters.
Excretion	
Primary excretion pathways (% dose) ±SD	- No human mass balance study was conducted. - In vitro, MAAA-1181a is a substrate of OATP1B1, OATP1B3, MATE2-K, P-gp, MRP1 and BCRP. - In rats, biliary secretion accounts ~ 71%, most of which are unchanged.

6.3.2. Clinical Pharmacology Questions

6.3.2.1. Does the clinical pharmacology program provide supportive evidence of effectiveness?

Data:

Exposure-response analyses for efficacy were conducted using logistic regression modeling for efficacy (ORR, DoR, and PFS) using data from Studies DS8201-A-J101 and DS8201-A-U201. The relationship between fam-trastuzumab deruxtecan average concentration (C_{avg}) from beginning of treatment to the time of ORR (C_{avg}ORR) was positive and statistically significant (P-value = 0.028) and best described by a linear logistic regression model. There was a trend of improved PFS with higher fam-trastuzumab deruxtecan exposures (both C_{avg} at PFS and minimum observed serum concentration at steady-state [C_{min,ss}]), which was not statistically significant (P-value >0.05). Based on the exposure-efficacy models, the mean probability of ORR was predicted to be 0.63 (90% CI: 0.55, 0.70) for the 5.4 mg/kg dose and 0.68 (95% CI: 0.58, 0.76) for the 6.4 mg/kg dose, which agrees with observed data. Results of a covariate search (including baseline demographics, country, and disease status factors) demonstrated that no covariates were significant for exposure-ORR relationship.

Exposure-response analyses for safety were conducted using data across all doses tested (0.8 mg/kg to 8.0 mg/kg) across the 5 clinical studies included in the submission. Exposure-

response analyses for safety endpoints were conducted using logistic regression modeling and Cox proportional regression (for interstitial lung disease [ILD] only). The ER analysis for safety (N = 639) was performed on the following treatment-emergent safety endpoints:

- Discontinuation associated with AEs
- Dose reduction associated with AEs
- Drug interruption associated with AEs
- AEs of $\geq$ Grade 3
- SAEs
- Anemia, neutropenia, or thrombocytopenia; severity grades derived using Common Terminology Criteria for Adverse Events (CTCAE) grading criteria
- AEs of special interest (AESIs), ie, ILD/pneumonitis (any grade and $\geq$ Grade 3), and decreased left ventricular ejection fraction (LVEF; $\geq$ Grade 2); severity grades derived using CTCAE grading criteria

In these multivariate analyses, all evaluated safety endpoints showed a statistically significant (P-value ≤ 0.05) ER relationship with 1 or more of the exposure parameters evaluated for fam-trastuzumab deruxtecan or MAAA-1181a. Fam-trastuzumab deruxtecan AUC_{ss} was a statistically significant predictor for discontinuation associated with AEs (P-value < 0.001) and ILD of any grade (P-value < 0.001); fam-trastuzumab deruxtecan C_{max} at steady-state (C_{max,ss}) was a statistically significant predictor for ILD $\geq$ Grade 3 (P-value < 0.001) and $\geq$ Grade 2 LVEF decrease (P-value < 0.001), while C_{avg} through the event cycle for MAAA-1181a was a significant predictor for all other evaluated safety endpoints.

In these analyses, race-country combination (Asian-Japan, Asian non-Japan, and non-Asian) and number of prior non-hormonal cancer therapies ≥ 6 were statistically significant covariates for discontinuation associated with AEs and drug interruptions associated with AEs. Baseline body weight, number of prior non-hormonal cancer therapies ≥ 6 , and HER2 status were statistically significant covariates for dose reductions associated with AEs. Eastern Cooperative Oncology Group performance status (ECOG PS) and baseline body weight were statistically significant covariates for SAEs and AEs $\geq$ Grade 3, respectively. Race-country combination and baseline neutrophils were statistically significant covariates for neutropenia (all grades or $\geq$ Grade 3). Baseline hemoglobin, sex, and body weight were statistically significant covariates for anemia (all grades), and race-country combination, age, ECOG PS, and baseline hemoglobin were statistically significant covariates for anemia ($\geq$ Grade 3). Race-country combination and baseline platelets were statistically significant covariates for thrombocytopenia (all grades), and race-country combination and number of prior non-hormonal cancer therapies ≥ 6 were

statistically significant covariates for thrombocytopenia ($\geq$ Grade 3). Race-country combination was a statistically significant covariate for ILD (any grade).

The Applicant's Position:

Yes, the clinical pharmacology program provides supportive evidence of effectiveness. Results from the Phase 1 and Phase 2 exposure-efficacy (ORR) analyses show that fam-trastuzumab deruxtecan exposure is a significant predictor of ORR, supporting the use of the 5.4 mg/kg dose. Results from the exposure-safety analyses also indicate correlation between fam-trastuzumab deruxtecan or MAAA-1181a exposures and safety endpoints. These safety endpoints included: treatment discontinuation, dose reduction, or drug interruption associated with AEs; AEs of $\geq$ Grade 3; SAEs; AEs of anemia, neutropenia, or thrombocytopenia; AESIs, ie, ILD/pneumonitis (any grade and $\geq$ Grade 3) and decreased LVEF ($\geq$ Grade 2). Overall, these analyses support that fam-trastuzumab deruxtecan has a positive benefit-risk ratio at a 5.4 mg/kg dose administered Q3W.

The FDA's Assessment:

FDA agrees with the applicant's position that the dose of 5.4 mg/kg Q3W support positive benefit/risk profile of fam-trastuzumab deruxtecan based on currently available data. as stated above. However, FDA is uncertain whether the proposed dose of 5.4 mg/kg Q3W is optimal due to a lack of anticancer activity and safety data at dose levels < 5.4 mg/kg.

6.3.2.2. Is the proposed dosing regimen appropriate for the general patient population for which the indication is being sought?

Data:

Refer to Sections 6.2.2.1 and 6.3.2.1.

The Applicant's Position:

Yes. The recommended dosing regimen of fam-trastuzumab deruxtecan for the treatment of subjects with unresectable or metastatic HER2-positive breast cancer is 5.4 mg/kg administered once Q3W. This regimen was evaluated in the FIH Phase 1 study (DS8201-A-J101) and the Phase 2 study (DS8201-A-U201) in subjects with metastatic breast cancer (see [Table 19](#)). The cumulative efficacy data across these 2 studies and safety data across all 5 clinical studies included in this application support a positive benefit/risk profile for fam-trastuzumab deruxtecan at the 5.4 mg/kg dose for subjects with unresectable or metastatic HER2-positive breast cancer. Additionally, the ER analyses for efficacy and key safety variables from clinical studies further support the positive benefit/risk profile for the 5.4 mg/kg dose.

The FDA's Assessment:

The FDA agrees.

6.3.2.3. Is an alternative dosing regimen or management strategy required for subpopulations based on intrinsic patient factors?

Data:

The effect of intrinsic factors (age, sex, race/ethnicity, body weight, hepatic impairment, renal impairment, baseline albumin, baseline tumor size, and tumor type) on the exposures of fam-trastuzumab deruxtecan and MAAA-1181a was assessed by studying covariate effects in the PopPK analysis, with a focus on those that may differ between dose groups and studies.

Based on the PopPK analysis, body weight, baseline albumin, sex, baseline tumor size, and country (Japan and non-Japan) were identified as statistically significant covariates on clearance of fam-trastuzumab deruxtecan; body weight and sex were identified as significant covariates on V_c , and country was identified as a significant covariate on peripheral compartment volume. The model suggests that intact fam-trastuzumab deruxtecan clearance increased with increasing body weight, baseline tumor size, and being a male, and decreased with increasing albumin and in subjects from Japan; V_c increased with increasing body weight and being male; and peripheral volume of distribution decreased in subjects from Japan.

In the final model for MAAA-1181a, baseline body weight, baseline AST, and TBL were identified as statistically significant covariates on the clearance of MAAA-1181a, and age as statistically significant covariates on volume of distribution. The model suggested that the clearance of MAAA-1181a increased with increasing body weight and decreased with increasing TBL and AST; and the volume of distribution of MAAA-1181a increased with increasing age. While these covariates were statistically significant, the magnitude of these effects indicates that no dose adjustment is required for any of the identified covariates.

The Applicant's Position:

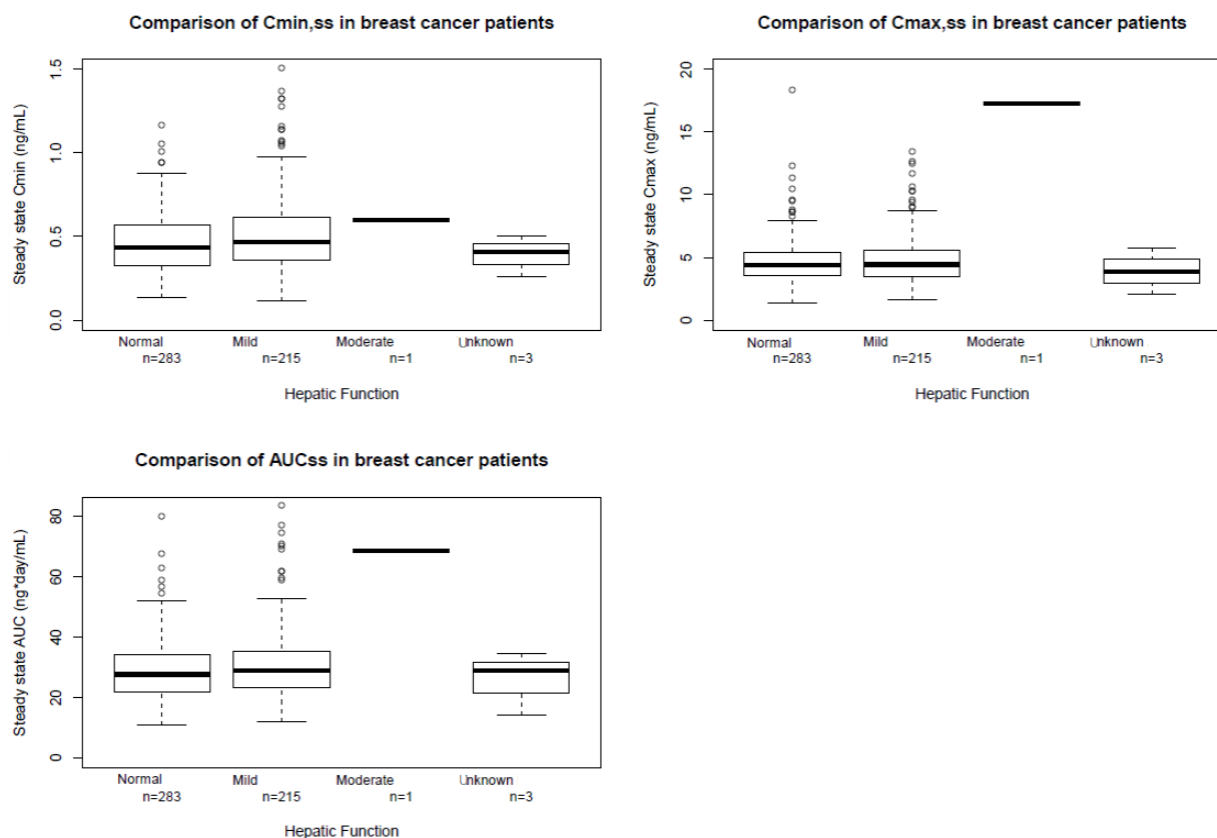
No. Based on the PopPK analysis, no dose adjustment is required for intrinsic factors (age, sex, race/ethnicity, body weight, baseline albumin, baseline tumor size, and tumor type) and for patients with mild or moderate renal impairment and for patients with mild hepatic impairment. Limited data preclude a recommendation for patients with moderate hepatic impairment and severe renal impairment. Fam-trastuzumab deruxtecan was not studied in subjects with severe hepatic impairment.

The FDA's Assessment:

The FDA generally agrees with the Applicant's position. However, FDA noted that ethnicity was not evaluated in the PopPK analysis. In addition, FDA generally agree with the applicant's assessment on the limited data in patient with moderate hepatic impairment. There is only 1 patient with moderate hepatic impairment. Based on FDA's PopPK analysis (Figure 4), this

patient appears to have higher C_{max} and AUC values of MAAA-1181a (DXd) than those in patients with normal hepatic function and mild hepatic impairment. This patient received 5 cycles of 5.4 mg/kg fam-trastuzumab deruxtecan. There is one dose interruption at Cycle 5, but no dose reduction.

Figure 4: Comparison of MAAA-1181a (DXd) and patients with hepatic impairment (NCI-ODWG criteria)



FDA concluded that the proposed dose can be administered to patients with moderate hepatic impairment with closely safety monitoring, based on the following rationales:

1. There is no human mass balance study to determine the main elimination pathway of MAAA-118a. However, based on rat study, over 70% of MAAA-1181a is eliminated in the liver, suggesting that the liver is the main eliminating pathway. However, in the DDI study (Study A104), coadministration of itraconazole (strong CYP3A inhibitor) or ritonavir (CYP3A and OATP1B inhibitor) did not increase the exposure of MAAA-1181a. The data indirectly suggests that the risk of increased MAAA-1181a exposure in patients with moderate hepatic impairment is low.

2. There is a lack of efficacy data to support dose adjustment for patient with moderate hepatic impairment. There is limited data at doses < 5.4 mg/kg to evaluate whether efficacy is maintained.
3. Fam-trastuzumab deruxtecan appears tolerable at the proposed recommended dose, and the label contains a sufficient safety monitoring plan. Increased liver AST/ALT level was observed at doses of 5.4 mg/kg (12%) and ≥ 6.4 mg/kg (23%) but was lower than that of the other approved similar antibody drug conjugate, T-DM1, which is 79%. There is sufficient dose modification plan in the current package insert to monitor for safety.

Based on the above rationales, FDA recommends that no dose adjustment for patient with moderate hepatic impairment. However, FDA recommends that the applicant to collect safety data in at least 10 patients with moderate hepatic impairment in ongoing trials and submit the data in the final analysis reports.

FDA also explored the relationship between baseline serum HER2 extracellular domain (ECD) levels and treatment response in Study DS8201-A-U201. No significant differences between responders and non-responders were observed. See Appendix 19.4 for additional details.

6.3.2.4. Are there clinically relevant food-drug or drug-drug interactions, and what is the appropriate management strategy?

Data:

Human biomaterial studies indicated that released drug, MAAA-1181a, is a substrate of P-gp, MATE2-K, OATP1B1, OATP1B3, BCRP, MRP1, and CYP3A4. Based on in vitro data, MAAA-1181a is not a reversible or time-dependent inhibitor of CYP isoforms, and also did not show induction potential on mRNA expression or metabolic activity of CYP isoforms.

A clinical DDI study (DS8201-A-A104) was conducted to evaluate the effect of a dual inhibitor of OATP1B/CYP3A (ritonavir) and a strong inhibitor of CYP3A (itraconazole). Results from this study indicated that there was no clinically meaningful effect of these inhibitors on the exposures of fam-trastuzumab deruxtecan or MAAA-1181a.

In human biomaterial studies of transporters, MAAA-1181a inhibited OAT1 and OATP1B1 with IC₅₀ values of 12.7 µmol/L and 14.4 µmol/L, respectively. MAAA-1181a did not inhibit OAT3, OCT1, OCT2, OATP1B3, MATE1, MATE2-K, P-gp, BCRP, or BSEP.

Although MAAA-1181a in vitro inhibited OAT1 or OATP1B1, no clinically meaningful DDI is expected with drugs that are substrates of OAT1 or OATP1B1 transporters as the mean MAAA-1181a C_{max} at the highest fam-trastuzumab deruxtecan dose of 8.0 mg/kg used in clinical studies or the clinical recommended fam-trastuzumab deruxtecan dose of 5.4 mg/kg

dose were 9.65 ng/mL (0.019 μ mol/L) and 10.8 ng/mL (0.02 μ mol/L), respectively, which are very low compared with the IC₅₀ values (Section 6.3.1).

The Applicant's Position:

Fam-trastuzumab deruxtecan is a biologics product administered by IV infusion; food-drug interaction is not anticipated.

Strong inhibitors of CYP3A and OATP1B can be administered without fam-trastuzumab deruxtecan dose adjustment.

Although a dedicated study with a P-gp inhibitor was not conducted, since itraconazole and ritonavir have been reported to be P-gp inhibitors, it can be inferred from the results of Study DS8201-A-A104 that P-gp inhibitors are also not expected to cause clinically meaningful increases in exposure of fam-trastuzumab deruxtecan or MAAA-1181a. Similarly, although MAAA-1181a was identified in vitro to be a substrate of the efflux transporter BCRP, no dedicated study was conducted to evaluate the effects of a BCRP inhibitor. Ritonavir has been reported to be an inhibitor of BCRP, supporting the conclusion that BCRP inhibitors will also have no clinically meaningful effects on exposures of fam-trastuzumab deruxtecan or MAAA-1181a. Based on these data, no dose adjustment is needed for fam-trastuzumab deruxtecan with drugs that are inhibitors of P-gp, OATP1B1, OATP1B3, BCRP, and CYP3A4.

The FDA's Assessment:

The FDA agrees with the applicant's position.

The drug interaction study (Study A104) is comprised of 2 parallel cohorts. Cohort 1 was non-randomized cohort to evaluate the effect of ritonavir (CYP3A and OATP1B inhibitor) on PK of fam-trastuzumab deruxtecan and MAAA-1181a (DXd). Cohort 2 was non-randomized cohort to evaluate the effect of itraconazole (strong CYP3A inhibitor) on PK of fam-trastuzumab deruxtecan and MAAA-1181a. Patients with HER2-expressing advanced solid malignant tumors were enrolled and treated with 5.4 mg/kg Q3W (one cycle is Q3W). Ritonavir (200 mg BID) or itraconazole (200 mg BID) were administered PO between Day 17 of Cycle 2 and Day 21 of Cycle 3. The PK parameters were evaluated during Cycle 2 and Cycle 3. Those Cycle 2 and Cycle 3 timepoints were chosen because Cycle 1 had a different infusion rate which may result in different PK profiles. The geometric mean ratio and the 90% confidence intervals of C_{max} and AUC_{0-17days} were compared. This applicant's rationale appears reasonable as fam-trastuzumab deruxtecan and MAAA-1181a has limited accumulation in the proposed dosing regimen, and therefore, the exposure with and without inhibitor can be compared. The PK results showed no clinically meaningful changes (Table 18).

Table 18: Summary of statistical analysis of assessment of PK parameters with or without inhibitors

	+/- Ritonavir (GMR, 90% CI)	+/- Itraconazole (GMR, 90% CI)
Fam-trastuzumab deruxtecan		
Ratio Cmax	1.05 (0.98-1.13)	1.02 (0.96-1.09)
Ratio AUC _{0-17days}	1.19 (1.13-1.25)	1.10 (1.07-1.15)
MAAA-1181a		
Ratio Cmax	0.99 (0.85-1.14)	1.03 (0.92-1.18)
Ratio AUC _{0-17days}	1.22 (1.08-1.37)	1.18 (1.11-1.25)

Source: 5.3.3.4 DS8201-a-A104; Study Report Body; Tables 9.7 and 9.9

FDA agrees that the drug interaction of MAAA-1181a (DXd) with inhibitors of P-gp, OATP1B1, OATP1B3, BCRP, and CYP3A4 is unlikely based on clinical DDI and in vitro data and the applicant's rationale mentioned above.

X

X

Clinical Pharmacology Team

Edwin Chiu Yuen Chow

Pengfei Song

Pharmacometrics Team

Junshan Qiu

Jingyu (Jerry) Yu

Genomics Team

Jielin (Jillian) Sun

Rosane Charlab Orbach

Primary Reviewer

Team Leader

7 Sources of Clinical Data

This application is based primarily on the totality of efficacy and safety data from subjects in the target population of unresectable or metastatic HER2-positive (defined as IHC 3+ or ISH+) breast cancer treated at the target dose of 5.4 mg/kg from 2 clinical studies, Phase 2 study DS8201-A-U201 and Phase 1 study DS8201-A-J101 ([Table 19](#)). Three Phase 1 clinical studies (DS8201-A-J102, DS8201-A-A103, and DS8201-A-A104) are also included in this application to support safety assessments.

7.1. Table of Clinical Studies

Data:

All studies pertinent to the evaluation of efficacy and safety are summarized in [Table 19](#).

Table 19: Listing of Clinical Trials Relevant to this BLA

Trial Identity	NCT No.	Trial Design	Regimen/ schedule/ route	Study Endpoints	Treatment Duration/ Duration of Follow-up	No. of patients enrolled and treated	Study Population	No. of Centers and Countries
Studies to Support Efficacy and Safety								
DS8201-A-J101	02564900	Phase 1, multicenter, non-randomized, open-label, multiple dose, first-in-human study in 2 parts: Dose Escalation (Part 1) and Dose Expansion (Part 2)	Fam-trastuzumab deruxtecan IV infusion Q3W <u>Dose Escalation (Part 1):</u> 0.8 mg/kg; 1.6 mg/kg; 3.2 mg/kg; 5.4 mg/kg; 6.4 mg/kg; 8.0 mg/kg <u>Dose Expansion (Part 2):</u> 5.4 mg/kg; 6.4 mg/kg	<u>Primary:</u> • Dose Escalation: safety and tolerability of fam-trastuzumab deruxtecan; determination of MTD or RP2D • Dose Expansion: safety and tolerability; efficacy at the MTD/RP2D <u>Secondary:</u> • Dose Escalation: PK profile of DS-8201a, total anti-HER2 antibody, and MAAA-11811; efficacy; incidence of ADAs against fam-trastuzumab deruxtecan • Dose Expansion: PK profile of DS-8201a, total anti-HER2 antibody, and MAAA-11811, and incidence of ADAs	DCO: 01 Feb 2019 <u>Median treatment duration</u> Overall: 7.4 months (range: 0.7, 40.4) HER2-positive breast cancer cohort : 8.8 months (range: 0.7-35.0) <u>Median duration of follow-up</u> Overall: 8.4 months (range: 0.1-39.8) HER2-positive breast cancer cohort: 10.2 months (range: 0.1, 34.4)	<u>Enrolled:</u> Overall: 292 HER2-positive breast cancer: 118 <u>Treated:</u> Overall: 289 HER2-pos breast cancer: 116	<u>Part 1:</u> advanced unresectable or metastatic breast cancer and gastric or GEJ adenocarcinoma <u>Part 2a:</u> T-DM1-treated HER2-positive breast cancer <u>Part 2b:</u> trastuzumab-treated HER2-positive gastric or GEJ adenocarcinoma <u>Part 2c:</u> HER2 low expressing breast cancer <u>Part 2d:</u> HER2-expressing other solid	US (8 centers) and Japan (6)

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Trial Identity	NCT No.	Trial Design	Regimen/ schedule/ route	Study Endpoints	Treatment Duration/ Duration of Follow-up	No. of patients enrolled and treated	Study Population	No. of Centers and Countries
							malignant tumor/HER2- mutated tumor <u>Part 2e:</u> HER2 expressing breast cancer	
DS8201- A-U201	03248492	Phase 2, multicenter, open-label, multiple- dose, 2-part study	Fam-trastuzumab deruxtecan IV infusion Q3W <u>Part 1:</u> 5.4 mg/kg 6.4 mg/kg 7.4 mg/kg <u>Part 2:</u> 5.4 mg/kg	<u>Primary:</u> ORR by ICR <u>Secondary:</u> - Efficacy: investigator-assessed ORR; DCR, CBR, DoR, and best percent change in sum of diameters of measurable tumors per RECIST v1.1 per ICR; and PFS and OS - Safety - Serum concentration and PK parameters for DS-8201a, total anti-HER2 antibody, and MAAA-1181a	<u>DCO:</u> 21 Mar 2019 <u>Median treatment duration</u> 7.0 months (range: 0.7-16.2) <u>Median duration of follow-up:</u> 7.8 months (range: 0.7-17.2)	<u>Enrolled:</u> Part 1: 119 Part 2: 134 Overall: 253 <u>Treated:</u> Part 1: 119 Part 2: 134 Overall: 253	Subjects with HER2-positive unresectable or metastatic breast cancer previously treated with T-DM1	US (24 centers), Japan (10), France (9), Spain (8), South Korea (7), Belgium (5), United Kingdom (5), and Italy (4)

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Trial Identity	NCT No.	Trial Design	Regimen/ schedule/ route	Study Endpoints	Treatment Duration/ Duration of Follow-up	No. of patients enrolled and treated	Study Population	No. of Centers and Countries
Studies to Support Safety								
DS8201-A-J102	03366428	Phase 1, multicenter, non-randomized, open-label, multiple dose study	Fam-trastuzumab deruxtecan 6.4 mg/kg IV infusion Q3W	<u>Primary:</u> - Baseline-adjusted QTcF interval - Serum concentrations and PK parameters of DS-8201a, total anti-HER2 antibody, and MAAA-1181a after multiple dosing <u>Secondary:</u> - Safety - Serum concentrations of fam-trastuzumab deruxtecan at the time of ECG measurements - Efficacy	DCO: 05 Dec 2018 <u>Median treatment duration</u> 4.6 months (range: 0.7-9.0) <u>Median duration of follow-up:</u> 147.0 days (range: 44-272 days) ^a	<u>Enrolled:</u> 51 <u>Treated:</u> 51	Unresectable or metastatic breast cancer with HER2 expression	Japan (7 centers)
DS8201-A-A103	03368196	Phase 1, multicenter, non-randomized, open-label study	Fam-trastuzumab deruxtecan 6.4 mg/kg IV infusion Q3W	<u>Primary:</u> Safety and tolerability in subjects of Chinese descent <u>Secondary:</u> - PK profile of DS-8201a, total anti-HER2 antibody, and MAAA-1181a - Anti-tumor activity of	DCO: 14 Sep 2018 <u>Median treatment duration:</u> 140.0 days (range: 84-168 days) ^b	<u>Enrolled:</u> 12 <u>Treated:</u> 12	HER2-positive advanced/unresectable or metastatic gastric, GEJ adenocarcinoma or breast cancer with both HER2-positive disease and	Taiwan (2 centers)

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Trial Identity	NCT No.	Trial Design	Regimen/ schedule/ route	Study Endpoints	Treatment Duration/ Duration of Follow-up	No. of patients enrolled and treated	Study Population	No. of Centers and Countries
				fam-trastuzumab deruxtecan • Incidence of ADAs against fam- trastuzumab deruxtecan	Median duration of follow-up: 3.95 months (range: 2.8, 4.9)		documented disease progression in subjects of Chinese descent	
DS8201- A-A104	03383692	Phase 1, multicenter, non- randomized, open-label, single- sequence crossover study	<u>Cohort 1:</u> Fam-trastuzumab deruxtecan 5.4 mg/kg IV infusion Q3W in combination with ritonavir 200 mg BID on Day 17 of Cycle 2 until Day 21 of Cycle 3 <u>Cohort 2:</u> Fam-trastuzumab deruxtecan 5.4 mg/kg IV infusion Q3W in combination with itraconazole 200 mg BID on Day 17 of Cycle 2 followed by 200 mg QD until Day 21 of Cycle 3	<u>Primary:</u> • Cohort 1: effect of ritonavir (dual inhibitor of CYP3A [strong] and OATP1B) on PK parameters of DS-8201a and MAAA-1181a • Cohort 2: effect of itraconazole (strong CYP3A inhibitor) on PK parameters of DS-8201a and MAAA-1181a <u>Secondary:</u> • Safety • Efficacy • ADAs	<u>DCO:</u> 26 Sep 2018 <u>Cohort 1:</u> Median treatment duration: 211.0 days (range: 42-254) ^c Median duration of follow-up: 192.0 days (range: 69-234) ^d <u>Cohort 2:</u> Median treatment duration: 148.0 days (range: 21-213) ^c Median duration of follow-up: 134 days (range: 3-193) ^d	<u>Enrolled:</u> 40 <u>Treated:</u> 40 Cohort 1: 17 Cohort 2: 23	Unresectable or metastatic solid malignant tumors with HER2 expression	Japan (7 centers), South Korea (2), Taiwan (1)
Other Studies (All Ongoing)								

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Trial Identity	NCT No.	Trial Design	Regimen/ schedule/ route	Study Endpoints	Treatment Duration/ Duration of Follow-up	No. of patients enrolled and treated	Study Population	No. of Centers and Countries
DS8201-A-U301	03523585	Phase 3 multicenter, randomized, controlled, open-label versus investigator's choice	Fam-trastuzumab deruxtecan 5.4 mg/kg IV infusion Q3W versus investigator's choice of trastuzumab/ capecitabine or lapatinib/ capecitabine	• To compare PFS benefit of fam-trastuzumab deruxtecan vs investigator's choice • To further assess efficacy and evaluate safety compared to investigator's choice • To determine PK of DS-8201a	Ongoing study	Enrolled as of 28 Jun 2019: 116 (Planned: ~600)	HER2-positive unresectable and/or metastatic breast cancer who progressed after T-DM1	Global, multicenter
DS8201-A-U302	03529110	Phase 3 multicenter, randomized, controlled, open-label versus T-DM1	Fam-trastuzumab deruxtecan 5.4 mg/kg IV infusion Q3W versus T-DM1: 3.6 mg/kg IV Q3W	• To compare PFS benefit of fam-trastuzumab deruxtecan vs T-DM1 • To further assess efficacy and evaluate safety compared to T-DM1 • To determine PK of DS-8201a	Ongoing study	Enrolled as of 28 Jun 2019: 99 (Planned: ~500)	HER2-positive unresectable and/or metastatic breast cancer previously treated with trastuzumab and a taxane	Global, multicenter
DS8201-A-U303	03734029	Phase 3 multicenter, randomized, controlled, open-label versus treatment of	Fam-trastuzumab deruxtecan 5.4 mg/kg IV infusion Q3W versus	• To compare PFS benefit of fam-trastuzumab deruxtecan vs treatment of physician's choice • To further assess	Ongoing study	Enrolled as of 28 Jun 2019: 12 (Planned:	HER2 low unresectable and/or metastatic breast cancer previously treated with 1	Global, multicenter

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Trial Identity	NCT No.	Trial Design	Regimen/ schedule/ route	Study Endpoints	Treatment Duration/ Duration of Follow-up	No. of patients enrolled and treated	Study Population	No. of Centers and Countries
		physician's choice	1 of the following choices: capecitabine eribulin gemcitabine paclitaxel nab-paclitaxel	efficacy and evaluate safety compared to treatment of physician's choice • To determine PK of DS-8201a		~540)	or 2 lines of chemotherapy	

DCO = data cut-off

For treatment duration, 1 month = 365.25/12 days = 30.44 days

Duration of follow-up (or study follow-up) was defined as: (last visit date – randomization/registration date +1)/365.25*12

^a For DS8201-A-J102, median treatment duration = 4.8 months (range: 1.4-8.9)

^b For DS8201-A-A103, median treatment duration = 4.6 months (range: 2.8-5.5)

^c For DS8201-A-104 median treatment duration = 6.9 months (range: 1.4-8.3) for Cohort 1 and 4.9 months (range: 0.7-7.0) for Cohort 2

^d For DS8201-A-104, median duration of follow-up = 6.3 months (range: 2.3-7.7) for Cohort 1 and 4.4 months (0.1-6.3) for Cohort 2

The Applicant's Position:

All studies listed above are pertinent to the evaluation of efficacy and safety fam-trastuzumab deruxtecan to support the proposed indication.

The FDA's Assessment:

For this BLA application, the clinical data for the FDA's analysis of efficacy was based on the data for the 184 HER2-positive breast cancer patients from study DS8201-A-U201 who received fam-trastuzumab deruxtecan at a dose of 5.4mg/kg.

The FDA's safety analysis was primarily based on the pooled 234 patients with HER2-positive breast cancer from clinical studies DS8201-A-J101 and DS8201-A-U201 who received fam-trastuzumab deruxtecan at a dose of 5.4mg/kg. Data from the other studies were not used for the clinical efficacy and safety analysis due to the heterogeneity of the patient populations enrolled.

Clinical studies DS8201-A-U301, -U302 and U303 are ongoing studies that are still accruing patients and no data from these studies was submitted by the applicant for this BLA application.

8 Statistical and Clinical Evaluation

8.1. Review of Relevant Individual Trials Used to Support Efficacy

Efficacy data supporting the overall efficacy of fam-trastuzumab deruxtecan monotherapy at the target dose of 5.4 mg/kg IV Q3W in patients with unresectable or metastatic HER2-positive breast cancer are based on data from 2 studies, Phase 2 study DS8201-A-U201 and Phase 1 study DS8201-A-J101 (see [Table 19](#)). Study DS8201-A-U201 reflects the treatment type (monotherapy), proposed dosing regimen (5.4 mg/kg Q3W), and proposed indication

(b) (4)

and is using the intended commercial drug formulation (Lyo-DP). The FIH Phase 1 study DS8201-A-J101 contributes subjects with HER2-positive breast cancer with longer exposure data, supporting durability of confirmed response. Both studies, by replicating efficacy results in the HER2-positive unresectable or metastatic breast cancer population, provide a robust estimate of the ORR.

8.1.1. Studies DS8201-A-U201 and DS8201-A-J101

8.1.1.1. Study DS8201-A-U201

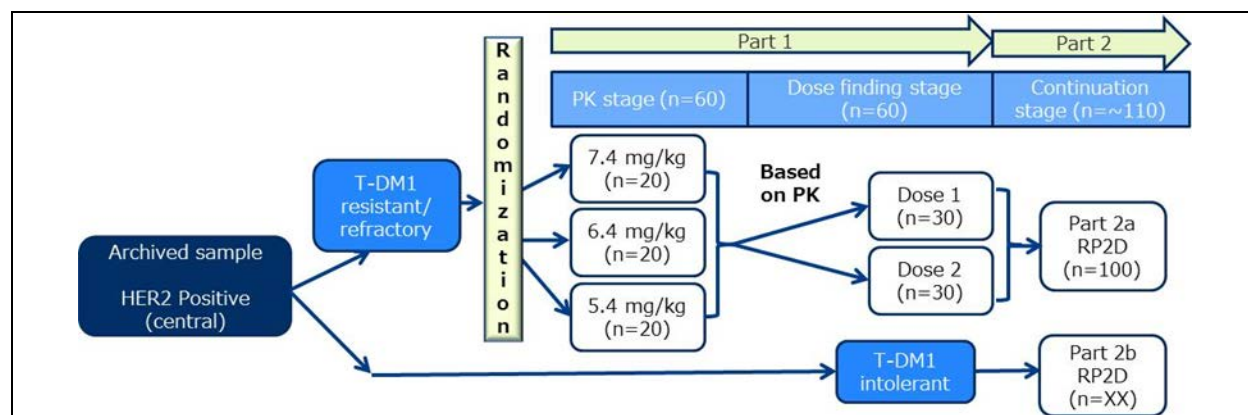
Trial Design

The Applicant's Description:

Overview of Study Design:

Study DS8201-A-U201 was a Phase 2, open-label, multicenter, 2-part study designed to justify the recommended dose of fam-trastuzumab deruxtecan and investigate its safety and efficacy in subjects with unresectable and/or metastatic HER2-positive breast cancer previously treated with T-DM1, with study sites in the US, Canada, Japan, South Korea, Belgium, France, Italy, Spain, and the United Kingdom. Determination of tumor HER2 positivity according to American Society of Clinical Oncology – College of American Pathologists guidelines ([Wolff et al 2013](#)) expression was based on archival tissue tested at a central laboratory prior to enrolment. The study design is depicted in [Figure 5](#).

Figure 5: DS8201-A-U201 Study Design



Note: for Part 2b, “XX” indicated the number of subjects was open-ended with no fixed enrolment target.

Part 1 of the study consisted of 2 stages: a PK stage and a Dose Finding stage. In the PK stage, subjects were randomized by region in a 1:1:1 ratio to 1 of 3 doses: 5.4 mg/kg, 6.4 mg/kg, and 7.4 mg/kg, designed to provide clinical data to bridge between 2 dosing formulations, FL-DP2 and Lyo-DP. After review of the PK findings, subjects were randomized by region in a 1:1 ratio to 1 of the 2 dose levels that were selected for further evaluation in the Dose Finding stage: 5.4 mg/kg and 6.4 mg/kg. Subsequently, a dose-selection efficacy and safety analysis to justify the dose was conducted to determine the optimal dose of fam-trastuzumab deruxtecan (5.4 mg/kg).

Part 2 of the study was not randomized, and all subjects received fam-trastuzumab deruxtecan at the recommended 5.4 mg/kg dose. Part 2 was subdivided into 2 cohorts: Part 2a, the main study part, and Part 2b, which was designed to explore the efficacy of fam-trastuzumab deruxtecan in subjects who discontinued T-DM1 for reasons other than PD. The duration of enrollment in Part 2b was determined by the enrollment in Part 2a, with completion of enrollment in Part 2b when the target number of approximately 100 subjects was reached in Part 2a.

The subjects with HER2-positive breast cancer treated at 5.4 mg/kg who are part of the analyses were enrolled in Part 1, Part 2a, and Part 2b of Study DS8201-A-U201.

Key Inclusion Criteria:

- Men or women ≥18 years old in the US/Europe or ≥20 years old in Japan and South Korea with pathologically documented breast cancer that was unresectable or metastatic and had HER2-positive expression
- Breast cancer resistant or refractory to T-DM1 with documentation of progression during or after T-DM1 (for Part 2b, subjects were to have discontinued T-DM1 for reasons other than PD)

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

- Presence of at least 1 measurable lesion per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST v1.1)
- LVEF $\geq$ 50%
- ECOG PS of 0 or 1
- Adequate bone marrow, renal, hepatic, and blood clotting functions

Key Exclusion Criteria:

- Medical history of myocardial infarction within 6 months before randomization/registration, symptomatic congestive heart failure (CHF), troponin levels consistent with myocardial infarction as defined according to the manufacturer, unstable angina or serious cardiac arrhythmia requiring treatment within 28 days before randomization/registration
- Prolongation of QTc $>$ 470 milliseconds (females) or $>$ 450 milliseconds (males)
- History of noninfectious ILD/pneumonitis that required treatment, or current or suspected ILD/pneumonitis
- Brain metastases that were untreated, symptomatic or required treatment, or history of radiation, surgery or other therapy to control symptoms from brain metastases within 60 days prior to randomization/registration
- Pregnancy or breastfeeding

Dose Selection:

The dose selection for each stage of the study was based on clinical data from the FIH study DS8201-A-J101 and on the prespecified interim ER analysis during the conduct of Study DS8201-A-U201 itself (see dose justification in Section [6.2.2.1](#)).

Study Treatments:

After a screening period of up to 28 days, eligible subjects were administered fam-trastuzumab deruxtecan as an IV solution Q3W. The first IV dose of fam-trastuzumab deruxtecan was administered over 90 minutes ($\pm$ 10 minutes). If there was no infusion-related reaction (IRR) after the first dose, subsequent doses were administered over 30 minutes ($\pm$ 5 minutes). The duration of treatment was not fixed. Treatment with fam-trastuzumab deruxtecan continued as long as the subjects derived clinical benefit until they experienced PD, unacceptable toxicity, or withdrew consent.

Administrative Structure:

An Interstitial Lung Disease Adjudication Committee (ILD AC) reviewed and adjudicated potential cases of ILD on an ongoing basis. An independent radiology review committee (ICR) was instated to assess tumor scans.

Dietary Restrictions/Instructions:

During the treatment period, foods or beverages containing grapefruit, a CYP3A4 inhibitor, were prohibited.

Concurrent Medications/Rescue Medications:

During the treatment period, the following medications, treatments, and procedures were prohibited: other anticancer therapies or investigational therapeutic agents; radiotherapy (except for palliative radiation to known metastatic sites); chronic systemic corticosteroids or other immunosuppressive medications; CYP3A4 strong inhibitors; and OATP1B inhibitors.

Hematopoietic growth factors could be used for prophylaxis or treatment based on the clinical judgment of the investigator. Prophylactic or supportive treatment of study-drug induced AEs were otherwise to be as per investigator's discretion and institutional guidelines.

Concomitant use of dietary supplements, medications not prescribed by the investigator, and alternative/complementary treatments was discouraged, but not prohibited.

Treatment Compliance:

Fam-trastuzumab deruxtecan was administered IV only to subjects participating in the study and under the supervision of the clinical study personnel at the study site. Start and stop times of injection and amount of drug administered were recorded by the clinical study personnel.

Treatment Discontinuation or Study Withdrawal:

Subjects could discontinue study drug after signing the informed consent form (ICF) for the following reasons: PD per RECIST v1.1 criteria; clinical progression (ie, definitive clinical signs of progression but a recent radiographic assessment did not meet the RECIST criteria for PD); AE; consent withdrawal by subject; physician's decision; death; pregnancy; study terminated by the Sponsor; and subject lost to follow-up. If there was evidence that the subject was receiving benefit from treatment, even though the subject had met 1 of the criteria for study drug discontinuation listed above, the subject could remain on study drug after discussion with the Sponsor. After study drug discontinuation, subjects were followed for survival status either through direct contacts or by collecting public records (e.g., death certificates) as allowed by local laws.

Subjects could withdraw from the study for the following reasons: consent withdrawal by subject, death, termination of study, loss to follow-up. All subjects were followed for survival status even after they withdrew consent for study procedures. Subjects discontinued from the study because of withdrawal of consent were followed for survival by collecting public records (ego, death certificates) every 3 months from the last study visit, unless prohibited by local laws.

Subjects withdrawn from the study were not replaced if they had been administered study medication. It was allowable to replace a subject who was enrolled but was not administered any study medication, although this situation did not arise during the course of the study.

Procedures and schedule:

Figure 6: Schedule of Events for Part 1 Pharmacokinetic Stage

	Tissue SCR	SCR	Cycle 1							Cycle 2			Cycle 3			Cycle 4 and Subsequent Cycles		Q6W ± 7 days	EOT ^a	40- day F/U ^b	Q3M F/U ± 14 days
			D1		D2	D4	D8	D15	D22 ^c	D1		D22 ^c	D1		D22 ^c	D1					
			BI	EOI			± 1 day	± 1 day	± 2 days	BI	EOI	± 2 days	BI	EOI	± 2 days	BI	EOI				
-14 days or as noted																					
Informed consent (could sign >14 days prior)	●	●																			
Administer DS-8201a			●						●			●			●						
Medical history		●																			
Demographics		●																			
Pregnancy test		● ^c																●			
Vital signs		●	● ^a	●		●	●		● ^d	●		● ^d	●		● ^d			●	●		
Physical examination		●	● ^d						● ^d			● ^d			● ^d			●	●		
SpO2		●	● ^d						● ^d			● ^d			● ^d			●	●		
Inclusion/ exclusion		●																			
Height		●																			
Weight, ECOG PS		●	● ^d						● ^d			● ^d			● ^d			●	●		
Urinalysis and coagulation		●																			
Hematology and blood chemistry		●	● ^d			●	●		● ^d			● ^d			● ^d			●	●		
Troponin ^e		●		●						●			●			●		●	●		

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	Tissue SCR	SCR -14 days or as noted	Cycle 1							Cycle 2			Cycle 3		Cycle 4 and Subsequent Cycles		Q6W ± 7 days	EOT ^a	40- day F/U ^b	Q3M F/U ± 14 days
			D1		D2	D4	D8	D15	D22 ^c	D1		D22 ^c	D1		D22 ^c	D1				
			BI	EOI			± 1 day	± 1 day	± 2 days	BI	EOI	± 2 days	BI	EOI	± 2 days	BI	EOI			
Biomarker blood samples			•													•			•	
Pharmaco-genomic blood sample			•																	
PK blood (serum) sample			•	•	•	•	•	•	•	•	•	•	•	•	•	•	•			
ADA blood sample			•							•						•			•	•
HIV antibody test (as required by local regulations)		• ^f																		
Blood sample for HER2 ECD		•											• ^g			• ^g		•		
ECHO or MUGA (LVEF) ^h		• ^f														•		•		
12-Lead ECG in triplicate		•	• ^d							• ^d			• ^d			• ^d		•		
CT/MRI of brain		• ^f																• ⁱ	• ⁱ	
Tumor assessment		• ^f																•	•	
Ophthalmologic assessment		• ^f								• ^d						• ^d		•		

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

	Tissue SCR	SCR	Cycle 1							Cycle 2			Cycle 3			Cycle 4 and Subsequent Cycles		Q6W ± 7 days	EOT ^a	40- day F/U ^b	Q3M F/U ± 14 days
			D1		D2	D4	D8	D15	D22 ^c	D1		D22 ^c	D1		D22 ^c	D1					
		-14 days or as noted	BI	EOI				± 1 day	± 1 day	± 2 days	BI	EOI	± 2 days	BI	EOI	± 2 days	BI				
Tumor sample for HER2 status and optional biomarkers	•																				
Concomitant medications		•																			
Adverse events	•	•																			
Survival F/U																				•	

^a The date the investigator decided to discontinue study drug (+ 7 days).

^b 40 days (+ 7 days) after the last study drug administration or before starting new anticancer treatment, whichever came first.

^c Within 72 hours before randomization.

^d Within 3 days before administration. The blood collections were to be done after ECG (if performed on the same day as ECG).

^e Blood samples for troponin (preferably high-sensitivity troponin-T) were to be collected 2-3 hours after EOI.

^f Within 28 days before randomization/registration.

^g Before administration at every 2 cycles from Cycle 3 (Cycles 3, 5, 7, 9, etc.).

^h ECHO or MUGA scan assessments (Note: the same test had to be used for the subject throughout the study) were to be performed at screening and every 4 cycles (± 7 days) (Cycles 5, 9, 13, etc.).

ⁱ An MRI of the brain was mandatory for all subjects who were enrolled with baseline stable brain metastases. Subjects without brain metastases did not need additional brain scans for tumor assessment unless clinically indicated.

Figure 7: Schedule of Events for Part 1 Dose Finding Stage and Part 2

	Tissue SCR	SCR -14 days or as noted	Cycle 1					Cycle 2			Cycle 3			Cycle 4 and Subsequent Cycles		Q6W ± 7 days	EOT ^a	40- day F/U ^b	Q3M F/U ± 14 days
			D1		D8	D15	D22 ^c	D1		D22 ^c	D1		D22 ^c	D1					
			BI	EOI	± 1 day	± 1 day	± 2 days	BI	EOI	± 2 days	BI	EOI	± 2 days	BI	EOI				
Informed consent (could sign >14 days prior)	●	●																	
Administer DS-8201a			●					●			●		●						
Medical history		●																	
Demographics		●																	
Pregnancy test		● ^c															●		
Vital signs		●	● ^d	●	●	●		● ^d	●		● ^d	●		● ^d			●	●	
Physical examination		●	● ^d					● ^d			● ^d			● ^d			●	●	
SpO2		●	● ^d					● ^d			● ^d			● ^d			●	●	
Inclusion/exclusion		●																	
Height		●																	
Weight, ECOG PS		●	● ^d					● ^d			● ^d			● ^d			●	●	
Urinalysis and coagulation		●																	
Hematology and blood chemistry		●	● ^d		●	●		● ^d			● ^d			● ^d			●	●	
Troponin ^e		●		●					●			●			●		●	●	
Biomarker blood samples			●											●			●		

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	Tissue SCR	SCR -14 days or as noted	Cycle 1					Cycle 2			Cycle 3		Cycle 4 and Subsequent Cycles		Q6W ± 7 days	EOT ^a	40- day F/U ^b	Q3M F/U ± 14 days	
			D1		D8	D15	D22 ^c	D1		D22 ^c	D1		D22 ^c	D1					
			BI	EOI	± 1 day	± 1 day	± 2 days	BI	EOI	± 2 days	BI	EOI	± 2 days	BI					EOI
Pharmacogenomic blood sample			●																
PK blood (serum) sample			●	●	●	●	●	●	●	●	●	●	●	●					
ADA blood sample			●					●						●				●	●
HIV antibody test (as required by local regulations)		● ^f																	
Blood sample for HER2 ECD		●									● ^g			● ^g			●		
ECHO/MUGA (LVEF) ^h		● ^f												●			●		
12-lead ECG in triplicate		●	● ^d					● ^d			● ^d			● ^d			●		
CT/MRI of brain		● ^f														● ⁱ	● ⁱ		
Tumor assessment		● ^f														●	●		
Ophthalmologic assessment		● ^f						●						●			●		
Tumor sample for HER2 status and optional biomarkers	●																		
Concomitant medications			●																

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	Tissue SCR	SCR -14 days or as noted	Cycle 1				Cycle 2			Cycle 3			Cycle 4 and Subsequent Cycles		Q6W ± 7 days	EOT ^a	40- day F/U ^b	Q3M F/U ± 14 days	
			D1		D8	D15	D22 ^c	D1		D22 ^c	D1		D22 ^c	D1					
			BI	EOI	± 1 day	± 1 day	± 2 days	BI	EOI	± 2 days	BI	EOI	± 2 days	BI					EOI
Adverse events		●																	
Survival F/U																		●	

^a The date the investigator decided to discontinue study drug (+ 7 days).

^b 40 days (+ 7 days) after the last study drug administration or before starting new anticancer treatment, whichever came first.

^c Within 72 hours before randomization.

^d Within 3 days before administration. The blood collections were to be done after ECG (if performed on the same day as ECG).

^e Blood samples for troponin (preferably high-sensitivity troponin-T) were to be collected 2-3 hours after EOI.

^f Within 28 days before randomization/registration.

^g Before administration at every 2 cycles from Cycle 3 (Cycles 3, 5, 7, 9, etc.).

^h ECHO or MUGA scan assessments (Note: the same test had to be used for the subject throughout the study) were to be performed at screening and every 4 cycles (± 7 days) (Cycles 5, 9, 13, etc.).

ⁱ An MRI of the brain was mandatory for all subjects who were enrolled with baseline stable brain metastases. Subjects without brain metastases did not need additional brain scans for tumor assessment unless clinically indicated.

The FDA's Assessment:

FDA agrees with the summary of the study design.

Study Endpoints

The Applicant's Description:

Primary Efficacy Endpoint:

The primary efficacy endpoint of the study was the confirmed ORR based on RECIST v1.1 as assessed by an ICR of tumor scans. All responses had to be confirmed at the subsequent scan. Endpoints were discussed and agreed with the Food and Drug Administration (FDA) prior to the conduct of study (refer to Section 3).

Because ORR provides a direct measure of the antitumor activity in an objective manner and can be directly attributable to drug effect, using ORR as the primary evidence of efficacy is appropriate in single-arm studies conducted in patients with tumors where no available therapy exists. The magnitude of ORR along with the DoR provides an accurate assessment of antitumor activity of an investigational agent and could have the potential to reasonably predict clinical benefit.

Tumor assessments were performed at screening and every 6 weeks thereafter. Assessments were conducted using computed tomography (CT) or magnetic resonance imaging (MRI) (spiral CT or MRI with ≤ 5 mm cuts). Imaging was assessed by the investigators and sent out for ICR assessment. Two trained radiologists read each study subject's images independently, according to the ICR Charter. In case of discordance, adjudication was performed by a third radiologist. The adjudication variables were best confirmed response and date of progression.

Secondary Efficacy Endpoints:

Secondary efficacy endpoints included:

- Investigator-assessed ORR based on RECIST v1.1.
- DoR, defined as the time interval between the date of first documentation of objective response (complete response [CR] or partial response [PR]) and the date of the first objective documentation of disease progression or death due to any cause.
- PFS, defined as the time interval between the date of randomization/registration and the first documentation of PD or death due to any cause. Disease progression was determined through an ICR of tumor scans using RECIST v1.1. Clinical progression without objective documentation of disease progression per RECIST v1.1 was not considered to be progression while deriving the PFS endpoint.

- Best change from baseline in sum of diameters of measurable tumors based on RECIST v1.1.

Exploratory efficacy endpoints included time to response (TTR).

The FDA's Assessment:

FDA agrees with the applicant's summary of the study's efficacy endpoints; however, DS8201-A-U201 was a single arm trial and because there is not a comparator arm, time-to-event endpoints (e.g., progression-free survival, time to response) results are uninterpretable. The effectiveness evaluation relied on the demonstration of an improvement over available therapy based on the magnitude of the response rate and an adequate duration of response.

The Applicant's Description:

The statistical analysis plan (SAP) was finalized prior to the database lock.

A data cut-off (DCO) corresponding to 6 months of follow-up from the last subject randomized in the trial was established and all analyses were performed based on the data up-to this DCO date. Data in Part 1 and Part 2 were combined by the starting dose level. Data from all centers were pooled.

The primary efficacy analyses were based on ICR review of tumor scans. Primary and secondary efficacy analyses were based on data using the Enrolled Analysis Set (EAS), which, following the intent-to-treat (ITT) principles, included all subjects who signed an ICF and were randomized in Part 1 or registered in Part 2. As a measure of sensitivity, efficacy analyses were also performed based on the data from the Response Evaluable Set (RES), which included all subjects enrolled in Part 1 or Part 2 who received at least 1 dose of fam-trastuzumab deruxtecan and had measurable tumors as assessed by the ICR at baseline.

Subgroup analyses by baseline demographic and key disease characteristics were prespecified to assess the overall consistency of efficacy results. The primary efficacy endpoint, ORR, was summarized through descriptive statistics (N, %) and using exact binomial confidence intervals (Clopper-Pearson method). The secondary endpoint of DCR was also summarized using the same methodology as the primary analysis. The distribution of time-to-event endpoint of PFS and DoR were estimated using Kaplan-Meier method. Estimates of the quartiles of the time to event endpoints, along with the corresponding 95% confidence intervals (Brookmeyer-Crowley method), were provided. There was no formal interim analysis of efficacy in this trial. An interim dose selection analysis was conducted as per protocol during the study (see description in Section 6.2.2.1).

The FDA's Assessment:

The FDA considers the description of the SAP for Study DS8201-A-U201 acceptable in general.

FDA's efficacy evaluation was based on patients enrolled in Part 1, Part 2a and Part 2b of DS8201-A-U201. The efficacy evaluation was based on the magnitude of response rate and adequate duration of response. Please note no statistical inference for the time to event endpoints of progression free survival and time to response can be made from a single-arm study and results presented are considered descriptive only. Subgroup analyses are considered as exploratory. Subgroup analysis were performed to evaluate whether the treatment effect was consistent across subgroups.

Protocol Amendments

The Applicant's Description:

The initial protocol was amended globally 3 times prior to the DCO for the primary analysis. These changes are not expected to have had an impact on the integrity of the trial or interpretation of the results. Key changes were as follows:

Amendment 1 (dated 19 October 2017):

- Added the following endpoints: determination of the RP2D, treatment-emergent AEs (TEAEs) leading to discontinuation, AESIs, and increased troponin levels
- Clarified that there was no randomization in Part 2; instead, subjects enrolled in Part 2 were registered using the interactive web/voice response system
- Clarified that a sensitivity analysis of the primary efficacy analysis was also to be performed for all subjects who received the RP2D of fam-trastuzumab deruxtecan and had measurable tumors as assessed by ICR at baseline (RES). Analyses of ORR and DCR were also to be performed on the RES
- Added guidance on AESIs (ILD/pneumonitis, QT prolongation, LVEF decrease, and IRRs)
- Added definitions of mild and moderate hepatic dysfunction

Amendment 2 (dated 22 January 2018):

- Updated the exclusion criteria to add history of noninfectious ILD/pneumonitis that required steroids, current ILD/pneumonitis, or suspected ILD/pneumonitis that could not be ruled out
- Updated the management guidance and reporting procedures for AESIs

Amendment 3 (dated 27 July 2018):

- Updated the starting dose of fam-trastuzumab deruxtecan for Part 2 to 5.4 mg/kg, which was determined to be the RP2D

The FDA's Assessment:

FDA agrees with the applicant's summary of the protocol amendments for protocol DS8201-A-U201. Based on Amendment 3 (dated 27 July 2018), the starting dose of fam-trastuzumab of 5.4 mg/kg was determined to be the RP2D in Part 2. Therefore, patients with unresectable and/or metastatic HER2-positive breast cancer treated at the target dose of 5.4 mg/kg were the focus of the efficacy review in this application.

8.1.1.2. Study DS8201-A-J101

Trial Design

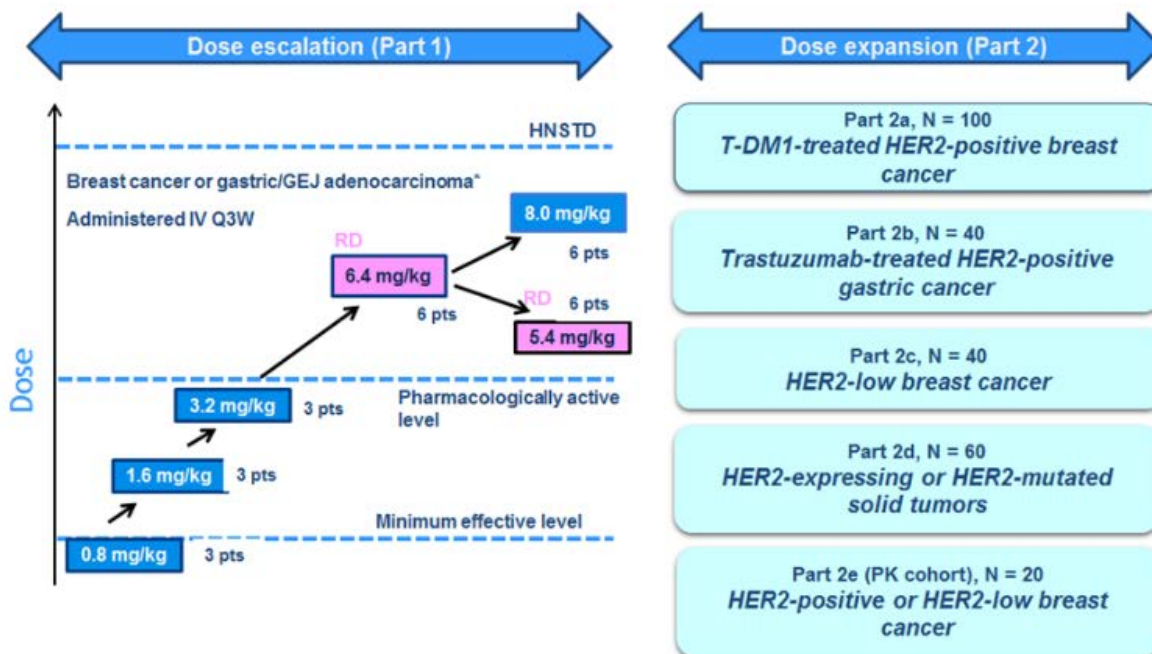
The Applicant's Description:

Overview of Study Design:

Study DS8201-A-J101 was a Phase 1, 2-part, multicenter, nonrandomized, open-label, multiple-dose, FIH study of fam-trastuzumab deruxtecan, with study sites in Japan and the US (see [Table 19](#)). Part 1 (Dose Escalation) was intended to identify the MTD or the recommended dose of fam-trastuzumab deruxtecan and evaluated doses ranging from 0.8 mg/kg to 8.0 mg/kg. Part 2 (Dose Expansion) was intended to further assess the safety, tolerability, and efficacy of fam-trastuzumab deruxtecan at the MTD/recommended doses of 5.4 mg/kg and 6.4 mg/kg. Dose Expansion was conducted in 5 cohorts of subjects with various tumor types (breast cancer, gastric or GEJ adenocarcinoma) and levels of HER2 expression ([Figure 8](#)). The target population of subjects with unresectable or metastatic HER2-positive breast cancer was enrolled in Part 1 and Part 2a and all subjects in the target population were administered fam-trastuzumab deruxtecan as the FL-DP1 formulation.

Unlike Study DS8201-A-U201, HER2 expression status was based on local laboratory reports of archival tissue samples. Otherwise, eligibility criteria for the HER2-positive breast cancer patient population, study treatments, dose modifications, schedule of events, and overall study conduct were similar across the 2 studies. Similar to Study DS8201-A-U201, ORR assessed by ICR and based on RECIST v1.1 was the primary efficacy endpoint.

Figure 8: Study DS8201-A-J101 study design



- HER2 status was assessed on archival tissue in both parts 1 and 2

Statistical Analysis Plan

The SAP was finalized prior to the database lock.

The DCO for the primary analysis occurred after the last subject enrolled in Part 2 of the study had completed approximately 6 months of study drug treatment or had discontinued study drug. All analyses were performed based on the data up to the DCO date. Data from all centers were pooled.

Primary and secondary efficacy analyses were performed on the data from the EAS, which included all patients who signed an ICF and were deemed eligible for the study. Sensitivity analyses of ORR and other response-based RECIST endpoints were performed based on data from the RES, which included all subjects enrolled in Part 1 or Part 2 who received at least 1 dose of fam-trastuzumab deruxtecan and had measurable tumors as assessed by the ICR at baseline. Efficacy data from subjects dosed at the RP2D in the Dose Escalation phase were pooled with the efficacy data from the corresponding tumor type cohorts in the Dose Expansion phase. Efficacy analyses were performed based on both ICR and investigator review of tumor

scans, with ICR-based response rates in the EAS considered to be the main efficacy variables of interest.

Subgroup analyses by baseline demographic and key disease characteristics were prespecified to assess the overall consistency of efficacy results. The primary efficacy endpoint, ORR, was summarized through descriptive statistics (N, %) and using exact binomial confidence intervals (Clopper-Pearson method). The distribution of time-to-event endpoint of PFS and DoR were estimated using Kaplan-Meier method. Estimates of the quartiles of the time-to-event endpoints, along with the corresponding 95% confidence intervals (Brookmeyer-Crowley method), were provided. There was no formal interim analysis of efficacy in this trial.

The FDA's Assessment:

The FDA agrees with the applicant's description of the SAP for study DS8201-A-J101 above. FDA's efficacy evaluation was on the magnitude of response rate and adequate duration of response. Please note no statistical inference for the time to event endpoints of progression free survival and time to response can be made from a single-arm study and results presented are considered descriptive only. Subgroup analyses are considered as exploratory. Subgroup analysis were performed to evaluate whether the treatment effect was consistent across subgroups.

Study Results of DS8201-A-U201 and DS8201-A-J101

Compliance with Good Clinical Practices

The Applicant's Position:

These studies were conducted in compliance with the protocol, the ethical principles that have their origin in the Declaration of Helsinki, the International Council for Harmonisation (ICH) consolidated Guideline E6 for Good Clinical Practice (GCP) (CPMP/ICH/135/95), and applicable regulatory requirement(s), including the following: US FDA GCP Regulations, Code of Federal Regulations Title 21, parts 11, 50, 54, 56, and 312 as appropriate; and/or Japanese Ministry of Health, Labor and Welfare Ordinance No. 28 of 27 Mar 1997; and/or Japanese Act on Securing Quality, Efficacy and Safety of Pharmaceuticals, Medical Devices, Regenerative and Cellular Therapy Products, Gene Therapy Products, and Cosmetics No. 1 of 25 Nov 2014; and/or European Commission Directive (2001/20/EC Apr 2001); and/or European Commission Directive (2005/28/EC Apr 2005), and/or; other applicable local regulations.

The FDA's Assessment:

The FDA agrees with the applicant's position that there is no evidence that compliance with good clinical practices was violated during conduct of studies DS8201-A-U201 and DS8201-A-

J101.

Financial Disclosure

Data:

Financial disclosure information was provided for investigators involved in the covered clinical studies DS8201-A-J101 and DS8201-A-U201. The summary of financial disclosure information is provided in Appendix 19.2. There were 5 clinical investigators with disclosable financial arrangements.

The Applicant's Position:

No concerns were raised regarding the overall integrity of study data.

The FDA's Assessment:

FDA agrees with the applicant's assessment.

8.1.1.3. Study DS8201-A-U201

Patient Disposition

Data:

A total of 253 subjects across all doses were enrolled and treated in Study DS8201-A-U201. Enrollment was proportional across geographic regions (Asia [Japan and Korea], US, Europe), with each region treating over 30% of subjects: 96 (37.9%) subjects in Asia; 77 (30.4%) subjects in the US; and 80 (31.6%) subjects in Europe (Table 20). At DCO, 139 (54.9%) subjects remained on study drug while 114 (45.1%) subjects had discontinued study drug. Of those who discontinued from study treatment, the primary reasons were PD per RECIST v1.1 (59 [23.3%] subjects) and AEs (29 [11.5%] subjects).

Table 20: Subject Disposition (Enrolled Analysis Set)

	Part 1 (PK + Dose Finding Stages)		Part 1 + Part 2a	Part 1 + Parts 2a and 2b	Overall
	6.4 mg/kg N = 48 n (%)	7.4 mg/kg N = 21 n (%)	5.4 mg/kg N = 180 n (%)	5.4 mg/kg N = 184 n (%)	All Doses N = 253 n (%)
Subjects enrolled	48 (100.0)	21 (100.0)	180 (100.0)	184 (100.0)	253 (100.0)
Europe	12 (25.0)	0	67 (37.2)	68 (37.0)	80 (31.6)
Japan	15 (31.3)	11 (52.4)	30 (16.7)	30 (16.3)	56 (22.1)
South Korea	7 (14.6)	0	32 (17.8)	33 (17.9)	40 (15.8)

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	Part 1 (PK + Dose Finding Stages)		Part 1 + Part 2a	Part 1 + Parts 2a and 2b	Overall
	6.4 mg/kg N = 48 n (%)	7.4 mg/kg N = 21 n (%)	5.4 mg/kg N = 180 n (%)	5.4 mg/kg N = 184 n (%)	All Doses N = 253 n (%)
United States	14 (29.2)	10 (47.6)	51 (28.3)	53 (28.8)	77 (30.4)
Subjects treated	48 (100.0)	21 (100.0)	180 (100.0)	184 (100.0)	253 (100.0)
Subjects ongoing on study drug at DCO	26 (54.2)	3 (14.3)	108 (60.0)	110 (59.8)	139 (54.9)
Subjects by reasons for discontinuation of study drug	22 (45.8)	18 (85.7)	72 (40.0)	74 (40.2)	114 (45.1)
PD per RECIST v1.1	12 (25.0)	9 (42.9)	38 (21.1)	38 (20.7)	59 (23.3)
AE	6 (12.5)	8 (38.1)	14 (7.8)	15 (8.2)	29 (11.5)
Withdrawal by subject	3 (6.3)	0	6 (3.3)	7 (3.8)	10 (4.0)
Physician decision	0	1 (4.8)	3 (1.7)	3 (1.6)	4 (1.6)
Death	0	0	7 (3.9)	7 (3.8)	7 (2.8)
Other ^a	1 (2.1)	0	4 (2.2)	4 (2.2)	5 (2.0)

^a “Other” reasons for discontinuation of study drug included: patient decision (2 subjects), physical deterioration (1 subject), Grade 3 troponin (1 subject), and decision “that the progression of her disease and the toxicity of therapy results in a compromised quality of life and as such chose to terminate cancer directed treatment” (1 subject).

DCO = 21 Mar 2019

The Applicant’s Position:

The most common cause of discontinuation of study drug was clinical or radiological progression, as was expected.

The FDA’s Assessment:

The recommended dosing regimen of fam-trastuzumab deruxtecan for the treatment of patients with unresectable or metastatic HER2-positive breast cancer is 5.4 mg/kg administered intravenously once every 3 weeks (Q3W). Therefore, patients with unresectable and/or metastatic HER2-positive breast cancer treated at the target dose of 5.4 mg/kg were the focus of the efficacy review in this application.

The applicant included 253 patients across all doses in their evaluation of efficacy; however, FDA’s efficacy evaluation includes only the 184 patients enrolled in Part 1, Part 2a and Part 2b of DS8201-A-U201 who received the target dose of 5.4 mg/kg. Part 2b of the study included 4 additional patients who had discontinued T-DM1 for reasons other than disease progression. The FDA included these 4 patients in the efficacy analysis because they had all received prior T-DM1, and despite discontinuing T-DM1 for reasons other than progressive disease, these 4

patients are anticipated to have similar clinical outcomes compared to patients whose disease had progression on T-DM1.

Protocol Violations/Deviations

Data:

Overall, 46.2% of subjects met major protocol deviations criteria, the 2 most common of which were related to laboratory assessments (13.8%) and study procedures (18.2%) (Table 21). The categories of major protocol deviations were as follows:

- Study procedures criteria (18.2% of subjects):
The most common deviation in this category was for vital signs not taken or not documented. Other reported deviations included: other examinations or tests not performed or conducted out of the visit window; triplicate ECGs not performed 3 minutes apart; and subjects not coming for planned visits
- Laboratory assessment criteria (13.8% of subjects):
Most common deviations in this category included missing or inappropriately collected PK samples or missing local troponin laboratory samples. Other reported deviations included: additional laboratory samples not collected or sampling out of the visit window; central laboratory tests results not collected as planned; and technical issues with samples.
- Informed consent (9.5% of subjects):
Most common deviations in this category included: submission of pharmacogenomics sample without documentation of consent; incorrect ICF version used to obtain signature; and source documents not reflecting verbal notification of safety risk. Other reported deviations included: signatures of principal investigator and subject not obtained on the same day; subject consented by physicians not on the log; and screening procedures done prior to main ICF consent.
- Investigational product compliance (7.1% of subjects):
Reported deviations included: use of incorrect weight to calculate the dose; dosage incorrectly calculated or rounded; dosing not reduced or interrupted as required for AEs; dose reduced inadvertently; delays in stopping study drug administration; dose administration out of window; deviation of interval between study drug administration; and incorrect allocation of study drug after temperature excursion.
- Eligibility and entry criteria (6.3% of subjects):
The most common deviation in this category was missing brain CT/MRI scan at entry. Other reported deviations included: history of other malignancy (untreated thyroid cancer); missing screening international normalized ratio (INR) or troponin; INR out of

range; screening tests out of window; no measurable tumor lesions (TLs); and enrollment of a subject with ongoing Grade 3 toxicity.

- Efficacy criteria (4.0% of subjects):
Reported deviations included inactive treated brain metastases categorized as TLs (together with other organ TLs) instead of non-tumor lesions per RECIST v1.1; continuation of treatment after confirmed PD per RECIST v1.1; CT scan not performed or performed out of window; missing tumor assessments; subject with baseline stable brain metastases with MRI of the brain not performed before later cycle; and missing brain CT scan at entry.
- Concomitant or prohibited medications (0.8% of subjects):
2 subjects took prohibited concomitant medications (prednisone and hydrocortisone, respectively) during the study. Although intermittent steroid use was permitted, chronic steroid use was not.
- SAE criteria (0.8% of subjects):
For the 2 subjects, the study sites did not report an SAE within 24 hours as required.

Table 21: Major Protocol Deviations (Enrolled Analysis Set)

Protocol Deviation Category	Part 1 (PK + Dose Finding Stages)		Part 1 + Part 2a	Part 1 + Parts 2a and 2b	Overall
	6.4 mg/kg N = 48 n (%)	7.4 mg/kg N = 21 n (%)	5.4 mg/kg N = 180 n (%)	5.4 mg/kg N = 184 n (%)	All Doses N = 253 n (%)
Subjects with any major protocol deviations	21 (43.8)	9 (42.9)	85 (47.2)	87 (47.3)	117 (46.2)
Concomitant or prohibited medications or non-drug therapy	0	0	2 (1.1)	2 (1.1)	2 (0.8)
Efficacy criteria	1 (2.1)	2 (9.5)	7 (3.9)	7 (3.8)	10 (4.0)
Eligibility and entry criteria	5 (10.4)	0	11 (6.1)	11 (6.0)	16 (6.3)
IP compliance	1 (2.1)	2 (9.5)	13 (7.2)	15 (8.2)	18 (7.1)
Informed consent	4 (8.3)	4 (19.0)	16 (8.9)	16 (8.7)	24 (9.5)
Laboratory assessment criteria	7 (14.6)	3 (14.3)	23 (12.8)	25 (13.6)	35 (13.8)
SAE criteria	0	1 (4.8)	1 (0.6)	1 (0.5)	2 (0.8)
Study procedures criteria	11 (22.9)	0	34 (18.9)	35 (19.0)	46 (18.2)

Subjects may have had protocol deviations in multiple categories.
DCO = 21 Mar 2019

The Applicant's Position:

Review of all the individual protocol deviations did not identify deviations that would affect the interpretation of results or conclusions of the primary analysis. Because the primary analysis set was the EAS (ITT), no subjects were removed from the analysis due to protocol deviations.

The FDA's Assessment:

Details of the protocol violations/deviations are shown in the applicant's assessment above. While the number of total violations is a concern, the majority (81 of 117) protocol violations were related to study procedure criteria and laboratory assessment criteria. There were few protocol violations related to eligibility criteria and prohibited medications which would have a greater impact on efficacy and safety outcomes; therefore, FDA agrees these did not significantly affect the efficacy analysis as no subjects were removed from the analysis due to protocol deviations.

Table of Demographic Characteristics

Data:

Key demographic characteristics are summarized in [Table 22](#).

Table 22: Key Demographic Characteristics (Enrolled Analysis Set)

Demographic Characteristics	Part 1 (PK + Dose Finding Stages)		Part 1 + Part 2a	Part 1 + Parts 2a and 2b	Overall
	6.4 mg/kg N = 48	7.4 mg/kg N = 21	5.4 mg/kg N = 180	5.4 mg/kg N = 184	All Doses N = 253
Age at informed consent, years ^a					
Mean (Std Dev)	55.8 (12.98)	54.4 (10.47)	56.1 (11.75)	56.0 (11.72)	55.8 (11.83)
Median	57.0	54.0	55.5	55.0	56.0
Range	29-79	32-69	28-96	28-96	28-96
Age group, n (%)					
<65 years	33 (68.8)	16 (76.2)	136 (75.6)	140 (76.1)	189 (74.7)
≥65 years	15 (31.3)	5 (23.8)	44 (24.4)	44 (23.9)	64 (25.3)
<75 years	45 (93.8)	21 (100.0)	171 (95.0)	175 (95.1)	241 (95.3)
≥75 years	3 (6.3)	0	9 (5.0)	9 (4.9)	12 (4.7)
Sex, n (%)					
Female	48 (100.0)	21 (100.0)	180 (100.0)	184 (100.0)	253 (100.0)

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

Demographic Characteristics	Part 1 (PK + Dose Finding Stages)		Part 1 + Part 2a	Part 1 + Parts 2a and 2b	Overall
	6.4 mg/kg N = 48	7.4 mg/kg N = 21	5.4 mg/kg N = 180	5.4 mg/kg N = 184	All Doses N = 253
Race, n (%)					
Asian	22 (45.8)	12 (57.1)	69 (38.3)	70 (38.0)	104 (41.1)
White	23 (47.9)	8 (38.1)	100 (55.6)	101 (54.9)	132 (52.2)
Black or African American	0	1 (4.8)	3 (1.7)	4 (2.2)	5 (2.0)
American Indian or Alaskan Native	1 (2.1)	0	0	1 (0.5)	2 (0.8)
Native Hawaiian or Pacific Islander	0	0	1 (0.6)	1 (0.5)	1 (0.4)
Other	1 (2.1)	0	3 (1.7)	3 (1.6)	4 (1.6)
Missing	1 (2.1)	0	4 (2.2)	4 (2.2)	5 (2.0)

^a Age in years was calculated using the informed consent date and the birth date.

DCO = 21 Mar 2019

The Applicant's Position:

The subject population enrolled in this study was representative of the pretreated patient population with unresectable or metastatic HER2-positive breast cancer.

The FDA's Assessment:

FDA agrees with the data presented in the baseline demographics in Table 22. All patients in the study had received at least 2 lines of prior HER2-directed therapy in the metastatic setting. In the 184 patients in study DS8201-A-U201, 53 (29%) were from the US, 68 (37%) were from the European Union, and 63 were from Asia (34%). Overall the demographics from study DS8201-A-U201 were consistent with a US population that will have received at least 2 lines of prior HER2-directed therapy in the indicated metastatic population.

Other Baseline Characteristics (e.g., Disease Characteristics, Important Concomitant Drugs)

Data:

Key baseline disease characteristics are summarized in [Table 23](#) and prior anti-cancer therapy is summarized in [Table 24](#).

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

Table 23: Baseline Disease Characteristics (Enrolled Analysis Set)

Baseline Disease Characteristics	Part 1 (PK + Dose Finding Stages)		Part 1 + Part 2a	Part 1 + Parts 2a and 2b	Overall
	6.4 mg/kg N = 48	7.4 mg/kg N = 21	5.4 mg/kg N = 180	5.4 mg/kg N = 184	All Doses N = 253
ECOG performance status, n (%) ^a					
0	30 (62.5)	14 (66.7)	101 (56.1)	102 (55.4)	146 (57.7)
1	18 (37.5)	7 (33.3)	78 (43.3)	81 (44.0)	106 (41.9)
2	0	0	1 (0.6)	1 (0.5)	1 (0.4)
Hormone receptor status, n (%)					
Positive	22 (45.8)	8 (38.1)	94 (52.2)	97 (52.7)	127 (50.2)
Negative	26 (54.2)	13 (61.9)	83 (46.1)	83 (45.1)	122 (48.2)
Unknown	0	0	3 (1.7)	4 (2.2)	4 (1.6)
HER2 expression (IHC) by central laboratory, n (%)					
1+	0	0	2 (1.1)	2 (1.1)	2 (0.8)
ISH positive	0	0	2 (1.1)	2 (1.1)	2 (0.8)
2+	7 (14.6)	3 (14.3)	28 (15.6)	28 (15.2)	38 (15.0)
ISH positive	7 (14.6)	3 (14.3)	26 (14.4)	26 (14.1)	36 (14.2)
ISH equivocal	0	0	1 (0.6)	1 (0.5)	1 (0.4)
ISH examined but not evaluable	0	0	1 (0.6)	1 (0.5)	1 (0.4)
3+	41 (85.4)	18 (85.7)	150 (83.3)	154 (83.7)	213 (84.2)
Presence of metastases, n (%) ^b					
Yes	46 (95.8)	19 (90.5)	168 (93.3)	172 (93.5)	237 (93.7)
Brain metastases	12 (25.0)	2 (9.5)	24 (13.3)	24 (13.0)	38 (15.0)
Bone metastases	19 (39.6)	4 (19.0)	51 (28.3)	53 (28.8)	76 (30.0)
Lung metastases	24 (50.0)	8 (38.1)	103 (57.2)	105 (57.1)	137 (54.2)
Liver metastases	13 (27.1)	5 (23.8)	55 (30.6)	56 (30.4)	74 (29.2)
Visceral disease ^c	44 (91.7)	17 (81.0)	165 (91.7)	169 (91.8)	230 (90.9)
Sum of diameters of target lesions, cm					
n	44	21	167	170	235
Mean (Std Dev)	7.10 (4.097)	6.37 (5.627)	6.68 (4.530)	6.67 (4.524)	6.72 (4.540)
Median	6.15	4.50	5.40	5.40	5.40
Range	1.5-17.4	1.1-23.8	1.2-24.5	1.2-24.5	1.1-24.5

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

Baseline Disease Characteristics	Part 1 (PK + Dose Finding Stages)		Part 1 + Part 2a	Part 1 + Parts 2a and 2b	Overall
	6.4 mg/kg N = 48	7.4 mg/kg N = 21	5.4 mg/kg N = 180	5.4 mg/kg N = 184	All Doses N = 253
Renal function at baseline, ^d n (%)					
Normal	23 (47.9)	9 (42.9)	90 (50.0)	90 (48.9)	122 (48.2)
Mild impairment	15 (31.3)	11 (52.4)	66 (36.7)	69 (37.5)	95 (37.5)
Moderate impairment	9 (18.8)	1 (4.8)	24 (13.3)	25 (13.6)	35 (13.8)
Missing	1 (2.1)	0	0	0	1 (0.4)
Hepatic function at baseline, ^e n (%)					
Normal	25 (52.1)	14 (66.7)	102 (56.7)	105 (57.1)	144 (56.9)
Mild impairment	22 (45.8)	7 (33.3)	75 (41.7)	76 (41.3)	105 (41.5)
Moderate impairment	0	0	1 (0.6)	1 (0.5)	1 (0.4)
Missing	1 (2.1)	0	2 (1.1)	2 (1.1)	3 (1.2)
Time from initial diagnosis to study treatment, months					
n	48	21	180	184	253
Mean (Std Dev)	93.87 (61.864)	80.21 (82.046)	88.89 (66.479)	88.70 (65.900)	88.97 (66.429)
Median	75.68	49.87	74.17	74.17	73.79
Range	17.2-279.3	10.3-288.5	1.6-431.4	1.6-431.4	1.6-431.4

^a ECOG performance status based on last value prior to first infusion.

^b Lung, liver and bone metastases were derived from baseline target or non-target lesions; brain metastases were from the eCRF collection.

^c Visceral disease included target or non-target lesions except ones in skin, breast, lymph nodes, and bone.

^d Renal function based on central laboratory results; enrollment was permitted based on local laboratory results. Renal impairment was determined by baseline creatinine clearance calculated using the Cockcroft-Gault equation. Normal renal function: creatinine clearance ≥ 90 mL/min; mild renal impairment: creatinine clearance ≥ 60 mL/min and < 90 mL/min; moderate renal impairment: creatinine clearance 30 mL/min and < 60 mL/min.

^e Hepatic function based on central laboratory results; enrollment was permitted based on local laboratory results. Normal hepatic function defined as: TBL $\leq$ ULN and AST $\leq$ ULN except for subjects with Gilbert syndrome; or TBL $\leq 3 \times$ ULN and AST $\leq$ ULN for subjects with Gilbert syndrome. Mild dysfunction defined as: TBL $>$ ULN, $\leq 1.5 \times$ ULN and any AST except for subjects with Gilbert syndrome; or TBL $>$ ULN, $\leq 3 \times$ ULN and AST $>$ ULN for subjects with Gilbert syndrome; or TBL $\leq$ ULN and AST $>$ ULN regardless of Gilbert syndrome. Moderate dysfunction defined as: TBL $> 1.5 \times$ ULN, $\leq 3 \times$ ULN and any AST except for subjects with Gilbert syndrome.

DCO = 21 Mar 2019

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

Table 24: Prior Anti-cancer Therapy

Prior Cancer Therapy	Part 1 (PK + Dose Finding Stages)		Part 1 + Part 2a	Part 1 + Parts 2a and 2b	Overall
	6.4 mg/kg N = 48	7.4 mg/kg N = 21	5.4 mg/kg N = 180	5.4 mg/kg N = 184	All Doses N = 253
Lines of prior systemic therapy not including hormone therapy, n (%)					
<3	1 (2.1)	4 (19.0)	17 (9.4)	17 (9.2)	22 (8.7)
≥3	47 (97.9)	17 (81.0)	163 (90.6)	167 (90.8)	231 (91.3)
Prior pertuzumab, n (%)					
Yes	34 (70.8)	20 (95.2)	118 (65.6)	121 (65.8)	175 (69.2)
No	14 (29.2)	1 (4.8)	62 (34.4)	63 (34.2)	78 (30.8)
Prior pertuzumab in first or second line in advanced/metastatic breast cancer, n (%)					
Yes	15 (31.3)	9 (42.9)	50 (27.8)	51 (27.7)	75 (29.6)
No	33 (68.8)	12 (57.1)	130 (72.2)	133 (72.3)	178 (70.4)
Prior cancer systemic therapy, n (%)					
Yes	48 (100.0)	21 (100.0)	180 (100.0)	184 (100.0)	253 (100.0)
Trastuzumab	48 (100.0)	21 (100.0)	180 (100.0)	184 (100.0)	253 (100.0)
T-DM1	48 (100.0)	21 (100.0)	180 (100.0)	184 (100.0)	253 (100.0)
Pertuzumab	34 (70.8)	20 (95.2)	118 (65.6)	121 (65.8)	175 (69.2)
Other anti-HER2	26 (54.2)	3 (14.3)	97 (53.9)	100 (54.3)	129 (51.0)
Hormone therapy	22 (45.8)	7 (33.3)	87 (48.3)	90 (48.9)	119 (47.0)
Other systemic therapy	48 (100.0)	21 (100.0)	179 (99.4)	183 (99.5)	252 (99.6)
Best response to T-DM1 therapy, ^a n (%)					
CR/PR	15 (31.3)	2 (9.5)	39 (21.7)	40 (21.7)	57 (22.5)
Stable disease	9 (18.8)	5 (23.8)	38 (21.1)	39 (21.2)	53 (20.9)
Not evaluable	3 (6.3)	2 (9.5)	38 (21.1)	39 (21.2)	44 (17.4)
PD	21 (43.8)	12 (57.1)	65 (36.1)	66 (35.9)	99 (39.1)
Number of regimens of prior cancer systemic therapy including hormone therapy, n (%)					
1	0	0	0	0	0
2	0	3 (14.3)	15 (8.3)	15 (8.2)	18 (7.1)
3	8 (16.7)	1 (4.8)	16 (8.9)	16 (8.7)	25 (9.9)
4	6 (12.5)	4 (19.0)	21 (11.7)	22 (12.0)	32 (12.6)
5	6 (12.5)	4 (19.0)	16 (8.9)	16 (8.7)	26 (10.3)
>5	28 (58.3)	9 (42.9)	112 (62.2)	115 (62.5)	152 (60.1)

^a The eCRF entry did not specify whether the response to T-DM1 was confirmed or not.

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

The Applicant's Position:

Overall, the subject population enrolled in this study was representative of the pretreated patient population with unresectable or metastatic HER2-positive breast cancer.

The FDA's Assessment:

The FDA agrees with the baseline disease characteristics presented by the applicant in this section.

FDA sent an information request regarding the prior lines of therapies on October 23, 2019. The sponsor has retrospectively reviewed the data for the 184 patients in Study DS8201-A-U201 with alterations of the definition for locally advanced/metastatic to the following revised definition:

1. Any regimens intended for Locally Advanced/Metastatic or Palliative setting as entered in eCRF pages.
2. Any regimens intended for "Neo-Adjuvant", "Adjuvant", or "Maintenance" setting as entered in eCRF pages, but with progression occurring within 6 months from end of the therapy (12 months for pertuzumab).
3. Any regimens in sequential lines of therapy according to eCRF entry with identical combination of agent names will be subtracted.

Based on this revised definition, for the 184 patients enrolled at the 5.4 mg/kg dose, the median lines of prior systemic therapy for locally advanced/metastatic was 5, with range of 2 to 17.

Table 25. Number of Prior Anti-Cancer Regimens in the Advanced/Metastatic Setting in DS8201-A-U201 Study (5.4 mg/kg)

	Part 1 + Parts 2a and 2b 5.4 mg/kg
Number of Regimens in Locally Advanced/Metastatic Setting	
n	184
Median	5
Range	2, 17
Number of Regimens in Locally Advanced/Metastatic Setting, n (%)	
1	0
2	30 (16.3)

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

3	24 (13.0)
4	26 (14.1)
5	28 (15.2)
>5	76 (41.3)

Treatment Compliance, Concomitant Medications, and Rescue Medication Use

Data:

The vast majority of subjects (248/253 [98.0%]) took concomitant medications during the study. Data collection setup did not allow for differentiation between prophylactic use of concomitant drugs as opposed to their use for toxicity management.

Protocol deviations due to the use of prohibited medications were reported in 2 (0.8%) subjects. In addition, 5 subjects received either a strong CYP3A4 inhibitor (4 subjects) or an OATP1B inhibitor (1 subject) prohibited per protocol.

The Applicant's Position:

Compliance was ensured, as fam-trastuzumab deruxtecan was infused at the study sites. There were a few cases of prohibited concomitant medication usage, the majority of which were usage of a strong CYP3A4 inhibitor or an OATP1B inhibitor. Given the results of the DDI study, this was considered unlikely to affect the efficacy results.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment.

Efficacy Results – Primary Endpoint (Including Sensitivity Analyses)

Data:

Confirmed Objective Response Rate

The confirmed ORR based on ICR in the primary 5.4 mg/kg dose cohort of 180 subjects enrolled in Part 1 and Part 2a was 60.6% (95% CI: 53.0, 67.8) (Table 26). In the sensitivity analysis conducted on the RES, the confirmed ORR by ICR for the primary 5.4 mg/kg dose cohort (N = 167) was 64.1% (95% CI: 56.3, 71.3), consistent with that in the EAS. The confirmed ORR for the EAS based on investigator assessment was consistent with the ORR per ICR.

Table 26: Objective Response Rate in the 5.4 mg/kg Dose Cohort (Enrolled Analysis Set)

Efficacy Parameters	DS-8201a 5.4 mg/kg	
	Part 1 + Part 2a N = 180	Part 1 + Parts 2a and 2b N = 184
Confirmed ORR by ICR		
n (%)	109 (60.6)	111 (60.3)
95% CI	53.0, 67.8	52.9, 67.5
Confirmed ORR by investigator		
n (%)	116 (64.4)	118 (64.1)
95% CI	57.0, 71.4	56.7, 71.1

The 2-sided 95% CI were based on the exact (Clopper-Pearson) binomial distribution.

Percentages were based on the number of subjects in the Enrolled Analysis Set.

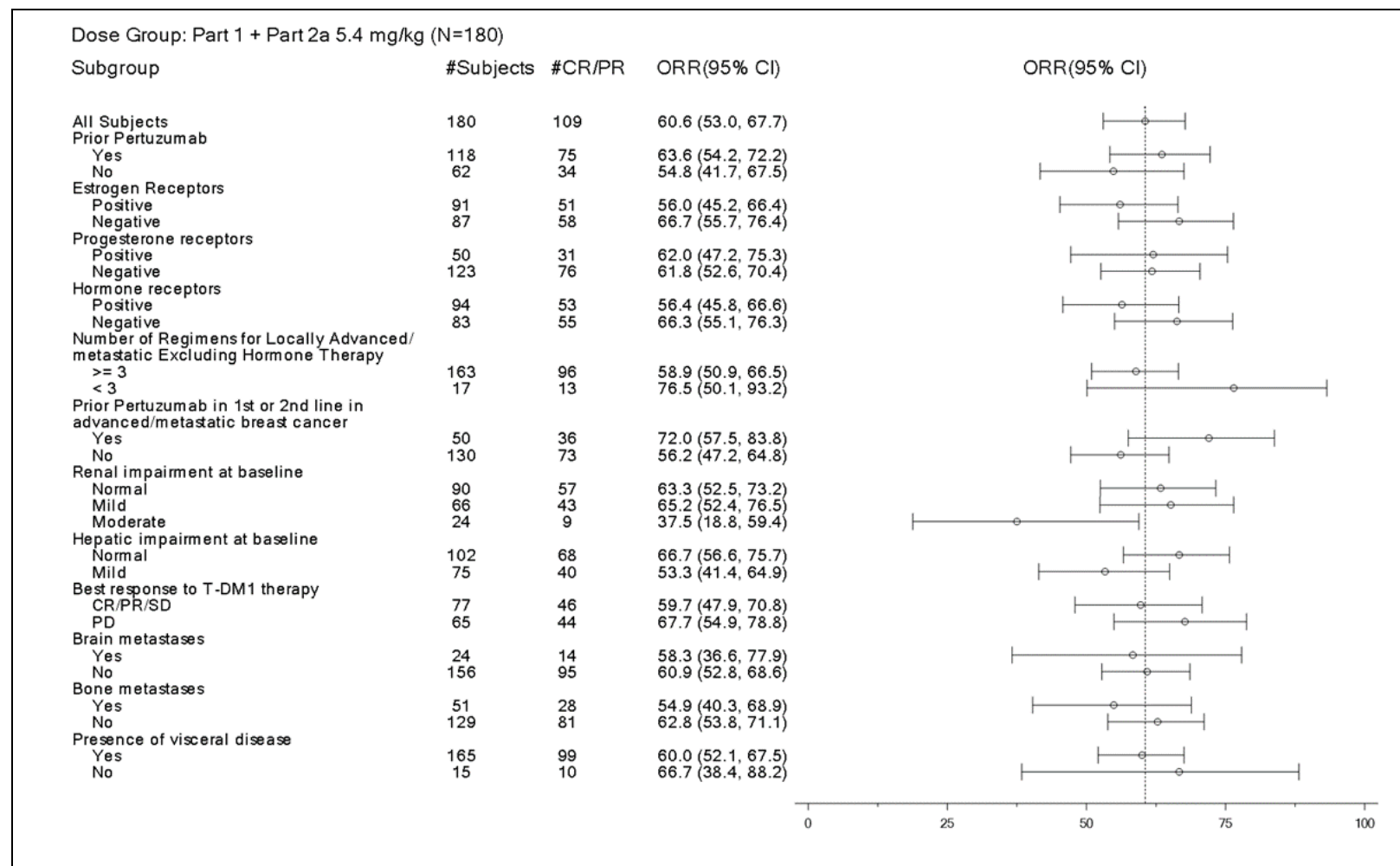
DCO = 21 Mar 2019

Subgroup Analyses of Objective Response Rate for the 5.4 mg/kg Dose Cohort

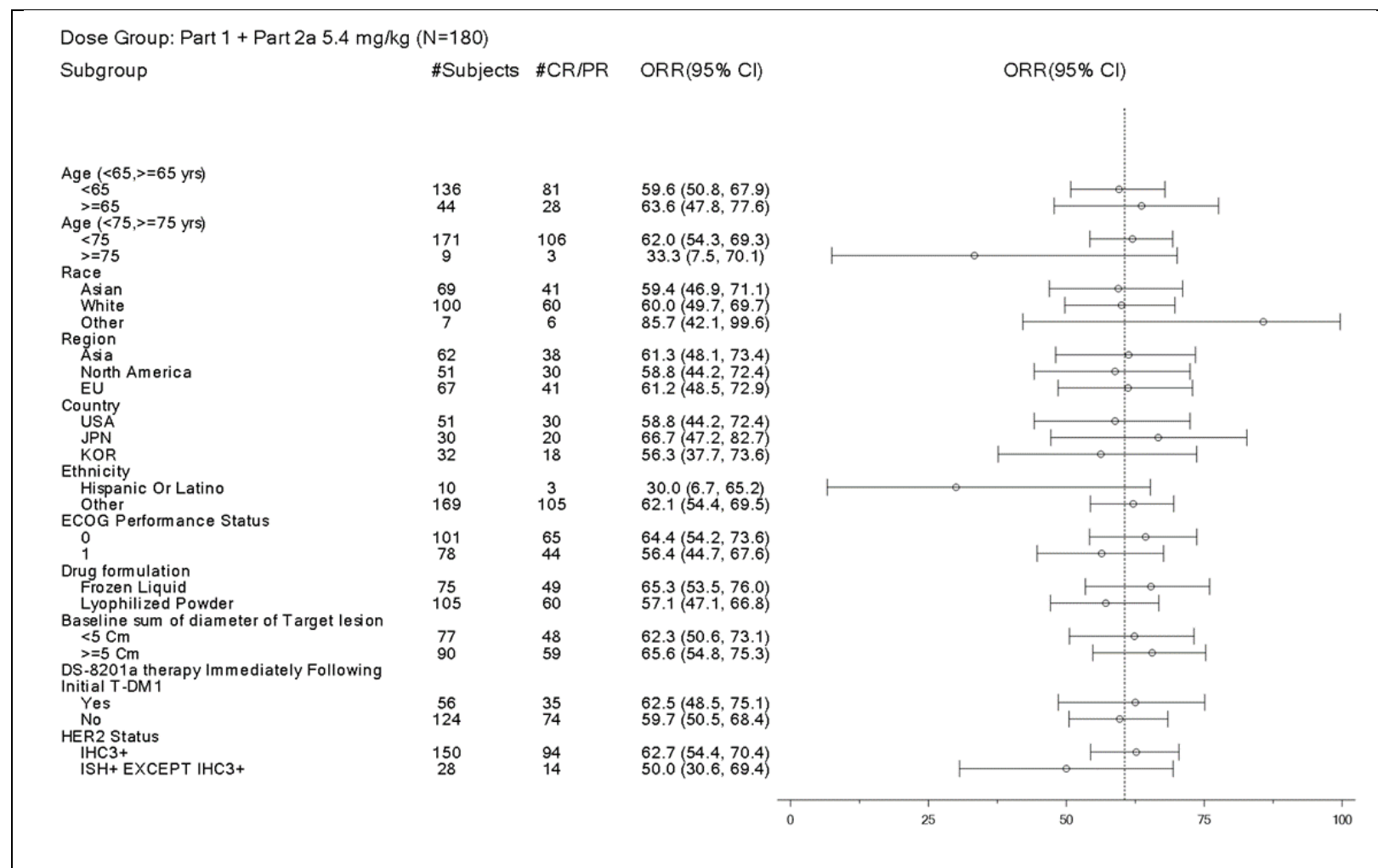
Subgroup analyses by baseline disease characteristics and baseline demographics for the primary 5.4 mg/kg dose cohort showed a confirmed ORR by ICR of at least 50.0% in most subgroups (Figure 9).

- In the subgroup of subjects who had received prior pertuzumab, the confirmed ORR by ICR (63.6% [95% CI: 54.2, 72.2]) was similar to that for the primary 5.4 mg/kg dose cohort (60.6% [95% CI: 53.0, 67.8]). The confirmed ORR by ICR was 54.8% (95% CI: 41.7, 67.5) in the subgroup of subjects who were pertuzumab-naïve.
- In the hormone receptor-positive and hormone receptor-negative subgroups, the confirmed ORRs by ICR were 56.4% (95% CI: 45.8, 66.6) and 66.3% (95% CI: 55.1, 76.3), respectively.
- In the subgroups based on presence or absence of brain metastases at baseline, the confirmed ORR by ICR was 58.3% (95% CI: 36.6, 77.9) in subjects with brain metastases and 60.9% (95% CI: 52.8, 68.6) in subjects without brain metastases.

Figure 9: Forest Plot of Objective Response Rate by Independent Central Review for the Primary 5.4 mg/kg Dose Cohort by Subgroup (Enrolled Analysis Set)



NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)



ORR defined as percentage of subjects with confirmed BOR of CR and PR by ICR based on RECIST 1.1 criteria
 The 2-sided 95% CI were based on the exact (Clopper-Pearson) binomial distribution.

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

The Applicant's Position:

Fam-trastuzumab deruxtecan demonstrated substantial antitumor activity in an extensively pretreated population of subjects with unresectable or metastatic HER2-positive breast cancer.

The FDA's Assessment:

The primary endpoint for this study was confirmed objective response rate (ORR) by Independent Central Review (ICR) based on RECIST 1.1. The Applicant reported the results of the confirmed ORR and subgroup analyses mainly based on the patients enrolled in Part 1 and Part 2a (N=180).

FDA's primary efficacy analysis was based on 184 patients who received 5.4 mg/kg dose in Part 1, Parts 2a and 2b (N=184). The FDA included 4 additional patients from Part 2b in the efficacy analysis as they had all received prior T-DM1 even though they had not progressed on T-DM1. From a clinical perspective these 4 patients were anticipated to have similar clinical outcomes compared to patients who progressed on T-DM1. The confirmed ORR based on ICR in these 184 patients was 60.3% (95% CI: 52.9, 67.4).

Table 27 shows the concordance between confirmed responses determined by IRC assessment compared to the ORR as determined by the investigator. Of 184 patients, there were 143 patients (77.7%) whose overall responses were agreed upon completely by both ICR and investigator. Of the 111 patients with confirmed response (CR or PR) by IRC, 94 (84.7%) also had confirmed response (CR or PR) by investigator, and the remaining 17 patients were considered to have SD. The discordance rate was 22.3% and is acceptable for ORR assessment in a single arm study.

Table 27: Discordance Rate of Response Assessment between Independent Central Review and Investigator

		Investigator Assessment		Discordance Rate
		Responder	Non-Responder	
Independent Central Review	Responder	94	17	22.3%
	Non-Responder	24	49	

FDA conducted the subgroup analyses on ORR and the results are summarized in Table 28. No apparent outliers were observed in these subgroup analyses.

Table 28: FDA Subgroup Analyses of ORR per ICR assessment

	Number of patients	Number of Responders	ORR (%) 95% CI
Overall	184	111	60.3 (52.9, 67.5)
Age			
<65	140	83	59.3 (50.7, 67.5)
≥65	44	28	63.6 (47.8, 77.6)
Region			
USA	53	32	60.4 (46.0, 73.6)
Asian	63	38	60.3 (47.2, 72.4)
EU	68	41	60.3 (47.7, 72.0)
Race			
Asian	70	41	58.6 (46.2, 70.2)
White	101	60	59.4 (49.2, 69.1)
Other	9	8	88.9 (51.8, 99.7)
Missing	4	2	50.0 (6.8, 93.2)
Hormone Status			
Positive	97	55	56.7 (46.3, 66.7)
Negative	83	55	66.3 (55.1, 76.3)
Unknown	4	1	25.0 (0.6, 80.6)
Visceral Disease			
Yes	169	101	59.8 (52.0, 67.2)
No	15	10	66.7 (38.4, 88.2)
Brain Metastases			
Yes	24	14	58.3 (36.6, 77.9)
No	160	97	60.6 (52.6, 68.3)
Number of Prior Targeted Therapies			
2	30	23	76.7 (57.7, 90.1)
3	24	15	62.5 (40.6, 81.2)
4	26	14	53.8 (33.4, 73.4)
5	28	17	60.7 (40.6, 78.5)
>5	76	42	55.3 (43.4, 66.7)
Renal Impairment at Baseline			
Normal	90	57	63.3 (52.5, 73.2)
Mild	69	45	65.2 (52.8, 76.3)
Moderate	25	9	36.0 (18.0, 57.5)
PD as Discontinuation Reason on T-DM1			
Yes	180	109	60.6 (53.0, 67.8)
No	4	2	50.0 (6.8, 93.2)

Data Quality and Integrity

Data:

None

The Applicant's Position:

No meaningful concerns are anticipated in the quality and integrity of the submitted datasets; these were sufficiently complete to allow for a thorough review of efficacy. Furthermore, no data integrity concerns were reported following completion of site inspections by the Applicant.

The FDA's Assessment:

FDA agrees with the applicant's assessment.

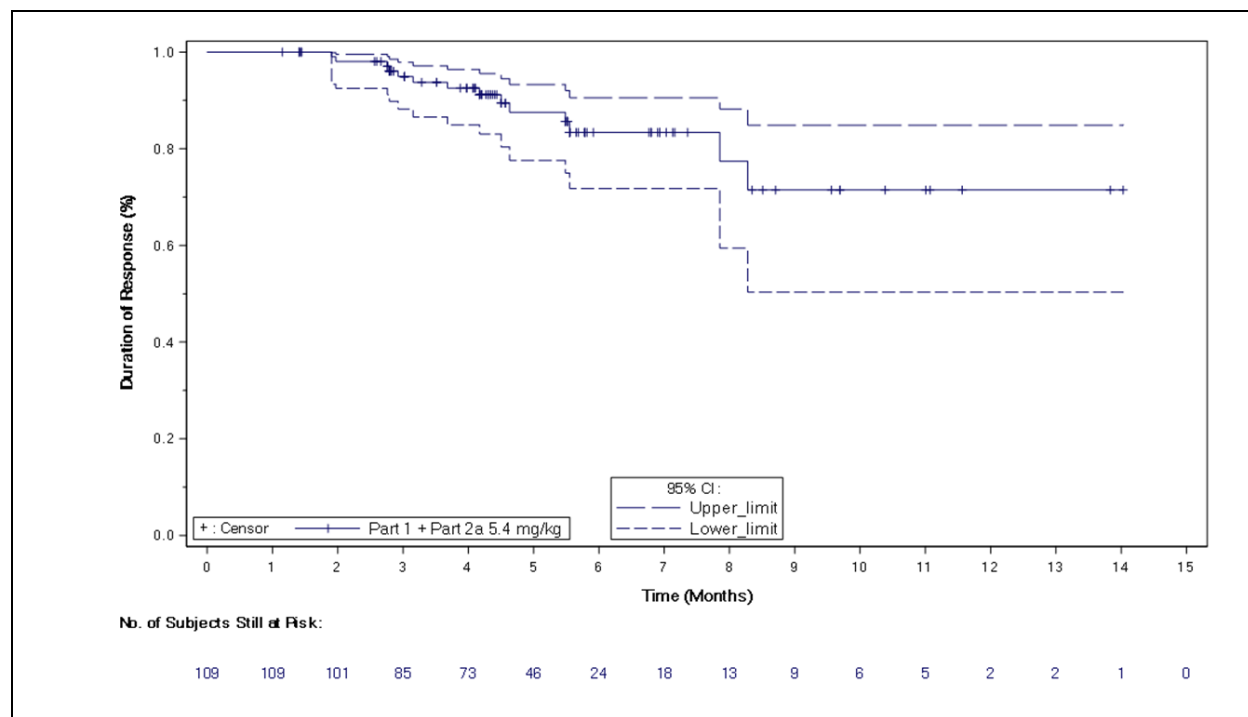
Efficacy Results – Secondary and Other Relevant Endpoints

Data:

Durability of Confirmed Response:

At DCO, the median duration of follow-up was 7.8 months (range: 0.7-17.2) across all doses. In the primary 5.4 mg/kg dose cohort, of the 109 subjects with best overall responses (BORs) of CR or PR by ICR, 79 (72.5%) subjects had ongoing responses without PD at DCO. The median DoR for confirmed responses for the primary 5.4 mg/kg dose cohort was not estimable. Based on a Kaplan-Meier estimate, 83.4% (95% CI: 71.8, 90.5) of responders in the primary 5.4 mg/kg dose cohort were progression-free and continuing to respond at 6 months (Figure 10). Subgroup analyses of DoR showed that most responders in subgroups were still in response at DCO. The median DoR was not estimable in any subgroup.

Figure 10: Kaplan-Meier Plot of Duration of Response by Independent Central Review for the Primary 5.4 mg/kg Dose Cohort (Enrolled Analysis Set)



Best overall response with confirmation was assessed by independent central review using Response Evaluation Criteria in Solid Tumors v1.1.

DCO = 21 Mar 2019

Progression-free Survival:

At DCO, in the primary 5.4 mg/kg dose cohort, 22.2% of subjects had PFS events; 77.8% of subjects were censored for the PFS analysis: 75.0% had no events of PD or death, 2.2% received new anticancer therapy, and 0.6% had no post-baseline tumor assessment. The median PFS based on ICR was not estimable. Based on a Kaplan-Meier analysis, the proportion of subjects in the primary 5.4 mg/kg dose cohort estimated to be progression-free at 6 months was 81% (95% CI: 73, 86).

Time to Response:

The median TTR in responders in the primary 5.4 mg/kg dose cohort was 1.5 months (95% CI: 1.4, 2.6), corresponding to the first scheduled tumor assessment.

The FDA's Assessment:

At data cut-off (DCO) of March 21, 2019, the median DoR for confirmed responses for the primary 5.4 mg/kg dose cohort was not estimable. The Applicant submitted an efficacy update

for DoR based on a DCO of August 1, 2019. The DCO for the efficacy update was 10.3 months after the last patient with HER2-positive breast cancer assigned to the 5.4 mg/kg dose group. The median duration of treatment was 9.97 months. The median DoR of the 111 responders was 14.8 months (95% CI: 13.8, 16.9).

Please note the results of progression-free survival and time to response are uninterpretable due to lack of a comparator arm in a single-arm study.

8.1.1.4. Study DS8201-A-J101

Patient Disposition

Data:

A total of 51 subjects (21 [41.2%] from Japan, 30 [58.8%] from the US) with unresectable or metastatic HER2-positive breast cancer were enrolled in the 5.4 mg/kg dose cohort in Part 1 or Part 2a of the study and constitute the EAS. One of the 51 subjects was not dosed, for a total of 50 subjects who received at least 1 dose of study drug (Safety Analysis Set). The DCO for the primary analysis (01 February 2019) was set to occur after approximately 6 months of follow-up from the date of first dosing of the last subject enrolled.

At DCO, with a median study follow-up of 9.76 months (range: 0.8-30.4), 20.0% of subjects were ongoing on treatment; 80.0% of subjects had discontinued study drug, with the most common reasons for discontinuation being PD (40%), AEs (18.0%), and clinical progression (10.0%).

The Applicant's Position:

The most common cause of discontinuation of study drug was clinical or radiological progression, as was expected.

The FDA's Assessment:

One patient (b) (6) was not dosed. FDA conducted sensitivity analysis based on dataset excluding this patient (N=50).

Demographics and Baseline Characteristics:

Data:

Overall, demographic and baseline characteristics of the 51 subjects in the HER2-positive breast cancer cohort enrolled at 5.4 mg/kg were consistent with those of subjects in the DS8201-A-U201 study and similar to those reported in the metastatic breast cancer population. Of the 51 subjects with HER2-positive breast cancer in the 5.4 mg/kg dose cohort, 98.0% were female, 66.7% were <65 years of age, 58.8% were from the US, and 64.7% had an ECOG PS of 0.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

There were 62.7% of subjects who were hormone receptor-positive, with 60.8% estrogen-receptor positive and 41.2% progesterone-receptor positive. Subjects were extensively pretreated, with a median number of 7 prior anti-cancer regimens (range: 2 to 17 lines) in the locally advanced/metastatic setting.

The Applicant's Position:

The subject population enrolled in this study was representative of the pretreated patient population with unresectable or metastatic HER2-positive breast cancer.

The FDA's Assessment:

Of the 50 patients with HER2-positive breast cancer received at least 1 dose in the 5.4 mg/kg dose cohort, 98.0% were female, 66% were <65 years of age, 58% were from the US, and 66% had an ECOG PS of 0. There were 64% of patients who were hormone receptor-positive, with 62% estrogen-receptor positive and 42% progesterone-receptor positive.

Study DS8201-A-J101 was not used for the primary efficacy analysis by the FDA and therefore, a retrospective review of the prior lines of therapy was not performed.

Confirmed Overall Response Rate, Disease Control Rate, and Duration of Response:

Data:

Fam-trastuzumab deruxtecan 5.4 mg/kg demonstrated substantial antitumor activity, with a confirmed ORR by ICR in the extensively-pretreated subjects with HER2-positive breast cancer in the 5.4 mg/kg dose cohort of 51.0% (95% CI: 36.6, 65.2). The DCR was 88.2% (95% CI: 76.1, 95.6).

The median TTR of 2.7 months (95% CI: 1.5, 2.9) corresponded to the second tumor assessment scan.

Confirmed responses were durable with a median duration of confirmed response at DCO of 12.7 months (95% CI: 6.7, non-estimable). Based on a Kaplan-Meier estimate of confirmed CR/PR by ICR assessment, the proportion of subjects remaining in response at 6 months was 77.6% (95% CI: 54.2, 90.0).

Median Progression-free Survival and Overall Survival:

The median PFS was 13.7 months (95% CI: 8.5, 19.6). Based on a Kaplan-Meier analysis, the estimated proportion of subjects alive and progression-free by ICR at 6 months was 84.6% (95% CI: 70.4, 92.4). The median OS was non-estimable at DCO due to the limited number of subjects who died.

The Applicant's Position:

Data from 2 studies, DS8201-A-U201 and DS8201-A-J101, demonstrate the efficacy of fam-trastuzumab deruxtecan for subjects with metastatic HER2-positive breast cancer. All subjects previously received T-DM1, and all but 1 subject previously received trastuzumab. Although subjects were extensively pretreated, 60.3% (95% CI: 52.9, 67.4) of subjects in DS8201-A-U201 and 51.0% (95% CI: 36.6, 65.2) of subjects in DS8201-A-J101 achieved a confirmed response. Similar response rates were observed independent of whether or not pertuzumab was previously given and in both hormone receptor-positive and hormone receptor-negative subgroups.

In addition, these responses were durable, with a median DoR for subjects with HER2-positive breast cancer who were assigned to 5.4 mg/kg in DS8201-A-J101 reaching 12.7 months (95% CI: 6.7, non-estimable). Longer follow-up is needed in Study DS8201-A-U201.

Together, these data demonstrate that fam-trastuzumab deruxtecan results in clinically meaningful and durable responses in subjects with unresectable or metastatic HER2-positive breast cancer who have received 2 or more prior anti-HER2 therapies.

The FDA's Assessment:

For DS8201-A-J101, the confirmed ORR based on ICR was 51.0% (95% CI: 36.6, 65.2) and the median DOR was 12.7 months (95% CI: 6.7, NE) at the time of DCO of February 1, 2019. The applicant submitted an efficacy update for DoR based on a DCO of August 1, 2019. The DCO for the efficacy update was 15.7 months after the last patient with HER2-positive breast cancer assigned to the 5.4 mg/kg dose group. The median duration of treatment was 8.54 months. The median DoR of the 26 responders was 10.8 months (95% CI: 6.7, NE).

Please note the results of progression-free survival, time-to-response and overall survival are uninterpretable due to lack of a comparator arm in a single-arm study.

8.1.2. Assessment of Efficacy Across Trials

Pooled Efficacy Analysis: Population

Data:

In order to provide a robust analysis of the efficacy endpoints in support of a clinically meaningful benefit from treatment with fam-trastuzumab deruxtecan in patients with unresectable or metastatic HER2-positive breast cancer who have received T-DM1, efficacy data from the 235 subjects in the target population enrolled at the target dose of 5.4 mg/kg in DS8201-A-J101 and DS8201-A-U201 were pooled. The primary population for the pooled efficacy analysis was the ITT Analysis Set (referred to as the EAS in the individual studies) and included all 235 enrolled subjects with the target condition at the target dose of 5.4 mg/kg. The primary efficacy analysis of the pooled population was based on confirmed responses as determined by ICR.

In both studies, the DCO dates for the primary analysis were set to occur after approximately 6 months of follow-up from the date of first dosing of the last subject enrolled. At DCO, the median duration of follow-up for the 235 subjects in the pooled analysis was 7.40 months (range: 0.7-30.4). Approximately half (51.3%) of the subjects in the pooled analysis remained on study treatment at the time of DCO.

The Applicant's Position:

The subject population treated in the 2 studies was a clearly defined and homogenous population of 235 extensively pretreated subjects with unresectable or metastatic HER2-positive breast cancer and was consistent with the population intended for treatment with fam-trastuzumab deruxtecan in the proposed indication. The entry criteria in both studies ensured subjects treated with fam-trastuzumab deruxtecan would be likely to benefit from treatment based on its mechanism of action. Demographics were similar across the 2 studies. The pooled population was generally representative of the broad population of patients with HER2-positive metastatic breast cancer who had received at least 2 prior anti-HER2 therapies seen in medical practice. The pooled population included similar proportions of subjects across geographies (US, 35.3%; Asia, 35.7%; Europe, 28.9%).

The FDA's Assessment:

FDA generally agrees with the applicant's assessment. The pooled 234 patients with HER2-positive disease in studies DS8201-A-U201 and DS8201-A-J101 dosed at 5.4mg/kg were generally similar patient populations to each other and to the indicated US population in regards in demographics.

HER2 testing was performed differently in each of the pooled studies. In study DS8201-A-J101, HER2 expression status was determined by local laboratory reports of archival tissue samples. In study DS8201-A-U201, HER2 expression was determined according to American Society of Clinical Oncology – College of American Pathologists (ASCO/CAP) guidelines 2013. Expression was based on archival tissue tested at a central laboratory prior to enrollment.

FDA's primary analysis of efficacy was based on the 184 patients from study DS8201-A-U201. Efficacy data from study DS8201-A-J101 was used as supportive evidence.

Pooled Efficacy Analysis: Primary Endpoint

Data:

In the pooled population of subjects with HER2-positive unresectable or metastatic breast cancer enrolled for treatment with fam-trastuzumab deruxtecan 5.4 mg/kg, the confirmed ORR based on ICR was 58.3% (95% CI: 51.7, 64.7) (Table 29).

Table 29: Confirmed Objective Response Rate and Best Objective Response by Independent Central Review (Intent-to-Treat Analysis Set)

	DS8201-A-J101 (5.4 mg/kg) (Part 1+2a) (N=51)	DS8201-A-U201 (5.4 mg/kg) Part 1 + Part 2a + Part 2b (N=184)	Pooled (5.4 mg/kg) (N=235)
Subjects with measurable tumors at baseline			
n (%)	45 (88.2)	170 (92.4)	215 (91.5)
Best objective response (confirmed), n (%)			
Complete Response	2 (3.9)	8 (4.3)	10 (4.3)
Partial Response	24 (47.1)	103 (56.0)	127 (54.0)
Stable Disease	19 (37.3)	68 (37.0)	87 (37.0)
Progressive Disease	4 (7.8)	3 (1.6)	7 (3.0)
Not Evaluable	2 (3.9)	2 (1.1)	4 (1.7)
Confirmed Objective Response Rate			
n (%)	26 (51.0)	111 (60.3)	137 (58.3)
95% CI ^a	(36.6, 65.2)	(52.9, 67.4)	(51.7, 64.7)

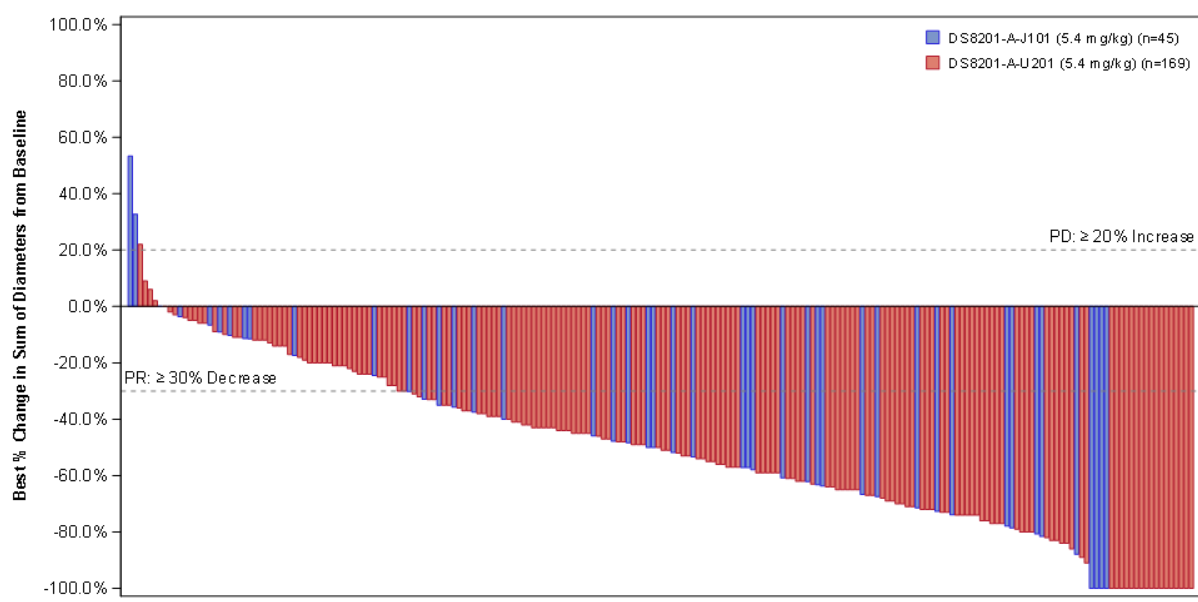
Percentages were calculated using the number of subjects in the ITT Analysis Set as the denominator.

Overall response was determined by independent central review based on RECIST v1.1 criteria.

^aThe 2-sided 95% CIs are based on the exact (Clopper-Pearson) binomial distribution.

As depicted in [Figure 11](#), evidence of shrinkage of tumor lesions was seen in most subjects in the pooled population.

Figure 11: Waterfall Plot of Best (Minimum) Percent Change from Baseline in Sum of Diameters of target Lesions Based on Independent Central Review (Intent-to-Treat Analysis Set)



For each subject, the best percent change from baseline in the sum of diameters for all target lesions is represented by a vertical line

Individual target lesions that were nonevaluable on time point scans were imputed as 0 mm in DS8201-A-J101. In case all target lesions were nonevaluable, the sum of diameter was imputed as nonevaluable.

Sum of diameters included some time point scans that occurred after progressive disease as defined by independent central review in DS8201-A-U201.

The Applicant's Position:

In the pooled analysis of 235 subjects with unresectable or metastatic HER2-positive breast cancer enrolled at the 5.4 mg/kg dose in DS8201-A-J101 and DS8201-A-U201, fam-trastuzumab deruxtecan demonstrated substantial antitumor activity in the extensively pretreated population. The high confirmed ORR in the pooled population of 58.3% (95% CI: 51.7, 64.7) was consistent across both DS8201-A-J101 and DS8201-A-U201, demonstrating the strong antitumor activity of fam-trastuzumab deruxtecan. The lower bound of the 95% CI for ORR was higher than the upper bound of the 95% CI observed with other therapeutic regimens, and even higher than the upper bound of the 95% CI for ORR with T-DM1 when given after a single anti-HER2 therapy in the EMILIA study (Table 1). The lower bound of the 95% CI for ORR was also higher than the reported ORRs and reported upper bounds of the 95% CIs from a matched historical cohort obtained from the French Unicancer database, with an ORR of 12.2% (95% CI: 6.2, 18.2), and a literature-based analysis of 37 studies requiring only prior trastuzumab and chemotherapy for inclusion, with an overall mean ORR across studies estimated to be 25.5% (95% prediction range: 17.1% to 36.1%).

The FDA's Assessment:

Generally, the patients enrolled in Studies DS8201-A-U201 and DS8201-A-J101 were similar. In study DS8201-A-J101 patients were enrolled based on historical HER2 status based on local laboratory assessment and later retrospectively tested centrally on archival tissue while HER2 testing in DS8201-A-U201 was performed centrally based on the most recently submitted adequate archival tissue. Therefore, FDA's primary efficacy assessment was based on the data from DS8201-A-U201 and data from DS8201-A-J101 is considered as supportive evidence only; and FDA considered the pooled analysis as exploratory only.

The efficacy evaluation was based on the magnitude of response rate and duration of this response. FDA agrees with the ORR of 60.3% (95% CI: 52.9, 67.4) in 184 patients enrolled in Study DS8201-A-U201 presented by the applicant.

The applicant also provided the waterfall plots of the best percent change from baseline in sum of diameters of target lesions based on IRC. The use of waterfall plots to visually convey the benefit seen in cancer clinical trials has gained popularity over time; however, it is noted that the waterfall plot has its own limitations. Not every bar below -30% line is a response and may represent stable disease, partial response, etc. In addition, the waterfall plot only includes patients assessed for response and not the entire ITT population. Therefore, waterfall plots may visually bias the estimate of response rate upward and misrepresent response rate, given that neither confirmatory scan results nor non-target lesion and new lesion status are reflected.

Pooled Efficacy Analysis: Secondary and Other Endpoints

Data:

Duration of Response:

In the pooled population, the preliminary estimate of the median DoR was 16.9 months (95% CI: 9.5, non-estimable) (Table 30). Based on a Kaplan-Meier analysis, the proportion of subjects with ongoing response at 6 months was estimated to be 80.1% (95% CI: 69.2, 87.5) (Figure 12).

Table 30: Duration of Confirmed Response by Independent Central Review

	DS8201-A-J101 5.4 mg/kg (Part 1+2a) (N=51)	DS8201-A-U201 5.4 mg/kg Part 1 + Part 2a + Part 2b (N=184)	Pooled 5.4 mg/kg (N=235)
Subjects with Complete or Partial Response, n	26	111	137

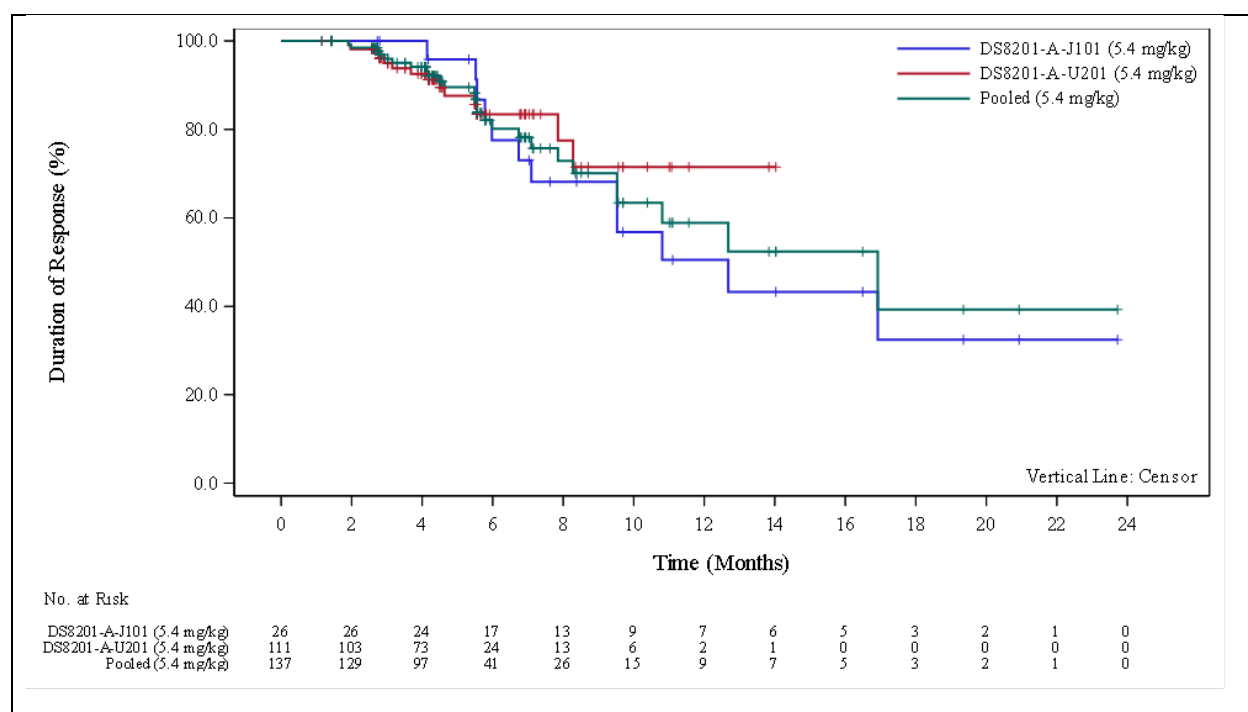
NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	DS8201-A-J101 5.4 mg/kg (Part 1+2a) (N=51)	DS8201-A-U201 5.4 mg/kg Part 1 + Part 2a + Part 2b (N=184)	Pooled 5.4 mg/kg (N=235)
Subjects with Progressive Disease, n (%)	11 (42.3)	9 (8.1)	20 (14.6)
Subjects with Death, n (%)	1 (3.8)	5 (4.5)	6 (4.4)
Subjects Censored, n (%)	14 (53.8)	97 (87.4)	111 (81.0)
Median (months)	12.7	NE	16.9
95% CI ^a	(6.7, NE)	(NE, NE)	(9.5, NE)

Percentages were calculated using the number of subjects with CR/PR as the denominator.

^aThe 2-sided 95% CIs for quartile survival times were computed using the Brookmeyer-Crowley method.

Figure 12: Kaplan-Meier Plot of Pooled Duration of Response Based on Independent Central Review



Progression-free Survival:

The median PFS for the pooled population was 13.9 months (95% CI: 10.9, non-estimable), similar to the median PFS in Study DS8201-A-J101 (13.7 months [95% CI: 8.5, 19.6]). The median PFS was non-estimable in DS8201-A-U201.

Time to Response:

For the pooled analysis, TTR was defined as the time from the date of randomization (date of registration for nonrandomized subjects in DS8201-A-U201 and date of registration in DS8201-A-J101) to the date of the first documentation of objective response (CR or PR). The TTR was measured for responding subjects (confirmed CR/PR) only. For the pooled analysis, the median TTR was 2.3 months (95% CI: 1.4, 2.7). For DS8201-A-J101, the median TTR was closer to the second post-baseline scan scheduled for Week 12 (2.7 months [95% CI: 1.5, 2.9]) than to the first post-baseline scan scheduled for Week 6 as for DS8201-A-U201 (1.5 months [95% CI: 1.4, 2.6]).

The Applicant's Position:

The observed ORR, DoR, DCR, and PFS with fam-trastuzumab deruxtecan supported the meaningful clinical benefit of treatment with fam-trastuzumab deruxtecan 5.4 mg/kg and represent a substantial improvement in a population of patients with a high unmet medical need. Additional follow-up in Study DS8201-A-U201 is needed to ensure a more stable estimate of DoR and PFS.

The FDA's Assessment:

For Study DS8201-A-U201, the applicant submitted an efficacy update for DoR based on a DCO of August 1, 2019. The median DoR of the 111 responders was 14.8 months (95% CI: 13.8, 16.9).

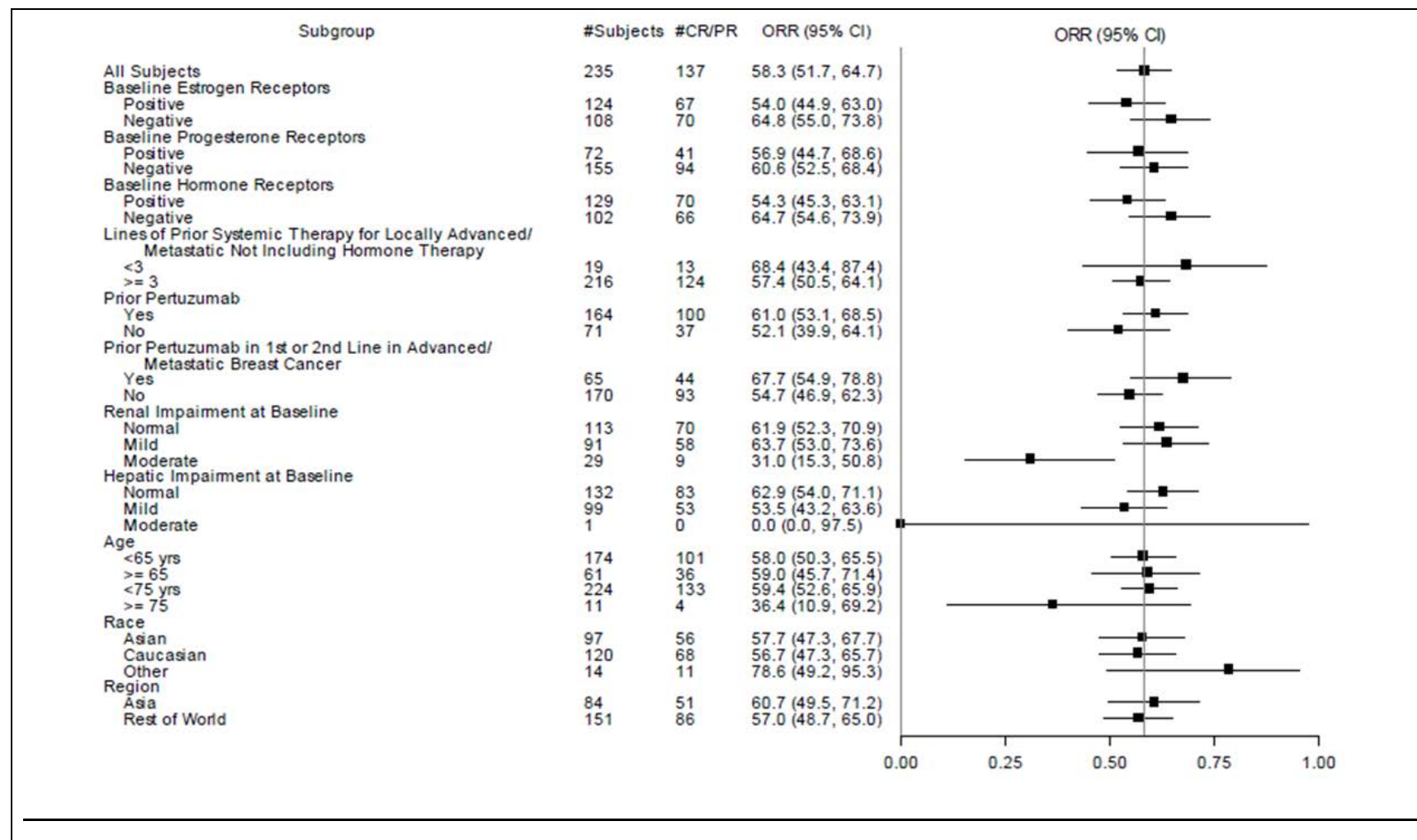
FDA considered the pooled analysis as exploratory. Please note the results of progression-free survival and time-to-response are uninterpretable due to lack of a comparator arm in a single-arm study.

Pooled Efficacy Analysis: Subpopulations

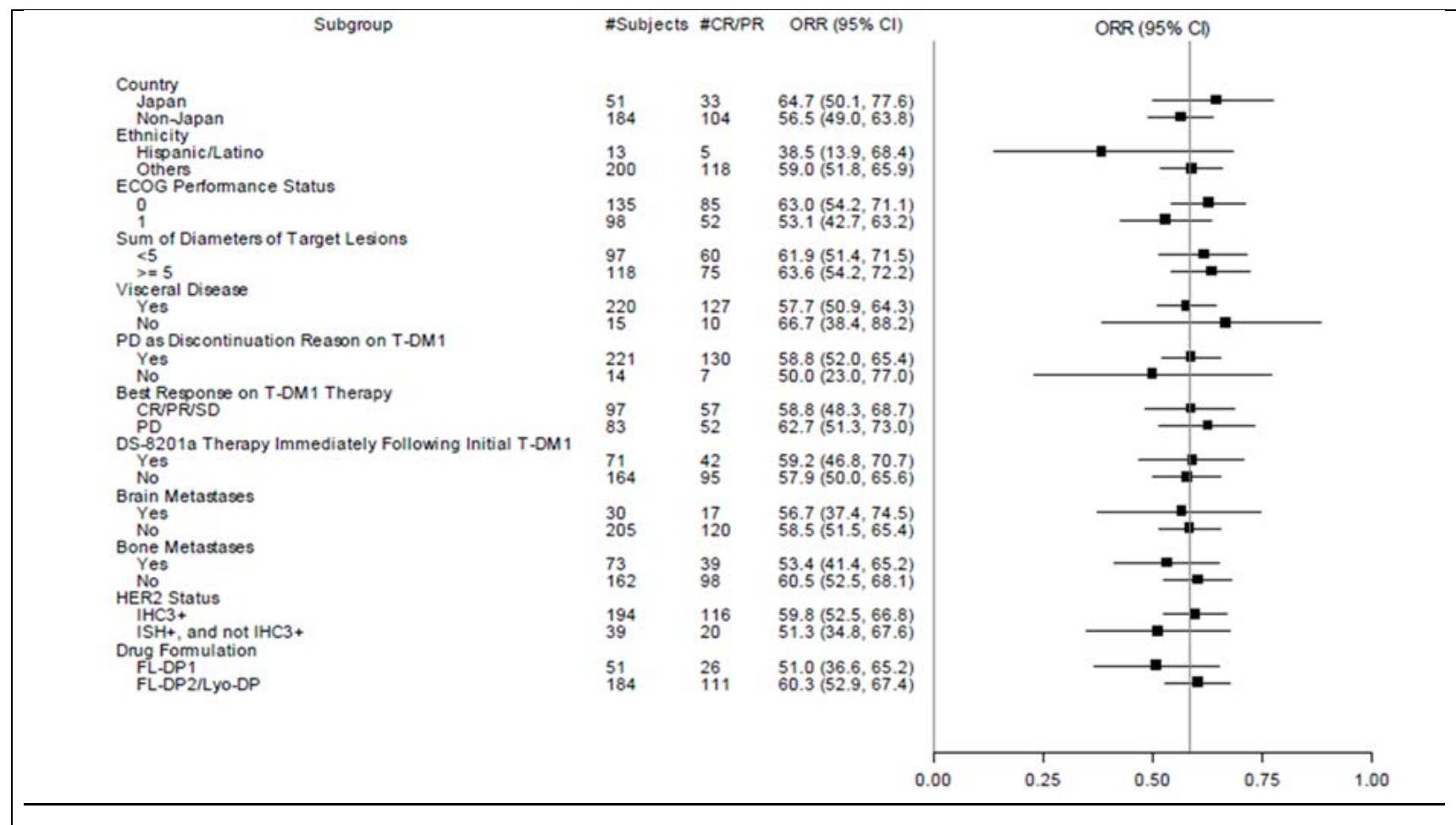
Data:

Subgroup analyses of confirmed ORR based on ICR were conducted in subgroups of interest (see forest plot in [Figure 9](#)). Among the 129 subjects who were hormone receptor-positive in the pooled analysis, the confirmed ORR was 54.3% (95% CI: 45.3, 63.1). For the 102 subjects who were hormone receptor-negative, the confirmed ORR was 64.7% (95% CI: 54.6, 73.9). For the 164 subjects who had previously received pertuzumab in the pooled analysis, the confirmed ORR was 61.0% (95% CI: 53.1, 68.5).

Figure 13: Forest Plot of Objective Response Rate by Independent Central Review by Subgroup for Pooled Population at 5.4 mg/kg Fam-Trastuzumab Deruxtecan



NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)



Percentages were calculated using the number of subjects in the Intent-to-Treat Analysis Set for each subgroup as the denominator.

Overall response was determined by independent central review based on RECIST v1.1.

The 2-sided 95% CIs were based on the exact (Clopper-Pearson) binomial distribution.

The dashed vertical line represents the overall ORR estimate, calculated separately for each study and pooled.

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

The Applicant's Position:

The efficacy results were consistent across subgroups, including those based on subjects' hormonal receptor status at baseline and prior treatment with pertuzumab. Overall, results across subgroups were generally consistent with those from the pooled analysis, with high ORR seen across most subgroups.

The FDA's Assessment:

FDA considered the pooled subgroup analyses as exploratory although in general it was supportive as there were no outliers among the subgroups.

Additional Efficacy Considerations

The FDA's Assessment:

FDA performed an independent analysis of the complete responses (CRs) reported in both studies DS8201-A-J101 and DS8201-A-U201 reported in the first cut-off date. The following details these cases.

Study DS8201-A-U201:

Overall, there were 8 confirmed CRs in study U201 in the ITT population of 184 patients. Out of the 8 cases, 1/8 were concordant reads by IRC and investigator. The remaining 7 cases were discordant between the IRC read of CR and investigator read with PR. In 1 of these 7 (subject (b) (6)), the investigator did note a CR in the last week evaluated before the cutoff date for the data (week 48). Subject (b) (6) was also a discordant case, however both IRC and investigator noted PD after confirmation of CR by IRC and PR by investigator.

All 8 CRs in this study had sum of diameters of all target lesions < 5cm (two cases had no target lesions selected by the IRC). Therefore, all the CRs were in patients with low volume disease.

Subject (b) (6): IRC (based on 1 target and 1 non-target LN lesions) found patient with a confirmed CR starting week 12 of treatment and then progression (PD) by week 42 in both the target and non-target lesion. Investigator noted a confirmed PR starting week 6 based on 1 target and 1 non-target LN that continued through week 42 due to the persistence on the non-target LN. PD was noted week 48.

Subject (b) (6): Both IRC and investigator reads were confirmed CRs. Of note, the IRC chose no target lesions and 2 non-target lesions while the investigator had 1 target and 2 non-target lesions.

Subject (b) (6): The IRC read was a CR based on 1 target and 1 non-target lesion

(both lymph nodes) and the investigator read was a PR also based on 1 target and 1 non-target lesion (both lymph nodes). The non-target lesion persisted for the investigator read.

Subject (b) (6): The IRC read was a confirmed CR starting at week 24 while the confirmed read of the investigator was a PR due to persistence of a target liver lesion until week 42. Of note, the investigator did note a CR in week 48 which was the last date of evaluation before data cut-off. Therefore, it remains to be seen whether this will become a confirmed CR.

Subject (b) (6): The IRC read was a CR based on 1 target and 1 non-target lesion. The investigator read was a PR based on 2 target and 2 non-target lesions. FDA would call this response overall as a PR given that the investigator did not select the second target lesion despite it being the same size as the other target lesion. An IR was sent to the sponsor....

Subject (b) (6): The IRC read was a CR based on 2 target and 2 non-target lesions (all lymph nodes). The investigator read was a PR and also had 2 target and 2 non-target lesions however 1 of the target lesions was in the liver. An IR was sent to the sponsor to see if the second IRC read used the liver lesion and whether this subject was an adjudicated case by radiologist #3.

Subject (b) (6): The IRC read was a CR, however, this was based on 2 non-target lesions. The investigator read was a PR based on 1 target and 2 non-target lesions. Both non-target lesions persisted by investigator read.

Subject (b) (6): The IRC read was a CR based on 1 target lesion. The investigator read was a PR based on a target and 2 non-target lesions. Both IRC and investigator chose a lung lesion as their target lesion, which persisted in the investigator read.

Study DS8201-A-J101:

Overall, in the ITT population of 51 patients, there were 2 confirmed complete responses reported. One patient had a sum of diameter of all target lesions $\geq 5\text{cm}$.

Subject (b) (6): Both IRC and Investigator reads were concordant based on target lesions that included at least 1 non-lymph node lesion.

Subject (b) (6): IRC confirmed response was a CR while the investigator response was a PR. Both identified lymph nodes as both target and non-target lesions (IRC with 1

target and non-target lesion each; and investigator with 2 target and 1 non-target lesion). Per IRC, the patient achieved a CR by week 6. By investigator, the best response was a PR due to the persistence of the non-target lymph node through Week 48 which was the last week of data included prior to the cut-off date.

8.1.3. Integrated Assessment of Effectiveness

Data:

In DS8201-A-J101 and DS8201-A-U201, all subjects previously received T-DM1, and all but 1 subject previously received trastuzumab. Although subjects were extensively pretreated, with a median number of 7 prior cancer regimens (range: 2 to 27 lines) in the locally advanced/metastatic setting, 58.3% (95% CI: 51.7, 64.7) of subjects in the pooled analysis achieved a confirmed response. The lower bound of the 95% CI for ORR of 51.7% is higher than the upper bound of the 95% CI for ORR from the Unicancer analysis, the literature review, or (b) (4). The lower bound of the 95% CI for fam-trastuzumab deruxtecan is also above the upper bound of the 95% CI for T-DM1 in the second-line setting in the EMILIA trial (ORR: 43.6% [95% CI, 38.6, 48.6]).

In a pooled analysis of DS8201-A-U201 and DS8201-A-J101, the subgroup of patients who received prior pertuzumab therapy had a confirmed ORR of 61% (95% CI: 53, 69) and those who did not receive prior pertuzumab therapy had a confirmed ORR of 52% (95% CI: 40, 64). The subgroup of patients who were hormone receptor-positive at baseline had a confirmed ORR of 54% (95% CI: 45, 63) and those who were hormone receptor-negative at baseline had a confirmed ORR of 65% (95% CI: 55, 74). Similar response rates were observed independent of whether or not pertuzumab was previously given and in both hormone receptor-positive and hormone receptor-negative subgroups.

The median DoR for subjects with HER2-positive breast cancer who were assigned to 5.4 mg/kg in DS8201-A-J101 was 12.7 months (95% CI: 6.7, non-estimable). The preliminary median DoR for the pooled population was 16.9 months (95% CI: 9.5, non-estimable). Based on a Kaplan-Meier estimate of the pooled population, 80.1% (95% CI: 69.2, 87.5) of responders would be expected to have an ongoing response at 6 months.

The Applicant's Position:

The current treatment options for previously treated HER2-positive breast cancer are summarized in Table 1. There are no approved therapies for the treatment of patients with unresectable or metastatic HER2-positive breast cancer who have received 2 or more prior anti-HER2 therapies and the lack of clearly preferential treatment options highlights the unmet medical need for these patients. To supplement the clinical data package with real-world

evidence and to estimate the expected clinical benefit (PFS, ORR) of fam-trastuzumab deruxtecan with a comparable patient population, individual patient data were collected and analyzed by Unicancer, a French network of 18 private, nonprofit hospitals. In 137 patients matched with subjects enrolled in the DS8201-A-U201 study, the overall ORR was 12.2% (95% CI: 6.2, 18.2).

To obtain a broad overview of the reported efficacy of regimens used for metastatic HER2-positive breast cancer, a literature-based analysis was conducted of 37 studies requiring only prior trastuzumab and chemotherapy for inclusion. The overall mean ORR across studies was estimated to be 25.5% (95% prediction range: 17.1% to 36.1%). The response in the subgroup of studies conducted in third- or later-lines of treatment was broadly similar to modeled values from the second-line or later population with an ORR of 15% (95% CI: 9, 30).

Evidence of efficacy and benefit of fam-trastuzumab deruxtecan for the treatment of patients with unresectable or metastatic HER2-positive breast cancer who have received 2 or more prior anti-HER2 therapies is based on the totality of data from DS8201-A-U201 and DS8201-A-J101. These studies demonstrated clinically meaningful and durable responses in extensively pretreated patients with unresectable or metastatic HER2-positive breast cancer.

These results represent a significant improvement over available therapies and provide benefit of fam-trastuzumab deruxtecan in patients in this setting.

The FDA's Assessment:

The FDA agrees with the applicant that fam-trastuzumab deruxtecan demonstrated efficacy in study DS8201-A-U201 with supportive evidence from study DS8201-A-J101 for the treatment of patients with unresectable or metastatic HER2-positive breast cancer who have received two or more prior anti-HER2-based regimens in the metastatic setting. Currently, treatment options for HER2-positive patients after two lines of therapy in the metastatic setting are limited, and this is an area of unmet need requiring further effective therapies. The confirmed ORR based on ICR in 184 patients in study DS8201-A-U201 was 60.3% (95% CI: 52.9, 67.4), with a 4.3% complete response rate and a 56% partial response rate. Median response duration was 14.8 months (95% CI: 13.8, 16.9). Generally, the ORR was consistent across subgroups and supportive of the overall efficacy, even though these analyses are exploratory. This durable objective response rate is of sufficient magnitude to serve as a surrogate endpoint that improved over that of available therapies and is reasonably likely to predict clinical benefit (i.e., improved survival or progression free survival) of fam-trastuzumab in patients with unresectable or metastatic HER2-positive breast cancer who have received two or more prior anti-HER2-based regimens in the metastatic setting and is supportive of accelerated approval.

Additionally, FDA queried the applicant to retrospectively review prior lines of therapy to not include dose reductions or interruptions as a new line of therapy, and to only consider a regimen as a line of therapy if a new agent was added or removed. Based on these guidelines,

the applicant revised the prior lines of therapy in study DS8201-A-U201 to a median of 5 therapies in the metastatic setting with a range from 2 to 17 lines.

Study DS8201-A-J101 had a confirmed ORR based on 26 out of 51 patients which was 51% (95% CI: 36.6, 65.2), with a 3.9% complete response rate and a 47.1% partial response rate. Median response duration was 12.7 months (95% CI: 6.7, NE). Although this is not the primary evidence for efficacy, the durable ORR observed in study DS8201-A-J101 is supportive of the results from study DS8201-A-U201.

Continued approval for this indication may be contingent upon verification and description of clinical benefit in a confirmatory trial. As noted in prior sections of the efficacy review, the results of progression-free survival, time-to-response and overall survival are uninterpretable due to lack of a comparator arm in a single-arm study.

8.2. Review of Safety

Data:

The safety profile of fam-trastuzumab deruxtecan is based on data from 645 subjects with various tumor types treated at various dose levels in 5 clinical studies. The primary focus is on data from 234 subjects with HER2-positive breast cancer who were treated with at least 1 dose of fam-trastuzumab deruxtecan 5.4 mg/kg in the Phase 1, open-label, non-randomized Study DS8201-A-J101 and the Phase 2, open-label, 2-part (1 non-randomized, 1 randomized) Study DS8201-A-U201. Additional support is provided by DS8201-A-J102 (a study to evaluate effects of fam-trastuzumab deruxtecan on QTc interval), DS8201-A-A103 (a safety/PK study), and DS8201-A-A104 (a DDI study).

The Applicant's Position:

Data from these studies allowed for a comprehensive and adequate assessment of the safety profile of fam-trastuzumab deruxtecan and an evaluation of the overall benefit-risk in adult patients with unresectable or metastatic HER2-positive breast cancer at a dose of 5.4 mg/kg. This safety population is also considered adequate for the detection and characterization of common AEs and to provide guidance on toxicity management. With 234 subjects in the pooled safety database of subjects with HER2-positive breast cancer who received 5.4 mg/kg fam-trastuzumab deruxtecan, there is at least a 90% chance of observing an AE with a true incidence rate of 1% or higher.

The FDA's Assessment:

For this BLA, the applicant submitted safety data from 5 different clinical studies that enrolled various populations of patients, various tumor types, and multiple dose levels of fam-trastuzumab deruxtecan.

The FDA's independent safety analysis for this BLA was based on a total of 234 patients with HER2-positive breast cancer in studies DS8201-A-J101 (50 patients) and DS8201-A-U201 (184 patients) that were treated with at least 1 dose of fam-trastuzumab deruxtecan at 5.4mg/kg as this was the dose that was being reviewed for regulatory approval.

The FDA also reviewed the 90-day safety analysis and datasets provided by the applicant and did not identify any new safety signals not already discussed in this safety review.

8.2.1. Safety Review Approach

Data:

The data presented here are a comprehensive analysis of safety data relevant to the use of 5.4 mg/kg Q3W fam-trastuzumab deruxtecan in the treatment of adult patients with unresectable or metastatic HER2-positive breast cancer. Data from other doses and other tumors were also included to evaluate the robustness and consistency of safety findings. The most common AEs were gastrointestinal (GI) and hematologic. The events of ILD, LVEF decrease, QT prolongation, and IRRs were identified as AESIs during the development of fam-trastuzumab deruxtecan, based on nonclinical and clinical experience, epidemiologic information, and literature review of products of similar class, and known class effects of other HER2-targeting mAbs.

The pooling strategy described below was agreed with FDA as part of a Type B pre-BLA meeting (see Section 3).

To allow for evaluation of the safety of the target dose in the target population, data from 2 studies (DS8201-A-J101 and DS8201-A-U201) were combined to create the HER2-positive Breast Cancer 5.4 mg/kg Pool (N = 234). Three additional pooled groups were formed to allow evaluation of possible dose-response effects and safety in other tumor types:

- HER2-positive breast cancer treated with ≥ 6.4 mg/kg (6.4, 7.4 or 8.0 mg/kg) (N = 137)
- All tumor types treated with 5.4 mg/kg (N = 275)
- All tumor types treated with ≥ 6.4 mg/kg (6.4, 7.4, or 8.0 mg/kg) (N = 258)

The integrated safety analyses focus primarily on safety data from the target indication of HER2-positive breast cancer and the target dosing regimen of 5.4 mg/kg Q3W, derived from the pooled data in Study DS8201-A-J101 or DS8201-A-U201 (hereafter referred to as the HER2-positive Breast Cancer 5.4 mg/kg Pool).

Three additional completed studies were not included in the pooled safety analyses due to the heterogeneity of study design and study population, and relatively short duration of exposure

in these studies (median: 4.6-5.5 months): DS8201-A-J102, DS8201-A-A103, and DS8201-A-A104 (see [Table 19](#) in Section 7.1).

The Applicant's Position:

Based on the potential mechanism of action and observed data from clinical studies, the expected common toxicities of fam-trastuzumab deruxtecan were primarily GI or hematologic in nature, with AESIs of ILD, QT prolongation, LVEF decrease, and IRR identified as part of the comprehensive safety review during the development of fam-trastuzumab deruxtecan.

The FDA's Assessment:

The FDA generally agrees with the applicant's assessment. The FDA conducted an independent analysis of the pooled safety data for the 234 patients in studies DS8201-A-J101 and DS8201-A-U201 as noted above. FDA did not pool data from the other 3 studies noted in the applicant's assessment due to the differences in study design and patient populations. Study DS8201-A-J102 was a phase 1 study of fam-trastuzumab deruxtecan to assess the effect on the QT interval in HER2-positive breast cancer in which fam-trastuzumab was dosed at 6.4mg/kg. Study DS8201-A-A103 was a phase 1 PK study of fam-trastuzumab deruxtecan in HER2-positive gastric, GEJ adenocarcinoma, and breast cancer also at a dose of 6.4mg/kg. Study DS8201-A-A104 was a phase 1, drug-drug interaction study of OATP1B/CYP3A inhibitor on the PK of fam-trastuzumab in patients with HER2-expressing solid tumors. Although the dose used in this study was 5.4mg/kg, there was heterogeneity of tumor types, line of therapy, and intent of therapy as not all patients had metastatic disease. Therefore, these 3 phase studies were not included in the pooled analysis of safety for this BLA.

8.2.2. Review of the Safety Database

Overall Exposure

Data:

The Safety Analysis Set was defined as all subjects who received at least 1 dose of fam-trastuzumab deruxtecan. The safety database for this submission included 645 subjects who received at least 1 dose of fam-trastuzumab deruxtecan from the 5 completed studies, including 234 subjects with HER2-positive breast cancer who were treated with 5.4 mg/kg.

The median duration of exposure to study drug was 6.96 months in the HER2-positive Breast Cancer 5.4 mg/kg Pool, compared to 8.97 months in the HER2-positive Breast Cancer ≥6.4 mg/kg Pool ([Table 31](#)).

Table 31: Summary of Duration of Exposure (Safety Analysis Set)

	Pool				Study	
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)	DS8201-A-J101 HER2-positive Breast Cancer 5.4 mg/kg (N = 50)	DS8201-A-U201 HER2-positive Breast Cancer 5.4 mg/kg (N = 184)
Treatment duration (months)						
Mean (Std Dev)	7.83 (4.709)	9.70 (6.109)	7.64 (4.973)	8.70 (6.110)	10.95 (7.555)	6.98 (3.092)
Median	6.96	8.97	6.90	7.84	8.54	6.90
Range	0.7, 31.0	0.7, 35.0	0.7, 32.0	0.7, 35.0	0.7, 31.0	0.7, 16.1
Treatment duration (categories) n (%)						
0 to ≤3 months	35 (15.0)	14 (10.2)	48 (17.5)	52 (20.2)	8 (16.0)	27 (14.7)
>3 to ≤6 months	35 (15.0)	24 (17.5)	46 (16.7)	47 (18.2)	6 (12.0)	29 (15.8)
>6 to ≤9 months	104 (44.4)	32 (23.4)	115 (41.8)	48 (18.6)	12 (24.0)	92 (50.0)
>9 to ≤12 months	30 (12.8)	31 (22.6)	32 (11.6)	49 (19.0)	7 (14.0)	23 (12.5)
>12 to ≤24 months	28 (12.0)	32 (23.4)	30 (10.9)	56 (21.7)	15 (30.0)	13 (7.1)
>24 months	2 (0.9)	4 (2.9)	4 (1.5)	6 (2.3)	2 (4.0)	0
Relative dose intensity						
Median	97.80	89.70	97.70	91.50	96.15	98.20
Range	47.2, 104.1	51.4, 106.0	47.2, 104.8	50.4, 106.0	55.3, 104.1	47.2, 103.3

The Applicant's Position:

The exposure to study drug was considered adequate to allow for a comprehensive assessment of safety in subjects who were representative of the intended target population.

The FDA's Assessment:

The FDA generally agrees with the applicant's assessment of durations of exposure in Table 31.

Relevant Characteristics of the Safety Population:

Data:

The demographic and baseline disease characteristics, based on the Safety Analysis Set, are summarized in [Table 32](#). In subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool (N = 234), the median age was 56 years (range: 28 to 96); 99.6% were female; 50.9% were White, 41.5% were Asian, 3.0% were Black or African American; and 57.7% had an ECOG PS of 0 and 41.9% had an ECOG PS of 1. The studies excluded patients with a history of treated ILD or ILD at screening and patients with a history of clinically significant cardiac disease.

Table 32: Summary of Demographics

Parameter	Pool				Study	
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)	DS8201-A-J101 HER2-positive Breast Cancer 5.4 mg/kg (N = 50)	DS8201-A-U201 HER2-positive Breast Cancer 5.4 mg/kg (N = 184)
Sex, n (%)						
Male	1 (0.4)	0	13 (4.7)	50 (19.4)	1 (2.0)	0
Female	233 (99.6)	137 (100.0)	262 (95.3)	208 (80.6)	49 (98.0)	184 (100.0)
Age (years) ^a						
Mean (Std Dev)	56.1 (11.77)	55.2 (11.15)	56.9 (11.87)	57.4 (11.54)	56.6 (12.05)	56.0 (11.72)
Median	56.0	55.0	57.0	58.0	58.0	55.0
Range	28, 96	29, 79	28, 96	23, 83	28, 77	28, 96
Age Group, n (%)						
<65 years	173 (73.9)	102 (74.5)	194 (70.5)	179 (69.4)	33 (66.0)	140 (76.1)
≥65 years	61 (26.1)	35 (25.5)	81 (29.5)	79 (30.6)	17 (34.0)	44 (23.9)
<75 years	223 (95.3)	134 (97.8)	260 (94.5)	246 (95.3)	48 (96.0)	175 (95.1)
≥75 years	11 (4.7)	3 (2.2)	15 (5.5)	12 (4.7)	2 (4.0)	9 (4.9)
Race, n (%)						
White	119 (50.9)	50 (36.5)	136 (49.5)	78 (30.2)	18 (36.0)	101 (54.9)
Black or African American	7 (3.0)	3 (2.2)	8 (2.9)	5 (1.9)	3 (6.0)	4 (2.2)
Asian	97 (41.5)	80 (58.4)	118 (42.9)	168 (65.1)	27 (54.0)	70 (38.0)

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Parameter	Pool				Study	
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)	DS8201-A-J101 HER2-positive Breast Cancer 5.4 mg/kg (N = 50)	DS8201-A-U201 HER2-positive Breast Cancer 5.4 mg/kg (N = 184)
American Indian or Alaska Native	2 (0.9)	1 (0.7)	2 (0.7)	1 (0.4)	1 (2.0)	1 (0.5)
Native Hawaiian or Other Pacific Islander	1 (0.4)	1 (0.7)	1 (0.4)	1 (0.4)	0	1 (0.5)
Other	4 (1.7)	1 (0.7)	6 (2.2)	4 (1.6)	1 (2.0)	3 (1.6)
Missing	4 (1.7)	1 (0.7)	4 (1.5)	1 (0.4)	0	4 (2.2)
Region/Country of Enrollment, n (%)						
North America	82 (35.0)	49 (35.8)	103 (37.5)	85 (32.9)	29 (58.0)	53 (28.8)
US	82 (35.0)	49 (35.8)	103 (37.5)	85 (32.9)	29 (58.0)	53 (28.8)
Asia	84 (35.9)	76 (55.5)	104 (37.8)	161 (62.4)	21 (42.0)	63 (34.2)
Japan	51 (21.8)	69 (50.4)	71 (25.8)	154 (59.7)	21 (42.0)	30 (16.3)
Korea	33 (14.1)	7 (5.1)	33 (12.0)	7 (2.7)	0	33 (17.9)
Europe	68 (29.1)	12 (8.8)	68 (24.7)	12 (4.7)	0	68 (37.0)
Belgium	7 (3.0)	2 (1.5)	7 (2.5)	2 (0.8)	0	7 (3.8)
France	19 (8.1)	1 (0.7)	19 (6.9)	1 (0.4)	0	19 (10.3)
Italy	9 (3.8)	1 (0.7)	9 (3.3)	1 (0.4)	0	9 (4.9)
Spain	21 (9.0)	8 (5.8)	21 (7.6)	8 (3.1)	0	21 (11.4)
United Kingdom	12 (5.1)	0	12 (4.4)	0	0	12 (6.5)

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Parameter	Pool				Study	
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)	DS8201-A-J101 HER2-positive Breast Cancer 5.4 mg/kg (N = 50)	DS8201-A-U201 HER2-positive Breast Cancer 5.4 mg/kg (N = 184)
Ethnicity, n (%) ^b						
Hispanic or Latino	13 (5.6)	8 (5.8)	17 (6.2)	10 (3.9)	3 (6.0)	10 (5.4)
Not Hispanic/Not Latino	167 (71.4)	80 (58.4)	184 (66.9)	114 (44.2)	26 (52.0)	141 (76.6)
Not Applicable	32 (13.7)	6 (4.4)	32 (11.6)	6 (2.3)	0	32 (17.4)
Missing	22 (9.4)	43 (31.4)	42 (15.3)	128 (49.6)	21 (42.0)	1 (0.5)
Median (range) number of prior regimens	7.0 (2-27)	7.0 (2-21)	6.0 (2-27)	5.5 (1-21)	7.0 (2-17)	6.0 (2-27)

The baseline value was defined as the last non-missing value before initial administration of study drug.

Percentages were calculated using the number of subjects in the Safety Analysis Set as the denominator.

The 4 pooled analysis groups were based on tumor type and first dose received for subjects in DS8201-A-J101 and DS8201-A-U201.

^a Age in years was calculated using the informed consent date and the birth date

^b Ethnicity was not required to be collected in all countries.

The Applicant's Position:

The eligibility criteria in the 2 primary studies were clinically relevant for the target population of subjects who would receive fam-trastuzumab deruxtecan following regulatory approval in the proposed indication.

The subject population treated in the 2 studies was a clearly defined and homogenous population of extensively pretreated subjects with HER2-positive unresectable or metastatic breast cancer. Demographic and baseline characteristics were similar across the 2 studies, and the pooled population was generally representative of the broad population of patients with HER2-positive metastatic breast cancer seen in medical practice who had received at least 2 prior anti-HER2 therapies.

The FDA's Assessment:

FDA agrees with the demographic table the applicant has presented and agrees that the population characteristics of the 234 patients in both studies DS8201-A-J101 and DS8201-A-U201 were generally representative of a US population for the intended indication and therefore appropriate to pool for the safety analysis for this BLA.

Adequacy of the safety database:

Data:

The safety database for this submission included 645 subjects who received at least 1 dose of fam-trastuzumab deruxtecan from the 5 completed studies (including 9 subjects who received doses <5.4 mg/kg in DS8201-A-J101). Among these 645 subjects, 234 had HER2-positive unresectable or metastatic breast cancer and were treated with 5.4 mg/kg. Among these 234 subjects, the median treatment duration was 6.96 months (range: 0.7-31.0); and 30 (12.9%) subjects had a treatment duration over 12 months and received a median of 10 cycles (range: 1-38). With 234 subjects in the pooled safety database, there was a $\geq 90\%$ chance of observing an AE with a true incidence rate of 1% or higher.

The Applicant's Position:

The safety database was considered adequate to evaluate the safety of fam-trastuzumab deruxtecan in the intended population at the target dose.

The FDA's Assessment:

The FDA generally agrees with the applicant's assessment. The FDA did not independently verify the probability of observing an AE with a true incidence rate of 1% or higher but believes the safety data from the pool of 234 patients is adequate to evaluate the safety profile of fam-trastuzumab deruxtecan.

8.2.3. Adequacy of Applicant's Clinical Safety Assessments

Issues Regarding Data Integrity and Submission Quality

Data:

Cardiotoxicity in association with fam-trastuzumab deruxtecan is considered to be an important potential risk based on the available nonclinical data, literature, and available safety information for drugs of similar class. Therefore, in addition to routine clinical safety assessment, troponin was measured after each infusion, and an echocardiogram (ECHO)/multigated acquisition (MUGA) scan was done to evaluate LVEF every 2 cycles in Study DS8201-A-J101 and every 4 cycles in Study DS8201-A-U201. Ophthalmological examination was done in both studies at Day 1 in Cycle 2 and then every 4 cycles, as agreed with FDA (see Section 3).

Additional data on LVEF, QT, and ILD were collected via targeted questionnaires built into the electronic data capture (EDC). An independent ILD AC was established to adjudicate all potential ILD cases (see Section 8.2.5.1).

The Applicant's Position:

No meaningful concerns are anticipated in the quality and integrity of the submitted datasets and individual case narratives; these were sufficiently complete to allow for a thorough review of safety. Furthermore, no data integrity concerns were reported following completion of site inspections by the Applicant; data in the eCRFs and AE databases were consistent.

The FDA's Assessment:

FDA agrees with applicant's assessment.

Categorization of Adverse Event

Data:

The safety of treatment with fam-trastuzumab deruxtecan was evaluated on the basis of the following:

- Frequency, type, severity, and causal relationship of AEs to study treatment
- Frequency of deaths, SAEs, and other clinically significant AEs (including AEs leading to discontinuation and AEs requiring dose interruption and/or reduction)
- Frequency and type of AEs in key intrinsic and extrinsic factors (eg, age, race, country, and region)
- Changes in laboratory variables, with particular attention to Grade 3/4 abnormalities

- Certain AEs under the same medical concept grouped to represent the true incidence of that AE; specifically, the grouped term of:
 - neutrophil count decrease: preferred terms (PTs) of neutrophil count decreased and neutropenia
 - platelet count decrease: PTs of platelet count decreased and thrombocytopenia
 - anemia: PTs of anaemia, red blood cell count decreased, and hemoglobin count decreased
 - white blood cell (WBC) count decrease: PTs of WBC count decreased and leukopenia
 - lymphocyte count decrease: PTs of lymphocyte count decreased and lymphopenia
 - abdominal pain: PTs of abdominal discomfort, abdominal pain, abdominal pain lower, and abdominal pain upper
 - stomatitis: PTs of stomatitis, aphthous ulcer, mouth ulceration, oral mucosa erosion, and oral mucosal blistering
- For AESIs, relevant PTs selected for review to evaluate whether a case was a true AESI, including 44 PTs for ILD adjudication (see Section 8.2.5).

A treatment-emergent AE (TEAE) was defined as an AE that was absent before the first dose of study drug and occurred or worsened in severity or seriousness after the first dose of study drug, up to 28 days (for DS8201-A-J101) or 47 days (for DS8201-A-U201) after last dose of study drug. If considered related to study drug per investigator, SAEs with an onset or worsening 29 days or 48 days (respectively) or more after the last dose of study drug were also considered to be TEAEs.

Adverse events could be directly observed, reported spontaneously by the subject or elicited by questioning the subject at each study visit. Per protocol, subjects were to be questioned in a general way, without asking about the occurrence of any specific symptoms.

Adverse events were classified according to the Medical Dictionary for Regulatory Activities (MedDRA) Version 20.1.

Severity of AEs and laboratory parameters was graded using NCI-CTCAE Version 4.03.

The Applicant's Position:

The approach of categorization of AEs was appropriate in evaluating the safety data.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment.

Routine Clinical Tests

Data:

The tables of study procedures and scheduled assessments in the protocols for the individual studies specified the time points at which laboratory evaluations were conducted.

Data from all sources (central and local laboratories) were combined. The summaries included all laboratory assessments collected within 28 days (for DS8201-A-J101) or 47 days (for DS8201-A-U201) after last dose of study drug. Per protocol, abnormal laboratory values considered clinically significant by the investigator were to be reported as AEs.

The Applicant's Position:

The clinical monitoring of subject safety (methods and time points) described in the protocols were reasonable and adequate for the population, disease, and indication being investigated.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment.

8.2.4. Safety Results

Deaths

Data:

Deaths due to any cause (including deaths that occurred during follow-up) were reported in 28 (12.0%) subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool, which included any death in DS8201-A-J101 as of the DCO of 1 Feb 2019 and any death in DS8201-A-U201 as of the DCO of 21 Mar 2019 (Table 33). Sixteen (6.8%) subjects had PD as a primary cause of death and 7 (3.0%) subjects had TEAEs as a primary cause of death (including 2 subjects who had a TEAE associated with an outcome of death that was not considered an on-study AE). In all pools, the highest proportion of all deaths was due to PD.

"On-study death due to any cause" refers to death, based on the investigator assessment of the primary cause of death, that occurred during study treatment or within 28 days after the last dose of study drug in DS8201-A-J101 or 47 days after the last dose in DS8201-A-U201. On-study deaths in the HER2-positive Breast Cancer 5.4 mg/kg Pool were reported in 10 (4.3%) subjects, which was similar to the other pools, including 4 (1.7%) subjects having PD as the primary cause and 5 (2.1%) subjects having AEs listed as the primary cause.

Table 33: Summary of Deaths in the Primary Studies

Primary Cause	Number (%) of Subjects in Each Pool			
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)
Any Death	28 (12.0)	28 (20.4)	40 (14.5)	70 (27.1)
Disease Progression	16 (6.8)	20 (14.6)	24 (8.7)	46 (17.8)
Adverse Event	7 (3.0)	3 (2.2)	8 (2.9)	11 (4.3)
Other	2 (0.9)	2 (1.5)	3 (1.1)	4 (1.6)
Unknown	3 (1.3)	3 (2.2)	5 (1.8)	9 (3.5)
On-study Death	10 (4.3)	3 (2.2)	10 (3.6)	9 (3.5)
Disease Progression	4 (1.7)	1 (0.7)	4 (1.5)	3 (1.2)
Adverse Event	5 (2.1)	2 (1.5)	5 (1.8)	6 (2.3)
Other	1 (0.4)	0	1 (0.4)	0

In the 3 other completed studies, 3 subjects had died as of the respective DCO for each study, with 1 on-study death ([Table 34](#)).

Table 34: Summary of Deaths in Other Completed Studies

Primary Cause	Number (%) of Subjects in Each Study		
	DS8201-A-J102 (N = 51)	DS8201-A-A103 (N = 12)	DS8201-A-A104 (N = 40)
Any Death	2 (3.9)	0	1 (2.5)
On-study Death	0	0	1 (2.5)
Disease Progression	2 (3.9)	0	0
Adverse Event	0	0	1 (2.5)

The Applicant's Position:

A review of the deaths across the 5 completed studies did not identify any pattern and most deaths were associated with PD.

The FDA's Assessment:

The FDA conducted an independent analysis of deaths in studies DS8201-A-J101 and DS8201-A-U201 based on the death narratives submitted by the applicant. All cause death occurred in 12% of the 234 patients with HER2-positive breast cancer treated at 5.4mg/kg at the time of

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

data cutoff. The majority (6.8%) of these all-cause deaths were due to disease progression. The FDA reviewed the on-study deaths that occurred during study treatment or within 28 days after the last dose of study drug in DS8201-A-J101 or 47 days after the last dose in DS8201-A-U201. There were 10 (4.3%) on-study deaths, 3 (1.3%) occurred from study J101 and 7 (3.0%) occurred from study U201. FDA reviewed the on-study deaths and generally agreed with the Applicant with the following comments:

Study DS8201-A-J101: Three (1.3%) on-study deaths occurred in this study, 1 reported due to disease progression and 2 due to adverse events.

1. Patient (b) (6): The 56-year-old Asian female received a total of 15 doses of study drug. The patient developed potential pneumonitis noted on a CT scan 3 months after first dose of study drug. She was later diagnosed with pneumocystis jirovecii pneumonia and received antibiotics and steroids. The patient was reported recovered from this event and received 11 more doses of study drug. Her death was reported due to disease progression with CT at the time showing lymphangitic carcinomatosis. This death was not adjudicated by the independent ILD committee (refer to Section 8.2.5.1). FDA believes that a component of potential ILD/pneumonitis cannot be excluded at the time of the PNA diagnosis as the CT scan showed potential pneumonitis at the time and the patient received steroids. There may also have been a component of ILD at the time of reported disease progression.
2. Patient (b) (6): The 66-year-old White female received a total of 8 doses of study drug. The reported adverse event leading to death was respiratory failure. This case was adjudicated by the independent ILD committee as ILD related to study drug. This patient is one of six deaths attributed to ILD in the USPI. ILD accounted for 2.6% of deaths in the pool of 234 patients.
3. Patient (b) (6): The 58-year-old Asian female received a total of 14 doses of study drug. Prior to death, the patient developed grade 4 pneumonitis and was started on antibiotics and steroids. The adverse events leading to death was respiratory failure. This case was adjudicated by the independent ILD committee as ILD related to study drug. This is one of the 6 deaths attributed to ILD in the USPI.

Study DS8201-A-U201: Seven (3.0%) on-study deaths occurred in this study, 3 reported due to disease progression, 3 due to adverse events, and 1 due to "other."

1. Patient (b) (6): The 45-year-old White female received a total of 4 doses of study drug. The patient developed respiratory failure and was hospitalized, decompensated and died. The investigator assessed the respiratory failure as unrelated to study drug and death as a result of disease progression. The FDA disagrees with the investigator's assessment as there was no clear evidence of disease progression presented in the narrative. This case was also adjudicated by the ILD committee as ILD related to study

- drug. This is one of the 6 deaths attributed to ILD in the USPI.
2. Patient (b) (6): The 62-year-old White female received a total of 3 doses of study drug. The patient developed suspected pneumonitis and lymphangitic metastasis and died shortly after. The investigator considered the case unrelated to study drug as the patients did not respond to steroids and died due to disease progression. The FDA disagrees with the investigator's assessment as the patient was only on steroids a very short period of time before death and any underlying pneumonitis may have been severe by the time steroids could be initiated. This case was also adjudicated by the ILD committee as ILD related to study drug. This is one of the 6 deaths attributed to ILD in the USPI.
 3. Patient (b) (6): The 39-year-old White female received a total of 2 doses of study drug. FDA agrees with the narrative that patient experienced disease progression as there is a CT scan reported with clear progression of disease in the lung and liver, patient died shortly after.
 4. Patient (b) (6): The 69-year-old White female received a total of 3 doses of study drug. This patient died due to hemorrhagic shock potentially due to tumor erosion into the esophagus per the investigator. Hemorrhagic shock was listed as the adverse event (AE) leading to death. The FDA queried the applicant whether this case was potentially a result of disease progression as opposed to an AE. As there was no imaging or endoscopy evidence in this case per the applicant confirming progression due to tumor erosion, the cause of the bleeding is unknown. The patient's last platelet count prior to the event was reported normal. Therefore, the FDA agrees this case should be classified as an AE.
 5. Patient (b) (6): The 66-year-old White female received a total of 10 doses of study drug. The narrative notes the patient had episodes of recurrent ascites require multiple drainages over the course of treatment. Ascites was not reported at baseline prior to the study. The patient progressively decompensated and developed pulmonary sepsis eventually leading to death after a prolonged hospital course. The investigator considered all events unrelated to study drug. FDA disagrees as it cannot be excluded that the drug contributed to the AEs.
 6. Patient (b) (6): The 48-year-old-female received a total of 8 doses of study drug. The patient developed pneumococcal pneumonia which evolved into septic shock before death. The investigator listed the pneumonia as unrelated to study drug. This case was adjudicated by the ILD committee as ILD related to study drug. This is one of the 6 deaths attributed to ILD in the USPI. FDA agrees with the ILD committee assessment.
 7. Patient (b) (6): The 53-year-old White female received a total of 5 doses of study drug. This patient had reported extensive baseline hepatic metastasis prior to initiating the study drug with elevated LFTs. These LFTs continued to trend up after initiation of study drug. The patient developed hepatic failure and acute kidney injury eventually leading to death. This case was listed as "other" leading to fatality. The FDA queried the applicant whether there was clear evidence of progression in this case of the hepatic mets. The applicant stated that imaging performed at the time showed "extensive

hepatic metastatic disease”. As it is unclear whether there was clear progression prior to death, FDA would consider this case an AE leading to death.

Serious Adverse Events

Data:

Treatment-emergent SAEs were reported in 47 (20.1%) subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool, with 7 PTs and the AESI-defined ILD (which includes the PTs of ILD, pneumonitis, organising pneumonia, and acute interstitial pneumonitis) reported in ≥1% of subjects (Table 35).

Overall, SAEs were reported for similar proportions of subjects in the 4 pools. A higher proportion of subjects with an SAE of AESI-defined ILD was observed in the higher-dose pools.

Table 35: Serious Adverse Events in the Primary Studies, by Preferred Term

MedDRA Preferred Term	Number (%) of Subjects in Each Pool			
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)
Subjects with Any Serious TEAE	47 (20.1)	31 (22.6)	54 (19.6)	69 (26.7)
Pneumonia	5 (2.1)	1 (0.7)	7 (2.5)	2 (0.8)
Respiratory failure	5 (2.1)	0	5 (1.8)	2 (0.8)
Interstitial lung disease ^a	4 (1.7)	6 (4.4)	4 (1.5)	13 (5.0)
Vomiting	4 (1.7)	2 (1.5)	4 (1.5)	3 (1.2)
Cellulitis	3 (1.3)	0	4 (1.5)	1 (0.4)
Hypokalaemia	3 (1.3)	0	3 (1.1)	0
Intestinal obstruction	3 (1.3)	0	3 (1.1)	0
Nausea	3 (1.3)	3 (2.2)	3 (1.1)	3 (1.2)

Includes treatment-emergent SAEs reported in at least 1% of subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool.

^a AESI-defined ILD includes preferred terms of interstitial lung disease, pneumonitis, organising pneumonia, and acute interstitial pneumonitis.

Serious AEs were reported for 9 subjects across the 3 other completed studies, with only nausea reported in more than 1 subject (Table 36).

Table 36: Serious Adverse Events in Other Completed Studies, by Preferred Term

MedDRA Preferred Term	Number (%) of Subjects in Each Study		
	DS8201-A-J102 ^a (N = 51)	DS8201-A-A103 ^b (N = 12)	DS8201-A-A104 ^c (N =40)
Subjects with Any Serious Adverse Event	2 (3.9)	2 (16.7)	5 (12.5)
Diarrhea	0	0	1 (2.5)
Disease progression	0	0	1 (2.5)
Febrile neutropenia	0	1 (8.3)	0
Femur fracture	1 (2.0)	0	0
Ileus	0	0	1 (2.5)
Nausea	1 (2.0)	0	1 (2.5)
Pneumonitis	0	0	1 (2.5)
Pyrexia	0	0	1 (2.5)
Sepsis	0	0	1 (2.5)
Urinary tract infection	0	1 (8.3)	0
Vomiting	0	0	1 (2.5)

The Applicant's Position:

The most common SAEs in the HER2-positive Breast Cancer 5.4 mg/kg Pool (ie, those reported in at least 1% of subjects) were pneumonia, respiratory failure, ILD (grouped term of 4 PTs), vomiting, cellulitis, hypokalaemia, intestinal obstruction, and nausea.

The FDA's Assessment:

The FDA generally agrees with the applicant's assessment of SAEs. Forty-seven (20%) of the pooled 234 patients with HER2-positive metastatic breast cancer treated with 5.4mg/kg of fam-trastuzumab deruxtecan experienced SAEs. The FDA pooled all SAEs that were adjudicated as ILD by the independent ILD committee which included 44 preferred terms (PTs) compared to the four PT terms of interstitial lung disease, pneumonitis, organizing pneumonia, and acute interstitial pneumonitis used by the applicant as the AESI-defined ILD. Based on the definition of ILD using the 44 PTs used by the independent adjudication committee, SAEs in >1% of patients who received fam-trastuzumab deruxtecan were interstitial lung disease, pneumonia, vomiting, nausea, cellulitis, hypokalemia, and intestinal obstruction.

Dropouts and/or Discontinuations Due to Adverse Effects

Data:

In the HER2-positive Breast Cancer 5.4 mg/kg Pool, 114 (48.7%) subjects discontinued study drug, the majority due to PD (per RECIST or clinical progression per the investigator). Twelve PTs were associated with discontinuation of study drug in 22 (9.4%) subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool, with only AESI-defined ILD reported in more than 1 subject (13 [5.6%] subjects) (Table 37). Protocol requirements for discontinuation of subjects due to ILD changed during the course of trial conduct.

These results were similar those in the All Tumor Types 5.4 mg/kg Pool and were lower than those in the ≥6.4 mg/kg Pools. For each pool, the frequencies were driven primarily by events of AESI-defined ILD.

Table 37: Adverse Events Associated with Study Drug Discontinuation, by Preferred Term or Grouped Term

MedDRA Preferred Term or Grouped Term	Number (%) of Subjects in Each Pool			
	HER2-positive Breast cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)
Any TEAE Associated with Study Drug Discontinuation	22 (9.4)	38 (27.7)	22 (8.0)	59 (22.9)
ILD (AESI-defined) ^a	13 (5.6)	25 (18.2)	13 (4.7)	39 (15.1)
Neuropathy peripheral	1 (0.4)	1 (0.7)	1 (0.4)	1 (0.4)
Pneumonia	1 (0.4)	1 (0.7)	1 (0.4)	2 (0.8)
Thrombocytopenia	1 (0.4)	1 (0.7)	1 (0.4)	1 (0.4)
Alveolitis	1 (0.4)	0	1 (0.4)	0
Cardiac failure congestive	1 (0.4)	0	1 (0.4)	0
Dyspnoea	1 (0.4)	0	1 (0.4)	0
Osteonecrosis of jaw	1 (0.4)	0	1 (0.4)	0
Performance status decreased	1 (0.4)	0	1 (0.4)	0
Pleural effusion	1 (0.4)	0	1 (0.4)	0
Troponin increased	1 (0.4)	0	1 (0.4)	0

^a AESI-defined ILD includes preferred terms of interstitial lung disease, pneumonitis, organising pneumonia, and acute interstitial pneumonitis.

The Applicant's Position:

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

There was a notable difference in the proportion of subjects with AEs associated with discontinuation between subjects who received an initial dose of 5.4 mg/kg compared to those who received an initial dose of ≥ 6.4 mg/kg. The difference was primarily due to ILD, which was also the main AE associated with discontinuation.

The FDA's Assessment:

The FDA generally agrees with the applicant's assessment of adverse events leading to dropouts and discontinuations. Fam-trastuzumab deruxtecan was permanently discontinued in 9.4% of patients, of which ILD accounted for 6%. Similar to SAEs, the PTs that were adjudicated as ILD by the independent adjudication committee were pooled to define the incidence of discontinuations due to ILD.

Dose Interruption/Reduction Due to Adverse Effects

Data:

Treatment-emergent AEs were associated with dose interruption in 78 (33.3%) subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool (Table 38). This was similar to the proportion of subjects in the All Tumor Types 5.4 mg/kg Pool and was lower than the proportion of subjects in either of the ≥ 6.4 mg/kg pools.

The most commonly reported TEAE associated with dose interruption in the HER2-positive Breast Cancer 5.4 mg/kg Pool was neutrophil count decrease (grouped term), which was reported in 12.8% of subjects. This was similar to the All Tumor Types 5.4 mg/kg Pool and lower than either of the ≥ 6.4 mg/kg pools. Similar differences across pools were seen for other hematologic TEAEs including anemia (grouped term) and WBC count decrease (grouped term).

Table 38: Adverse Events Associated with Dose Interruption, by Preferred or Grouped Term

MedDRA Preferred Term or Grouped Term	Number (%) of Subjects in Each Pool			
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)
Subjects with Any TEAE Associated with Dose Interruption	78 (33.3)	65 (47.4)	96 (34.9)	115 (44.6)
Neutrophil count decrease ^a	30 (12.8)	22 (16.1)	32 (11.6)	42 (16.3)
Anemia ^b	8 (3.4)	13 (9.5)	9 (3.3)	23 (8.9)
Interstitial lung disease ^c	7 (3.0)	7 (5.1)	7 (2.5)	11 (4.3)
Platelet count decrease ^d	7 (3.0)	5 (3.6)	8 (2.9)	9 (3.5)

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

MedDRA Preferred Term or Grouped Term	Number (%) of Subjects in Each Pool			
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)
Nausea	5 (2.1)	4 (2.9)	7 (2.5)	6 (2.3)
WBC count decrease ^e	5 (2.1)	6 (4.4)	5 (1.8)	10 (3.9)
Fatigue	4 (1.7)	4 (2.9)	5 (1.8)	7 (2.7)
Blood bilirubin increased	3 (1.3)	3 (2.2)	3 (1.1)	3 (1.2)
Upper respiratory tract infection	3 (1.3)	1 (0.7)	4 (1.5)	4 (1.6)
Vomiting	3 (1.3)	1 (0.7)	3 (1.1)	2 (0.8)

Includes treatment-emergent AEs associated with dose interruption in at least 1% of subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool.

^a Neutrophil count decrease (grouped term) includes PTs of neutrophil count decreased and neutropenia

^b Anemia (grouped term) includes PTs of anaemia and hemoglobin count decreased

^c ILD (grouped term) includes PTs of interstitial lung disease, pneumonitis, organising pneumonia, and acute interstitial pneumonitis

^d Platelet count decrease (grouped term) includes PTs of platelet count decreased and thrombocytopenia

^e White blood cell count decrease (grouped term) includes PTs of white blood cell count decreased and leukopenia

Treatment-emergent AEs were associated with dose reduction in 42 (17.9%) subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool, with 6 PTs each reported in ≥1% of subjects (Table 39). This was similar to results in the All Tumor Types 5.4 mg/kg Pool, and was lower than in either of the higher-dose pools.

Table 39: Adverse Events Associated with Dose Reduction, by Preferred Term

MedDRA Preferred Term or Grouped Term	Number (%) of Subjects			
	Pool			
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)
Subjects with Any TEAE Associated with Dose Reduction	42 (17.9)	49 (35.8)	50 (18.2)	82 (31.8)
Fatigue	7 (3.0)	12 (8.8)	8 (2.9)	17 (6.6)
Nausea	7 (3.0)	10 (7.3)	8 (2.9)	13 (5.0)
Neutrophil count decrease ^a	5 (2.1)	6 (4.4)	6 (2.2)	9 (3.5)
Blood bilirubin increased	4 (1.7)	1 (0.7)	4 (1.5)	1 (0.4)
Vomiting	3 (1.3)	1 (0.7)	3 (1.1)	1 (0.4)
Weight decreased	3 (1.3)	1 (0.7)	4 (1.5)	2 (0.8)

Includes treatment-emergent AEs associated with dose reduction in at least 1% of subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool.

^a Neutrophil count decrease (grouped term) includes preferred terms of neutrophil count decreased and neutropenia.

The Applicant's Position:

Observed AEs associated with dose reduction or interruption were as expected in this patient population and this class of drugs. As expected, there was a notable dose response.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment of adverse events leading to dose interruptions and dose reductions.

Dose reductions occurred in 18% of patients treated with fam-trastuzumab deruxtecan. The most frequent adverse reactions (>2%) associated with dose reduction were fatigue, nausea, and neutropenia.

Dose interruptions due to adverse reactions occurred in 33% of patients treated with fam-trastuzumab deruxtecan. The most frequent adverse reactions (>2%) associated with dose interruption were neutropenia, anemia, thrombocytopenia, leukopenia, upper respiratory tract infection, fatigue, nausea, and ILD.

Significant Adverse Events

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

Data:

In addition to the AESIs of ILD, LVEF decrease, QT prolongation, and IRR, which are discussed in detail in Section 8.2.5, 9 TEAEs were selected as other potentially significant AEs: nausea, vomiting, constipation, diarrhoea, anemia (grouped term including PTs of anaemia, red blood cell count decreased, and hemoglobin decreased), neutrophil count decrease (grouped term including PTs of neutrophil count decreased and neutropenia), febrile neutropenia, platelet count decrease (grouped term including PTs of platelet count decrease and thrombocytopenia), fatigue, and asthenia. These TEAEs were selected because they were expected in this class of drugs and they were the most commonly observed among TEAEs.

Results for selected PTs shown below for the HER2-positive Breast Cancer 5.4 mg/kg Pool were similar to those reported in the All Tumor Types 5.4 mg/kg Pool (Table 40). For most TEAEs, within each tumor type and across tumor types, the proportion of subjects with these TEAEs was higher at doses ≥ 6.4 mg/kg than at the 5.4 mg/kg dose.

Table 40: Selected Adverse Events Reported in the HER2-positive Breast Cancer 5.4 mg/kg Pool

MedDRA PT/Grouped Term	Number (%) of Subjects with Event (N=234)			Median (Range) Time to Onset of First Event (Days)	Median (Range) Duration of First Event (Days)	Number (%) of Subjects with Action Taken with Respect to Study Drug		
	Overall	At Least Grade 3	SAE			Discontinuation	Dose Reduction	Dose Interruption
Nausea ^a	185 (79.1)	16 (6.8)	3 (1.3)	2 (1-464)	9 (1-323)	0	7 (3.0)	5 (2.1)
Vomiting	111 (47.4)	9 (3.8)	4 (1.7)	7 (1-238)	2 (1-215)	0	3 (1.3)	3 (1.3)
Constipation	81 (34.6)	2 (0.9)	0	NA	NA	0	0	0
Diarrhoea	67 (28.6)	4 (1.7)	0	24 (1-325)	4 (1-289)	0	1 (0.4)	2 (0.9)
Anemia	72 (30.8)	17 (7.3)	1 (0.4)	35 (2-519)	21 (1-198)	0	1 (0.4)	8 (3.4)
Neutrophil count decrease	69 (29.5)	38 (16.2)	0	43 (8-547)	21.5 (2-210)	0	5 (2.1)	30 (12.8)
Febrile neutropenia	4 (1.7)	4 (1.7)	1 (0.4)	NA	NA	0	1 (0.4)	1 (0.4)
Platelet count decrease ^b	47 (20.1)	8 (3.4)	0	66 (2-324)	15 (4-203)	1 (0.4)	2 (0.9)	7 (30.0)
Fatigue	112 (47.9)	12 (5.1)	0	8 (1-407)	12 (1-256)	0	7 (3.0)	4 (1.7)
Asthenia	29 (12.4)	2 (0.9)	0	NA	NA	0	1 (0.4)	1 (0.4)

NA = not available

The Applicant's Position:

The most common AEs were GI, hematologic, or general disorders that are expected with this class of drugs and were manageable by dose modification and routine clinical practice. The available safety database was adequate to categorize and characterize these events.

The FDA's Assessment:

FDA generally agrees with the data presented by the applicant in Table 40 for select adverse events with the following comments:

Table 40 for select adverse events with the following comments:

Based in FDA analysis, the PT terms of fatigue and asthenia were combined as fatigue for the USPI. Based on this, all grade fatigue was 59% with 6% of patients with $\geq$ Grade 3 in the pooled 234 patients.

Treatment Emergent Adverse Events and Adverse Reactions

Data

Most Frequent Adverse Events by Preferred Term

In decreasing order of frequency within the HER2-positive 5.4 mg/kg Pool, the most frequently reported (>20% of subjects) TEAEs (MedDRA PTs or grouped terms) in this pool were nausea, fatigue, vomiting, alopecia, constipation, decreased appetite, anemia (grouped term), neutrophil count decrease (grouped term), diarrhoea, and platelet count decrease (grouped term) (Table 41).

Compared to the HER2-positive breast cancer ≥ 6.4 mg/kg Pool, the HER2-positive Breast Cancer 5.4 mg/kg Pool had a notably lower proportion of subjects with the following TEAEs: alopecia, decreased appetite, anemia (grouped term), neutrophil count decrease (grouped term), platelet count decrease (grouped term), WBC count decrease (grouped term), AST increased, stomatitis, and alanine aminotransferase (ALT) increased.

The proportion of subjects with each PT was similar between the HER2-positive Breast Cancer 5.4 mg/kg Pool and the All Tumor Types 5.4 mg/kg Pool.

Table 41: Most Common Adverse Events, by Preferred Term

MedDRA Preferred Term	Number (%) of Subjects in Each Pool			
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)
Subjects with Any TEAE	233 (99.6)	137 (100.0)	273 (99.3)	258 (100.0)
Nausea	185 (79.1)	106 (77.4)	213 (77.5)	198 (76.7)
Fatigue	112 (47.9)	68 (49.6)	127 (46.2)	104 (40.3)
Vomiting	111 (47.4)	61 (44.5)	124 (45.1)	114 (44.2)
Alopecia	107 (45.7)	81 (59.1)	117 (42.5)	126 (48.8)
Constipation	81 (34.6)	57 (41.6)	94 (34.2)	95 (36.8)
Decreased appetite	76 (32.5)	74 (54.0)	91 (33.1)	153 (59.3)
Anemia ^a	72 (30.8)	60 (43.8)	83 (30.2)	112 (43.4)
Neutrophil count decrease ^b	69 (29.5)	62 (45.3)	75 (27.3)	109 (42.2)
Diarrhoea	67 (28.6)	45 (32.8)	79 (28.7)	88 (34.1)
Platelet count decrease ^c	47 (20.1)	48 (35.0)	56 (20.4)	98 (38.0)
Cough	46 (19.7)	23 (16.8)	50 (18.2)	36 (14.0)
White blood cell count decrease ^d	45 (19.2)	50 (36.5)	53 (19.3)	92 (35.7)
Abdominal pain ^e	42 (17.9)	20 (14.6)	44 (16.0)	32 (12.4)
Headache	42 (17.9)	23 (16.8)	46 (16.7)	31 (12.0)
Aspartate aminotransferase increased	32 (13.7)	35 (25.5)	36 (13.1)	60 (23.3)
Stomatitis ^f	32 (13.7)	35 (25.5)	37 (13.5)	63 (24.4)
Dyspnoea	31 (13.2)	10 (7.3)	35 (12.7)	20 (7.8)
Epistaxis	30 (12.8)	10 (7.3)	33 (12.0)	23 (8.9)
Asthenia	29 (12.4)	8 (5.8)	29 (10.5)	8 (3.1)
Dyspepsia	29 (12.4)	14 (10.2)	32 (11.6)	19 (7.4)
Hypokalaemia	28 (12.0)	20 (14.6)	33 (12.0)	38 (14.7)
Dry eye	26 (11.1)	16 (11.7)	29 (10.5)	19 (7.4)
Upper respiratory tract infection	25 (10.7)	9 (6.6)	26 (9.5)	18 (7.0)
Alanine aminotransferase increased	24 (10.3)	28 (20.4)	26 (9.5)	46 (17.8)
Dizziness	24 (10.3)	10 (7.3)	27 (9.8)	23 (8.9)

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139} ENHERTU (fam-trastuzumab deruxtecan-nxki)

Includes treatment-emergent AEs reported in at least 10% of subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool.

- ^a Anemia (grouped term) includes PTs of anaemia, red blood cell count decreased, and hemoglobin count decreased
- ^b Neutrophil count decrease (grouped term) includes PTs of neutrophil count decreased and neutropenia
- ^c Platelet count decrease (grouped term) includes PTs of platelet count decreased, and thrombocytopenia
- ^d White blood cell count decrease (grouped term) includes PTs of leukopenia and white blood cell count decreased
- ^e Abdominal pain (grouped term) includes PTs of abdominal discomfort, abdominal pain, abdominal pain lower, and abdominal pain upper
- ^f Stomatitis (grouped term) includes PTs of stomatitis, aphthous stomatitis, mouth ulceration, oral mucosa erosion, and oral mucosal blistering

The FDA's Assessment:

The FDA independently reviewed all adverse event preferred terms. While the FDA generally agrees with the applicant's data in Table 41 for most common adverse events, there were additional PT's that were included in the following grouped terms for the pooled 234 patients that received 5.4mg/kg in the USPI:

Fatigue: The PT's fatigue and asthenia were grouped with all grade fatigue 59% from 48%. Grade 3-4 fatigue changed to 6% from 5%.

Anemia: The FDA added hematocrit decreased for the grouped term of anemia which included PTs of anemia, hemoglobin decreased, and red blood cell count decreased. All grade anemia remained 31%. Grade 3-4 anemia remained 7%.

White blood cell count decrease: The FDA added lymphopenia for the grouped term of white blood cell count decrease (renamed leukopenia) which included PT's of leukopenia, and white blood cell count decreased. All grade leukopenia is 22% from 19%. Grade 3-4 leukopenia changed to 6% from 4.3%.

Abdominal pain: The FDA added gastrointestinal pain for the grouped term of abdominal pain which included PTs of abdominal discomfort, gastrointestinal pain, abdominal pain, abdominal pain lower, and abdominal pain upper. All grade abdominal pain is 19% from 18%. Grade 3-4 abdominal pain remained 1.3%.

Headache: The FDA added migraine for the grouped term of headache which includes PTs headache, and sinus headache. All grade headache is 19% from 18%. Grade 3-4 headache remained 0%.

Upper respiratory tract infection: The FDA added influenza and influenza like illness for the grouped term of upper respiratory tract infection. All grade upper respiratory tract infection is 15% from 11%. Grade 3-4 upper respiratory tract infection remained 0%.

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

Severity of Adverse Events

The most frequently reported ($\geq 5\%$ of subjects) $\geq$ Grade 3 TEAEs in the HER2-positive Breast Cancer 5.4 mg/kg Pool were neutrophil count decrease (grouped term) (16.2%), anemia (grouped term) (7.3%), nausea (6.8%), and fatigue (5.1%) (Table 42).

The HER2-positive Breast Cancer 5.4 mg/kg Pool had a notably lower proportion of subjects with $\geq$ Grade 3 TEAEs overall and of subjects with $\geq$ Grade 3 events of anemia (grouped term) and fatigue, compared to the HER2-positive breast cancer ≥ 6.4 mg/kg Pool.

Table 42: Adverse Events of at Least Grade 3, by Preferred Term

MedDRA Preferred Term / Grouped Term	Number (%) of Subjects			
	HER2-positive Breast Cancer 5.4 mg/kg (N = 234)	HER2-positive Breast Cancer 6.4, 7.4, 8.0 mg/kg (N = 137)	All Tumor Types 5.4 mg/kg (N = 275)	All Tumor Types 6.4, 7.4, 8.0 mg/kg (N = 258)
Subjects with Any TEAE CTCAE at Least Grade 3	117 (50.0)	92 (67.2)	138 (50.2)	174 (67.4)
Neutrophil count decrease ^a	38 (16.2)	33 (24.1)	41 (14.9)	65 (25.2)
Anemia ^b	17 (7.3)	29 (21.2)	23 (8.4)	60 (23.3)
Nausea	16 (6.8)	8 (5.8)	17 (6.2)	12 (4.7)
Fatigue	12 (5.1)	19 (13.9)	13 (4.7)	22 (8.5)

Includes treatment-emergent AEs $\geq$ Grade 3 reported in at least 5% of subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool.

^a Neutrophil count decrease (grouped term) includes PTs of neutrophil count decreased and neutropenia

^b Anemia (grouped term) includes PTs of anaemia, red blood cell count decreased, and hemoglobin count decreased

Adverse Drug Reactions

Adverse drug reactions (ADRs) for this application in the target dose at the target indication were determined after reviewing all relevant safety data.

The events shown in Table 43 were determined to have reasonable possibility of causal association with fam-trastuzumab deruxtecan and were, therefore, considered to be ADRs. Adverse drug reactions of $\geq$ Grade 3 that were reported in $>5\%$ of subjects were neutrophil count decrease (grouped term; 16.2%), anemia (grouped term; 7.3%), nausea (6.8%), and fatigue (5.1%).

Table 43: All Adverse Drug Reactions in HER2-positive Breast Cancer 5.4 mg/kg Pool (N = 234), by MedDRA System Organ Class and Preferred Term/Grouped Term

MedDRA System Organ Class	Preferred Term/ Grouped Term	Any CTCAE Grade	CTCAE Grade 3-4
Blood and lymphatic system disorders	Anemia ^a	72 (30.8)	17 (7.3)
	Neutrophil count decrease ^b	69 (29.5)	38 (16.2)
	Platelet count decrease ^c	47 (20.1)	8 (3.4)
	White blood cell count decrease ^d	45 (19.2)	10 (4.3)
	Febrile neutropenia	4 (1.7)	4 (1.7)
Eye disorders	Dry eye	26 (11.1)	1 (0.4)
Gastrointestinal disorders	Nausea	185 (79.1)	16 (6.8)
	Vomiting	111 (47.4)	9 (3.8)
	Constipation	81 (34.6)	2 (0.9)
	Diarrhoea	67 (28.6)	4 (1.7)
	Abdominal pain ^e	42 (17.9)	3 (1.3)
	Stomatitis ^f	32 (13.7)	2 (0.9)
	Dyspepsia	29 (12.4)	0
General disorders and administration site conditions	Fatigue	112 (47.9)	12 (5.1)
	Asthenia	29 (12.4)	2 (0.9)
Infections and infestations	Upper respiratory tract infection	25 (10.7)	0
Injury, poisoning and procedural complications	Infusion related reaction ^g	6 (2.6%)	0
Investigations	Aspartate aminotransferase increased	32 (13.7)	2 (0.9)
	Alanine aminotransferase increased	24 (10.3)	2 (0.9)
Metabolism and nutrition disorders	Decreased appetite	76 (32.5)	3 (1.3)
	Hypokalaemia	28 (12.0)	8 (3.4)
Nervous system disorders	Headache	42 (17.9)	0
	Dizziness	24 (10.3)	0
Respiratory, thoracic, and mediastinal disorders	Cough	46 (19.7)	0
	Dyspnoea	31 (13.2)	3 (1.3)
	Epistaxis	30 (12.8)	0

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

MedDRA System Organ Class	Preferred Term/ Grouped Term	Any CTCAE Grade	CTCAE Grade 3-4
	Interstitial lung disease ^h	22 (9.4)	6 (2.6) ⁱ
Skin and subcutaneous tissue disorders	Alopecia	107 (45.7)	1 (0.4) ^j

^a Anemia (grouped term) includes PTs of anaemia, hemoglobin count decreased, and red blood cell count decreased.

^b Neutrophil count decrease (grouped term) includes PTs of neutropenia and neutrophil count decreased.

^c Platelet count decrease (grouped term) includes PTs of thrombocytopenia and platelet count decreased.

^d WBC count decrease (grouped term) includes PTs of leukopenia and white blood cell count decreased.

^e Abdominal pain (grouped term) includes PTs of abdominal discomfort, abdominal pain, abdominal pain lower, and abdominal pain upper.

^f Stomatitis (grouped term) includes PTs of stomatitis, aphthous ulcer, mouth ulceration, oral mucosa erosion, and oral mucosa blistering.

^g Cases of IRR include PTs of infusion related reaction (n=4), hypersensitivity (n=1), and flushing (n=1).

^h Interstitial lung disease includes events that were adjudicated as ILD: pneumonitis, interstitial lung disease, respiratory failure, organising pneumonia, acute respiratory failure, lung infiltration, lymphangitis, alveolitis.

ⁱ Includes 5 subjects with Grade 5 events and 1 subject with a Grade 4 event. After the database lock, the ILD Adjudication Committee updated the severity of this Grade 4 TEAE to Grade 5.

^j This Grade 3 event was reported by the investigator. Per CTCAE version 4.03, the highest CTCAE grade for alopecia is Grade 2.

The Applicant's Position:

The observed AEs were as expected in this population. The list of ADRs after a comprehensive review was as expected for this class of drugs.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment.

Laboratory Findings

Data:

Hematology parameters to be summarized included platelets, hemoglobin, WBC count, absolute neutrophil count, and lymphocytes; blood chemistry parameters included sodium, magnesium, potassium, calcium, AST, ALT, serum creatinine, TBL, and alkaline phosphatase. Shift tables using CTCAE grades to compare baseline to the worst post-baseline CTCAE grade were produced for applicable hematology and biochemistry laboratory parameters. The individual laboratory parameters were recorded on the shift tables using the appropriate CTCAE terms (e.g., "anemia" for hemoglobin decreased). The grades for laboratory parameters are based on the numeric component of the CTCAE grading scale v4.03. Per protocol, abnormal laboratory values considered clinically significant by the investigator were to be reported as AEs.

Changes in hematological parameters were analyzed for subjects with both baseline and post-baseline values. Overall, the observed changes were consistent with the reported hematologic TEAEs. The results in the HER2-positive Breast Cancer 5.4 mg/kg Pool (shown below) were qualitatively similar to those from the All Tumor Types 5.4 mg/kg Pool:

- Hemoglobin: 132/232 (56.9%) subjects had a shift of 1 grade toward worsening CTCAE grade for anemia (hemoglobin decreased), with 27/232 (11.6%) subjects shifting by 2 grades.
- Neutrophil Count: 46/231 (19.9%) subjects had a shift of 1 grade toward worsening CTCAE grade for neutrophil count decreased, with 65/231 (28.1%) subjects shifting by 2 grades, 31/231 (13.4%) shifting by 3 grades, and 2/231 (0.9%) shifting by 4 grades.
- Platelet Count: 71/231 (30.7%) subjects had a shift of 1 grade toward worsening CTCAE grade for platelet count decreased, with 10/231 (4.3%) subjects shifting by 2 grades, and 5/231 (2.2%) subjects shifting by 3 grades.

Overall, shifts in clinical chemistry parameters of magnesium and calcium were observed infrequently among subjects in any pool, and even fewer subjects had shifts to values $\geq$ Grade 3.

- Potassium: In the HER2-positive Breast Cancer 5.4 mg/kg Pool, 2/230 (0.9%) subjects had a shift of 1 grade for hypokalemia, with 54/230 (23.5%) subjects shifting by 2 grades and 5/230 (2.2%) subjects shifting by 3 grades. These results were similar to those from the All Tumor Types 5.4 mg/kg Pool. A total of 59 (25.7%) subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool had Grade 2 hypokalemia and 7 (3.0%) had Grade 3 hypokalemia.
- Liver Function Tests: Mild, transient, and reversible increases in ALT and AST were observed. Elevated liver function tests were less common and less severe in the HER2-positive Breast Cancer 5.4 mg/kg Pool than in the HER2-positive Breast Cancer ≥ 6.4 mg/kg Pool, with 1 (0.4%) subject and 3 (2.2%) subjects, respectively, meeting the biochemical criteria for potential Hy's Law (ALT or AST $\geq 3 \times$ upper limit of normal [ULN] and TBL $> 2 \times$ ULN). All events of potential Hy's Law were determined not to be causally associated with study drug due to alternative etiologies at the time of occurrence of the events. No events were determined to be Hy's law.
- Renal Function Tests: No clinically meaningful shifts to worse creatinine values were observed. No dose response was observed. Two subjects in the All Tumor Types ≥ 6.4 mg/kg Pool had a $\geq$ Grade 3 serum creatinine increase.

Laboratory-based abnormalities that were considered ADRs are summarized in [Table 44](#).

Table 44: New or Worsening Laboratory Abnormalities in the HER2-positive Breast Cancer 5.4 mg/kg Pool

Laboratory Abnormality	Number (%) of Subjects in HER2-positive Breast Cancer 5.4 mg/kg Pool (N = 234)	
	CTCAE Grade	
	Any CTCAE Grade	CTCAE Grade 3 or 4
Chemistry		
Aspartate aminotransferase increased	96 (41.0)	2 (0.9)
Alanine aminotransferase increased	89 (38.0)	1 (0.4)
Hypokalaemia	61 (26.1)	7 (3.0)
Hematology		
White blood cell count decreased	164 (70.1)	17 (7.3)
Hemoglobin decreased	163 (69.7)	17 (7.3)
Neutrophil count decreased	144 (61.5)	38 (16.2)
Platelet count decreased	86 (36.8)	8 (3.4)

Percentages are calculated using the number of subjects with both baseline and post-treatment measurements (n in the Total column) as the denominator.

^a Frequencies are based on CTCAE v4.03 grade-derived laboratory abnormalities.

The Applicant's Position:

The observed laboratory abnormalities were as expected in this population. Laboratory abnormalities were consistent with the observation of TEAEs, but at a higher incidence in general.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment. The FDA noted multiple entries in the ADLB datasets for both studies DS8201-A-U201 and DS8201-A-J101 that were missing grade values for the individual lab results that were gradable. The sponsor corrected the dataset based on an FDA information request. FDA included lab abnormalities in the USPI by including percentages of patients with worsening laboratory values from baseline by at least 1 grade using the updated ADLB datasets. Table 45 displays the laboratory abnormalities identified by the FDA that occurred in >20% of the 234 patients pooled for safety. In general, there was a higher incidence of laboratory abnormalities compared to the similar TEAEs in general.

Table 45: Selected Laboratory Abnormalities in Patients with Unresectable or Metastatic HER2-positive Breast Cancer Treated with ENHERTU

Laboratory Parameter	ENHERTU 5.4 mg/kg N = 234	
	All Grades %	Grades 3 or 4 %
Hematology		
White blood cell count decreased	70	7
Hemoglobin decreased	70	7
Neutrophil count decreased	62	16
Platelet count decreased	37	3.4
Chemistry		
Aspartate aminotransferase increased	41	0.9
Alanine aminotransferase increased	38	0.4
Hypokalemia	26	3.0

Percentages were calculated using patients with worsening laboratory values from baseline and the number of patients with both baseline and post-treatment measurements as the denominator.

Frequencies were based on NCI-CTCAE v.4.03 grade-derived laboratory abnormalities.

Vital Signs

Data:

Vital sign parameters included heart rate, systolic and diastolic blood pressure, respiratory rate, temperature, and peripheral oxygen saturation (SpO2). The data collected in different units were converted into standard international (SI) units for summary. Respiratory rate was collected in only DS8201-A-U201.

The number and percent of subjects with notable abnormalities (ie, individual outlier values based on prespecified criteria) were similar between the HER2-positive Breast Cancer 5.4 mg/kg Pool and the All Tumor Types 5.4 mg/kg Pool among subjects who had a baseline assessment and at least 1 post-baseline assessment .

The Applicant's Position:

No clinically relevant changes in vital signs or trends over time were observed.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment. The following criteria were used to define abnormalities in vital signs which was generally acceptable:

Notable elevated values:

- Systolic BP ≥ 180 mmHg and an increase ≥ 20 mmHg from baseline
- Diastolic BP ≥ 105 mmHg and an increase ≥ 15 mmHg from baseline
- HR ≥ 120 beats per minute (bpm) with increase from baseline of $\geq 15\%$
- Temperature $> 39^{\circ}\text{C}$

Notable below-normal values:

- Systolic BP ≤ 90 mmHg and a decrease ≥ 20 mmHg from baseline
- Diastolic BP ≤ 50 mmHg and a decrease ≥ 15 mmHg from baseline
- HR ≤ 50 bpm with decrease from baseline of $\geq 15\%$
- Temperature $< 36^{\circ}\text{C}$
- SpO₂ $< 90\%$

Electrocardiograms

Data:

Electrocardiograms were recorded in triplicate before dosing on Day 1 of each cycle. The ECG parameters included heart rate, PR Interval, RR Interval, QRS Duration, QT Interval, and QTcF Interval. An evaluation of QTc data is presented below. All ECGs were required to be evaluated by investigator or delegated physician for the presence of abnormalities.

If triplicate ECG readings were collected at any assessment time points, the average of the triplicates at the time point was used in the summary. If multiple ECG measurements were taken at baseline, the average of all baseline values was used as the baseline value. The summaries include all ECG assessments collected within 28 days after the last dose of study drug for DS8201-A-J101, or within 47 days after the last dose of study drug for DS8201-A-U201.

The proportions of subjects with prespecified notable changes from baseline in heart rate, PR interval, and QRS interval were similar across the 4 pools. Notable changes were seen in $\leq 3\%$ of subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool.

The Applicant's Position:

No clinically relevant changes in ECG parameters or trends over time were observed.

The FDA's Assessment:

Waveforms from the ECG warehouse were reviewed. Overall ECG acquisition and interpretation in this study appears acceptable. Refer to QT-IRT consult review for further details.

QT

Data:

See Section 8.2.5.3 for a discussion of results from the QT prolongation Study DS8201-A-J102 and pooled results among subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool.

The Applicant's Position:

Based on the available pooled data and results from the DS8201-A-J102 study that evaluated QT, fam-trastuzumab deruxtecan was shown not to be associated with a risk of QT prolongation.

The FDA's Assessment:

No large mean increases in QTc (i.e., >20 ms) were observed in this QT assessment of fam-trastuzumab deruxtecan. Additionally, none of the subjects had a QTcF >480 ms or an increase from baseline >60 ms. Refer to QT-IRT consult review for further details.

Immunogenicity

Data:

Administration of therapeutic protein has the potential to elicit an immune response. The Applicant has developed and validated sensitive and specific binding assays to characterize the immune responses against fam-trastuzumab deruxtecan in clinical studies.

Subjects were tested for ADAs against fam-trastuzumab deruxtecan in all 5 clinical studies using a validated electrochemiluminescence assay in a multi-tiered approach. Serum samples that screened positive in the screening assay were further analyzed by competitive binding immunodepletion with fam-trastuzumab deruxtecan to confirm and characterize the detected positive response. Study samples that were confirmed as positive were assayed to determine the titer.

The incidence of ADA was monitored during the respective clinical studies due to its potential association with safety and efficacy assessment of fam-trastuzumab deruxtecan. Since low ADA-positive incidence rates were observed across clinical studies, and there was no observed association of ADA positivity to safety or irregular PK, neutralizing activity of anti-ADC antibodies was not assessed.

The proportion of ADA-positive subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool was low (5/207 [2.4%] in DS8201-A-J101, and 10/184 [5.4%] in DS8201-A-U201), and there was no association between positive ADA and allergic-type reactions. Fam-trastuzumab deruxtecan was not associated with severe IRRs. A small percentage of subjects experienced mild infusion reaction, which led to dose interruption in 1 (0.4%) subject in the HER2-positive Breast Cancer 5.4 mg/kg Pool.

In the ADA evaluations, no subject shifted from ADA negative at baseline to ADA positive after baseline. One subject who received fam-trastuzumab deruxtecan and itraconazole shifted from ADA-positive at baseline to ADA-negative after baseline.

The immunogenicity assays developed were sensitive and drug-tolerant. In general, the rate of baseline and postbaseline immunogenicity was low with 3.5% and 1.1% incidence, respectively, of subjects demonstrating ADA-positive data. The rate of treatment-emergent ADA (defined as subjects who were ADA negative at baseline and became ADA-positive post-treatment or subjects who were ADA-positive at baseline and post-treatment, but had an increase in ADA titer from baseline to post-treatment, or subjects who had missing ADA data at baseline and were ADA-positive post treatment) continued to be low, with a total of 4 (0.6%) subjects having treatment-emergent ADA.

The Applicant's Position:

Given the low rate of ADA-positive data, no obvious effect of immunogenicity on PK and efficacy was observed. No relationship was identified between the occurrence of IRR and the incidence of ADA.

The FDA's Assessment:

Refer to Clinical Pharmacology Section (Section 6.2) in this Assessment Aid Review for further details on the immunogenicity review.

8.2.5. Analysis of Submission-Specific Safety Issues

The Applicant has identified the following as AESIs for the fam-trastuzumab deruxtecan program based on nonclinical and clinical experience, epidemiologic information, and literature review of products of similar class: ILD, LVEF decrease, QT prolongation, and IRR.

8.2.5.1. Interstitial Lung Disease

Data:

Data from toxicity studies in monkeys at a dose of ≥ 30 mg/kg (human equivalent dose of 9.7 mg/kg) suggested fam-trastuzumab deruxtecan could lead to noninfectious lung toxicity (see Section 5.5.1).

At the time of the start of the clinical program, ILD was considered to be an important potential risk for fam-trastuzumab deruxtecan. After the first suspected fatal ILD case occurred, the Applicant established an external ILD AC for the program that adjudicated all events of potential ILD reported by investigators on an ongoing basis. Forty-four PTs (42 from the MedDRA ILD Standardised MedDRA Queries (SMQ), plus PTs of respiratory failure and acute respiratory failure) were selected for adjudication. The ILD AC adjudicated each potential ILD event with regard to whether it was ILD and whether it was related to fam-trastuzumab

deruxtecan (regardless of the determination made by the investigator), as well as determining onset dates and adjudicating grades for events that the AC considered to be ILD. All subjects with any of the 44 PTs of potential ILD events were adjudicated similarly. Protocol-defined on-study death among any subjects who experienced a potential ILD event was also adjudicated as to whether death was due to ILD.

All events of potential ILD that were adjudicated by the ILD AC from DS8201-A-J101 and DS8201-A-U201. Pooled data were analyzed and ILD events were summarized by frequency and severity.

Among the 30/234 (12.8%) subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool who had potential ILD, the adjudicated outcome was as follows:

- 24 (10.3%) subjects had events that were adjudicated as ILD:
 - 22 (9.4%) subjects had events that were adjudicated as drug-related ILD (Table 46); the ILD AC adjudicated the events to be Grade 1 in 4 (1.7%) subjects, Grade 2 in 12 (5.1%), Grade 3 in 0, Grade 4 in 1 (0.4%), and Grade 5 in 5 (2.1%). The subject with an adjudicated Grade 4 ILD subsequently died, with the death adjudicated as related to ILD (after the database lock, the severity was updated from Grade 4 to Grade 5, following ILD AC re-adjudication of the event). As a result, a total of 6 (2.6%) fatal cases were adjudicated as drug-related ILD associated with an outcome of death.
 - 2 (0.9%) subjects had events adjudicated as not-drug-related ILD
- 6 (2.6%) subjects had events that were adjudicated as not ILD

Table 46: Events Adjudicated as Drug-related ILD, by Tumor Type/Dose, and Adjudicated Grade

Tumor Type/Dose	Number (%) of Subjects with Each CTCAE Grade Reported by Adjudication Committee ^a					
	1	2	3	4	5	Total
HER2-positive Breast Cancer 5.4 mg/kg (N=234)	4 (1.7)	12 (5.1) ^b	0	1 (0.4) ^c	5 (2.1)	22 (9.4)
HER2-positive Breast Cancer ≥6.4 mg/kg (N=137)	12 (8.8)	13 (9.5)	2 (1.5)	1 (0.7) ^d	2 (1.5)	30 (21.9)
All Tumor Types 5.4 mg/kg (N=275)	4 (1.5)	13 (4.7) ^b	0	1 (0.4) ^c	5 (1.8)	23 (8.4)
All Tumor Types ≥6.4 mg/kg (N=258)	17 (6.6)	19 (7.4) ^b	3 (1.2)	2 (0.8) ^{d,e}	5 (1.9)	46 (17.8)

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

^a The Adjudication Committee assigned grades to those events that were determined to be ILD

^b Includes 3 events in DS8201-A-J101 that occurred >28 days after the last dose, per investigator-reported onset date, including 1 subject from the HER2-positive Breast Cancer 5.4 mg/kg Pool

^c One subject had a TEAE of respiratory failure and subsequently died; the death was adjudicated as being due to ILD (after the database lock, the severity was updated from Grade 4 to Grade 5, following ILD AC re-adjudication of the event)

^d One subject died after DCO, with the death adjudicated as related to ILD.

^e One subject with HER2-low breast cancer subsequently died (39 days after last dose), with the death adjudicated as due to both ILD and disease progression.

The adjudicated ILD events by country (Japan and non-Japan) are presented for the HER2-positive Breast Cancer 5.4 mg/kg Pool in Table 47. A higher proportion of subjects in Japan experienced ILD than subjects from outside Japan with the rates driven primarily by Grade 1 or Grade 2 ILD, and no subjects in Japan experiencing Grade 5 events.

Table 47: ILD Events, by Country and Adjudicated Grade in the HER2-positive Breast Cancer 5.4 mg/kg Pool

Country	Number (%) of Subjects with Each CTCAE Grade Reported by Adjudication Committee ^a					
	1	2	3	4	5	Total
Japan (N=51)	3 (5.9)	7 (13.7)	0	0	0	10 (19.6)
Non-Japan (N=183)	1 (0.5)	5 (2.7)	0	1 (0.5) ^a	5 (2.7)	12 (6.6)

If a subject had multiple ILD events, the CTCAE grade was for the event with the worst grade.

^a One subject had a TEAE of respiratory failure and subsequently died, with the death adjudicated as related to ILD (after the database lock, the ILD Adjudication Committee updated the severity of this Grade 4 TEAE to Grade 5).

Additionally, a comprehensive exploratory analysis to evaluate the following 12 potential risk factors for ILD was performed based on the pooled safety data from the 5 clinical studies (N = 645): age, race, sex, country, region, tumor type, baseline lung cancer or metastasis, prior chest/lung radiotherapy, lung comorbidities, number of prior chemotherapy/targeted therapies, time since disease diagnosis, and dose of study drug. A multivariate stepwise regression model was used, with incidence of adjudicated drug-related ILD as the dependent variable in the 5-study data pool. Estimates of odds ratio and the corresponding 95% CIs for the final identified factors were reported and only the factors of country and number of prior regimens were statistically significant (Table 48):

- Subjects from Japan experienced a higher incidence of ILD compared to subjects from outside Japan (16.1% vs. 6.4%).
- Subjects with ≥10 prior regimens experienced a higher incidence of ILD (18.3% vs. 9.6%).

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

Table 48: Odds Ratio of Risk Factors from the Five Clinical Studies

Risk Factor	Odds Ratio (95% CI)	p-value
Country: Japan vs. Non-Japan	3.1 (1.8, 5.3)	<0.001
Number of Prior Targeted Therapies: ≥10 vs. <10	2.4 (1.4, 4.3)	0.002

Although the stepwise logistic regression model did not identify dose (5.4 mg/kg vs ≥6.4 mg/kg) as a statistically significant factor associated with the risk of ILD, a numerical difference was observed, in which a higher dose of fam-trastuzumab deruxtecan was associated with a higher incidence of ILD. This was supported by results of the ER analysis, in which higher exposure was associated with a higher risk of any-grade ILD.

The clinical studies were conducted using specific recommendations including close monitoring of signs/symptoms of ILD (e.g., cough, fever, and dyspnea) to identify potential ILD, and proactively managing ILD with dose modification and treatment (e.g., steroids) to reduce the occurrence of serious outcomes. The protocols were amended over time to include specific instructions, including dose modification.

The Applicant's Position:

Interstitial lung disease is an important identified risk for fam-trastuzumab deruxtecan and has been closely monitored from the beginning of the clinical study program. A well-established safety evaluation process was followed and robust data collection and adjudication by the independent ILD AC ensured a comprehensive assessment of the ILD events, and an adequate management plan for ongoing studies.

Higher doses of fam-trastuzumab deruxtecan were associated with a higher incidence of ILD regardless of tumor type. Subjects from Japan had a higher risk of ILD than non-Japan subjects. This difference may be related to differences in clinical practice. Starting in the early 2000s with the ILD cases related to gefitinib, oncology health-care providers in Japan have been more proactive in detecting, even at Grade 1 stage (ie, ILD manifesting only as abnormal high-resolution CT scan), and intensively managing ILD. In Western countries, these professionals (especially those treating patients with HER2-positive breast cancer) have focused primarily on proactively detecting and treating HER2-treatment-induced cardiac toxicity.

After ILD was recognized as a risk for the fam-trastuzumab deruxtecan program in early 2018, at the recommended dose of 5.4 mg/kg, there were no fatal cases of ILD reported in Japan while fatal cases were reported in Western countries, and there was a greater rate of Grade 1 and Grade 2 ILD in Japan than in the rest of the world, which is consistent with the idea that there is better early identification and management of ILD in Japan.

Interstitial lung disease requires proper monitoring, dose modification, and supportive care instituted in a timely fashion. A global enhanced patient safety program to educate both investigators and patients about ILD was initiated in early 2019, which may reduce the risk of severe ILD, including fatal cases.

Interstitial lung disease is a known serious risk of fam-trastuzumab deruxtecan and cases with fatal outcomes (2.6% in HER2-positive breast cancer subjects treated with 5.4 mg/kg) have been reported. These data warrant a boxed warning for the label and instructions to patients through a medication guide for early intervention.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment of ILD. The FDA has been actively engaged with the applicant regarding the assessment of ILD cases related to fam-trastuzumab deruxtecan.

Severe, life-threatening, or fatal interstitial lung disease (ILD), including pneumonitis, can occur in patients treated with fam-trastuzumab deruxtecan. Of the 234 patients HER2-positive breast cancer who received 5.4mg/kg, ILD occurred in 9.4% of patients that was adjudicated by the independent ILD committee as related to the drug. Fatal outcomes due to ILD and/or pneumonitis occurred in 6 (2.6%) patients treated with fam-trastuzumab deruxtecan. One of these deaths was initially listed as a grade 4 ILD however this patient died after the first data cut-off date from respiratory failure which was adjudicated as drug related ILD. Median time to first onset was 4.1 months (range: 1.2 to 8.3). ILD/pneumonitis is a black box warning in the USPI.

There were 6 cases of ILD related to fam-trastuzumab deruxtecan in the US versus 10 cases in Japan. All the Japanese cases were either G1 (3 cases) or Grade 2 (7 cases). The US cases included G1 (1 case), G2 (3 cases), G3 (0 cases), G4 (0 cases), and G5 (2 cases). In general, ILD is more closely monitored in Japan and this may account for why there were less severe cases of ILD reported in Japanese patients compared to US patients.

FDA also analyzed the number of responders (CR or PR) reported for each grade of ILD adjudicated as related to fam-trastuzumab deruxtecan. For grade 1 ILD, there was 3 responders out of 4 patients (75%). For grade 2, there was 7 out of 12 (58%), for grade 4, there was 1 out of 1 (100%). This patient that experienced grade 4 ILD died after the first data cutoff date and became a grade 5 ILD as noted above. Of the 5 remaining grade 5 cases, none were responders. In general, there were not enough cases to make any significant conclusions regarding incidence of ILD and tumor response, and there was no specific pattern to timing and ILD and response, therefore FDA considered this analysis exploratory.

8.2.5.2. Left Ventricular Ejection Fraction Decrease

Data:

Although LVEF decrease has been observed with anti-HER2 therapies, significant LVEF decrease was not observed with fam-trastuzumab deruxtecan. Events of CHF have been reported in clinical trials of T-DM1, with asymptomatic LVEF decline reported in 1.7% to 6.5% patients (Verma et al 2012, Chen et al 2011). Because of the differences in study population and duration of exposure, a direct comparison of these data to those for fam-trastuzumab deruxtecan is not feasible. Anthracycline therapies are known to cause cardiotoxicity and have been shown to have a synergistic effect of cardiotoxicity with trastuzumab. In the HER2-positive Breast Cancer 5.4 mg/kg Pool, 28.6% and 20.1% of subjects received prior therapy of doxorubicin and epirubicin, respectively.

Upon Applicant review of the 12 selected PTs in the AESI of LVEF decrease (acute left ventricular failure, acute right ventricular failure, cardiac failure, cardiac failure acute, cardiac failure chronic, cardiac failure congestive, chronic left ventricular failure, chronic right ventricular failure, ejection fraction decreased, left ventricular failure, right ventricular failure, and ventricular failure), only the PT of ejection fraction decreased (reported in 2 [0.9%] subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool) was determined to be actual LVEF decrease. Two additional subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool had events of PTs other than ejection fraction decreased: 1 subject with an SAE of Grade 2 cardiac failure congestive that led to the withdrawal of the study drug and resolved (LVEF value of 63% at baseline and 66% to 70% during the study); and 1 subject with Grade 1 cardiac failure on Day 1 that was not resolved at the DCO (LVEF value of 52% at baseline and 57% to 60% during the study).

In addition, measurements of LVEF from laboratory data (every 2 cycles in DS8201-A-J101 and every 4 cycles in DS8201-A-U201) were graded per CTCAE v4.03, with each LVEF measurement assigned a CTCAE grade.

Using these criteria, 32 (14.6%) subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool met the laboratory criteria for a Grade 2 value, with no subjects meeting the criteria for Grade 3 or Grade 4 LVEF decrease. No subject had a post-baseline resting LVEF of <40%. In an ER analysis, higher exposure of intact fam-trastuzumab deruxtecan was associated with higher incidence of LVEF decreased.

The Applicant's Position:

Fam-trastuzumab deruxtecan is not associated with a risk of LVEF decrease. Since LVEF decrease and CHF have been reported for drugs in a similar class, and considering the limited long-term exposure data of fam-trastuzumab deruxtecan, LVEF decrease is an important potential risk, and should be monitored regularly during treatment and managed as clinically indicated.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment.

In the 234 patients with unresectable or metastatic HER2-positive breast cancer who received 5.4 mg/kg fam-trastuzumab deruxtecan, two cases (0.9%) of asymptomatic LVEF decrease were reported as of the first clinical cut-off date. Treatment with fam-trastuzumab deruxtecan has not been studied in patients with a history of clinically significant cardiac disease or LVEF less than 50% prior to initiation of treatment. Left ventricular dysfunction is listed as a Warning and Precaution in the USPI as LVEF is a known class effect of HER2-directed therapies. Dose management guidelines for LVEF and symptomatic congestive heart failure are also in the USPI.

8.2.5.3. QT Prolongation

Data:

In a 6-week toxicity study in cynomolgus monkeys, reversible QTc prolongation was seen at 78.8 mg/kg (see Section 5.5.1).

Study DS8201-A-J102 evaluated the effect of fam-trastuzumab deruxtecan on QTcF (see Table 19). Based on the primary analysis of QTcF, fam-trastuzumab deruxtecan 6.4 mg/kg administration was not associated with a clinically meaningful QTcF prolongation (defined as change from baseline >10 milliseconds). A slight increase in the mean QTcF interval was observed for up to 7 hours post-dose at both Cycles 1 and 3. However, no prolongation of the mean Δ QTcF was observed on Cycle 1 and Cycle 3 Day 8 and Day 15 (Δ QTcF over time <0 millisecond). The upper bound of the 90% CI was under 10 milliseconds at all time points.

There was a slightly positive trend toward an increase in Δ QTcF with an increase in concentrations of fam-trastuzumab deruxtecan and MAAA-1181a. However, the upper bound of the 90% CI for Δ QTcF at the observed mean C_{max} for each analyte in the linear model of concentration vs change from baseline in mean QTcF was under 10 milliseconds for Cycle 1 and Cycle 3. Additionally, almost all observations of Δ QTcF around the median T_{max} for fam-trastuzumab deruxtecan were below 10 milliseconds.

Of the 12 selected PTs in the AESI of QT prolongation (ECG QT prolonged, ECG QT interval abnormal, torsade de pointes, sudden cardiac death, sudden death, syncope, ventricular arrhythmia, ventricular fibrillation, ventricular flutter, ventricular tachycardia, ventricular tachyarrhythmia, and seizure), events of ECG QT prolonged (13 [5.6%] subjects) and seizure (1 [0.4%] subject), were reported in the HER2-positive Breast Cancer 5.4 mg/kg Pool. The seizure was in a subject who had a history of brain metastases and brain irradiation; the event was not associated with QT prolongation. The proportion of subjects with events of ECG QT prolonged in the HER2-positive Breast Cancer 5.4 mg/kg Pool was similar to that of the other pools: 7 (5.1%) in HER2-positive Breast Cancer $\geq$ 6.4 mg/kg Pool, 18 (6.5%) in All Tumor Types 5.4 mg/kg Pool, and 9 (3.5%) in the All Tumor Types $\geq$ 6.4 mg/kg Pool. No symptomatic QT prolongation, torsade de pointes, or severe arrhythmic events were reported.

Among the 13 subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool who had a PT of ECG QT prolonged, 12 subjects had lower QTc value than baseline or <470 milliseconds. The other subject, who had a prior medical history of QT prolongation, had a QTc elevation of 493 milliseconds at Cycle 17 and end-of-treatment (EOT) value of 490 milliseconds.

In the HER2-positive Breast Cancer 5.4 mg/kg Pool, 2 (0.9%) subjects had an absolute maximum QTcF value of >500 milliseconds with concurrent >60 milliseconds change from baseline in QTcF, with values normalizing by the time of the DCO.

The Applicant's Position:

No clinically meaningful effect of study drug on the QTc interval was observed in the available pooled data or in the open-label, single-arm study in 51 patients with HER2-expressing metastatic breast cancer (where the upper bound of the 90% CI for change in QTcF was <10 milliseconds).

The FDA's Assessment:

FDA generally agrees with the applicant's assessment. Refer to sections 8.2.4 under QT subsection. Refer also to FDA QT-IRT consult review for further details.

8.2.5.4. Infusion-Related Reactions

Data:

All relevant PTs (infusion-related reaction, flushing, anaphylactic reaction, dyspnea, hypotension, wheezing, hypersensitivity, bronchospasm, pruritus, angioedema, urticaria, skin exfoliation, oedema, and rash) were reviewed for potential IRR, defined as any of these preselected PTs occurring within the same day of an infusion at any cycle. Among the 6 subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool who had events considered to be IRR (including PTs of IRR in 4 subjects, hypersensitivity in 1 subject, and flushing in 1 subject), all were CTCAE Grade 1 or 2, with 1 report of an SAE of Grade 1 hypersensitivity. Dosing was interrupted in 1 subject who had a PT of ORR. No event was associated with treatment discontinuation or dose reduction.

Premedication was allowed at investigator's discretion but was not mandatory per protocol.

The Applicant's Position:

Mild or moderate events of IRR were observed, which resolved with no study drug discontinuation or dose reduction required. No severe allergic reactions were observed.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment.

8.2.6. Clinical Outcome Assessment (COA) Analyses Informing Safety/Tolerability

Data:

Patient-reported outcomes were not included in the fam-trastuzumab deruxtecan studies included in the application.

The Applicant's Position:

Not applicable

The FDA's Assessment:

No patient reported outcome data was submitted by the applicant for this BLA application.

8.2.7. Safety Analyses by Demographic Subgroups

Subgroup evaluations of TEAEs were conducted to identify potential safety issues in particular subpopulations defined by intrinsic/extrinsic factors: age, sex, race, ethnicity, baseline renal function, baseline hepatic function, country, and geographic region. Findings in the HER2-positive Breast Cancer 5.4 mg/kg Pool are as follows:

Age

A higher proportion of subjects ≥65 had ≥Grade 3 TEAEs, TEAEs associated with dose interruption, and MedDRA PTs of anaemia, neutrophil count decreased, febrile neutropenia, and WBC count decreased.

Sex

Since there was only 1 male subject in the HER2-positive Breast Cancer 5.4 mg/kg Pool, male and female subjects were not compared.

Race

A higher proportion of White subjects than Asian subjects had SAEs, TEAEs associated with an outcome of death (driven primarily by deaths due to ILD), MedDRA PTs of nausea, vomiting, diarrhoea, dyspnoea, and fatigue, and the grouped term of abdominal pain.

A higher proportion of Asian subjects had TEAEs associated with dose interruption, MedDRA PTs of AST increased and ALT increased, and grouped terms of anemia, neutrophil count decrease, and platelet count decrease.

Ethnicity

Given the small number of Hispanic-Latino subjects (13 in the HER2-positive Breast Cancer 5.4 mg/kg Pool), TEAEs were not compared between Hispanic/Latino and non-Hispanic/non-Latino subjects.

Country

A higher proportion of subjects in Japan were ≥ 65 years of age, experienced TEAEs associated with dose interruption, treatment discontinuation, adjudicated drug-related ILD, MedDRA PTs of AST increased and ALT increased, and grouped terms of anemia, neutrophil count decrease, platelet count decrease, and lymphocyte count decrease.

Geographic Region

Differences among subjects from Asia (n = 84) compared to subjects from non-Asia regions (n = 150) in the HER2-positive Breast Cancer 5.4 mg/kg Pool were similar (although of a smaller magnitude) to the differences seen by country (Japan vs non-Japan).

Renal Function at Baseline

Subjects were categorized into 5 groups of baseline renal function, calculated using the Cockcroft-Gault equation of creatinine clearance.

In the HER2-positive Breast Cancer 5.4 mg/kg Pool, 113 subjects entered the study with normal renal function, 91 with mild impairment, 29 with moderate impairment, and 1 with severe impairment. No subject entered the study with end-stage renal disease.

In the HER2-positive Breast Cancer 5.4 mg/kg Pool, a higher proportion of subjects with moderate renal impairment at baseline (6 [20.7%]) had an action taken of study drug discontinuation for TEAEs than subjects with normal renal function at baseline (8 [7.1%]) or mild renal impairment (7 [7.7%]). This difference was driven primarily by the frequency of ILD among the groups. In the HER2-positive Breast Cancer 5.4 mg/kg Pool, a higher proportion of subjects with moderate renal impairment at baseline (7 [24.1%] subjects) had ILD events compared to subjects with normal renal function (6 [5.3%]) or mild renal impairment (7 [7.7%]) at baseline.

Hepatic Function at Baseline

Subjects were categorized into 4 groups based on baseline hepatic function per the NCI criteria. In the HER2-positive Breast Cancer 5.4 mg/kg Pool, 132 subjects entered the study with normal hepatic function, 99 with mild impairment, and 1 with moderate impairment. No subject entered the study with severe hepatic impairment.

The safety profile was similar between subjects with normal hepatic function and those with mild hepatic impairment, except for $\geq$ Grade 3 TEAEs, which were reported in 72/132 (54.5%)

subjects with normal hepatic function and 42/99 (42.4%) subjects with mild hepatic impairment.

The Applicant's Position:

The subgroup results were similar to those from the ER analysis for safety, which evaluated factors including age, sex, race (Asian, White, African American, other), country (Japan, non-Japan Asia, non-Japan Other), HER2 status, ECOG PS, lung metastases (present vs absent) at baseline, number of prior non-hormonal cancer therapies ≥ 6 , and baseline levels of platelets, hemoglobin, and neutrophils.

A higher proportion of subjects ≥ 65 years of age had $\geq$ Grade 3 TEAEs than younger subjects. Given that elderly patients generally have more comorbidities, a higher background incidence of AEs in this group is to be expected.

Asian subjects and subjects from Japan experienced a higher frequency and severity of hematologic TEAEs than White subjects or subjects from non-Japan countries, respectively. Since most of the Asian subjects in these 2 studies were from Japan, the observed difference in Asians was likely driven by country. Regardless of race or country, hematologic events were manageable and rarely led to dose discontinuation. It has been reported that T-DM1 has been associated with a higher incidence and severity of thrombocytopenia in Asians than non-Asians due to different genetic susceptibility ([Dieras et al 2014](#)). The observed differences in TEAEs were not observed (or were of a smaller magnitude) when comparing laboratory-derived data, with notable differences in only $\geq$ Grade 3 neutrophil count decreased and platelet count decreased. Therefore, the observed difference in hematologic TEAEs could be mainly due to different reporting practices by investigators in Japan.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment.

8.2.8. Specific Safety Studies/Clinical Trials

Data:

DS8201-A-A104 was a Phase 1 study of the potential DDI of fam-trastuzumab deruxtecan 5.4 mg/kg with CYP3A and OATP1B inhibitors in subjects with HER2-expressing tumors (40 treated subjects). The results of this study were as follows:

- Concomitant use of fam-trastuzumab deruxtecan with ritonavir (dual inhibitor of OATP1B/CYP3A) or itraconazole (strong inhibitor of CYP3A) resulted in increases of 22% and 18% in AUC of MAAA-1181a, respectively.
- Concomitant use of ritonavir and itraconazole did not affect the exposure of fam-trastuzumab deruxtecan.

- There was no clinically meaningful DDI with ritonavir or itraconazole.
- Fam-trastuzumab deruxtecan in combination with ritonavir or itraconazole showed a similar safety profile as observed in other fam-trastuzumab deruxtecan studies.

The Applicant's Position:

Pharmacokinetic data from this study did not indicate a need to limit concomitant administration of OATP1B or CYP3A inhibitors.

The FDA's Assessment:

Refer to clinical-pharmacology Section 6.3.2.4 of this assessment aid for further details on study DS8201-A-A104.

8.2.9. Additional Safety Explorations

Human Carcinogenicity or Tumor Development

Data:

Thirty subjects received treatment with fam-trastuzumab deruxtecan for more than 1 year.

Within the MedDRA system organ class (SOC) of Neoplasms Benign, Malignant and Unspecified, the following 6 PTs were reported among 5 subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool, with no PT reported in more than 1 subject: basal cell carcinoma, cancer pain, malignant pleural effusion, metastases to lymph nodes, neoplasm skin, and squamous cell carcinoma of skin.

The Applicant's Position:

Treatment-emergent AEs (primarily of skin cancer) were observed with fam-trastuzumab deruxtecan; however, carcinogenicity studies have not been conducted and limited long-term data preclude definitive evaluation of carcinogenicity.

The FDA's Assessment:

FDA agrees with the applicant's assessment.

Human Reproduction and Pregnancy

Data:

No clinical reproductive and developmental toxicity studies have been conducted to date and the teratogenic potential of fam-trastuzumab deruxtecan has not been established.

The Applicant's Position:

Based on the composition and mechanism of action of the product, as well as data from similar products of the same class, teratogenic potential cannot be excluded. In postmarketing reports, use of trastuzumab, a HER2-receptor antagonist, during pregnancy resulted in cases of oligohydramnios manifesting as fatal pulmonary hypoplasia, skeletal abnormalities, and neonatal death. Based on findings in animals and the fam-trastuzumab deruxtecan mechanism of action, the topoisomerase I inhibitor component of fam-trastuzumab deruxtecan can also cause embryo-fetal harm when administered to a pregnant woman. Fam-trastuzumab deruxtecan should not be administered to pregnant women, women attempting to conceive, or women who are breastfeeding.

Based on results from animal toxicity studies, fam-trastuzumab deruxtecan may impair male reproductive function and fertility.

The FDA's Assessment:

FDA generally agrees with the applicant's assessment. Please see section 5.5.4 of the nonclinical review for more information. The USPI includes the following information that females of reproductive potential should use effective contraception during treatment and for at least 7 months following the last dose of fam-trastuzumab deruxtecan. Male patients with female partners of reproductive potential should use effective contraception during treatment with fam-trastuzumab deruxtecan and for at least 4 months after the last dose of fam-trastuzumab deruxtecan. Embryo-fetal toxicity is included as a black-box warning in the USPI.

Pediatrics and Assessment of Effects on Growth

Data:

Breast cancer is not reported to occur in the pediatric population.

The Applicant's Position:

A full waiver from pediatric study requirements for fam-trastuzumab deruxtecan in the treatment of HER2-positive breast cancer is applicable because studies are impossible or highly impracticable to conduct.

The FDA's Assessment:

FDA agrees with the applicant's assessment.

Overdose, Drug Abuse Potential, Withdrawal, and Rebound

Data:

There is currently no information available regarding potential overdose with fam-trastuzumab deruxtecan. The highest dose that has been administered was 8.0 mg/kg and no dose-limiting toxicity was observed.

Drug abuse and dependence with fam-trastuzumab deruxtecan have not been studied.

No studies have been conducted that assess withdrawal or rebound effects of fam-trastuzumab deruxtecan.

The Applicant's Position:

In the event of overdose, subjects should be observed and appropriate supportive care should be given, if required. Given the nature of fam-trastuzumab deruxtecan and its IV administration by a licensed health care professional, no abuse or dependence is anticipated. Given the disease being treated and the nature of fam-trastuzumab deruxtecan, no withdrawal or rebound effects are anticipated.

The FDA's Assessment:

FDA agrees with the applicant's assessment.

Safety in the Postmarket Setting

Not applicable (fam-trastuzumab deruxtecan is not currently registered or approved in the US or any other part of the world).

Safety Concerns Identified Through Postmarket Experience

Data:

Not applicable

The Applicant's Position:

Not applicable

The FDA's Assessment:

Not applicable as fam-trastuzumab is not approved in any country.

Expectations on Safety in the Postmarket Setting

Data:

Toxicities have been adequately represented in the DS8201-A-J101 and DS8201-A-U201 studies.

The Applicant's Position:

Potential safety concerns beyond the risks conveyed in the proposed labeling are not expected. Routine pharmacovigilance (including using targeted questionnaires) will be conducted in the post-marketing setting to monitor for unexpected adverse events.

The FDA's Assessment:

The FDA will continue to monitor any post-marketing reports including safety reports that are submitted after accelerated approval. However, the FDA generally agrees that the safety of fam-trastuzumab deruxtecan has been adequately characterized in the DS8201-A-J101 and DS8201-A-U201 clinical studies. In general FDA expects that the safety of fam-trastuzumab deruxtecan will be similar to that observed on the clinical trials reviewed in this application.

8.2.10. Integrated Assessment of Safety

Data:

The 234 subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool received a median of 6.96 months of treatment (range: 0.7-31.0); 30 (12.9%) subjects had over 1 year of study treatment.

Most of the observed TEAEs were GI or hematologic in nature, were Grade 1-2, and could be managed via dose modification (including discontinuation of study drug, dose reduction, and dose interruption), as well as use of standard supportive therapies.

In the HER2-positive Breast Cancer 5.4 mg/kg Pool, 22/234 (9.4%) subjects discontinued study drug due to a TEAE, 13 (5.6%) due to AESI-defined ILD. A total of 42 (17.9%) subjects in this pool experienced dose reduction due to TEAE, mainly GI (MedDRA PTs of nausea and vomiting) or hematologic (grouped terms of neutrophil count decrease, platelet count decrease, and anemia) events. Among all of the GI or hematologic events, 1 event of thrombocytopenia led to study drug discontinuation.

Fam-trastuzumab deruxtecan was associated with a risk of ILD. An independent ILD AC adjudicated all potential ILD events, which included an extensive list of 42 PTs from the MedDRA ILD SMQ plus the PTs of respiratory failure and acute respiratory failure. Among the

234 subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool, 22 (9.4%) had events adjudicated as drug-related ILD; 16/22 were Grade 1 or Grade 2, and 6 (2.6%) subjects had events associated with a fatal outcome.

Higher doses of fam-trastuzumab deruxtecan were associated with a higher incidence of ILD regardless tumor type. Subjects from Japan had a higher risk of ILD than non-Japan subjects, a difference that has also been reported with other oncology drugs. However, no $\geq$ Grade 3 ILD cases were observed among subjects in Japan.

A multivariate logistic regression model identified only 2 statistically significant risk factors for ILD: country (Japan vs non-Japan) and number of prior chemotherapy/targeted regimens (≥ 10 vs < 10). Subjects from Japan and those with ≥ 10 prior regimens of systemic cancer therapies had a higher risk of ILD. Although the stepwise logistic regression model did not identify dose (5.4 mg/kg vs ≥ 6.4 mg/kg) as a statistically significant factor associated with the risk of ILD, a numerical difference was observed where a higher dose of fam-trastuzumab deruxtecan was associated with a higher incidence of ILD. In the ER analysis, higher exposure of fam-trastuzumab deruxtecan was associated with a higher risk of any grade ILD.

Events of CHF have been reported in clinical trials of T-DM1, with asymptomatic LVEF decline reported in 1.7% to 6.5% patients. Because of the differences in study population and duration of exposure, a direct comparison of these data to those for fam-trastuzumab deruxtecan is not feasible. Anthracycline therapies are known to cause cardiotoxicity and have been shown to have a synergistic effect of cardiotoxicity with trastuzumab. In the HER2-positive Breast Cancer 5.4 mg/kg Pool, 28.6% and 20.1% of subjects received prior therapy of doxorubicin and epirubicin, respectively.

Asymptomatic LVEF decrease (mostly Grade 2 events) has been reported with fam-trastuzumab deruxtecan, including in the HER2-positive Breast Cancer 5.4 mg/kg Pool, with most subjects recovering while continuing on study drug. In this pool, only the PT of ejection fraction decreased (reported in 2 subjects) was determined to be actual LVEF. Two subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool had events of PTs other than ejection fraction decreased: 1 subject with an SAE of Grade 2 cardiac failure congestive that led to the withdrawal of the study drug and resolved (LVEF value of 63% at baseline and 66% to 70% during the study); and 1 subject with Grade 1 cardiac failure on Day 1 that was not resolved at the DCO (LVEF value of 52% at baseline and 57% to 60% during the study).

Fam-trastuzumab deruxtecan was not associated with a risk of severe liver injury. Mild, transient and reversible ALT/AST increase was observed.

Based on the available data in the HER2-positive Breast Cancer 5.4 mg/kg Pool and results from the DS8201-A-J102 study that evaluated QT, fam-trastuzumab deruxtecan was shown not to be associated with a risk of QT prolongation.

The proportion of ADA-positive subjects in the HER2-positive Breast Cancer 5.4 mg/kg Pool was low, and there was no association between positive ADA and allergic type reactions. Fam-trastuzumab deruxtecan was not associated with severe IRRs. A small percentage of subjects

experienced mild infusion reaction, which led to dose interruption in 1 (0.4%) subject in the HER2-positive Breast Cancer 5.4 mg/kg Pool. Premedication was allowed at investigator's discretion but was not mandatory per protocol.

Asian subjects and subjects from Japan experienced a higher frequency and severity of hematologic TEAEs than White subjects or subjects from non-Japan countries, respectively. Regardless of race or country, hematologic events were manageable and rarely led to dose discontinuation.

A higher proportion of subjects ≥65 years of age had ≥Grade 3 TEAEs than younger subjects. Given that elderly patients generally have more comorbidities, a higher background incidence of AEs in this group is to be expected.

The fam-trastuzumab deruxtecan dose of 5.4 mg/kg was associated with a better safety profile than 6.4 mg/kg in HER2-positive breast cancer and the subgroups examined. This observation was further supported by the ER analysis for safety.

The Applicant's Position:

The safety and tolerability of fam-trastuzumab deruxtecan have been evaluated, including at the target dose of 5.4 mg/kg Q3W in the target population of patients with unresectable or metastatic HER2-positive breast cancer who have received 2 or more prior anti-HER2 therapies. A total of 645 subjects received at least 1 dose of fam-trastuzumab deruxtecan among the 5 studies. The relatively large safety database allowed detection and characterization of common risks. However, the relatively small number of subjects with long-term treatment duration prevented the evaluation of AEs with delayed onset or cumulative toxicity.

Fam-trastuzumab deruxtecan was generally tolerated, with a median relative dose intensity of 97.8% in the HER2-positive 5.4 mg/kg breast cancer Pool, indicating a dosing schedule and regimen manageable as planned for the majority of patients. Most of the observed TEAEs were gastrointestinal or hematologic in nature, were Grade 1 or Grade 2, and could be managed via dose modification (including discontinuation of study drug, dose reduction, and dose interruption) and standard clinical practice.

Based on clinical experience with ILD cases related to gefitinib, health care providers nationwide in Japan have had a high awareness of ILD and more experience in proactively detecting, even at Grade 1 stage (ie, ILD manifesting only as abnormal high-resolution CT scan), and intensively managing ILD. In contrast, breast cancer health care providers in Western countries are primarily focused in proactively detecting and treating anti-HER2 therapy-induced cardiac toxicity and were not equally proactive in monitoring for ILD cases. In the HER2-positive Breast Cancer 5.4 mg/kg Pool, a higher proportion of subjects from Japan were reported to have adjudicated drug-related ILD compared to subjects from non-Japan countries (19.6% vs. 6.6%, respectively). This difference was driven primarily by events of Grade 1 (5.9% vs. 0.5%) or Grade 2 (13.7% vs. 2.7%) in severity. After ILD was recognized as a risk for the fam-trastuzumab deruxtecan program in early 2018, at the recommended dose of 5.4 mg/kg, there were no fatal cases of ILD reported in Japan while fatal cases were reported in Western countries, and there

was a greater rate of Grade 1 and 2 ILD in Japan than in the rest of the world, which is consistent with the idea that there is better early identification and management of ILD in Japan. A global ILD awareness campaign to educate both investigators and patients was initiated in Spring 2019, and is currently well underway in clinical trials, with the goal of reducing the risk of severe ILD, including fatal cases, in ongoing and future clinical studies. The risk of ILD, as well as proper management guidelines, will be described in labeling and medication guide, which will be seminal in supporting the appropriate management of this risk.

Fam-trastuzumab deruxtecan is not associated with a risk of LVEF decrease. Since LVEF decrease and CHF have been reported for drugs in a similar class and considering the limited long-term exposure data of fam-trastuzumab deruxtecan, LVEF decrease is an important potential risk, and should be monitored regularly during treatment and managed as clinically indicated.

No relationship was identified between the occurrence of IRR and the incidence of ADA. Fam-trastuzumab deruxtecan did not lead to any de novo ADA in any subject in the HER2-positive Breast Cancer 5.4 mg/kg Pool.

Fam-trastuzumab deruxtecan can cause neutropenia, including febrile neutropenia. Complete blood counts should be monitored prior to initiation of fam-trastuzumab deruxtecan and prior to each dose, and as clinically indicated. Based on the severity of neutropenia, fam-trastuzumab deruxtecan may require dose interruption, dose reduction, or treatment discontinuation.

Fam-trastuzumab deruxtecan can cause fetal harm when administered to a pregnant woman. No data of fam-trastuzumab deruxtecan in pregnant women are available. However, trastuzumab is known to cause fetal toxicity. In addition, based on findings in animals and its mechanism of action, the topoisomerase I inhibitor component of fam-trastuzumab deruxtecan can cause embryo-fetal harm when administered to a pregnant woman. Pregnancy status of females of reproductive potential should be verified prior to the initiation of fam-trastuzumab deruxtecan. Patients should be informed of the potential risks to the fetus and effective contraception should be used.

Overall, fam-trastuzumab deruxtecan was generally tolerable at the proposed dose of 5.4 mg/kg Q3W in the target population of patients with unresectable or metastatic HER2-positive breast cancer who have received 2 or more prior anti-HER2 therapies. Adverse events were generally manageable by dose modification and standard clinical practice.

The FDA's Assessment:

Overall, the FDA agrees that studies DS8201-A-U201 and DS8201-A-J101 adequately characterized the safety profile of fam-trastuzumab deruxtecan.

The most common adverse reactions occurring in $\geq 20\%$ of patients were nausea, fatigue, vomiting, alopecia, constipation, decreased appetite, anemia, neutropenia, diarrhea, leukopenia, cough, and thrombocytopenia. Serious adverse reactions occurred in 20% of

patients receiving fam-trastuzumab deruxtecan, including interstitial lung disease, pneumonia, vomiting, nausea, cellulitis, hypokalemia, and intestinal obstruction. Fatalities due to adverse reactions occurred in 4.3% of patients including interstitial lung disease (2.6%), and the following events occurred in one patient each (0.4%): acute hepatic failure/acute kidney injury, general physical health deterioration, pneumonia, and hemorrhagic shock.

Fam-trastuzumab was permanently discontinued in 9% of patients, of which ILD accounted for 6%. Dose interruptions due to adverse reactions occurred in 33% of patients, the most frequent (>2%) were neutropenia, anemia, thrombocytopenia, leukopenia, upper respiratory tract infection, fatigue, nausea, and ILD. Dose reductions occurred in 18% of patients treated with fam-trastuzumab deruxtecan. The most frequent adverse reactions (>2%) associated with dose reduction were fatigue, nausea, and neutropenia.

ILD/pneumonitis is an important safety signal related to fam-trastuzumab deruxtecan. Of the 234 patients with HER2-positive breast cancer who received 5.4mg/kg of fam-trastuzumab deruxtecan, ILD (adjudicated by the independent ILD committee) occurred in 9.4% of patients. Fatal outcomes due to ILD and/or pneumonitis occurred in 6 (2.6%) patients treated with fam-trastuzumab deruxtecan. The prescribing information includes a Boxed Warning to advise health professionals of the risk of interstitial lung disease (ILD). Median time to first onset was 4.1 months (range: 1.2 to 8.3).

Therefore, fam-trastuzumab deruxtecan demonstrated acceptable tolerability for the indicated population with a serious and life-threatening disease.

SUMMARY AND CONCLUSIONS

8.3. Statistical Issues

The FDA's Assessment:

FDA efficacy evaluation was based on Study DS8201-A-U201. The estimate of confirmed ORR per blinded independent central review based on RECIST v1.1 was 60.3% (95% CI: 52.9%, 67.4%) with 6.0% of patients achieving a CR and 55.0% of patients achieving a PR. The median estimate of response duration was 14.8 months (95% CI: 13.8, 16.9). No inferential procedures can be used to evaluate results from this single arm study. Instead, the efficacy evaluation was based on the magnitude of response rate and adequate duration of response. Additionally, although progression-free survival and time to response results were summarized in the clinical study report, we noted that time-to-event endpoints are uninterpretable in a single arm study without a comparator arm. There are no major statistical issues identified for study DS8201-A-U201.

There were differences in HER2 testing between Study DS8201-A-U201 and Study DS8201-A-

J101. Given these differences, FDA considered the pooled analysis as exploratory.

8.4. Conclusions and Recommendations

The FDA's Assessment:

The clinical and statistical reviewers agree with accelerated approval for this BLA for the reasons stated in the FDA assessments above.

X

X

Wei Zhang, PhD
Primary Statistical Reviewer

Shenghui Tang, PhD
Statistical Team Leader

X

X

Preeti Narayan, MD
Primary Clinical Reviewer

Christy Osgood, MD
Clinical Team Leader

9 Advisory Committee Meeting and Other External Consultations

The FDA's Assessment:

No advisory committee discussion or consultations external to the FDA were deemed necessary for this BLA application.

10 Pediatrics

The Applicant's Position:

Pediatric studies of fam-trastuzumab deruxtecan for the treatment of HER2-positive breast cancer are impossible or highly impracticable to conduct as the condition is not reported to occur in the pediatric population.

The FDA's Assessment:

The FDA agrees with the applicant's assessment given the extremely low incidence of breast cancer in the pediatric population.

11 Labeling Recommendations

Data:

Summary of Significant Labeling Changes (High level changes and not direct quotations)		
Section	Applicant's Proposed Labeling	FDA's proposed Labeling
Highlights	Boxed warning for ILD and embryo-fetal toxicity.	Revised Highlights to reflect revisions to the Boxed Warning, Indications and Usage, Warnings and Precautions (W&Ps), and Adverse Reactions (ARs) Sections in the Full Prescribing Information (FPI). The established pharmacologic class (b) (4) was revised to "HER2-directed antibody and topoisomerase inhibitor conjugate" to be consistent with pharmacology data and related products.
Boxed Warning	Boxed warning for ILD and embryo-fetal toxicity.	Revised (b) (4) to "Permanently discontinue ENHERTU in all patients with Grade 2 or higher ILD/pneumonitis"
1 Indications and Usage	Indication included with accelerated approval statement.	Revised to specify ENHERTU is indicated in "adult" patients; and revised (b) (4) to "anti-HER2-based regimens in the metastatic setting".
2 Dosage and Administration	2.2 Dose modification and management table provided for ILD, LVEF decrease, neutropenia and	Revised the Dose Modification Table for Adverse Reactions (Table 2) to add "Consider corticosteroid treatment as soon as ILD/pneumonitis is suspected".

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

	febrile neutropenia. 2.3 Preparation for Administration	Added “ENHERTU (fam-trastuzumab deruxtecan-xxxx) is a cytotoxic drug. Follow applicable special handling and disposal procedures.” Added “Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration, whenever solution and container permit” [21 CFR 201.57(c)(3)(iv)]
3 Dosage Forms and Strengths		Added “a white to yellowish white” lyophilized powder. [21 CFR 201.57(c)(4)]
5 Warnings and Precautions, 17 Patient Counseling, Highlights	Included ILD, LVEF decrease, neutropenia, and embryo-fetal toxicity.	Reordered the Warnings and Precautions and Patient Counseling Information to reflect the clinical significance and incidence of ARs with ENHERTU (e.g., neutropenia moved from 5.3 to 5.2; Left Ventricular Dysfunction moved from 5.2 to 5.3). 5.1 ILD/pneumonitis: - Removed (b) (4) - Added information on the corticosteroid regimen for ILD management. - Removed (b) (4) 5.3 Left Ventricular Dysfunction: - Revised the title from (b) (4) to “Left Ventricular Dysfunction” to better communicate AR significance.

		- Added “Left ventricular ejection fraction (LVEF) decrease has been observed with anti-HER2 therapies, including ENHERTU.” to express ARs and sequelae observed with pharmacologically-related drugs - Added “Permanently discontinue ENHERTU if LVEF of less than 40% or absolute decrease from baseline of greater than 20% is confirmed. Permanently discontinue ENHERTU in patients with symptomatic congestive heart failure (CHF).” 5.4 Embryo-Fetal Toxicity: Revised to add clarification that Dxd (MAAA-118a) can cause fetal harm “because it is genotoxic and targets actively dividing cells.”
6 Adverse Reactions	Text provided above ADR table. Common adverse reactions table included. Select laboratory abnormalities table included.	6.1 Clinical Trials Experience: Added the following: - “Serious adverse reactions occurred in 20% of patients receiving ENHERTU. Serious adverse reactions in >1% of patients who received ENHERTU were interstitial lung disease, pneumonia, vomiting, nausea, cellulitis, hypokalemia, and intestinal obstruction.” - “Fatalities due to adverse reactions occurred in 4.3% of patients including interstitial lung disease (2.6%), and the following events occurred in one patient each (0.4%): acute hepatic failure/acute kidney injury, general physical health deterioration, pneumonia, and hemorrhagic shock.” - ILD accounted for 6% of permanent discontinuations due to ARs.

		-In the Common AR table (Table 3), revised the incidence rates of fatigue ((b) (4) % to 59%), upper respiratory infection ((b) (4) % to 15%), headache ((b) (4) % to 19%) to group terms for similar ARs and accuracy; and added rash as a common AR (10%) based on group term analysis. 6.2 Immunogenicity: Removed (b) (4) (b) (4) and added “Due to the limited number of patients who tested positive for ADA, no conclusions can be drawn concerning a potential effect of immunogenicity on efficacy or safety. In addition, neutralizing activity of anti-ENHERTU antibodies has not been assessed.” (b) (4)
8 Use in Specific Populations	Terminology and formatting consistent with current PLLR guidance and best labeling practices.	8.1 Pregnancy, <i>Risk Summary</i>: - Added information to the Embryo-Fetal W&P above and added “advise patients of the potential risks to a fetus”. - Removed (b) (4) (b) (4) (b) (4)

		no information on overdose with ENHERTU.”
11 Description		- Added the pharmacologic class [21CFR201.57(c)(12)] - Deleted (b) (4) - Added information regarding the topoisomerase inhibitor Dxd (MAAA-1181a)
12 Clinical Pharmacology	Details provided.	12.1 Mechanism of Action: Revised to provide a concise summary of the established mechanism of action and removed (b) (4) which may be false or misleading. 12.2 Pharmacodynamics, Cardiac Electrophysiology: - Added “which is 1.2 times the recommended dosage” - Removed (b) (4) ” and revised to “did not show large mean effect (i.e., > 20 ms)” on QT interval. 12.3 Pharmacokinetics: - Added additional detail regarding the coefficient of variation [CV]% and AUC. - In the <i>Elimination</i> subsection, revised the fam-trastuzumab deruxtecan t1/2 from (b) (4) to “5.7 days”), added the MAAA-1181a t1/2, and added clearance based on single dose PK data from Study J101. - Revised the <i>Specific Populations</i> subsection based on PopPK analysis

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

		and removed (b) (4) - Revised and reformatted the <i>Drug Interactions Section</i> to present and organize by the clinically relevant CYP inhibitors and transporters with data to support proposed labeling.
13 Nonclinical Toxicology	Details provided.	Removed (b) (4)
14 Clinical Studies	Included ORR and DoR from both studies. Subgroup data provided for pooled analysis.	14.1 Metastatic Breast Cancer: - Removed (b) (4) See the FDA's <i>Assessment, 8.1.1.2. Study DS8201-A-J101 and 8.1.3. Integrated Review of Effectiveness for more information.</i> - Revised to specify that patients with active brain metastases or ECOG performance status >1 were excluded from the trial. - Revised the demographic data to add "29% had bone metastases, and 13% had brain metastases". - Removed (b) (4)
Medication Guide	Consistent with Full Prescribing Information	The Medication Guide (MG) was reviewed by the FDA Patient Labeling Team to implement several format and content changes, and to align with original and revised

		content with in the USPI.
--	--	---------------------------

The Applicant's Position:

The final USPI for fam-trastuzumab deruxtecan reflects several changes from the version originally submitted by the Applicant on 29 August 2019.

The FDA's Assessment:

With the agreed upon revisions implemented during this review, the final labeling meets the regulatory requirements, is consistent with current FDA guidance and best labeling practices and includes a summary of the essential scientific information needed for the safe and effective use of ENHERTU.

12 Risk Evaluation and Mitigation Strategies (REMS)

The FDA's Assessment:

No REMS is recommended for fam-trastuzumab deruxtecan.

13 Postmarketing Requirements and Commitment

The FDA's Assessment:

The following Postmarketing requirements and commitments were agreed upon by FDA and the Applicant.

Postmarketing Requirements (PMRs)

3762-1: Submit the final progression free survival analysis and datasets with the final report from a confirmatory Phase 3, multicenter, randomized, open-label, active-controlled study of DS-8201a for HER2-positive, unresectable and/or metastatic breast cancer patients previously treated with trastuzumab, to confirm clinical benefit and provide additional efficacy data that may inform product labeling for fam-trastuzumab deruxtecan (DS-8201a).

Final Protocol Submission: 05/2019

Interim Report (OS) Submission: 06/2023

Trial Completion: 12/2022

Final Report Submission: 06/2023

Postmarketing Commitments (PMCs)

Immunogenicity:

3762-2: Submit the integrated immunogenicity summary report for all patients with solid tumors in clinical studies treated with DS-8201a, including the ongoing Phase 3 trials, having an immunogenicity component. The final report should include anti-drug antibody (ADA) results from screening, confirmatory, titering, domain specificity, and neutralization assays, the results of linear or non-linear correlation analyses between ADA status and titers with PK, PD, efficacy, and safety (adverse event) data. Submit the Integrated Immunogenicity Summary Report in accordance with Section VIII Documentation of the 2019 FDA Guidance for Industry: Immunogenicity Testing of Therapeutic Protein Products — Developing and Validating Assays for Anti-Drug Antibody Detection.

Final Report Submission: June 2023

3762-16: Develop and validate a neutralizing antibody assay to test confirmed anti-DS-8201a antibody positive samples from studies J101, J102, A103, A104, and U201 as well as the ongoing phase 3 clinical studies U301, U302, and U303.

Final report submission date: June 2020

3762-17: Develop and validate domain specificity assays to test confirmed anti-DS-8201a antibody positive samples from studies J101, J102, A103, A104, and U201 as well as the ongoing phase 3 clinical studies U301, U302, and U303. Specifically, the assays should determine the specificity of anti-DS-8201a antibodies for the monoclonal antibody MAAL-9001, the drug MAAA-1181a, and the linker.

Final report submission date: June 2020

Microbiology:

3762-3: Provide the bioburden test method qualification report for two additional batches of (b) (4) and DS-8201a drug substance in-process and release samples by May 2020.

Final Report Submission: May 2020.

3762-4: Perform the microbial ingress container closure integrity test to validate the maximum and minimum crimping pressures and include a positive control with a breach size of ≤ 20 microns. In addition, monitor the viability of the challenge microorganism at the end of testing.

Final Report Submission: January 2020.

3762-5: Provide data from a media fill run to support the use of the drug product specific container closure system.

Final Report Submission: January 2020.

3762-6: Provide endotoxin method qualification using two (2) additional drug product batches manufactured at (b) (4) Final Report Submission: January 2020

Final Report Submission: January 2020

3762-7: Provide bioburden method qualification using 100 mL sample volumes from three (3) batches of DP manufactured at (b) (4)

Final Report Submission: March 2020

3762-8: Perform the dye ingress method validation for container closure integrity

testing of the drug product stability samples using positive controls with a ≤ 20 micro breach size.

Final Report Submission: June 2020.

Product Quality:

3762-9: Improve the requalification procedures for the current (b) (4) primary reference standard (PRS) (b) (4) and secondary reference standard (SRS) (b) (4) by revising the requalification protocols to include numerical acceptance criteria for HER2 binding of the PRS and stability trending of additional quantitative quality attributes with pre-defined trending rules and criteria, to allow timely detection of changes in the quality attributes of the PRS and SRS and to inform when the current RSs should be replaced. The acceptance criteria for HER2 binding of the PRS, the selected quality attributes for trending, the trending rules and criteria, and the criteria for assessment of when the RSs should be replaced will be scientifically justified. The revised qualification protocol will be submitted to the Agency per 21 CFR 601.12.

Final report submission date: March 2020

3762-10: Improve the requalification procedures for the current DS-8201a primary reference standard (PRS) (b) (4) and secondary reference standard (SRS) (b) (4) by revising the requalification protocols to include numerical acceptance criteria for cell growth inhibition and HER2 binding activity and stability trending of additional quantitative quality attributes with pre-defined trending rules and criteria, to allow timely detection of changes in the quality attributes of the PRS and SRS and to inform when the current RSs should be replaced. The numerical acceptance criteria for cell growth inhibition and HER2 binding activity, the selected quality attributes for trending, the trending rules and criteria, and the criteria for assessment of when the RSs should be replaced will be scientifically justified. The revised qualification protocols will be submitted to the Agency per 21 CFR 601.12.

Final report submission date: March 2020

3762-11: Confirm that the potency of the current DS-8201a primary reference standard (b) (4) and secondary reference standard (b) (4) is precise and accurate by conducting additional qualification of potency for primary reference standard (b) (4) using a sufficient number of independent assays and replicates. The number of independent assays and replicates will be scientifically justified. The qualification data will be reported as per 21 CFR 601.12.

Final report submission date: February 2020.

3762-12: Confirm that the potency of (b) (4) primary reference standard (b) (4) is precise and accurate by conducting additional qualification of potency of primary reference standard (b) (4) using a sufficient number of independent assays and replicates. The number of independent assays and replicates will be scientifically justified. The qualification data will be reported as per 21 CFR 601.12.

Final report submission date: February 20, 2020.

3762-13: Strengthen the qualification of the current (b) (4) primary and secondary reference standards by conducting additional characterization studies including full glycan profile analysis and FcγRIIIA binding activity of (b) (4) primary reference standard (b) (4) and secondary reference standard (b) (4) to support the use of these reference standards in comparability assessments. The qualification data will be reported as per 21 CFR 601.12

Final report submission date: February 2020.

3762-14: Strengthen the qualification of the DS-8201a primary and secondary reference standards by conducting characterization of FcγRIIIA binding activity for primary reference standard (b) (4) to support the use of these reference standards in comparability assessments. The qualification data will be reported as per 21 CFR 601.12.

Final report submission date: February 2020.

3762-15: Re-evaluate (b) (4) precision for the (b) (4) protein concentration and glycan analysis methods at Daiichi Sankyo Tatebayashi Plant, and for protein concentration, non-proteinaceous impurities (NPI) and purity of payload (PoP) methods for DS-8201a drug substance at Daiichi Sankyo Onahama Plant and (b) (4). The final report will be reported as per 21 CFR 601.12.

Final report submission date: March 2020

14 Division Director (DHOT) (NME ONLY)

X

15 Division Director (OCP)

X

16 Division Director (OB)

X

17 Division Director (Clinical)

X

18 Office Director (or designated signatory authority)

This application was reviewed by the Oncology Center of Excellence (OCE) per the OCE Intercenter Agreement. My signature below represents an approval recommendation for the clinical portion of this application under the OCE.

X

19 Appendices

19.1. References

The Applicant's References:

Bartsch R, Wenzel C, Altorjai G, et al. Capecitabine and trastuzumab in heavily pretreated metastatic breast cancer. *J Clin Oncol*. 2007; 25(25):3853-8.

Baselga J, Cortes J, Kim SB, et al. Pertuzumab plus trastuzumab plus docetaxel for metastatic breast cancer. *N Engl J Med*. 2012; 366(2):109-19.

Bierman WF, Scheffer GL, Schoonderwoerd A, et al. Protease inhibitors atazanavir, lopinavir and ritonavir are potent blockers, but poor substrates, of ABC transporters in a broad panel of ABC transporter-overexpressing cell lines. *J Antimicrob Chemother*. 2010 Aug;65(8):1672-80. doi: 10.1093/jac/dkq209. Epub 2010 Jun 15.

Blackwell K, Burstein HJ, Storniolo AM, et al. Randomized study of lapatinib alone or in combination with trastuzumab in women with ErbB2-positive, trastuzumab-refractory metastatic breast cancer. *J Clin Oncol*. 2010; 28(7):1124-30.

Bray F, Ferlay J, Soerjomataram I, et al. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin*. 2018; 68(6):394-424.

Chan A, Delaloge S, Holmes FA, et al. Neratinib after trastuzumab-based adjuvant therapy in patients with HER2-positive breast cancer (ExteNET): a multicentre, randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet Oncol*. 2016; 17(3):367-77.

Chen T, Xu T, Li Y, et al. Risk of cardiac dysfunction with trastuzumab in breast cancer patients: a meta-analysis. *Cancer Treat Rev*. 2011;37(4):312-20.

Dieras V, Harbeck N, Budd GT, et al. Trastuzumab emtansine in human epidermal growth factor receptor 2-positive metastatic breast cancer: an integrated safety analysis. *J Clin Oncol*. 2014; 32(25):2750-7.

Dzimitrowicz H, Berger M, Vargo C, et al. T-DM1 activity in metastatic human epidermal growth factor receptor 2-positive breast cancers that received prior therapy with trastuzumab and pertuzumab. *J Clin Oncol*. 2016; 34:3511-7.

Ferlay J, Ervik M, Lam F, et al. Global Cancer Observatory: Cancer Today. Lyon, France: International Agency for Research on Cancer. 2018. Available from: <https://gco.iarc.fr/today>; Last accessed 02 May 2019.

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

Geyer CE, Forster J, Lindquist D, et al. Lapatinib plus capecitabine for HER2-positive advanced breast cancer. *N Engl J Med*. 2006; 355(26):2733-43.

Gupta A, Zhang Y, Unadkat JD, Mao Q J. HIV protease inhibitors are inhibitors but not substrates of the human breast cancer resistance protein (BCRP/ABCG2). *Pharmacol Exp Ther*. 2004 Jul;310(1):334-41. Epub 2004 Mar 8.

Howlader N, Altekruse SF, Li CI, et al. US incidence of breast cancer subtypes defined by joint hormone receptor and HER2 status. *J Natl Cancer Inst*. 2014; 106(5): dju055
doi:10.1093/jnci/dju055.

Howlader N, Noone AM, Krapcho M, et al. SEER Cancer Statistics Review, 1975-2016, National Cancer Institute. Bethesda, MD, https://seer.cancer.gov/csr/1975_2016/, based on November 2018 SEER data submission, posted to the SEER web site, April 2019.

Krop IE, Kim SB, González-Martín A, et al. Trastuzumab emtansine versus treatment of physician's choice for pretreated HER2-positive advanced breast cancer (TH3RESA): a randomised, open-label, phase 3 trial. *Lancet Oncol*. 2014; 15(7):689-99.

Mitri Z, Constantine T, O'Regan R. The HER2 receptor in breast cancer: Pathophysiology, clinical use, and new advances in therapy. *Chemotherapy Research and Practice*. 2012;
doi:10.1155/2012/743193.

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines). Breast cancer. Version 2.2019 – 02 Jul 2019. NCCN.org. Accessed: 02 Aug 2019.

Noda-Narita S, Shimomura A, Kawachi A, et al. Comparison of the efficacy of trastuzumab emtansine between patients with metastatic human epidermal growth factor receptor 2-positive breast cancers previously treated with combination trastuzumab and pertuzumab and with trastuzumab only in Japanese population. *Breast Cancer*. 2019; 26:492-8.

Ottini L, Capalbo C, Rizzolo P, et al. HER2-positive male breast cancer: an update. *Breast Cancer: Targets and Therapy*. 2010; 2:45-58.

(b) (4)

Taucher S, Rudas M, Mader RM, et al. Do we need HER-2/neu testing for all patients with

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

primary breast carcinoma? Cancer. 2003; 98(12):2547-53.

U.S. Food and Drug Administration. Center for Drug Evaluation and Research. The Voice of the Patient. A series of reports from the U.S. Food and Drug Administration (FDA's) patient-focused drug development initiative. Breast Cancer. Report Date: September 2015. Available at: <https://www.fda.gov/media/93924/download>.

Verma S, Miles D, Gianni L, et al. Trastuzumab emtansine for HER2-positive advanced breast cancer. New Engl J Med. 2012; 367(19):1783-91.

Von Minckwitz G, Procter M, de Azambuja E, et al. Adjuvant pertuzumab and trastuzumab in early HER2-positive breast cancer. N Engl J Med. 2017; 377(2):122-131.

Von Minckwitz G, Huang CS, Mano MS, et al for the KATHERINE investigators. Trastuzumab emtansine for residual invasive HER2-positive breast cancer. N Engl J Med. 2019; 380:617-28.

Wolff AC, Hammond ME, Hicks DG et al. Recommendations for human epidermal growth factor receptor 2 testing in breast cancer: American Society of Clinical Oncology/College of American Pathologists clinical practice guideline update. J Clin Oncol. 2013 Nov 1;31(31):3997-4013.

The FDA's References:

No additional references for the FDA.

19.2. Financial Disclosure

The Applicant's Position:

The Applicant provided financial disclosure information for clinical investigators involved in the covered clinical studies DS8201-A-J101 and DS8201-A-U201. A total of 5 investigators had disclosable financial interests or arrangements, of which 3 participated in both studies. All disclosures were reported under the significant payments of other sorts financial disclosure category. No concerns were raised regarding the overall integrity of study data.

Despite the Applicant's due diligence in attempting to obtain the information, the Applicant was unable to obtain financial disclosure information for 5 investigators for DS8201-A-U201. Of these 5 investigators, 2 were delegated at a site that did not screen any subjects, 2 did not see any patients while delegated on the study and 1 did not perform any study related activities.

The FDA's Assessment:

FDA agrees with the applicant's assessment.

Covered Clinical Study (Name and/or Number):* DS8201-A-J101

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Was a list of clinical investigators provided:	Yes ☒	No ☐ (Request list from Applicant)
Total number of investigators identified: <u>435</u>		
Number of investigators who are Sponsor employees (including both full-time and part-time employees): <u>0</u>		
Number of investigators with disclosable financial interests/arrangements (Form FDA 3455): <u>3</u>		
If there are investigators with disclosable financial interests/arrangements, identify the number of investigators with interests/arrangements in each category (as defined in 21 CFR 54.2(a), (b), (c) and (f)): Compensation to the investigator for conducting the study where the value could be influenced by the outcome of the study: <u>0</u> Significant payments of other sorts: <u>3</u> Proprietary interest in the product tested held by investigator: <u>0</u> Significant equity interest held by investigator in study: <u>0</u> Sponsor of covered study: <u>0</u>		
Is an attachment provided with details of the disclosable financial interests/arrangements:	Yes ☒	No ☐ (Request details from Applicant)
Is a description of the steps taken to minimize potential bias provided:	Yes ☒	No ☐ (Request information from Applicant)
Number of investigators with certification of due diligence (Form FDA 3454, box 3) <u>0</u>		
Is an attachment provided with the reason:	Yes ☐	No ☐ (Request explanation from Applicant)

*The table above should be filled by the applicant, and confirmed/edited by the FDA.

Covered Clinical Study (Name and/or Number):* DS8201-A-U201

Was a list of clinical investigators provided:	Yes ☒	No ☐ (Request list from Applicant)
Total number of investigators identified: <u>957</u>		
Number of investigators who are Sponsor employees (including both full-time and part-time employees): <u>0</u>		

Disclaimer: In this document, the sections labeled as "The Applicant's Position" are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

Number of investigators with disclosable financial interests/arrangements (Form FDA 3455): <u>5</u>		
If there are investigators with disclosable financial interests/arrangements, identify the number of investigators with interests/arrangements in each category (as defined in 21 CFR 54.2(a), (b), (c) and (f)): Compensation to the investigator for conducting the study where the value could be influenced by the outcome of the study: <u>0</u> Significant payments of other sorts: <u>5</u> Proprietary interest in the product tested held by investigator: <u>0</u> Significant equity interest held by investigator in study: <u>0</u> Sponsor of covered study: <u>0</u>		
Is an attachment provided with details of the disclosable financial interests/arrangements:	Yes ☒	No ☐ (Request details from Applicant)
Is a description of the steps taken to minimize potential bias provided:	Yes ☒	No ☐ (Request information from Applicant)
Number of investigators with certification of due diligence (Form FDA 3454, box 3) <u>5</u>		
Is an attachment provided with the reason:	Yes ☒	No ☐ (Request explanation from Applicant)

*The table above should be filled by the applicant, and confirmed/edited by the FDA.

19.3. Nonclinical Pharmacology/Toxicology

Not applicable

19.4. OCP Appendices (Technical documents supporting OCP recommendations)

The FDA's Assessment:

Summary of Bioanalytical Method Validation and Performance

In the pivotal studies, Study DS8201-A-J101 and DS8201-A-U201, serum concentrations of fam-trastuzumab deruxtecan and MAAL-9001 (total anti-HER2 antibody) were measured using two validated electrochemiluminescent methods, and serum concentrations of MAAA-1181a (ie. the payload) were measured using the validated LC/MS/MS method. The method performance is summarized in Table 48 (trastuzumab deruxtecan), Table 49 (total anti-HER2 antibody), and

Table 50 (MAAA-1181a). There were no modifications made to these three PK assays in other clinical trials.

Table 49: Summary method performance of a bioanalytical method to measure trastuzumab deruxtecan in human serum

Bioanalytical method validation report name, amendments, and hyperlinks	Validation of an Electrochemiluminescent Method for the Quantitation of Intact DS-8201a in Human Serum [REFE2] Report addendums: [REFE3] extended long-term frozen stability [REFE5] demonstrated equivalence of drug substance 1 (DS 1) vs drug substance 2 (DS 2) in trastuzumab deruxtecan PK assay	
Method description	Trastuzumab deruxtecan (ie. “Intact DS-8201a”, which was used as the analyte name in bioanalytical reports) is quantitatively measured from human serum using an electrochemiluminescent method. A high bind MSD plate is coated with capture antibody (anti-XAFG-5737, a mouse IgG monoclonal antibody) that specifically binds to the conjugated payload. Samples and controls undergo the MRD1000 and are then sonicated to de-aggregate trastuzumab deruxtecan in the sample. Trastuzumab deruxtecan in the samples binds to anti-XAFG-5737 on the plate. Detection solution containing complexes of biotinylated anti-MAAL-9001 (a mouse IgG monoclonal antibody) and sulfo-TAG streptavidin bind to the captured trastuzumab deruxtecan on the plate and produce a chemiluminescent signal that is proportional to the concentration of trastuzumab deruxtecan in the sample.	
Materials used for calibration curve & concentration	Human serum from healthy donors DS-8201a	
Validated assay range	0.4 to 25.6 µg/mL (anchor point at 0.2 µg/mL) for DS-8201a in human serum	
Material used for QCs & concentration	Human serum from healthy donors DS-8201a	
Minimum required dilutions (MRDs)	Two-step 1:1000	
Source & lot of reagents (LBA)	DS-8201a: Lot No. HA103-1-U; source: Daiichi Sankyo, Ltd. Anti-XAFG-5737: Lot No. 201501-1A-530; sourced from (b) (4) Anti-MAAL-9001: Lot No. 201411-1K-419; sourced from (b) (4) Biotin-anti-MAAL-9001: Lot No. NB10846-62-25; biotinylation performed by (b) (4)	
Regression model & weighting	Four-parameter logistic, 1/response ² , weighted least-squares regression	
Validation parameters	Method validation summary	Source Location

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Calibration curve performance during accuracy & precision	Number of standard calibrators from LLOQ to ULOQ trastuzumab deruxtecan	8	Table 2A in REFE2
	Cumulative accuracy (%bias) from LLOQ to ULOQ trastuzumab deruxtecan	-0.991 to 1.18%	Table 2A in REFE2
	Cumulative precision (%CV) from LLOQ to ULOQ trastuzumab deruxtecan	≤3.07%	Table 2A in REFE2
QCs performance during accuracy & precision	Cumulative accuracy (%bias) in 6 QCs QCs: QC1, QC2, QC3 (Table 4A) and QC5, QC6, C7 (Table 4C) for trastuzumab deruxtecan	-2.96 to 11.2%	Tables 4A, 4C in REFE2
	Inter-batch %CV QCs: QC1, QC2, QC3 (Table 4A) and QC5, QC6, C7 (Table 4C) for trastuzumab deruxtecan	≤ 12.6%	Tables 4A, 4C in REFE2
	Total error QCs: QC1, QC2, QC3 (Table 4A) and QC5, QC6, C7 (Table 4C) for trastuzumab deruxtecan	≤ 23.8%	Table 1.7 in m2.7.1 [Tables 4A, 4C in REFE2]
Selectivity & matrix effect	10 lots of blank matrix were tested for selectivity without any observation. All blank lots were quantified below quantitation limit (LLOQ). 10 lots of blank matrix were spiked with DS-8201a at LLOQ (0.4 µg/mL) and HQC (18 µg/mL). Bias% from -9.09 to 2.56 at LLOQ Bias% from -14.9 to -2.39 at HQC		Table 11A in REFE2 Table 11B in REFE2
Interference & specificity	Not applicable.		
Hemolysis effect	Hemolytic effect was tested with blank matrix, LQC (0.5 µg/mL) and HQC (18 µg/mL). The blank matrix sample was measured to be below quantitation limit. Bias% at LQC and HQC were 5.38 and 0.720 respectively, with CV% ≤ 4.00.		Table 12A and Table 12B in REFE2
Lipemic effect	Lipemic effect was tested with blank matrix, LQC (0.5 µg/mL) and HQC (18 µg/mL). The blank matrix sample was measured to be below quantitation limit. Bias% at LQC and HQC were -2.61 and 10.8, respectively, with CV% ≤ 2.89.		Table 13 in REFE2

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Dilution linearity & hook effect	A high concentration sample at 1025 µg/mL was serially diluted into the assay quantitation range (0.4 – 25.6 µg/mL). Bias% for back-calculated concentrations of three dilution levels (ie. samples diluted to be within the assay quantitation range) ranged from 4.99 to 8.51, with CV% ≤ 6.25. No apparent “hook effect” was observed at concentrations up to 1025 µg/mL.	Table 5 in REFE2
Bench-top/process stability	Thawed matrix maintained on ice for 24 hours; LQC, HQC, and dilution QCs evaluated. Bias% from -0.381 to 9.87, with CV% ≤ 8.78. Process stability is not applicable.	Table 8 in REFE2
Freeze-Thaw stability	Five freeze/thaw cycles; LQC, HQC, and dilution QCs evaluated. Bias% from 2.52 to 6.20, with CV% ≤ 5.63.	Table 7 in REFE2
Long-term storage	93 days at -25°C; LQC, HQC, and dilution QC evaluated. Bias% from -12.1 to 5.65, with CV% ≤ 4.38. 855 days at -80°C; LQC, HQC, and dilution QC evaluated. Bias% from -6.74 to 2.11, with CV% ≤ 6.22.	Tables 4A, 4B, 5E in REFE3
Parallelism	Not performed.	
Carry over	Not applicable.	
Method performance in DS8201-A-J101		
Bioanalytical Report: An Electrochemiluminescent Method for the Quantitation of Intact DS-8201a in Human Serum (b) (4) Version 1.01) for Clinical Study DS8201-A-J101 [RFGL/RFGL2]		
Assay passing rate	96.6%	Table 5 and 6 in RFGL/RFGL2
Standard curve performance	- Cumulative bias range: -2.33% to 2.47% - Cumulative precision: ≤ 4.27% CV	Tables 9 and 10 in RFGL/RFGL2
QC performance	- Cumulative bias range: -1.90% to 0.00792% - Cumulative precision: ≤ 10.8% CV	Tables 11 and 12 in RFGL/RFGL2
Method reproducibility	Incurred sample reanalysis (ISR) was performed in 9.48% of Part 1 study samples. ISR passing rate was 78.7% and met the pre-specified criteria. ISR was performed in 10.0% of Part 2 study samples. ISR passing rate was 96.4% and met the pre-specified criteria.	Appendix E and F in RFGL/RFGL2
Study sample analysis/stability	All samples were analyzed within the 855 days of demonstrated long-term storage stability in human serum at -80°C ± 10°C (section Sample Analysis in REFGL/REFGL2).	
Method performance in DS8201-A-U201		

Bioanalytical report: Analysis of Intact DS-8201a in Human Serum by a Validated Electrochemiluminescent Method ((b) (4)) for Clinical Study DS8201-A-U201 [RINX]		
Assay passing rate	96.3%	Table 4 in RINX
Standard curve performance	- Cumulative bias range: -1.06% to 1.69% - Cumulative precision: ≤ 4.23% CV	Tables 7 and 8 in RINX
QC performance	- Cumulative bias range: -4.67% to 2.58% - Cumulative precision: ≤ 43.9% CV * * The high cumulative precision (43.9% CV) is attributed to one run (no. 37) having a QC replicate position switch.	Tables 9 and 10 in RINX
Method reproducibility	Incurred sample reanalysis was performed in 9.58% of study samples and 94.7% of samples met the pre-specified criteria.	Appendix E in RINX
Study sample analysis/stability	All samples were analyzed within the 855 days of demonstrated long-term storage stability in human serum at -80°C (section Sample Analysis in RINX).	

Source: Response to FDA information request on 10/18/2019; Table 1.1

Table 50: Summary method performance of a bioanalytical method to measure total anti-HER2 antibody in human serum

Bioanalytical method validation report name, amendments, and hyperlinks	Validation of an Electrochemiluminescent Method for the Quantitation of the Antibody Component of DS-8201a (Total Antibody) in Human Serum [REFF2] Report addendums: [REFF3] extended long-term frozen stability [REFE5] demonstrated equivalence of drug substance 1 (DS 1) vs drug substance 2 (DS 2) in total anti-HER2 antibody PK assay
Method description	Total anti-HER2 antibody (ie. “Total Antibody”, which was used as the analyte name in bioanalytical reports) is quantitatively measured from human serum using an electrochemiluminescent method. A high bind MSD plate is coated with capture antibody (anti-MAAL-9001, a mouse IgG monoclonal antibody) that specifically binds to the variable region of the antibody component of anti-HER2 antibodies. Samples and controls undergo the MRD1000 and are then sonicated to de-aggregate anti-HER2 antibodies in the sample. Anti-HER2 antibody in the samples bind to anti-MAAL-9001 on the plate. Detection solution containing complexes of biotinylated anti-MAAL-9001 and sulfo-TAG streptavidin bind to the captured anti-HER2 antibodies on the plate and produce a chemiluminescent signal that is proportional to the concentration of total anti-HER2 antibody in the sample.
Materials used for calibration curve & concentration	Human serum from healthy donors DS-8201a
Validated assay range	0.4 to 25.6 µg/mL (anchor point at 0.2 µg/mL) for DS-8201a in human serum

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Material used for QCs & concentration	Human serum from healthy donors DS-8201a		
Minimum required dilutions (MRDs)	Two-step 1:1000		
Source & lot of reagents (LBA)	DS-8201a: Lot No. HA103-1-U; source: Daiichi Sankyo, Ltd. Anti-MAAL-9001: Lot No. 201411-1K-419; sourced from (b) (4) Biotin-anti-MAAL-9001: Lot No. NB10846-62-25; biotinylation performed by (b) (4)		
Regression model & weighting	Four-parameter logistic, 1/response ² , weighted least-squares regression		
Validation parameters	Method validation summary		Source Location
Calibration curve performance during accuracy & precision	Number of standard calibrators from LLOQ to ULOQ anti-HER2 antibody	8	Table 2A in REFF2
	Cumulative accuracy (%bias) from LLOQ to ULOQ anti-HER2 antibody	-3.21 to 3.45%	Table 2A in REFF2
	Cumulative precision (%CV) from LLOQ to ULOQ anti-HER2 antibody	≤3.60%	Table 2A in REFF2
QCs performance during accuracy & precision	Cumulative accuracy (%bias) in 7 QCs QCs: QC11, QC2, QC3 (Table 4A) and QC5, QC6, C7, C15 (Table 4C) for anti-HER2 antibody	-3.24 to 11.2%	Table 4A and Table 4C in REFF2
	Inter-batch %CV QCs: QC11, QC2, QC3 (Table 4A) and QC5, QC6, C7, C15 (Table 4C) for anti-HER2	≤ 9.25%	Table 4A and Table 4C in REFF2
	Total error QCs: QC11, QC2, QC3 (Table 4A) and QC5, QC6, C7, C15 (Table 4C) for anti-HER2	≤ 17.4%	Table 1.8 in m2.7.1 [Table 4A and Table 4C in REFF2]
Selectivity & matrix effect	10 lots of blank matrix were tested for selectivity without any observation. All blank lots were quantified below quantitation limit (LLOQ). 10 lots of blank matrix were spiked with DS-8201a at LLOQ (0.4 µg/mL) and HQC (18 µg/mL). Bias% from -18.2 to 4.91 at LLOQ. Bias% from -6.41 to 5.33 at HQC.		Table 11A in REFF2 Table 11B in REFF2

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Interference & specificity	Not applicable.	
Hemolysis effect	Hemolytic effect was tested with blank matrix, LQC (0.9 µg/mL) and HQC (18 µg/mL). The blank matrix sample was measured to be below quantitation limit. Bias% at LQC and HQC were -0.0160 and 2.91, respectively, with CV% ≤ 5.15.	Table 12 in REFF2
Lipemic effect	Lipemic effect was tested with blank matrix, LQC (0.9 µg/mL) and high QC (18 µg/mL). The blank matrix sample was measured to be below quantitation limit. Bias% at LQC and HQC were -6.75 and -3.36, respectively, with CV% ≤ 3.32.	Table 13 in REFF2
Dilution linearity & hook effect	A high concentration sample at 1025 µg/mL was serially diluted into the assay quantitation range (0.4 – 25.6 µg/mL). Bias% for back-calculated concentrations of three dilution levels (ie. samples diluted to be within the assay quantitation range) ranged from -18.0 to -9.11, with CV% ≤ 5.97. No apparent “hook effect” was observed at concentrations up to 1025 µg/mL.	Table 5 in REFF2
Bench-top/process stability	Thawed matrix maintained on ice for 24 hours; LQC, HQC, and dilution QCs evaluated. Bias% from -3.33 to 4.60, with CV% ≤ 4.70. Process stability is not applicable.	Table 8 in REFF2
Freeze-Thaw stability	Five freeze/thaw cycles; LQC, HQC, and dilution QCs evaluated. Bias% from -4.60 to -0.860, with CV% ≤ 4.52.	Table 7 in REFF2
Long-term storage	386 days at -25°C; LQC, HQC, and dilution QC evaluated. Bias% from -11.6 to 0.924, with CV% ≤ 8.74. 853 days at -80°C; LQC, HQC, and dilution QC evaluated. Bias% from -18.0 to -7.49, with CV% ≤ 5.68.	Table 4C in REFF3 Table 5E in REFF3
Parallelism	Not performed.	
Carry over	Not applicable.	
Method performance in DS8201-A-J101 Bioanalytical Report: An Electrochemiluminescent Method for the Quantitation of the Antibody Component of DS-8201a (Total Antibody) in Human Serum ((b) (4) Version 1.01) for clinical study DS8201-A-J101 [RFGM/RFGM2]		
Assay passing rate	96.4%	Table 7 and 8 in RFGM/RFGM2

Disclaimer: In this document, the sections labeled as “The Applicant’s Position” are completed by the Applicant and do not necessarily reflect the positions of the FDA.

This document shows original U.S. government data provided by the U.S. Food & Drug Administration and is available in the public domain. It has been processed to facilitate searching and data extraction and may be viewed at www.pharmapendium.com

Standard curve performance	- Cumulative bias range: -3.39% to 6.46% - Cumulative precision: $\leq 5.99\%$ CV	Tables 11 and 12 in RFGM/RFGM2
QC performance	- Cumulative bias range: -0.776% to 5.86% - Cumulative precision: $\leq 9.35\%$ CV	Tables 13 and 14 in RFGM/RFGM2
Method reproducibility	ISR was performed in 9.48% of Part 1 study samples. ISR passing rate was 85.1% and met the pre-specified criteria. ISR was performed in 10.1% of Part 2 study samples. ISR passing rate was 96.1% and met the pre-specified criteria.	Appendix E and F in RFGM/RFGM2
Study sample analysis/ stability	All samples were analyzed within the 853 days of demonstrated long-term storage stability in human serum at -80°C $\pm$ 10°C (section Sample Analysis of RFGM/RFGM2).	
Method performance in DS8201-A-U201		
Bioanalytical report: Analysis of Total DS-8201a in Human Serum by a Validated Electrochemiluminescent Method ((b) (4)) for Clinical Study DS8201-A-U201 [RINW]		
Assay passing rate	91.6%	Table 4 in RINW
Standard curve performance	- Cumulative bias range: -4.7% to 7.86% - Cumulative precision: $\leq 5.99\%$ CV	Tables 8 and 9 in RINW
QC performance	- Cumulative bias range: -2.35% to 6.70% - Cumulative precision: $\leq 10.7\%$ CV	Tables 10 and 11 in RINW
Method reproducibility	Incurring sample reanalysis was performed in 9.62% of study samples and 93.8% of samples met the pre-specified criteria.	Appendix E in RINW
Study sample analysis/ stability	All samples were analyzed within the 853 days of demonstrated long-term storage stability in human serum at -80°C (section Sample Analysis in RINW).	

Source: Response to FDA information request on 10/18/2019; Table 1.2

Table 51: Summary method performance of a bioanalytical method to measure MAAA-1181a in human serum

Bioanalytical method validation report name, amendments, and hyperlinks	Quantitation of MAAA-1181a in Human Serum via HPLC with MS/MS Detection [REFD2] Report addendums: [REFD3] extended long-term frozen storage stability of MAAA-1181a [REFD4] quantitation of MAAA-1181a in the presence of DS-8201a [REFD5] established long-term frozen storage stability of MAAA-1181a in presences of DS-8201a at -20°C
Method description	MAAA-1181a is quantitatively measured from human serum using a LC/MS/MS methodology. Serum samples containing analyte were mixed with internal standard and deproteinized by organic solvents and filtered; this was followed by evaporation to dryness of the eluate and then reconstitution of the

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	residue. Extracted samples were analyzed by reverse-phase chromatography with detection using a triple quadrupole mass spectrometer with an electrospray ion source operated in positive ionization mode.		
Materials used for calibration curve & concentration	Human serum from healthy donors MAAA-1181a (analyte) and MAAA-1454 (stable labeled internal standard)		
Validated assay range	10.0 to 2000 pg/mL for MAAA-1181a in human serum		
Material used for QCs & concentration	Human serum from healthy donors MAAA-1181a (analyte) and MAAA-1454 (stable labeled internal standard)		
Minimum required dilutions (MRDs)	Not applicable		
Source & lot of reagents (LBA)	Not applicable		
Regression model & weighting	Linear regression ($y = ax + b$), with $1/x^2$ as weighting factor		
Validation parameters	Method validation summary		Source Location
Calibration curve performance during accuracy & precision	Number of standard calibrators from LLOQ to ULOQ MAAA-1181a	8	Table 2 in REFD2
	Cumulative accuracy (%bias) from LLOQ to ULOQ MAAA-1181a	-1.57 to 1.10%	Table 2 in REFD2
	Cumulative precision (%CV) from LLOQ to ULOQ MAAA-1181a	≤5.82%	Table 2 in REFD2
QCs performance during accuracy & precision	Cumulative accuracy (%bias) in 6 QCs QCs: IA 0, IA 1, IA 2, IA 3, IA 4, IA 5 for MAAA-1181a	-5.09 to -0.244%	Table 4 in REFD2
	Inter-batch %CV QCs: IA 0, IA 1, IA 2, IA 3, IA 4, IA 5 for MAAA-1181a	≤ 14.8%	Table 4 in REFD2
	Total error QCs:	Not applicable	

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

Selectivity & matrix effect	6 lots of blank matrix were tested for selectivity without any observation. All blank lots were quantified below 20% of LLOQ calibrator's response. 6 lots of blank matrix were spiked with MAAA-1181a in replicates of 3 at LLOQ (10 pg/mL). Two-thirds of the replicates for a lot must quantitate within $\pm 20\%$ of the theoretical value (for five out of six fortified specificity sample lots). All 6 lots met the criteria. Matrix factor (MF) samples from six different individual matrix lots were fortified post-extraction to approximate low- and high-concentration QC levels and analyzed along with comparable external standard (ie. neat solution with no matrix) at each concentration level. The CV% observed was ≤ 4.18 (acceptance criteria ≤ 15).	Table 15A in REFD2 Table 15B in REFD2 Tables 16A, 16B, 16C and 16D in REFD2
Interference & specificity	Blank serum was fortified with 50 and 3200 pg/mL of MAAA-1509a and MAAA-1430a (metabolites for MAAA-1181a) separately and analyzed. LQC (20 pg/mL) and HQC (1600 pg/mL) of MAAA-1181a were each fortified with 50 pg/mL and 3200 pg/mL of MAAA-1509a and MAAA-1430a, respectively, and analyzed. The blank serum sample spiked with 3200 pg/mL MAAA-1509a showed $<30\%$ LLoQ response. The blank serum sample spiked with DS3200 pg/mL MAAA-1430a showed estimated <7 pg/mL level of interference (based on peak area comparison to LLOQ), attributable to impurities in the MAAA-1430a reference material. Both MAAA-1181a LQC and HQC levels (each spiked with low and high levels of metabolites, respectively) quantified within $\pm 15.0\%$ of expected concentration, indicating no impact from metabolites to quantitation of MAAA-1181a.	Tables 18A and 18B in REFD2
Hemolysis effect	Hemolytic effect was tested with blank matrix, LQC (20 pg/mL) and HQC (1600 pg/mL). No significant chromatographic peaks were detected for blank matrix sample. The bias for LQC and HQC in hemolytic matrix was 7.11% and 5.98%, respectively.	Table 14 in REFD2
Lipemic effect	Lipemic effect was not included as part of primary validation (REFD2).	
Dilution linearity & hook effect	Dilution linearity was evaluated with 4000 pg/mL (to demonstrate that above the upper limit of the calibration range samples can be accurately measured) and 150 pg/mL (to demonstrate that samples within assay range, but with insufficient volume, can be diluted and analyzed). The hook effect evaluate is not applicable for LC/MS/MS assay.	Table 6 in REFD2
Bench-top/process stability	24 hours stability on bench at chilled temperature (ie. remain on ice) for MAAA-1181a. Tested at LQC (20 pg/mL) and HQC (1600 pg/mL) levels.	Table 9 in REFD2

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
 ENHERTU (fam-trastuzumab deruxtecan-nxki)

	Bias% were 2.74 (LQC) and 2.16 (HQC) with CV% $\leq$ 4.79. 24 hours stability on bench at chilled temperature (ie. remain on ice) for MAAA-1181a, in the presence of DS-8201a. Tested at LQC (28.5 pg/mL) and HQC (1960 pg/mL) levels. Bias% were 10.2 (LQC) and 5.31 (HQC) with CV% $\leq$ 3.30.	Table 3 in REFD4
Freeze-Thaw stability	Five freeze-thaw cycles with storage at -20°C and -70°C. Tested for MAAA-1181a at LQC (20 pg/mL) and HQC (1600 pg/mL) levels. At -20°C, bias% were -0.197 (LQC) and -5.06 (HQC), with CV% $\leq$ 8.44. At -70°C, bias% were 4.84 (LQC) and 2.33(HQC), with CV% $\leq$ 8.33. Five freeze-thaw cycles with storage at -20°C and -70°C. Tested for MAAA-1181a, in the presence of DS-8201a, at LQC (28.5 pg/mL) and HQC (1960 pg/mL) levels. At -70°C, bias% were 7.56 (LQC) and 1.62(HQC), with CV% $\leq$ 2.78.	Table 8A and 8B in REFD2 Table 2 REFD4
Long-term storage	Frozen storage stability for MAAA-1181a. Tested at LQC (20 pg/mL) and HQC (1600 pg/mL) levels. 579 days at -20°C, bias% were -7.18 (LQC) and -5.59 (HQC), with CV% $\leq$ 3.53. 579 days at -70°C, bias% were -0.0143 (LQC) and -1.80 (HQC), with CV% $\leq$ 7.81. Frozen storage stability for MAAA-1181a, in the presence of DS-8201a. Tested at LQC (20 pg/mL) and HQC (1600 pg/mL) levels. 112 days at -20°C, bias% were 7.99 (LQC) and 8.58 (HQC), with CV% $\leq$ 2.66. 489 days at -70°C, bias% were -8.99 (LQC) and -4.44 (HQC), with CV% $\leq$ 3.94.	Tables 1C and 1D in REFD3 Table 2B in REFD5 Table 6 in REFD4
Parallelism	Not applicable.	
Carry over	Not applicable.	
Method performance in DS8201-A-J101 Bioanalytical report: Quantitation of MAAA-1181a in Human Serum via HPLC with MS/MS Detection [RFGK and RFGK2]		
Assay passing rate		Table 1B and Table 1C in RFGK and RFGK2
Standard curve performance	- Cumulative bias range: -1.7% to 1.1% - Cumulative precision: $\leq$ 7.9% CV	Table 3B and Table 4C in RFGK and RFGK2
QC performance	- Cumulative bias range: -2.4% to 0.5% - Cumulative precision: $\leq$ 9.2% CV	Table 4B and Table 5C in RFGK and RFGK2

NDA/BLA Multi-disciplinary Review and Evaluation {BLA 761139}
ENHERTU (fam-trastuzumab deruxtecan-nxki)

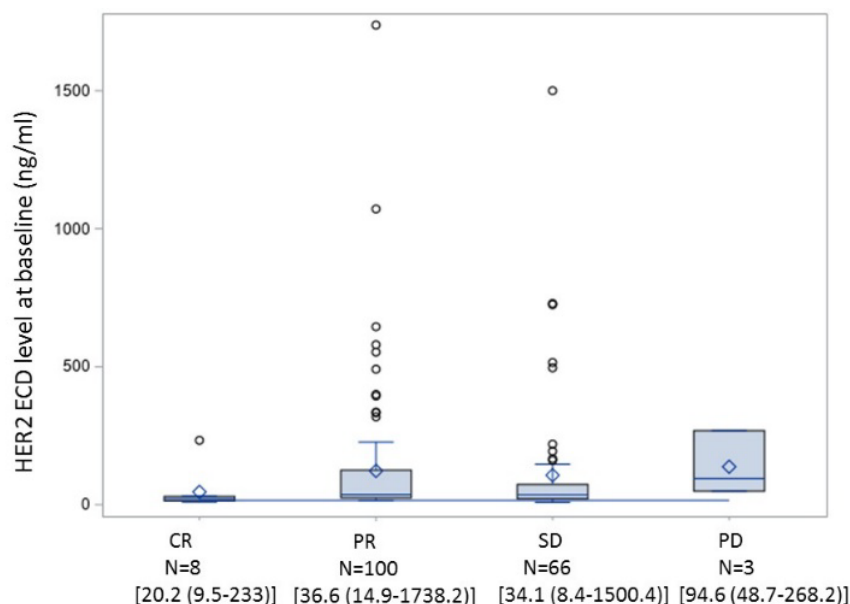
Method reproducibility	ISR was performed in 11.5% of Part 1 study samples. ISR passing rate was 96.5% and met the pre-specified criteria. ISR was performed in 9.7% of Part 2 study samples. ISR passing rate was 95.2% and met the pre-specified criteria.	Table 3A in RFGK and RFGK2
Study sample analysis/ stability	All but eight samples were analyzed within the 579 days (MAAA-1181a only). Long term stability for MAAA-1181a (toxin) in the presence of DS-8201a (ADC) has been established for 489 days. The samples outside of stability have been noted in the Sample Results table. Future stability will be conducted to cover the age of these samples (section Sample Analysis in report RFGK and RFGK2).	
Method performance in DS8201-A-U201 Bioanalytical report: Quantitation of MAAA-1181a in Human Serum via HPLC with MS/MS Detection for Protocol No. DS8201-A-U201 [RINV]		
Assay passing rate	92.4%	Table 6 in RINV
Standard curve performance	- Cumulative bias range: -1.2% to 1.2 % - Cumulative precision: ≤ 5.6% CV	Table 8 in RINV
QC performance	- Cumulative bias range: 0.1% to 1.9% - Cumulative precision: ≤ 20.0% CV	Table 9 in RINV
Method reproducibility	Incurred sample reanalysis was performed in 8.6% of study samples and 95.4% of samples met the pre-specified criteria.	Table 11 in RINV
Study sample analysis/ stability	All samples were analyzed within the 579 days of demonstrated long-term storage stability in human serum (section Sample Analysis in RINV).	

Source: Response to FDA information request on 10/18/2019; Table 1.3

Summary of HER2 ECD and treatment response

Baseline serum HER2 ECD levels were measured by a central lab in samples from 179 patients who received trastuzumab deruxtecan 5.4 mg/kg in Study DS8201-A-U201 (N=184). About 93% of patients had high levels (defined as >15.2 ng/ml; range 8.4 to 1738.2 ng/ml, Figure 14). Exploratory analysis (Wilcoxon two-sample test) conducted by the reviewer showed lower HER2 ECD levels in patients achieving CR (N =8), compared with all others (N=169) (P=0.02). No significant differences were found when responders (CR + PR) were compared with non-responders (SD+PD) (P = 0.50). HER2 ECD levels on and at the end of treatment were also available, however the number of samples was insufficient to conduct additional analyses.

Figure 14: Baseline HER2 ECD levels and confirmed best overall response by Independent Central Review in a subset of patients receiving trastuzumab deruxtecan 5.4 mg/kg enrolled in study DS8201-A-U201 (N=177*)



(Source: Reviewer's exploratory analysis. *Patients with non-evaluable responses (N=2) were not included in the analysis. ECD = extracellular domain; CR= complete response; PR= partial response; SD = stable disease; PD = progressive disease; N= Number of responses [Median HER2 ECD level (range)]).

Pharmacometrics Review

1. Population PK and exposure-response analysis

1.1 Introduction

Population pharmacokinetic analysis of intact DS-8201a and MAAA-1181a (released drug) uses data from DS8201-A-J101, DS8201-A-J102, DS8201-A-A103, DS8201-A-A104, and DS8201-A-U201 (hereafter referred to as Studies J101, J102, A103, A104, and U201, respectively).

The objectives of this population PK analysis are:

- To characterize population pharmacokinetics (PopPK) of both intact DS-8201a and released drug in subjects from Studies J101, J102, A103, A104, and U201;